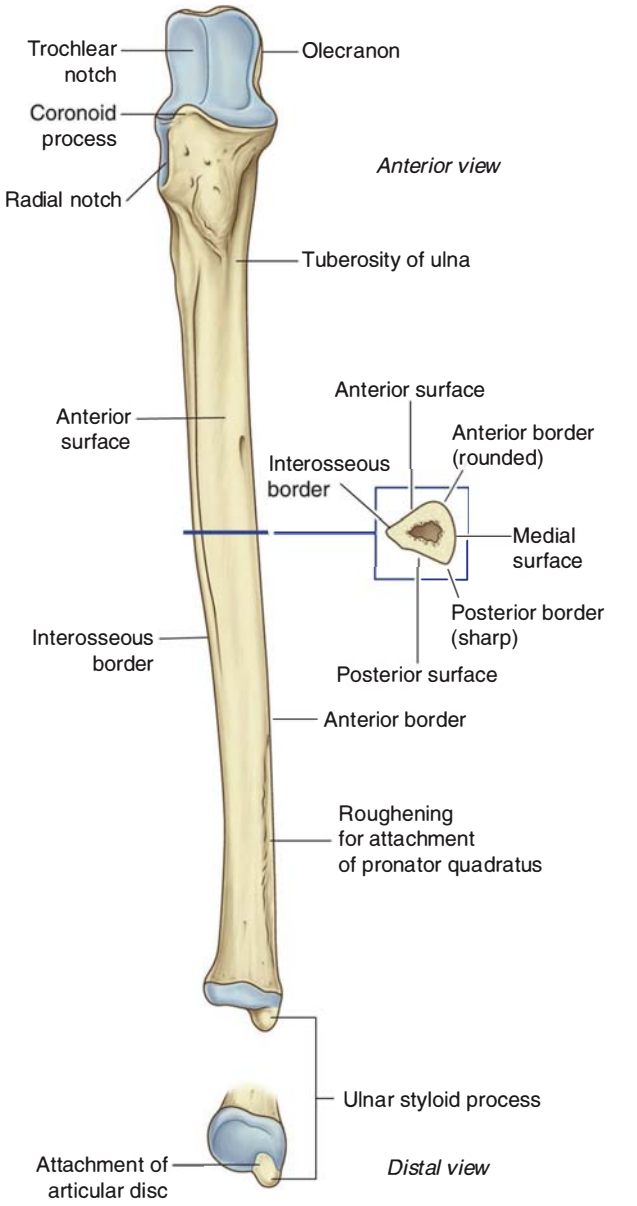


Fig. 7.79 Shaft and distal end of right ulna.

Muscles in the anterior compartment of the forearm flex the wrist and digits and pronate the hand. Muscles in the posterior compartment extend the wrist and digits and supinate the hand. Major nerves and vessels supply or pass through each compartment.

Bones

Shaft and distal end of radius

The shaft of the radius is narrow proximally, where it is continuous with the radial tuberosity and neck, and much broader distally, where it expands to form the distal end (Fig. 7.78).

Throughout most of its length, the shaft of the radius is triangular in cross section, with:

- three borders (anterior, posterior, and interosseous), and
- three surfaces (anterior, posterior, and lateral).

The **anterior border** begins on the medial side of the bone as a continuation of the radial tuberosity. In the superior third of the bone, it crosses the shaft diagonally, from medial to lateral, as the oblique line of the radius. The **posterior border** is distinct only in the middle third of the bone. The **interosseous border** is sharp and is the attachment site for the interosseous membrane, which links the radius to the ulna.

The anterior and posterior surfaces of the radius are generally smooth, whereas an oval roughening for the attachment of the pronator teres marks approximately the middle of the lateral surface of the radius.

Viewed anteriorly, the distal end of the radius is broad and somewhat flattened anteroposteriorly (Fig. 7.78). Consequently, the radius has expansive anterior and posterior surfaces and narrow medial and lateral surfaces. Its anterior surface is smooth and unremarkable, except for the prominent sharp ridge that forms its lateral margin.

The **posterior surface** of the radius is characterized by the presence of a large **dorsal tubercle**, which acts as a pulley for the tendon of one of the extensor muscles of the thumb (extensor pollicis longus). The medial surface is marked by a prominent facet for articulation with the distal end of the ulna (Fig. 7.78). The **lateral surface** of the radius is diamond shaped and extends distally as a **radial styloid process**.

The distal end of the bone is marked by two facets for articulation with two carpal bones (the scaphoid and lunate).

Shaft and distal end of ulna

The shaft of the ulna is broad superiorly where it is continuous with the large proximal end and narrow distally to form a small distal head (Fig. 7.79). Like the radius, the shaft of the ulna is triangular in cross section and has:

- three borders (anterior, posterior, and interosseous), and
- three surfaces (anterior, posterior, and medial).

The **anterior border** is smooth and rounded. The **posterior border** is sharp and palpable along its entire length. The **interosseous border** is also sharp and is the

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attachment site for the interosseous membrane, which joins the ulna to the radius.

The **anterior surface** of the ulna is smooth, except distally where there is a prominent linear roughening for the attachment of the pronator quadratus muscle. The **medial surface** is smooth and unremarkable. The **posterior surface** is marked by lines, which separate different regions of muscle attachments to bone.

The distal end of the ulna is small and characterized by a rounded head and the **ulnar styloid process** (Fig. 7.79). The anterolateral and distal part of the head is covered by articular cartilage. The ulnar styloid process originates from the posteromedial aspect of the ulna and projects distally.

In the clinic

Fractures of the radius and ulna

The radius and ulna are attached to the humerus proximally and the carpal bones distally by a complex series of ligaments. Although the bones are separate, they behave as one. When a severe injury occurs to the forearm it usually involves both bones, resulting in either fracture of both bones or more commonly a fracture of one bone and a dislocation of the other. Commonly, the mechanism of injury and the age of the patient determine which of these are likely to occur.

There are three classic injuries to the radius and ulna:

- Monteggia's fracture is a fracture of the proximal third of the ulna and an anterior dislocation of the head of the radius at the elbow.
- Galeazzi's fracture is a fracture of the distal third of the radius associated with subluxation (partial dislocation) of the head of the ulna at the wrist joint.
- Colles' fracture is a fracture, and posterior displacement, of the distal end of the radius.

Whenever a fracture of the radius or ulna is demonstrated radiographically, further images of the elbow and wrist should be obtained to exclude dislocations.

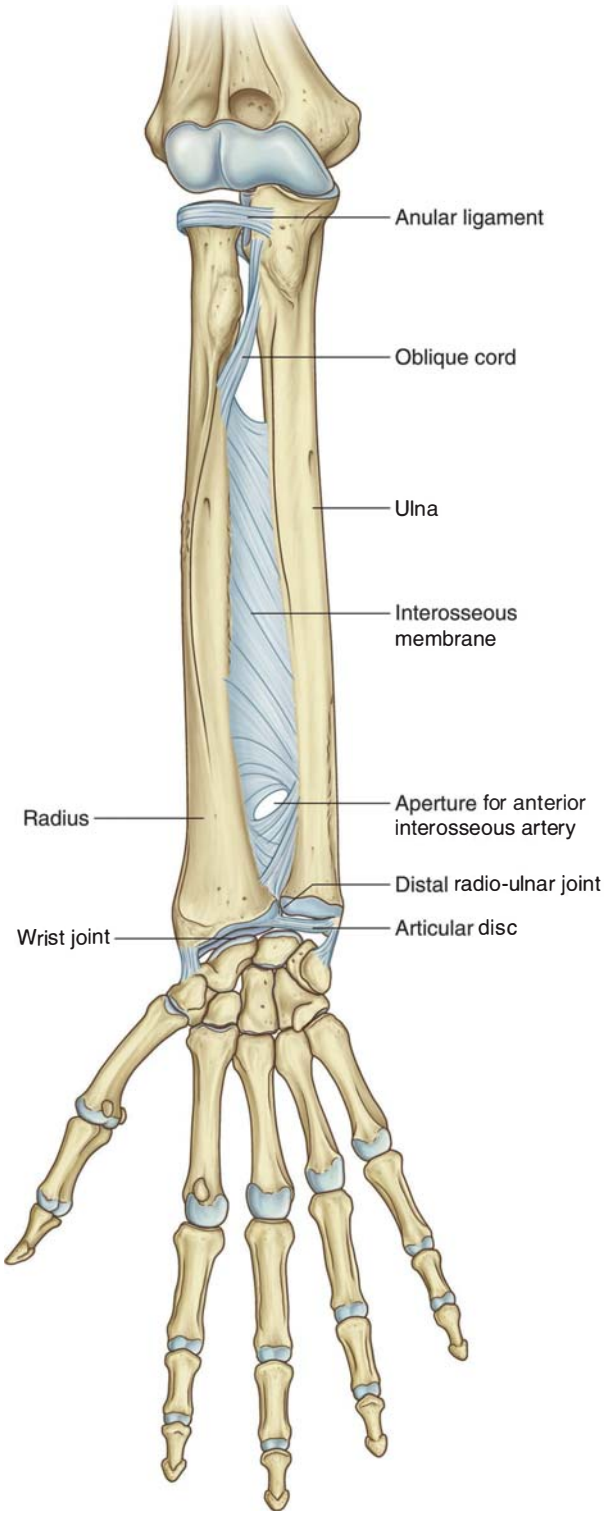


Fig. 7.80 Distal radio-ulnar joint and the interosseous membrane.

Joints

Distal radio-ulnar joint

The distal radio-ulnar joint occurs between the articular surface of the head of the ulna, with the ulnar notch on the end of the radius, and with a fibrous articular disc, which separates the radio-ulnar joint from the wrist joint (Fig. 7.80).

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The triangular-shaped articular disc is attached by its apex to a roughened depression on the ulna between the styloid process and the articular surface of the head, and by its base to the angular margin of the radius between the ulnar notch and the articular surface for the carpal bones.

The synovial membrane is attached to the margins of the distal radio-ulnar joint and is covered on its external surface by a fibrous joint capsule.

The distal radio-ulnar joint allows the distal end of the radius to move anteromedially over the ulna.

Interosseous membrane

The interosseous membrane is a thin fibrous sheet that connects the medial and lateral borders of the radius and ulna, respectively (Fig. 7.80). Collagen fibers within the sheet pass predominantly inferiorly from the radius to the ulna.

The interosseous membrane has a free upper margin, which is situated just inferior to the radial tuberosity, and a small circular aperture in its distal third. Vessels pass between the anterior and posterior compartments superior to the upper margin and through the inferior aperture.

The interosseous membrane connects the radius and ulna without restricting pronation and supination and provides attachment for muscles in the anterior and posterior compartments. The orientation of fibers in the membrane is also consistent with its role in transferring forces from the radius to the ulna and ultimately, therefore, from the hand to the humerus.

Pronation and supination

Pronation and supination of the hand occur entirely in the forearm and involve rotation of the radius at the elbow and movement of the distal end of the radius over the ulna (Fig. 7.81).

At the elbow, the superior articular surface of the radial head spins on the capitulum while, at the same time, the articular surface on the side of the head slides against the radial notch of the ulna and adjacent areas of the joint capsule and anular ligament of the radius. At the distal radio-ulnar joint, the ulnar notch of the radius slides anteriorly over the convex surface of the head of the ulna. During these movements, the bones are held together by:

- the anular ligament of the radius at the proximal radio-ulnar joint,
- the interosseous membrane along the lengths of the radius and ulna, and
- the articular disc at the distal radio-ulnar joint (Fig. 7.81).

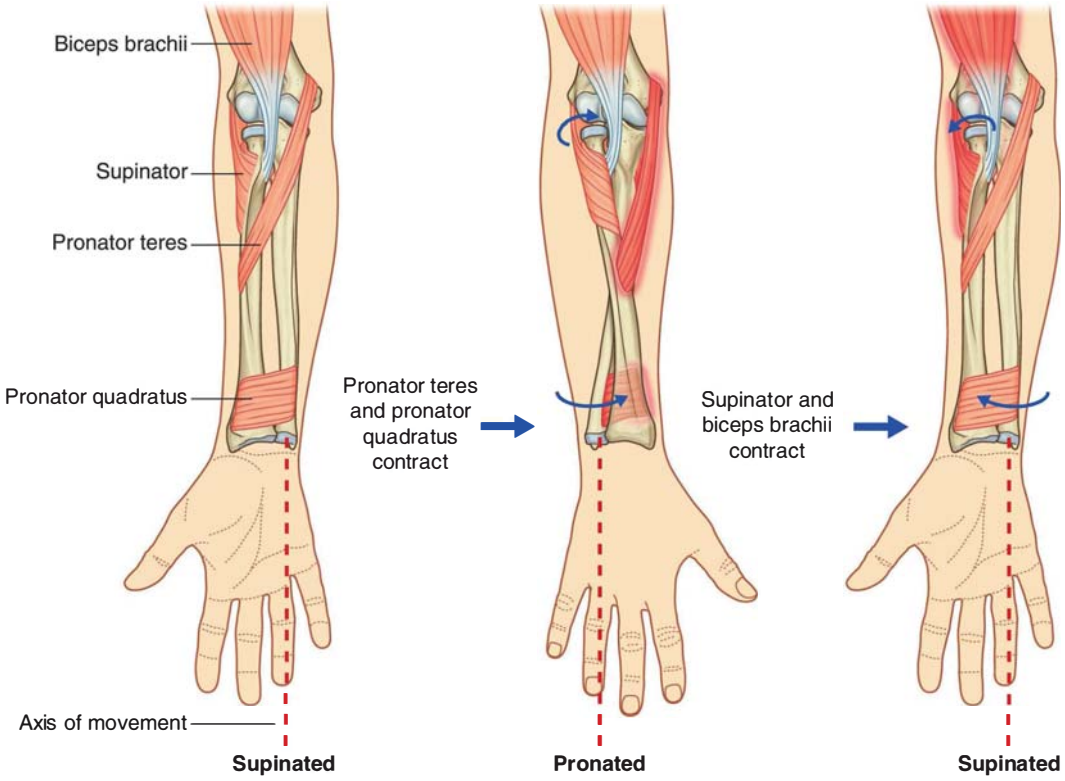


Fig. 7.81 Pronation and supination.

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Because the hand articulates predominantly with the radius, the translocation of the distal end of the radius medially over the ulna moves the hand from the palm-anterior (supinated) position to the palm-posterior (pronated) position.

Two muscles supinate and two muscles pronate the hand (Fig. 7.81).

Muscles involved in pronation and supination

Biceps brachii. The biceps brachii muscle, the largest of the four muscles that supinate and pronate the hand, is a powerful supinator as well as a flexor of the elbow joint. It is most effective as a supinator when the forearm is flexed.

Supinator. The second of the muscles involved with supination is the **supinator** muscle. Located in the posterior compartment of the forearm, it has a broad origin, from the supinator crest of the ulna and the lateral epicondyle of the humerus and from ligaments associated with the elbow joint.

The supinator muscle curves around the posterior surface and the lateral surface of the upper third of the radius to attach to the shaft of the radius superior to the oblique line.

The tendon of the biceps brachii muscle and the supinator muscle both become wrapped around the proximal end of the radius when the hand is pronated (Fig. 7.81). When they contract, they unwrap from the bone, producing supination of the hand.

Pronator teres and pronator quadratus. Pronation results from the action of the **pronator teres** and **pronator quadratus** muscles (Fig. 7.81). Both these muscles are in the anterior compartment of the forearm:

- The pronator teres runs from the medial epicondyle of the humerus to the lateral surface of the radius, approximately midway along the shaft.
- The pronator quadratus extends between the anterior surfaces of the distal ends of the radius and ulna.

When these muscles contract, they pull the distal end of the radius over the ulna, resulting in pronation of the hand (Fig. 7.81).

Anconeus. In addition to hinge-like flexion and extension at the elbow joint, some abduction of the distal end of the ulna also occurs and maintains the position of the palm of the hand over a central axis during pronation (Fig. 7.82). The muscle involved in this movement is the **anconeus muscle,** which is a triangular muscle in the posterior compartment of the forearm that runs from the lateral epicondyle to the lateral surface of the proximal end of the ulna.

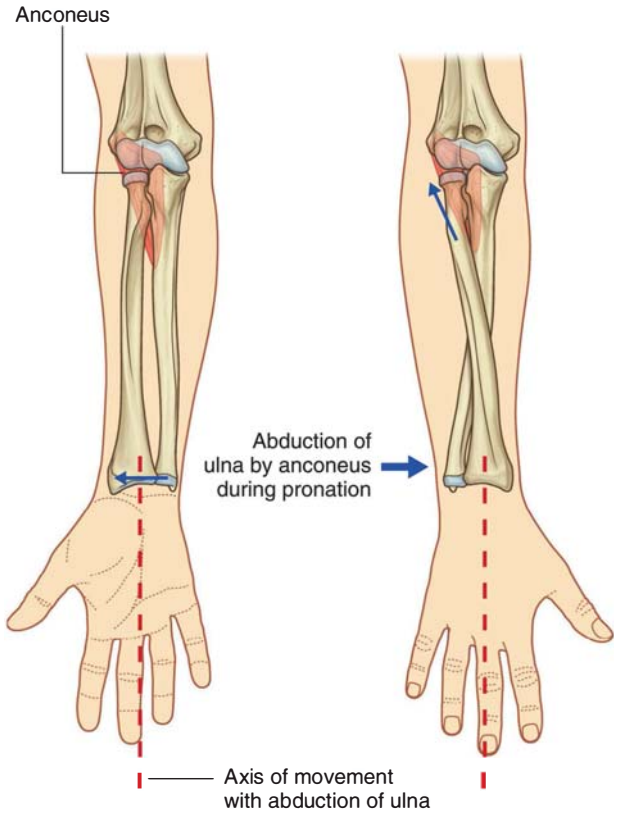


Fig. 7.82 Abduction of the distal end of the ulna by the anconeus during pronation.

ANTERIOR COMPARTMENT OF THE FOREARM

Muscles

Muscles in the anterior (flexor) compartment of the forearm occur in three layers: superficial, intermediate, and deep. Generally, these muscles are associated with:

- movements of the wrist joint,
- flexion of the fingers including the thumb, and
- pronation.

All muscles in the anterior compartment of the forearm are innervated by the median nerve, except for the flexor carpi ulnaris muscle and the medial half of the flexor digitorum profundus muscle, which are innervated by the ulnar nerve.

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Superficial layer

All four muscles in the superficial layer—the flexor carpi ulnaris, palmaris longus, flexor carpi radialis, and pronator teres—have a common origin from the medial epicondyle of the humerus, and, except for the pronator teres, extend distally from the forearm into the hand (Fig. 7.83 and Table 7.10).

Flexor carpi ulnaris

The **flexor carpi ulnaris** muscle is the most medial of the muscles in the superficial layer of flexors, having a long linear origin from the olecranon and posterior border of the ulna, in addition to an origin from the medial epicondyle of the humerus (Fig. 7.83A,B).

The ulnar nerve enters the anterior compartment of the forearm by passing through the triangular gap between the humeral and ulnar heads of the flexor carpi ulnaris (Fig. 7.83B). The muscle fibers converge on a tendon that passes distally and attaches to the pisiform bone of the wrist. From this point, force is transferred to the hamate bone of the wrist and to the base of metacarpal V by the **pisohamate** and **pisometacarpal ligaments**.

The flexor carpi ulnaris muscle is a powerful flexor and adductor of the wrist and is innervated by the ulnar nerve (Table 7.10).

Palmaris longus

The **palmaris longus** muscle, which is absent in about 15% of the population, lies between the flexor carpi ulnaris and the flexor carpi radialis muscles (Fig. 7.83A). It is a spindle-shaped muscle with a long tendon, which passes into the hand and attaches to the flexor retinaculum and to a thick layer of deep fascia, the palmar aponeurosis, which underlies and is attached to the skin of the palm and fingers.

In addition to its role as an accessory flexor of the wrist joint, the palmaris longus muscle also opposes shearing forces on the skin of the palm during gripping (Table 7.10).

Flexor carpi radialis

The **flexor carpi radialis** muscle is lateral to the palmaris longus and has a large and prominent tendon in the distal half of the forearm (Fig. 7.83A and Table 7.10). Unlike the tendon of the flexor carpi ulnaris, which forms the medial margin of the distal forearm, the tendon of the flexor carpi radialis muscle is positioned just lateral to the midline. In this position, the tendon can be easily palpated, making it an important landmark for finding the pulse in the radial artery, which lies immediately lateral to it.

The tendon of the flexor carpi radialis passes through a compartment formed by bone and fascia on the lateral side of the anterior surface of the wrist and attaches to the anterior surfaces of the bases of metacarpals II and III.

The flexor carpi radialis is a powerful flexor of the wrist and can also abduct the wrist.

Pronator teres

The **pronator teres** muscle originates from the medial epicondyle and supraepicondylar ridge of the humerus and from a small linear region on the medial edge of the coronoid process of the ulna (Fig. 7.83A). The median nerve often exits the cubital fossa by passing between the humeral and ulnar heads of this muscle. The pronator teres crosses the forearm and attaches to an oval roughened area on the lateral surface of the radius approximately midway along the bone.

Table 7.10 Superficial layer of muscles in the anterior compartment of the forearm (spinal segments indicated in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Flexor carpi ulnaris	Humeral head—medial epicondyle of humerus; ulnar head—olecranon and posterior border of ulna	Pisiform bone, and then via pisohamate and pisometacarpal ligaments into the hamate and base of metacarpal V	Ulnar nerve (C7, **C8**, T1)	Flexes and adducts the wrist joint
Palmaris longus	Medial epicondyle of humerus	Palmar aponeurosis of hand	Median nerve (**C7, C8**)	Flexes wrist joint; because the palmar aponeurosis anchors skin of the hand, contraction of the muscle resists shearing forces when gripping
Flexor carpi radialis	Medial epicondyle of humerus	Base of metacarpals II and III	Median nerve (**C6, C7**)	Flexes and abducts the wrist
Pronator teres	Humeral head—medial epicondyle and adjacent supra-epicondylar ridge; ulnar head—medial side of coronoid process	Roughening on lateral surface, midshaft, of radius	Median nerve (**C6, C7**)	Pronation

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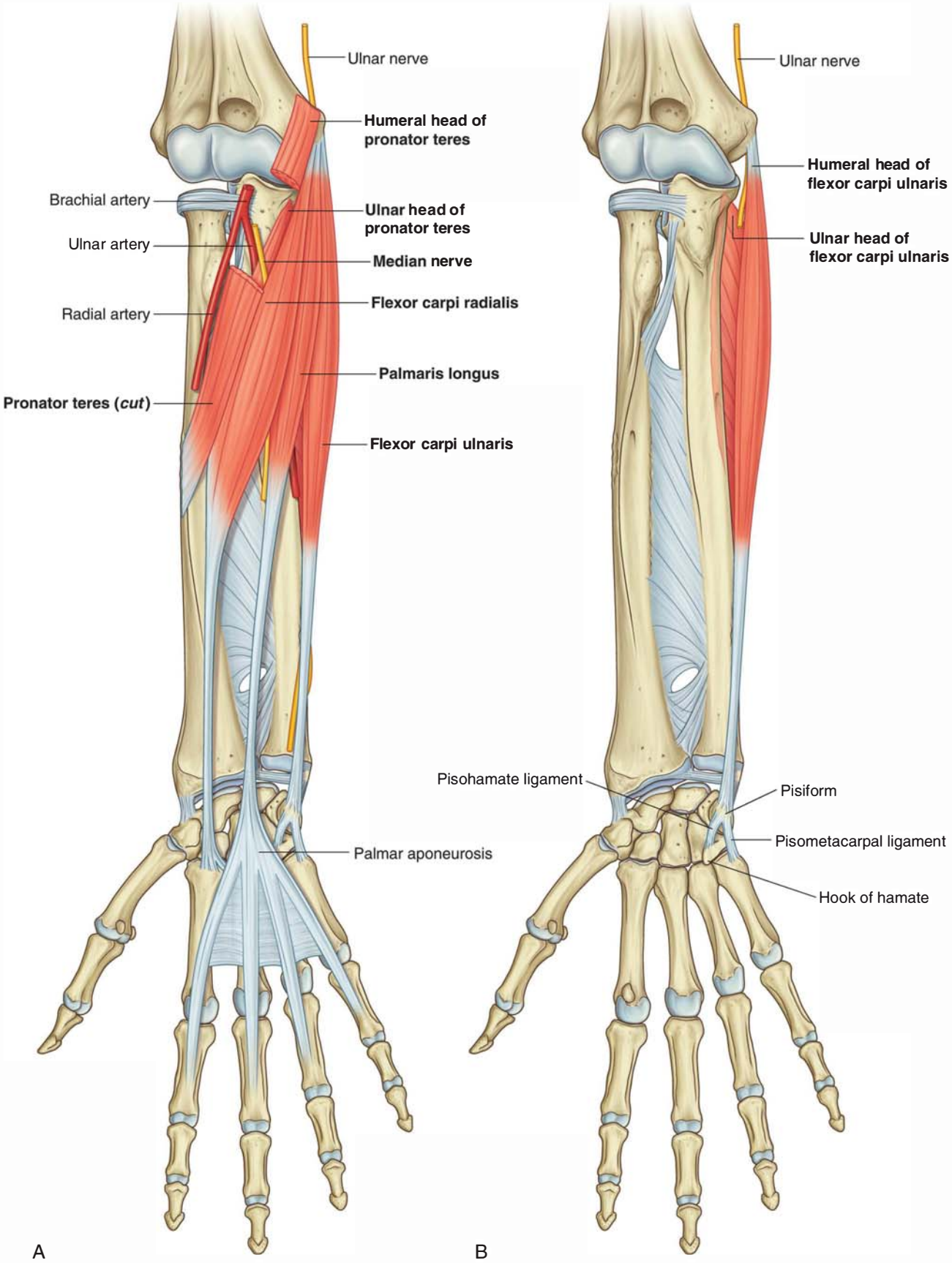


Fig. 7.83 Superficial layer of forearm muscles. **A.** Superficial muscles (flexor retinaculum not shown). **B.** Flexor carpi ulnaris muscle.

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The pronator teres forms the medial border of the cubital fossa and rotates the radius over the ulna during pronation (Table 7.10).

Intermediate layer

Flexor digitorum superficialis

The muscle in the intermediate layer of the anterior compartment of the forearm is the **flexor digitorum superficialis** muscle (Fig. 7.84). This large muscle has two heads:

- the humero-ulnar head, which originates mainly from the medial epicondyle of the humerus and from the adjacent medial edge of the coronoid process of the ulna; and
- the radial head, which originates from the anterior oblique line of the radius.

The median nerve and ulnar artery pass deep to the flexor digitorum superficialis between the two heads.

In the distal forearm, the flexor digitorum superficialis forms four tendons, which pass through the carpal tunnel of the wrist and into the four fingers. The tendons for the ring and middle fingers are superficial to the tendons for the index and little fingers.

In the forearm, carpal tunnel, and proximal regions of the four fingers, the tendons of the flexor digitorum superficialis are anterior to the tendons of the flexor digitorum profundus muscle.

Near the base of the proximal phalanx of each finger, the tendon of the flexor digitorum superficialis splits into two parts to pass posteriorly around each side of the tendon of the flexor digitorum profundus and ultimately attach to the margins of the middle phalanx (Fig. 7.84).

The flexor digitorum superficialis flexes the metacarpophalangeal joint and proximal interphalangeal joint of each finger; it also flexes the wrist joint (Table 7.11).

Table 7.11 Intermediate layer of muscles in the anterior compartment of the forearm (spinal segment indicated in bold is the major segment innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Flexor digitorum superficialis	Humero-ulnar head—medial epicondyle of humerus and adjacent margin of coronoid process; radial head—oblique line of radius	Four tendons, which attach to the palmar surfaces of the middle phalanges of the index, middle, ring, and little fingers	Median nerve (**C8**, T1)	Flexes proximal interphalangeal joints of the index, middle, ring, and little fingers; can also flex metacarpophalangeal joints of the same fingers and the wrist joint

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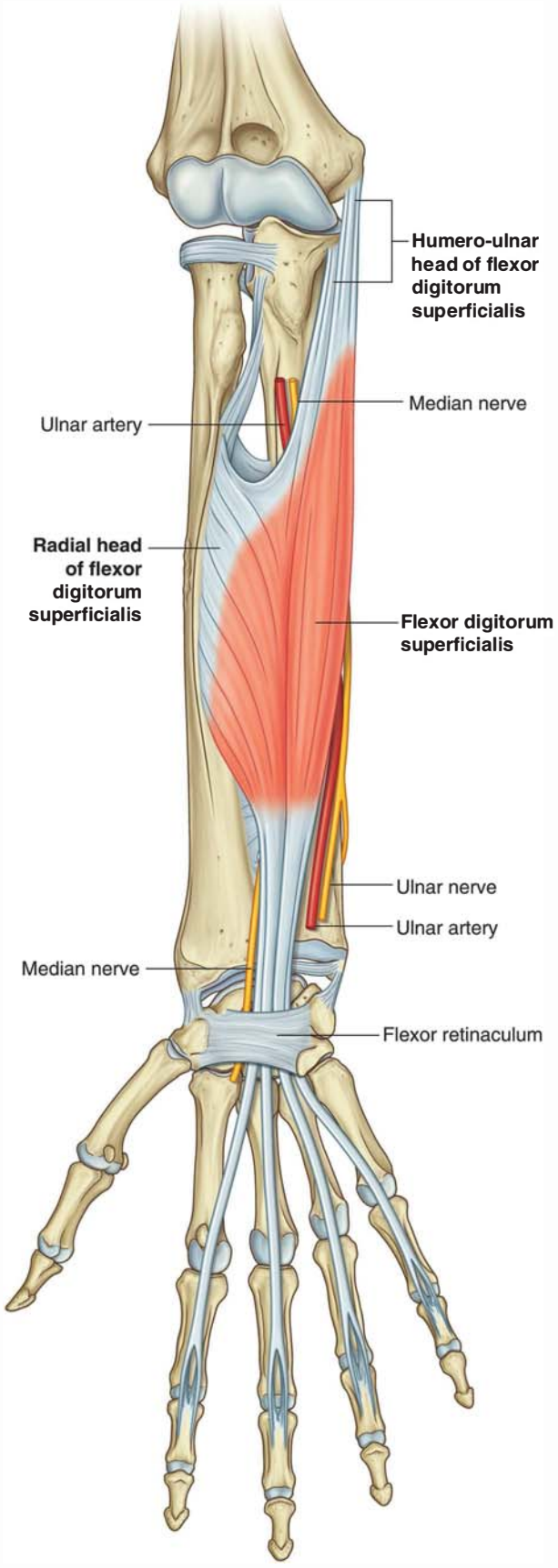


Fig. 7.84 Intermediate layer of forearm muscles.

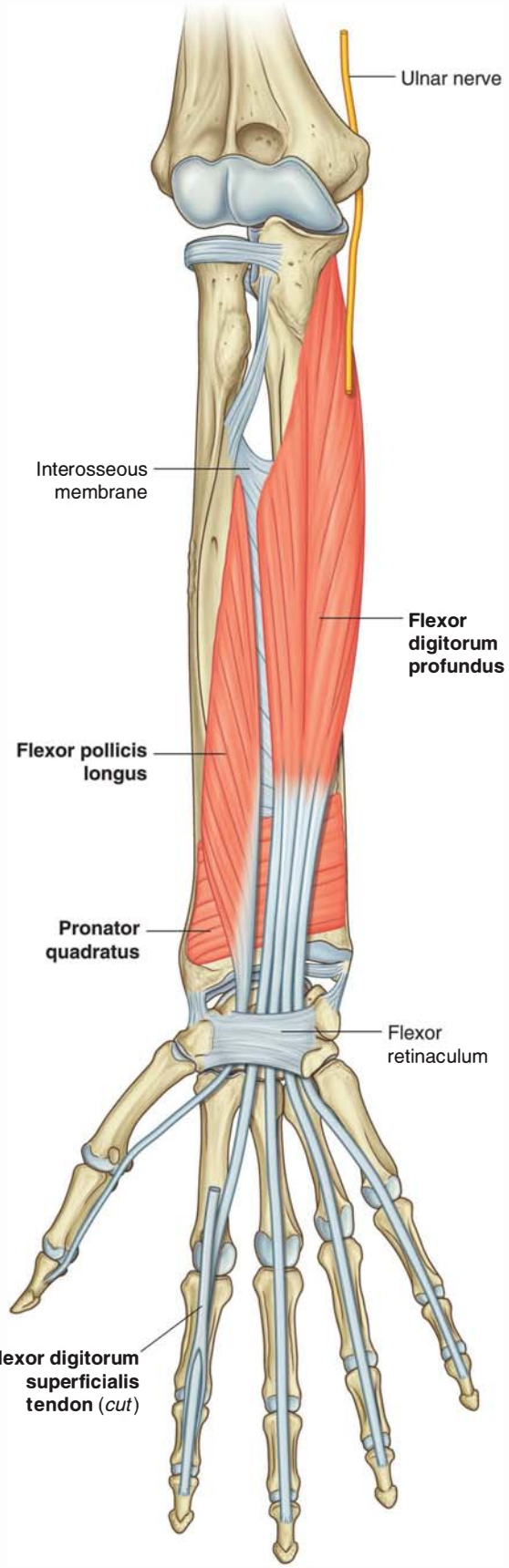


Fig. 7.85 Deep layer of forearm muscles.

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Deep layer

There are three deep muscles in the anterior compartment of the forearm: the flexor digitorum profundus, flexor pollicis longus, and pronator quadratus (Fig. 7.85).

Flexor digitorum profundus

The **flexor digitorum profundus** muscle originates from the anterior and medial surfaces of the ulna and from the adjacent half of the anterior surface of the interosseous membrane (Fig. 7.85). It gives rise to four tendons, which pass through the carpal tunnel into the four medial fingers. Throughout most of their course, the tendons are deep to the tendons of the flexor digitorum superficialis muscle.

Opposite the proximal phalanx of each finger, each tendon of the flexor digitorum profundus passes through a split formed in the overlying tendon of the flexor digitorum superficialis muscle and passes distally to insert into the anterior surface of the base of the distal phalanx.

In the palm, the lumbrical muscles originate from the sides of the tendons of the flexor digitorum profundus (see Fig. 7.104).

Innervation of the medial and lateral halves of the flexor digitorum profundus varies as follows:

- The lateral half (associated with the index and middle fingers) is innervated by the anterior interosseous nerve (branch of the median nerve).
- The medial half (the part associated with the ring and little fingers) is innervated by the ulnar nerve.

The flexor digitorum profundus flexes the metacarpophalangeal joints and the proximal and distal interphalangeal joints of the four fingers. Because the tendons cross the wrist, it can flex the wrist joint as well (Table 7.12).

Flexor pollicis longus

The **flexor pollicis longus** muscle originates from the anterior surface of the radius and the adjacent half of the anterior surface of the interosseous membrane (Fig. 7.85). It is a powerful muscle and forms a single large tendon, which passes through the carpal tunnel, lateral to the tendons of the flexor digitorum superficialis and flexor digitorum profundus muscles, and into the thumb where it attaches to the base of the distal phalanx.

The flexor pollicis longus flexes the thumb and is innervated by the anterior interosseous nerve (branch of the median nerve) (Table 7.12).

Pronator quadratus

The **pronator quadratus** muscle is a flat square-shaped muscle in the distal forearm (Fig. 7.85). It originates from a linear ridge on the anterior surface of the lower end of the ulna and passes laterally to insert onto the flat anterior surface of the radius. It lies deep to, and is crossed by, the tendons of the flexor digitorum profundus and flexor pollicis longus muscles.

The pronator quadratus muscle pulls the distal end of the radius anteriorly over the ulna during pronation and is innervated by the anterior interosseous nerve (branch of the median nerve) (Table 7.12).

Table 7.12 Deep layer of muscles in the anterior compartment of the forearm (spinal segments indicated in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Flexor digitorum profundus	Anterior and medial surfaces of ulna and anterior medial half of interosseous membrane	Four tendons, which attach to the palmar surfaces of the distal phalanges of the index, middle, ring, and little fingers	Lateral half by median nerve (anterior interosseous nerve); medial half by ulnar nerve (**C8**, T1)	Flexes distal interphalangeal joints of the index, middle, ring, and little fingers; can also flex metacarpophalangeal joints of the same fingers and the wrist joint
Flexor pollicis longus	Anterior surface of radius and radial half of interosseous membrane	Palmar surface of base of distal phalanx of thumb	Median nerve (anterior interosseous nerve) (C7, **C8**)	Flexes interphalangeal joint of the thumb; can also flex metacarpophalangeal joint of the thumb
Pronator quadratus	Linear ridge on distal anterior surface of ulna	Distal anterior surface of radius	Median nerve (anterior interosseous nerve) (C7, **C8**)	Pronation

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Arteries and veins

The largest arteries in the forearm are in the anterior compartment, pass distally to supply the hand, and give rise to vessels that supply the posterior compartment (Fig. 7.86).

The brachial artery enters the forearm from the arm by passing through the cubital fossa. At the apex of the cubital fossa, it divides into its two major branches, the radial and ulnar arteries.

Radial artery

The radial artery originates from the brachial artery at approximately the neck of the radius and passes along the lateral aspect of the forearm (Fig. 7.86). It is:

- just deep to the brachioradialis muscle in the proximal half of the forearm,
- related on its lateral side to the superficial branch of the radial nerve in the middle third of the forearm, and
- medial to the tendon of the brachioradialis muscle and covered only by deep fascia, superficial fascia, and skin in the distal forearm.

In the distal forearm, the radial artery lies immediately lateral to the large tendon of the flexor carpi radialis muscle and directly anterior to the pronator quadratus muscle and the distal end of the radius (Fig. 7.86). In the distal forearm, the radial artery can be located using the flexor carpi radialis muscle as a landmark. The radial pulse can be felt by gently palpating the radial artery against the underlying muscle and bone.

The radial artery leaves the forearm, passes around the lateral side of the wrist, and penetrates the posterolateral aspect of the hand between the bases of metacarpals I and II (Fig. 7.86). Branches of the radial artery in the hand often provide the major blood supply to the thumb and lateral side of the index finger.

Branches of the radial artery originating in the forearm include:

- a **radial recurrent artery**, which contributes to an anastomotic network around the elbow joint and to

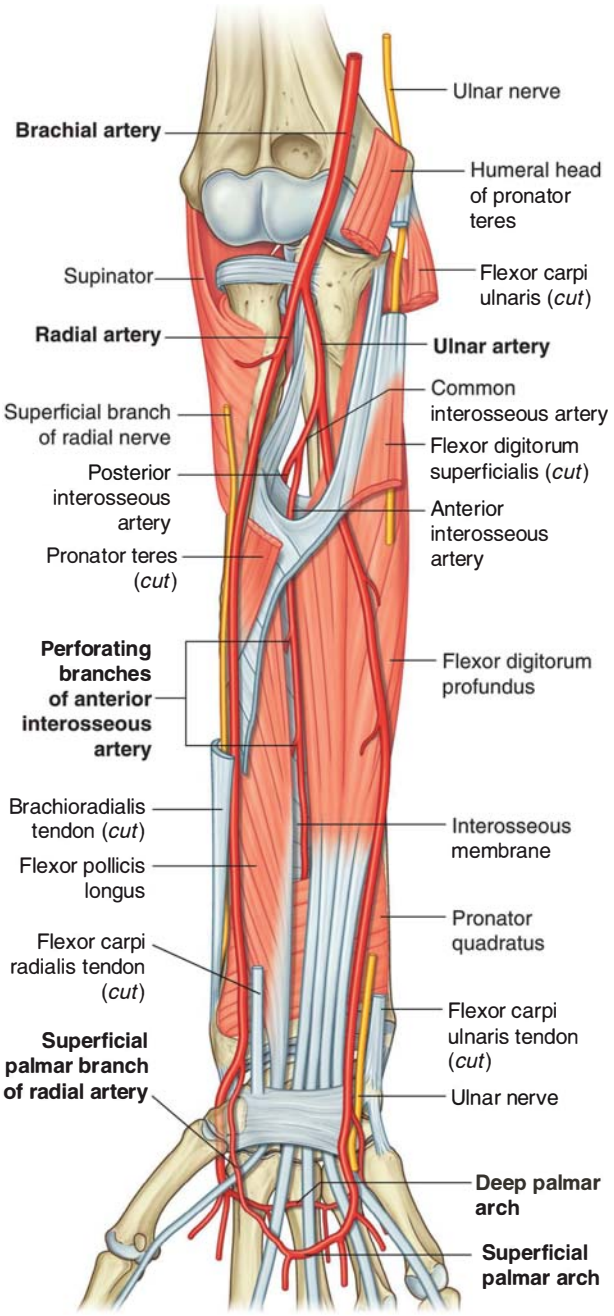


Fig. 7.86 Arteries of the anterior compartment of the forearm.

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numerous vessels that supply muscles on the lateral side of the forearm (see Fig. 7.66B);
- a small **palmar carpal branch,** which contributes to an anastomotic network of vessels that supply the carpal bones and joints;
- a somewhat larger branch, the **superficial palmar branch,** which enters the hand by passing through, or superficial to, the thenar muscles at the base of the thumb (Fig. 7.86) and anastomoses with the superficial palmar arch formed by the ulnar artery.

Ulnar artery

The ulnar artery is larger than the radial artery and passes down the medial side of the forearm (Fig. 7.86). It leaves the cubital fossa by passing deep to the pronator teres muscle, and then passes through the forearm in the fascial plane between the flexor carpi ulnaris and flexor digitorum profundus muscles.

In the distal forearm, the ulnar artery often remains tucked under the anterolateral lip of the flexor carpi ulnaris tendon, and is therefore not easily palpable.

In distal regions of the forearm, the ulnar nerve is immediately medial to the ulnar artery.

The ulnar artery leaves the forearm, enters the hand by passing lateral to the pisiform bone and superficial to the flexor retinaculum of the wrist, and arches over the palm (Fig. 7.86). It is often the major blood supply to the medial three and one-half digits.

Branches of the ulnar artery that arise in the forearm include:

- the **ulnar recurrent artery** with **anterior** and **posterior branches**, which contribute to an anastomotic network of vessels around the elbow joint;
- numerous muscular arteries, which supply surrounding muscles;
- the **common interosseous artery**, which divides into anterior and posterior interosseous arteries (Fig. 7.86); and
- two small carpal arteries (**dorsal carpal branch** and **palmar carpal branch**), which supply the wrist.

The **posterior interosseous artery** passes dorsally over the proximal margin of the interosseous membrane into the posterior compartment of the forearm.

The **anterior interosseous artery** passes distally along the anterior aspect of the interosseous membrane and supplies muscles of the deep compartment of the forearm and the radius and ulna. It has numerous branches, which perforate the interosseous membrane to supply deep muscles of the posterior compartment; it also has a small branch, which contributes to the vascular network around the carpal bones and joints. Perforating the interosseous membrane in the distal forearm, the anterior interosseous artery terminates by joining the posterior interosseous artery.

Veins

Deep veins of the anterior compartment generally accompany the arteries and ultimately drain into brachial veins associated with the brachial artery in the cubital fossa.

In the clinic

Transection of the radial or ulnar artery

Adult patients may transect the radial or ulnar artery because these vessels are relatively subcutaneous. A typical method of injury is when the hand is forced through a plate glass window. Fortunately, the dual supply to the hand usually enables the surgeon to tie off either the ulnar or the radial artery, without significant consequence.

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Nerves

Nerves in the anterior compartment of the forearm are the median and ulnar nerves and the superficial branch of the radial nerve (Fig. 7.87).

Median nerve

The median nerve innervates the muscles in the anterior compartment of the forearm except for the flexor carpi ulnaris and the medial part of the flexor digitorum profundus (ring and little fingers). It leaves the cubital fossa by passing between the two heads of the pronator teres muscle and passing between the humero-ulnar and radial heads of the flexor digitorum superficialis muscle (Fig. 7.87).

The median nerve continues a straight linear course distally down the forearm in the fascia on the deep surface of the flexor digitorum superficialis muscle. Just proximal to the wrist, it moves around the lateral side of the muscle and becomes more superficial in position, lying between the tendons of the palmaris longus and flexor carpi radialis muscles. It leaves the forearm and enters the palm of the hand by passing through the carpal tunnel deep to the flexor retinaculum.

Most branches to the muscles in the superficial and intermediate layers of the forearm originate medially from the nerve just distal to the elbow joint.

- The largest branch of the median nerve in the forearm is the **anterior interosseous nerve,** which originates between the two heads of the pronator teres, passes distally down the forearm with the anterior interosseous artery, innervates the muscles in the deep layer (the flexor pollicis longus, the lateral half of the flexor digitorum profundus, and the pronator quadratus) and terminates as articular branches to joints of the distal forearm and wrist.
- A small **palmar branch** originates from the median nerve in the distal forearm immediately proximal to the flexor retinaculum (Fig. 7.87), passes superficially into the hand, and innervates the skin over the base and central palm. This palmar branch is spared in carpal tunnel syndrome because it passes into the hand superficial to the flexor retinaculum of the wrist.

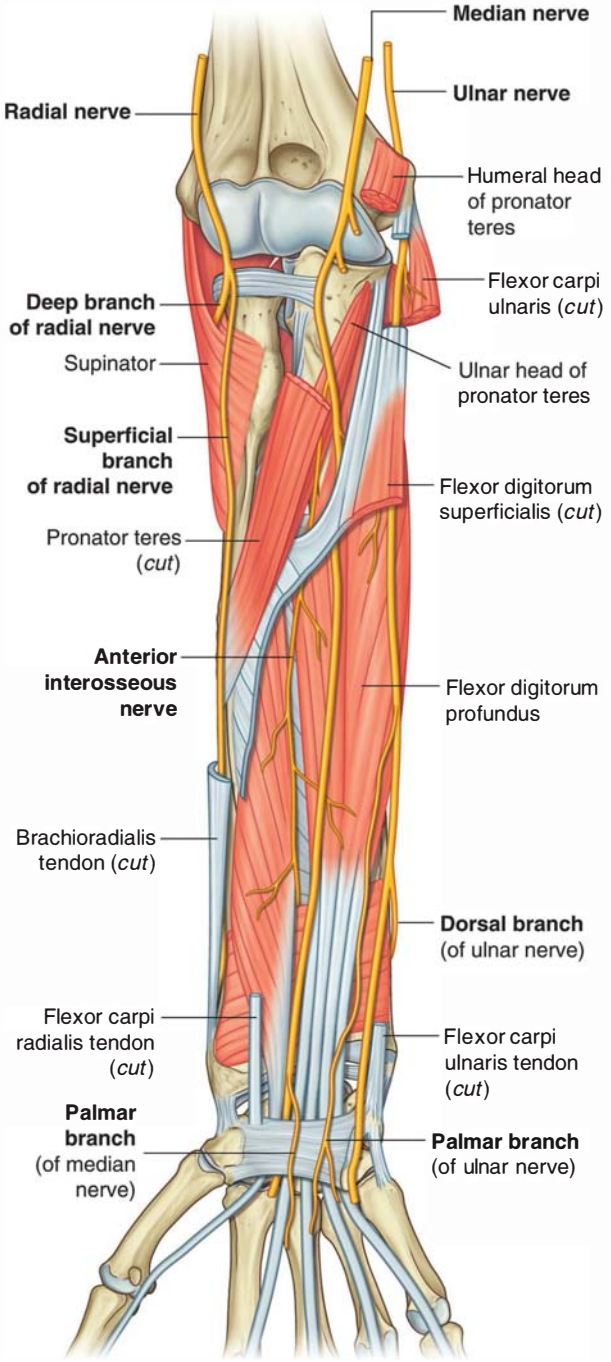


Fig. 7.87 Nerves of anterior forearm.

Ulnar nerve

The ulnar nerve passes through the forearm and into the hand, where most of its major branches occur. In the forearm, the ulnar nerve innervates only the flexor carpi ulnaris muscle and the medial part (ring and little fingers) of the flexor digitorum profundus muscle (Fig. 7.87).

The ulnar nerve enters the anterior compartment of the forearm by passing posteriorly around the medial epicondyle of the humerus and between the humeral and ulnar heads of the flexor carpi ulnaris muscle. After passing down the medial side of the forearm in the plane between the flexor carpi ulnaris and the flexor digitorum profundus muscles, it lies under the lateral lip of the tendon of the flexor carpi ulnaris proximal to the wrist.

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The ulnar artery is lateral to the ulnar nerve in the distal two-thirds of the forearm, and both the ulnar artery and nerve enter the hand by passing superficial to the flexor retinaculum and immediately lateral to the pisiform bone (Fig. 7.87).

In the forearm the ulnar nerve gives rise to:

- **muscular branches** to the flexor carpi ulnaris and to the medial half of the flexor digitorum profundus that arise soon after the ulnar nerve enters the forearm; and
- two small cutaneous branches—the **palmar branch** originates in the middle of the forearm and passes into the hand to supply skin on the medial side of the palm; the larger **dorsal branch** originates from the ulnar nerve in the distal forearm and passes posteriorly deep to the tendon of the flexor carpi ulnaris and innervates skin on the posteromedial side of the back of the hand and most skin on the posterior surfaces of the medial one and one-half digits.

Radial nerve

The radial nerve bifurcates into deep and superficial branches under the margin of the brachioradialis muscle in the lateral border of the cubital fossa (Fig. 7.87).

- The **deep branch** is predominantly motor and passes between the two heads of the supinator muscle to access and supply muscles in the posterior compartment of the forearm.
- The **superficial branch** of the radial nerve is sensory. It passes down the anterolateral aspect of the forearm deep to the brachioradialis muscle and in association with the radial artery. Approximately two-thirds of the way down the forearm, the superficial branch of the radial nerve passes laterally and posteriorly around the radial side of the forearm deep to the tendon of the brachioradialis. The nerve continues into the hand where it innervates skin on the posterolateral surface.

POSTERIOR COMPARTMENT OF THE FOREARM

Muscles

Muscles in the posterior compartment of the forearm occur in two layers: a superficial and a deep layer. The muscles are associated with:

- movement of the wrist joint,
- extension of the fingers and thumb, and
- supination.

All muscles in the posterior compartment of the forearm are innervated by the radial nerve.

Superficial layer

The seven muscles in the superficial layer are the brachioradialis, extensor carpi radialis longus, extensor carpi radialis brevis, extensor digitorum, extensor digiti minimi, extensor carpi ulnaris, and anconeus (Fig. 7.88). All have a common origin from the supraepicondylar ridge and lateral epicondyle of the humerus and, except for the brachioradialis and anconeus, extend as tendons into the hand.

Brachioradialis

The **brachioradialis** muscle originates from the proximal part of the supraepicondylar ridge of the humerus and passes through the forearm to insert on the lateral side of the distal end of the radius just proximal to the radial styloid process (Fig. 7.88).

In the anatomical position, the brachioradialis is part of the muscle mass overlying the anterolateral surface of the forearm and forms the lateral boundary of the cubital fossa.

Because the brachioradialis is anterior to the elbow joint, it acts as an accessory flexor of this joint even though it is in the posterior compartment of the forearm. Its action is most efficient when the forearm is midpronated and it forms a prominent bulge as it acts against resistance.

The radial nerve emerges from the posterior compartment of the arm just deep to the brachioradialis in the distal arm and innervates the brachioradialis. Lateral to the cubital fossa, the brachioradialis lies over the radial nerve and its bifurcation into deep and superficial branches. In more distal regions, the brachioradialis lies over the

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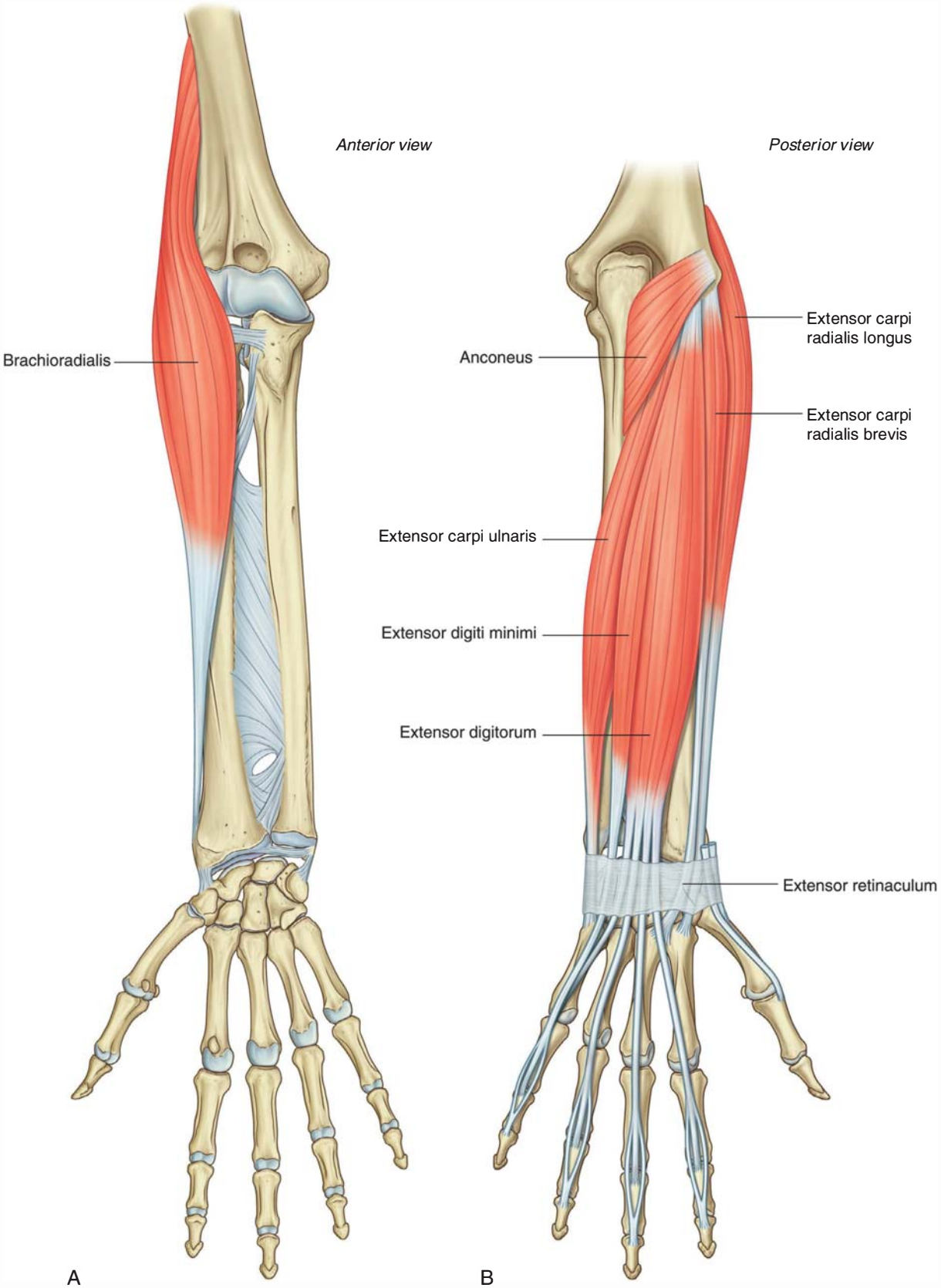


Fig. 7.88 Superficial layer of muscles in the posterior compartment of the forearm. **A.** Brachioradialis muscle (anterior view). **B.** Superficial muscles (posterior view).

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superficial branch of the radial nerve and radial artery (Table 7.13).

Extensor carpi radialis longus

The **extensor carpi radialis longus** muscle originates from the distal part of the supraepicondylar ridge and the lateral epicondyle of the humerus; its tendon inserts on the dorsal surface of the base of metacarpal II (Fig. 7.88). In proximal regions, it is deep to the brachioradialis muscle.

The extensor carpi radialis longus muscle extends and abducts the wrist, and is innervated by the radial nerve before the nerve divides into superficial and deep branches (Table 7.13).

Extensor carpi radialis brevis

The **extensor carpi radialis brevis** muscle originates from the lateral epicondyle of the humerus, and the tendon inserts onto adjacent dorsal surfaces of the bases of metacarpals II and III (Fig. 7.88). Along much of its course, the extensor carpi radialis brevis lies deep to the extensor carpi radialis longus.

The extensor carpi radialis brevis muscle extends and abducts the wrist, and is innervated by the deep branch of the radial nerve before the nerve passes between the two heads of the supinator muscle (Table 7.13).

Extensor digitorum

The **extensor digitorum** muscle is the major extensor of the four fingers (index, middle, ring, and little fingers). It originates from the lateral epicondyle of the humerus and forms four tendons, each of which passes into a finger (Fig. 7.88).

On the dorsal surface of the hand, adjacent tendons of the extensor digitorum are interconnected. In the fingers, each tendon inserts, via a triangular-shaped connective

Table 7.13 Superficial layer of muscles in the posterior compartment of the forearm (spinal segments indicated in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Brachioradialis	Proximal part of lateral supraepicondylar ridge of humerus and adjacent intermuscular septum	Lateral surface of distal end of radius	Radial nerve (C5, **C6**) before division into superficial and deep branches	Accessory flexor of elbow joint when forearm is midpronated
Extensor carpi radialis longus	Distal part of lateral supraepicondylar ridge of humerus and adjacent intermuscular septum	Dorsal surface of base of metacarpal II	Radial nerve (**C6**, C7) before division into superficial and deep branches	Extends and abducts the wrist
Extensor carpi radialis brevis	Lateral epicondyle of humerus and adjacent intermuscular septum	Dorsal surface of base of metacarpals II and III	Deep branch of radial nerve (**C7**, C8) before penetrating supinator muscle	Extends and abducts the wrist
Extensor digitorum	Lateral epicondyle of humerus and adjacent intermuscular septum and deep fascia	Four tendons, which insert via extensor hoods into the dorsal aspects of the bases of the middle and distal phalanges of the index, middle, ring, and little fingers	Posterior interosseous nerve (**C7**, C8)	Extends the index, middle, ring, and little fingers; can also extend the wrist
Extensor digiti minimi	Lateral epicondyle of humerus and adjacent intermuscular septum together with extensor digitorum	Extensor hood of the little finger	Posterior interosseous nerve (**C7**, C8)	Extends the little finger
Extensor carpi ulnaris	Lateral epicondyle of humerus and posterior border of ulna	Tubercle on the base of the medial side of metacarpal V	Posterior interosseous nerve (**C7**, C8)	Extends and adducts the wrist
Anconeus	Lateral epicondyle of humerus	Olecranon and proximal posterior surface of ulna	Radial nerve (**C6, C7, C8**) (via branch to medial head of triceps brachii)	Abduction of the ulna in pronation; accessory extensor of the elbow joint

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tissue aponeurosis (the extensor hood), into the base of the dorsal surfaces of the middle and distal phalanges.

The extensor digitorum muscle is innervated by the posterior interosseous nerve, which is the continuation of the deep branch of the radial nerve after it emerges from the supinator muscle (Table 7.13).

Extensor digiti minimi

The **extensor digiti minimi** muscle is an accessory extensor of the little finger and is medial to the extensor digitorum in the forearm (Fig. 7.88). It originates from the lateral epicondyle of the humerus and inserts, together with the tendon of the extensor digitorum, into the extensor hood of the little finger.

The extensor digiti minimi is innervated by the posterior interosseous nerve (Table 7.13).

Extensor carpi ulnaris

The **extensor carpi ulnaris** muscle is medial to the extensor digiti minimi (Fig. 7.88). It originates from the lateral epicondyle, and its tendon inserts into the medial side of the base of metacarpal V.

The extensor carpi ulnaris extends and adducts the wrist, and is innervated by the posterior interosseous nerve (Table 7.13).

Anconeus

The **anconeus muscle** is the most medial of the superficial extensors and has a triangular shape. It originates from the lateral epicondyle of the humerus and has a broad insertion into the posterolateral surface of the olecranon and related posterior surface of the ulna (see Fig. 7.82).

The anconeus abducts the ulna during pronation to maintain the center of the palm over the same point when the hand is flipped. It is also considered to be an accessory extensor of the elbow joint.

The anconeus is innervated by the branch of the radial nerve that innervates the medial head of the triceps brachii muscle (Table 7.13).

Deep layer

The deep layer of the posterior compartment of the forearm consists of five muscles: supinator, abductor pollicis longus, extensor pollicis brevis, extensor pollicis longus, and extensor indicis (Fig. 7.89).

Except for the supinator muscle, all these deep layer muscles originate from the posterior surfaces of the radius, ulna, and interosseous membrane and pass into the thumb and fingers.

- Three of these muscles—the abductor pollicis longus, extensor pollicis brevis, and extensor pollicis longus—emerge from between the extensor digitorum and the extensor carpi radialis brevis tendons of the superficial layer and pass into the thumb.
- Two of the three "outcropping" muscles (the abductor pollicis longus and extensor pollicis brevis) form a distinct muscular bulge in the distal posterolateral surface of the forearm.

All muscles of the deep layer are innervated by the posterior interosseous nerve, the continuation of the deep branch of the radial nerve.

Supinator

The **supinator** muscle has two heads of origin, which insert together on the proximal aspect of the radius (Fig. 7.89):

- The superficial (humeral) head originates mainly from the lateral epicondyle of the humerus and the related anular ligament and the radial collateral ligament of the elbow joint.
- The deep (ulnar) head originates mainly from the supinator crest on the posterolateral surface of the ulna.

From their sites of origin, the two heads wrap around the posterior and lateral aspect of the head, neck, and proximal shaft of the radius to insert on the lateral surface of the radius superior to the anterior oblique line and to the insertion of the pronator teres muscle.

The supinator muscle supinates the forearm and hand.

The deep branch of the radial nerve innervates the supinator muscle and passes to the posterior compartment of

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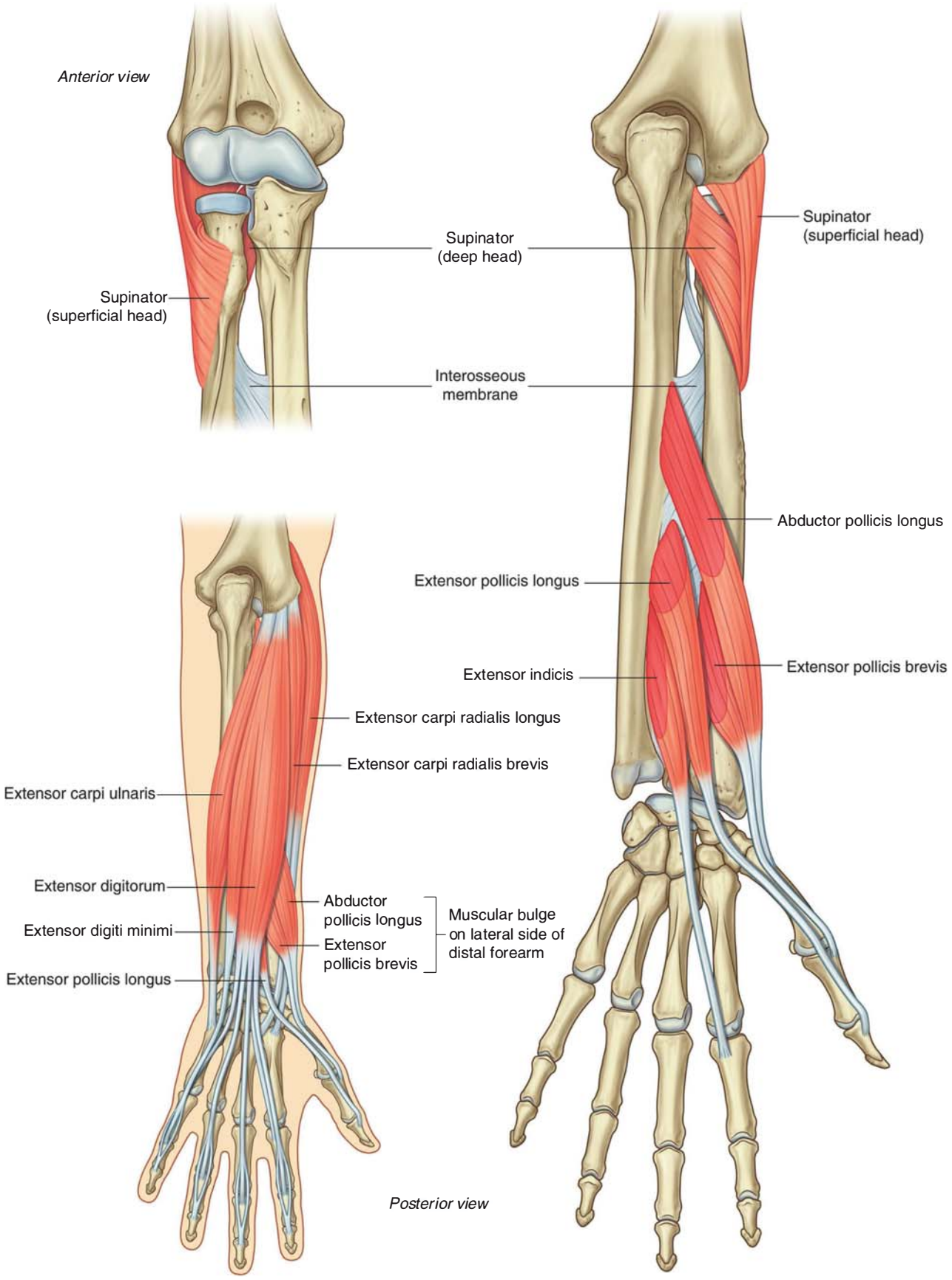


Fig. 7.89 Deep layer of muscles in the posterior compartment of the forearm.

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the forearm by passing between the two heads of this muscle (Table 7.14).

Abductor pollicis longus

The **abductor pollicis longus** muscle originates from the proximal posterior surfaces of the radius and the ulna and from the related interosseous membrane (Fig. 7.89). In the distal forearm, it emerges between the extensor digitorum and extensor carpi radialis brevis muscles to form a tendon that passes into the thumb and inserts on the lateral side of the base of metacarpal I. The tendon contributes to the lateral border of the anatomical snuffbox at the wrist.

The major function of the abductor pollicis longus is to abduct the thumb at the joint between the metacarpal I and trapezium bones (Table 7.14).

Extensor pollicis brevis

The **extensor pollicis brevis** muscle arises distal to the origin of the abductor pollicis longus from the posterior surface of the radius and interosseous membrane (Fig. 7.89). Together with the abductor pollicis longus, it emerges between the extensor digitorum and extensor carpi radialis brevis muscles to form a bulge on the posterolateral surface of the distal forearm. The tendon of the extensor pollicis brevis passes into the thumb and inserts on the dorsal surface of the base of the proximal phalanx. At the wrist, the tendon contributes to the lateral border of the anatomical snuffbox.

The extensor pollicis brevis extends the metacarpophalangeal and carpometacarpal joints of the thumb (Table 7.14).

Extensor pollicis longus

The **extensor pollicis longus** muscle originates from the posterior surface of the ulna and adjacent interosseous membrane and inserts via a long tendon into the dorsal surface of the distal phalanx of the thumb (Fig. 7.89). Like the abductor pollicis longus and extensor pollicis brevis, the tendon of this muscle emerges between the extensor digitorum and the extensor carpi radialis brevis muscles. However, it is held away from the other two deep muscles of the thumb by passing medially around the dorsal tubercle on the distal end of the radius. The tendon forms the medial margin of the anatomical snuffbox at the wrist.

The extensor pollicis longus extends all joints of the thumb (Table 7.14).

Extensor indicis

The **extensor indicis muscle** is an accessory extensor of the index finger. It originates distal to the extensor pollicis longus from the posterior surface of the ulna and adjacent interosseous membrane (Fig. 7.89). The tendon passes into the hand and inserts into the extensor hood of the index finger with the tendon of the extensor digitorum (Table 7.14).

Table 7.14 Deep layer of muscles in the posterior compartment of the forearm (spinal segments indicated in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Supinator	Superficial part—lateral epicondyle of humerus, radial collateral and anular ligaments; deep part—supinator crest of the ulna	Lateral surface of radius superior to the anterior oblique line	Posterior interosseous nerve (**C6**, C7)	Supination
Abductor pollicis longus	Posterior surfaces of ulna and radius (distal to the attachments of supinator and anconeus), and intervening interosseous membrane	Lateral side of base of metacarpal I	Posterior interosseous nerve (**C7**, C8)	Abducts carpometacarpal joint of thumb; accessory extensor of the thumb
Extensor pollicis brevis	Posterior surface of radius (distal to abductor pollicis longus) and the adjacent interosseous membrane	Dorsal surface of base of proximal phalanx of the thumb	Posterior interosseous nerve (**C7**, C8)	Extends metacarpophalangeal joint of the thumb; can also extend the carpometacarpal joint of the thumb
Extensor pollicis longus	Posterior surface of ulna (distal to the abductor pollicis longus) and the adjacent interosseous membrane	Dorsal surface of base of distal phalanx of thumb	Posterior interosseous nerve (**C7**, C8)	Extends interphalangeal joint of the thumb; can also extend carpometacarpal and metacarpophalangeal joints of the thumb
Extensor indicis	Posterior surface of ulna (distal to extensor pollicis longus) and adjacent interosseous membrane	Extensor hood of index finger	Posterior interosseous nerve (**C7**, C8)	Extends index finger

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Arteries and veins

The blood supply to the posterior compartment of the forearm occurs predominantly through branches of the radial, posterior interosseous, and anterior interosseous arteries (Fig. 7.90).

Posterior interosseous artery

The posterior interosseous artery originates in the anterior compartment from the common interosseous branch of the ulnar artery and passes posteriorly over the proximal margin of the interosseous membrane and into the posterior compartment of the forearm. It contributes a branch, the **recurrent interosseous artery** (see Fig. 7.66B), to the vascular network around the elbow joint and then passes between the supinator and abductor pollicis longus muscles to supply the superficial extensors. After receiving the terminal end of the anterior interosseous artery, the posterior interosseous artery terminates by joining the dorsal carpal arch of the wrist.

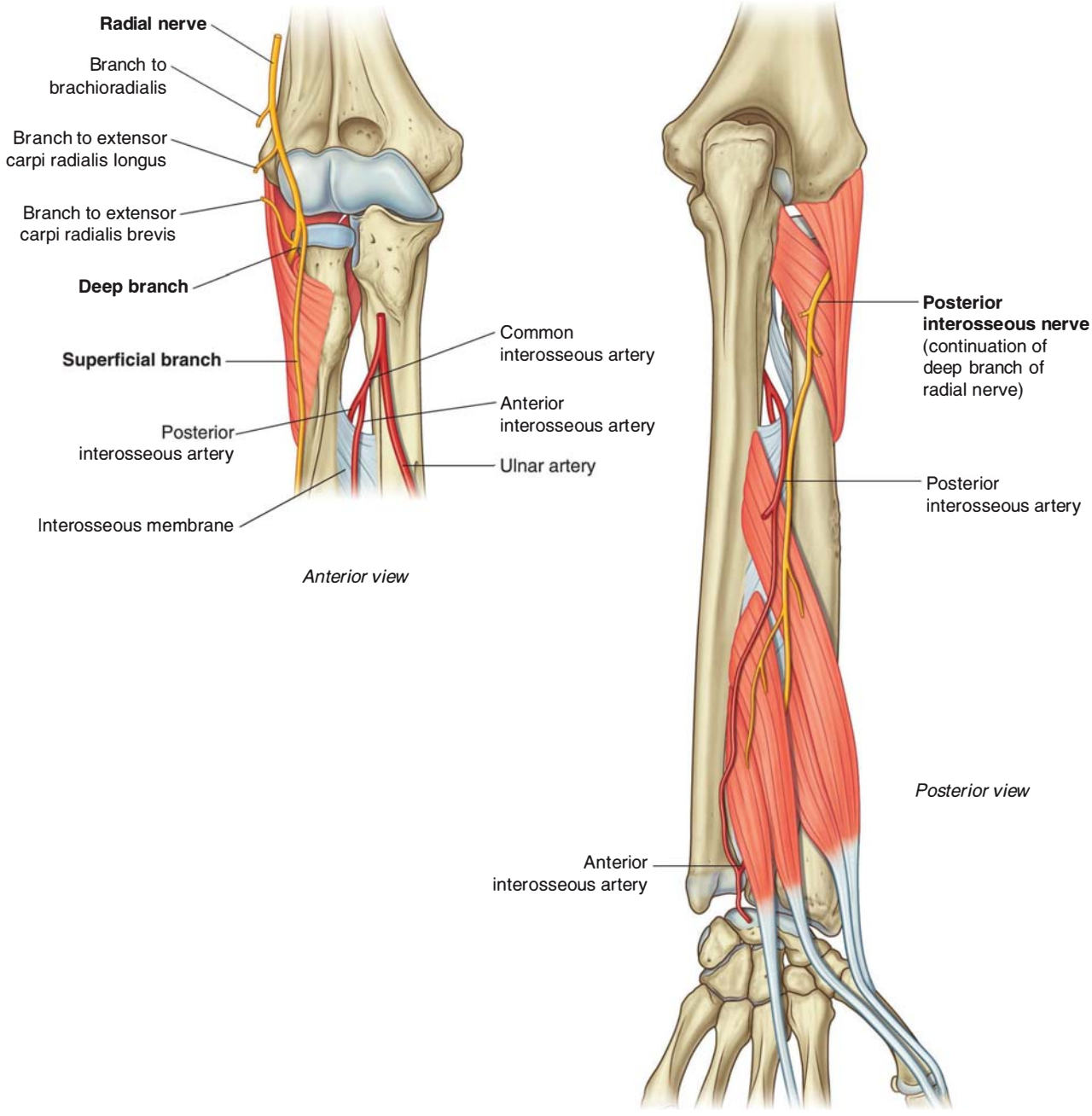


Fig. 7.90 Posterior interosseous artery and radial nerve in posterior compartment of forearm.

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Anterior interosseous artery

The anterior interosseous artery, also a branch of the common interosseous branch of the ulnar artery, is situated in the anterior compartment of the forearm on the interosseous membrane. It has numerous perforating branches, which pass directly through the interosseous membrane to supply deep muscles of the posterior compartment. The terminal end of the anterior interosseous artery passes posteriorly through an aperture in the interosseous membrane in distal regions of the forearm to join the posterior interosseous artery.

Radial artery

The radial artery has muscular branches, which contribute to the supply of the extensor muscles on the radial side of the forearm.

Veins

Deep veins of the posterior compartment generally accompany the arteries. They ultimately drain into brachial veins associated with the brachial artery in the cubital fossa.

Nerves

Radial nerve

The nerve of the posterior compartment of the forearm is the radial nerve (Fig. 7.90). Most of the muscles are innervated by the deep branch, which originates from the radial nerve in the lateral wall of the cubital fossa deep to the brachioradialis muscle and becomes the **posterior interosseous nerve** after emerging from between the two heads of the supinator muscle in the posterior compartment of the forearm.

In the lateral wall of the cubital fossa, and before dividing into **superficial** and **deep branches**, the radial nerve innervates the brachioradialis and extensor carpi radialis longus muscles.

The deep branch innervates the extensor carpi radialis brevis, then passes between the two heads of the supinator muscle and follows the plane of separation between the two heads dorsally and laterally around the proximal shaft of the radius to the posterior aspect of the forearm. It supplies the supinator muscle and then emerges, as the posterior interosseous nerve, from the muscle to lie between the superficial and deep layers of muscles.

The posterior interosseous nerve supplies the remaining muscles in the posterior compartment and terminates as articular branches, which pass deep to the extensor pollicis longus muscle to reach the wrist.

HAND

The hand (Fig. 7.91) is the region of the upper limb distal to the wrist joint. It is subdivided into three parts:

- the wrist (carpus),
- the metacarpus, and
- the digits (five fingers including the thumb).

The five digits consist of the laterally positioned thumb and, medial to the thumb, the four fingers—the index, middle, ring, and little fingers.

In the normal resting position, the fingers form a flexed arcade, with the little finger flexed most and the index finger flexed least. In the anatomical position, the fingers are extended.

The hand has an anterior surface (**palm**) and a dorsal surface (**dorsum of hand**).

Abduction and adduction of the fingers are defined with respect to the long axis of the middle finger (Fig. 7.91). In the anatomical position, the long axis of the thumb is

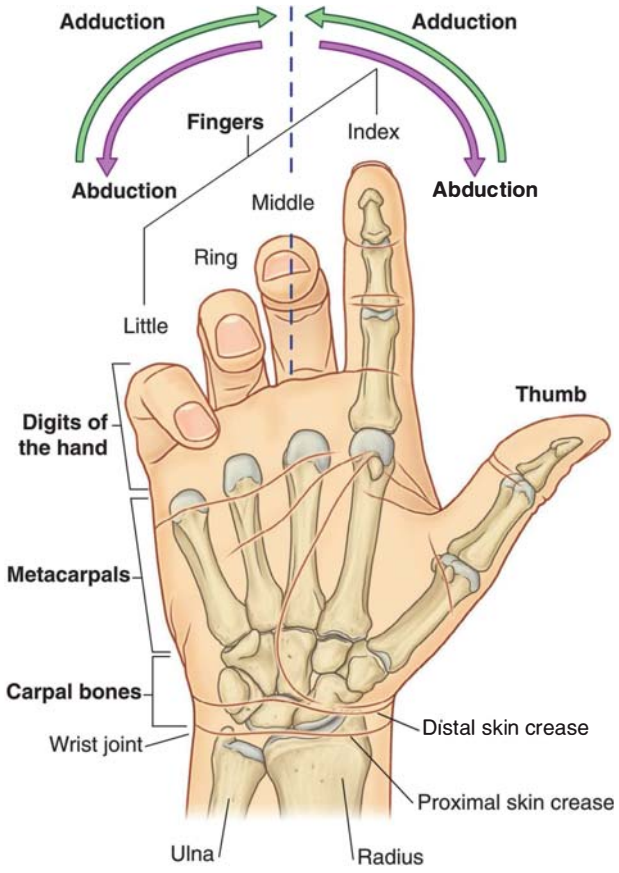


Fig. 7.91 Hand. The fingers are shown in a normal resting arcade in which they are flexed. In the anatomical position, the digits are straight and adducted.

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rotated 90° to the rest of the digits so that the pad of the thumb points medially; consequently, movements of the thumb are defined at right angles to the movements of the other digits of the hand.

The hand is a mechanical and sensory tool. Many of the features of the upper limb are designed to facilitate positioning the hand in space.

Bones

There are three groups of bones in the hand:

- The eight **carpal bones** are the bones of the wrist.
- The five **metacarpals** (**I to V**) are the bones of the metacarpus.
- The **phalanges** are the bones of the digits—the thumb has only two; the rest of the digits have three (Fig. 7.92).

The carpal bones and metacarpals of the index, middle, ring, and little fingers (metacarpals II to V) tend to function as a unit and form much of the bony framework of the palm. The metacarpal of the thumb functions independently and has increased flexibility at the carpometacarpal joint to provide opposition of the thumb to the fingers.

Carpal bones

The small carpal bones of the wrist are arranged in two rows, a proximal and a distal row, each consisting of four bones (Fig. 7.92).

Proximal row

From lateral to medial and when viewed from anteriorly, the proximal row of bones consists of:

- the boat-shaped **scaphoid**,
- the **lunate**, which has a crescent shape,
- the three-sided **triquetrum** bone, and
- the pea-shaped **pisiform** (Fig. 7.92).

The **pisiform** is a sesamoid bone in the tendon of the flexor carpi ulnaris and articulates with the anterior surface of the **triquetrum**.

The **scaphoid** has a prominent **tubercle** on its lateral palmar surface that is directed anteriorly.

Distal row

From lateral to medial and when viewed from anteriorly, the distal row of carpal bones consists of:

- the irregular four-sided **trapezium** bone,
- the four-sided **trapezoid**,
- the **capitate**, which has a head, and
- the **hamate**, which has a hook (Fig. 7.92).

The **trapezium** articulates with the metacarpal bone of the thumb and has a distinct **tubercle** on its palmar surface that projects anteriorly.

The largest of the carpal bones, the **capitate**, articulates with the base of metacarpal III.

The **hamate**, which is positioned just lateral and distal to the pisiform, has a prominent hook (**hook of hamate**) on its palmar surface that projects anteriorly.

Articular surfaces

The carpal bones have numerous articular surfaces (Fig. 7.92). All of them articulate with each other, and the carpal bones in the distal row articulate with the metacarpals of the digits. With the exception of the metacarpal of the thumb, all movements of the metacarpal bones on the carpal bones are limited.

The expansive proximal surfaces of the scaphoid and lunate articulate with the radius to form the wrist joint.

Carpal arch

The carpal bones do not lie in a flat plane; rather, they form an arch, whose base is directed anteriorly (Fig. 7.92). The lateral side of this base is formed by the tubercles of the scaphoid and trapezium. The medial side is formed by the pisiform and the hook of the hamate.

The flexor retinaculum attaches to, and spans the distance between, the medial and lateral sides of the base to form the anterior wall of the so-called carpal tunnel. The sides and roof of the carpal tunnel are formed by the arch of the carpal bones.

Metacarpals

Each of the five metacarpals is related to one digit:

- Metacarpal I is related to the thumb.
- Metacarpals II to V are related to the index, middle, ring, and little fingers, respectively (Fig. 7.92).

Each metacarpal consists of a **base**, a **shaft** (**body**), and distally, a **head**.

All of the bases of the metacarpals articulate with the carpal bones; in addition, the bases of the metacarpal bones of the fingers articulate with each other.

All of the heads of the metacarpals articulate with the proximal phalanges of the digits. The heads form the knuckles on the dorsal surface of the hand when the fingers are flexed.

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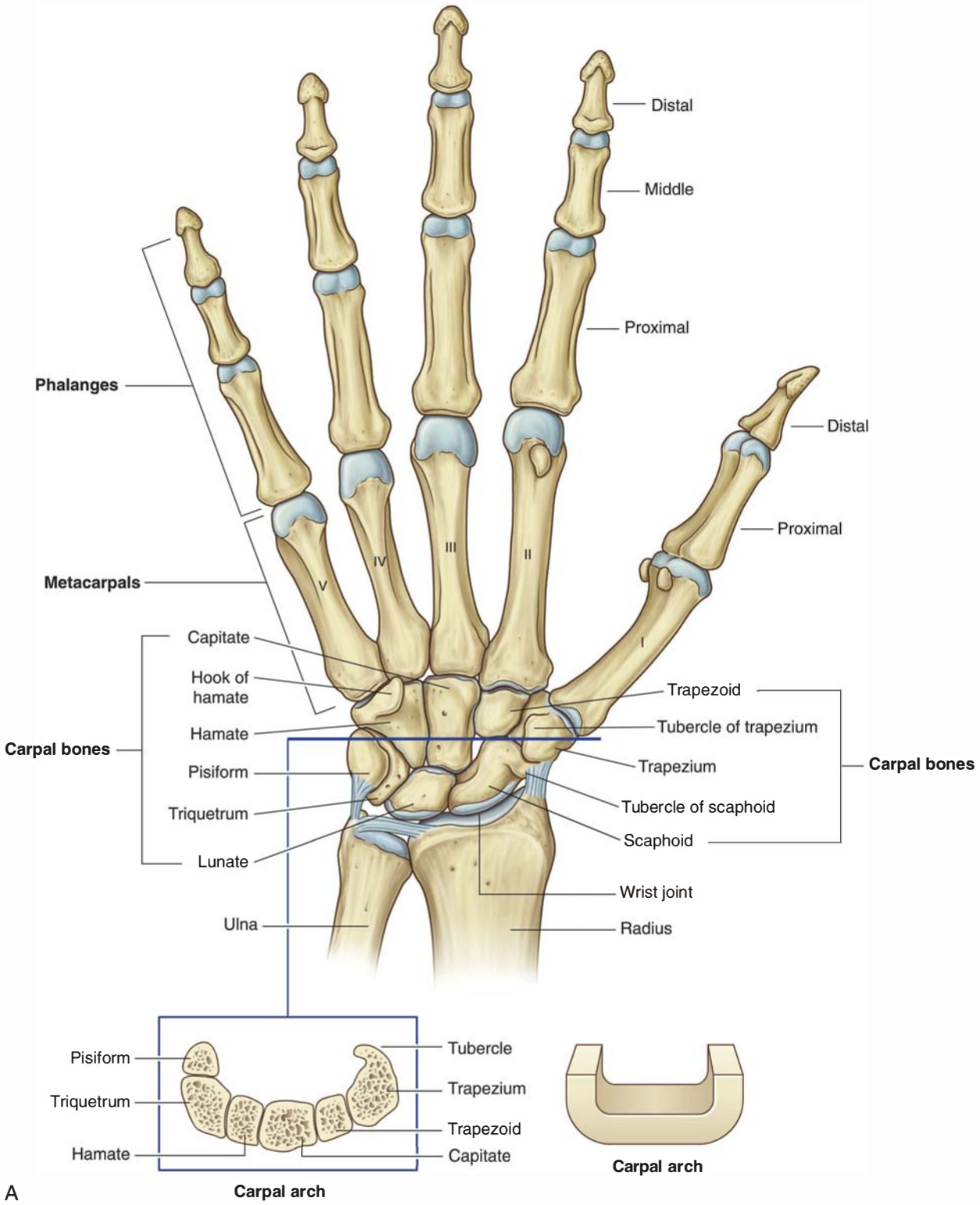


Fig. 7.92 Hand and wrist joint. **A.** Bones.

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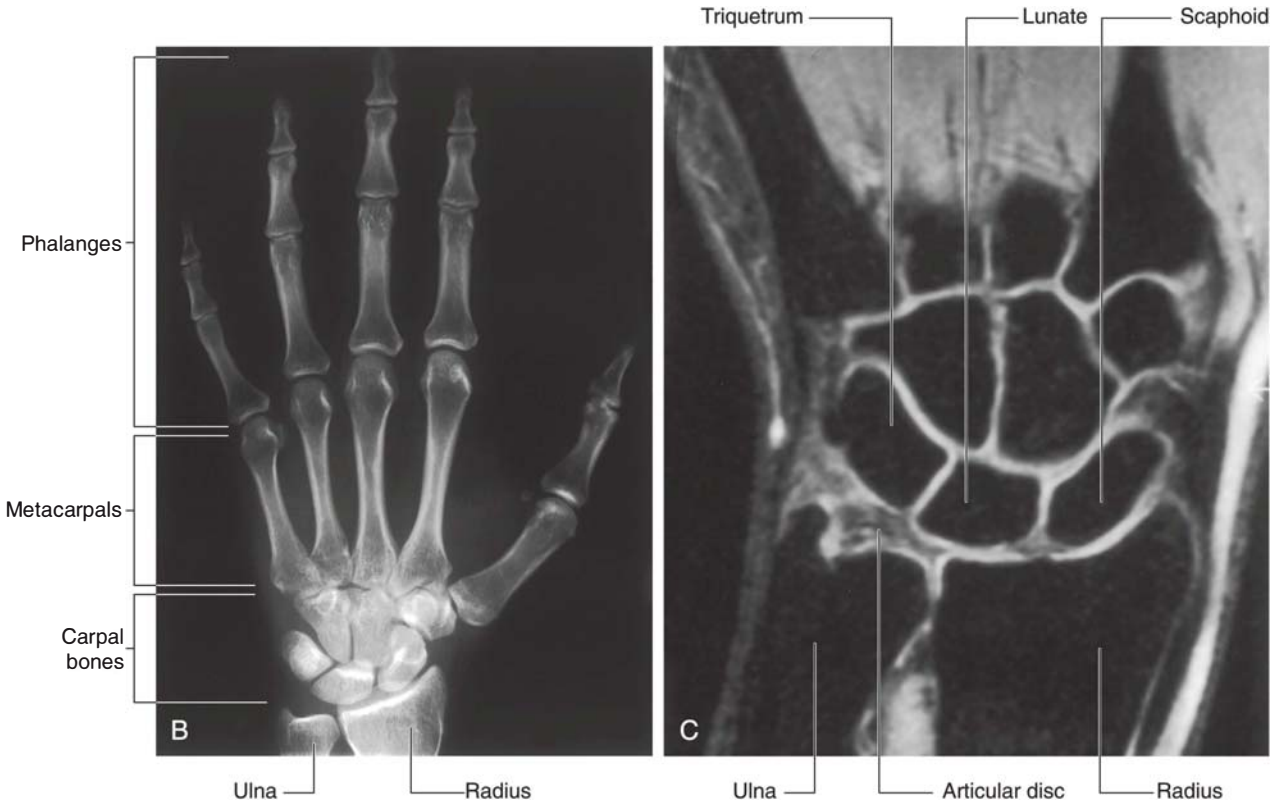


Fig. 7.92, cont'd Hand and wrist joint. **B.** Radiograph of a normal hand and wrist joint (anteroposterior view). **C.** Magnetic resonance image of a normal wrist joint in the coronal plane.

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Phalanges

The phalanges are the bones of the digits (Fig. 7.92):

- The thumb has two—a **proximal** and a **distal phalanx**.
- The rest of the digits have three—a **proximal**, a **middle**, and a **distal phalanx**.

Each phalanx has a **base**, a **shaft** (**body**), and distally, a **head**.

The base of each proximal phalanx articulates with the head of the related metacarpal bone.

The head of each distal phalanx is nonarticular and flattened into a crescent-shaped palmar tuberosity, which lies under the palmar pad at the end of the digit.

Joints

Wrist joint

The wrist joint is a synovial joint between the distal end of the radius and the articular disc overlying the distal end of the ulna, and the scaphoid, lunate, and triquetrum (Fig. 7.92). Together, the articular surfaces of the carpals form an oval shape with a convex contour, which articulates with the corresponding concave surface of the radius and articular disc.

The wrist joint allows movement around two axes. The hand can be abducted, adducted, flexed, and extended at the wrist joint.

Because the radial styloid process extends further distally than does the ulnar styloid process, the hand can be adducted to a greater degree than it can be abducted.

The capsule of the wrist joint is reinforced by **palmar radiocarpal**, **palmar ulnocarpal**, and **dorsal radiocarpal ligaments**. In addition, **radial** and **ulnar collateral ligaments of the wrist joint** span the distance between the styloid processes of the radius and ulna and the adjacent carpal bones. These ligaments reinforce the medial and lateral sides of the wrist joint and support them during flexion and extension.

Carpal joints

The synovial joints between the carpal bones share a common articular cavity. The joint capsule of the joints is reinforced by numerous ligaments.

Although movement at the **carpal joints** (**intercarpal joints**) is limited, the joints do contribute to the positioning of the hand in abduction, adduction, flexion, and, particularly, extension.

Carpometacarpal joints

There are five carpometacarpal joints between the metacarpals and the related distal row of carpal bones (Fig. 7.92).

The saddle joint, between metacarpal I and the trapezium, imparts a wide range of mobility to the thumb that is not a feature of the rest of the digits. Movements at this carpometacarpal joint are flexion, extension, abduction, adduction, rotation, and circumduction.

The carpometacarpal joints between metacarpals II to V and the carpal bones are much less mobile than the carpometacarpal joint of the thumb, allowing only limited gliding movements. Movement of the joints increases medially, so metacarpal V slides to the greatest degree. This can be best observed on the dorsal surface of the hand as it makes a fist.

Metacarpophalangeal joints

The joints between the distal heads of the metacarpals and the proximal phalanges of the digits are condylar joints, which allow flexion, extension, abduction, adduction, circumduction, and limited rotation (Fig. 7.92). The capsule of each joint is reinforced by the **palmar ligament** and by medial and lateral **collateral ligaments**.

Deep transverse metacarpal ligaments

The three deep **transverse metacarpal ligaments** (Fig. 7.93) are thick bands of connective tissue connecting the palmar ligaments of the metacarpophalangeal joints of the fingers to each other. They are important because, by linking the heads of the metacarpal bones together, they restrict the movement of these bones relative to each other. As a result, they help form a unified skeletal framework for the palm of the hand.

Significantly, a deep transverse metacarpal ligament does not occur between the palmar ligament of the metacarpophalangeal joint of the thumb and the palmar ligament of the index finger. The absence of this ligament, and the presence of a saddle joint between metacarpal I and the trapezium, are responsible for the increased mobility of the thumb relative to the rest of the digits of the hand.

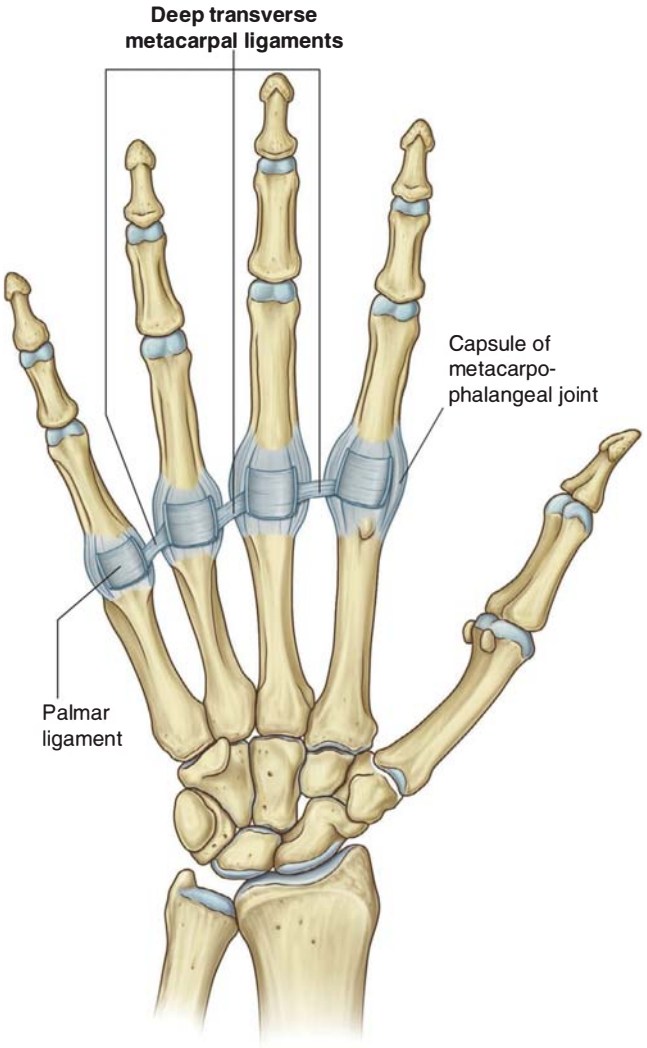


Fig. 7.93 Deep transverse metacarpal ligaments.

Interphalangeal joints of hand

The **interphalangeal joints of the hand** are hinge joints that allow mainly flexion and extension. They are reinforced by medial and lateral **collateral ligaments** and **palmar ligaments**.

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In the clinic

Fracture of the scaphoid and avascular necrosis of the proximal scaphoid

The commonest carpal injury is a fracture across the waist of the scaphoid bone (Fig. 7.94). It is uncommon to see other injuries. In approximately 10% of individuals, the scaphoid bone has a sole blood supply from the radial artery, which enters through the distal portion of the bone to supply the proximal portion. When a fracture occurs across the waist of the scaphoid, the proximal portion therefore undergoes avascular necrosis. It is impossible to predict which patients have this blood supply.

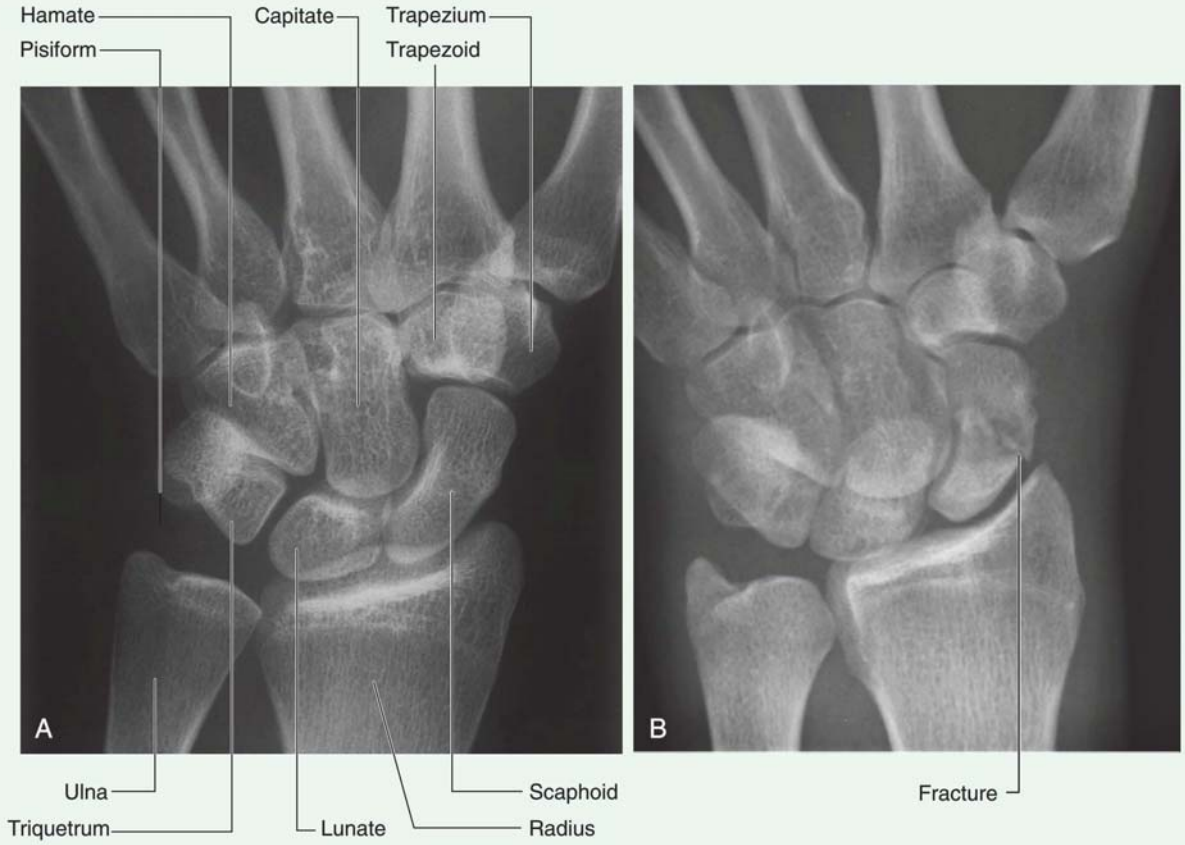


Fig. 7.94 Wrist radiographs (posteroanterior view). **A.** Normal. **B.** Scaphoid fracture.

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Carpal tunnel and structures at the wrist

The carpal tunnel is formed anteriorly at the wrist by a deep arch formed by the carpal bones and the flexor retinaculum (see Fig. 7.92).

The base of the carpal arch is formed medially by the pisiform and the hook of the hamate and laterally by the tubercles of the scaphoid and trapezium.

The flexor retinaculum is a thick connective tissue ligament that bridges the space between the medial and lateral sides of the base of the arch and converts the carpal arch into the carpal tunnel.

The four tendons of the flexor digitorum profundus, the four tendons of the flexor digitorum superficialis, and the tendon of the flexor pollicis longus pass through the carpal tunnel, as does the median nerve (Fig. 7.95).

The flexor retinaculum holds the tendons to the bony plane at the wrist and prevents them from "bowing."

Free movement of the tendons in the carpal tunnel is facilitated by synovial sheaths, which surround the tendons. All the tendons of the flexor digitorum profundus and flexor digitorum superficialis are surrounded by a single synovial sheath; a separate sheath surrounds the tendon of the flexor pollicis longus. The median nerve is anterior to the tendons in the carpal tunnel.

The tendon of the flexor carpi radialis is surrounded by a synovial sheath and passes through a tubular compartment formed by the attachment of the lateral aspect of the flexor retinaculum to the margins of a groove on the medial side of the tubercle of the trapezium.

The ulnar artery, ulnar nerve, and tendon of the palmaris longus pass into the hand anterior to the flexor retinaculum and therefore do not pass through the carpal tunnel (Fig. 7.95). The tendon of the palmaris longus is not surrounded by a synovial sheath.

The radial artery passes dorsally around the lateral side of the wrist and lies adjacent to the external surface of the scaphoid.

The extensor tendons pass into the hand on the medial, lateral, and posterior surfaces of the wrist in six compartments defined by an extensor retinaculum and lined by synovial sheaths (Fig. 7.95):

- The tendons of the extensor digitorum and extensor indicis share a compartment and synovial sheath on the posterior surface of the wrist.
- The tendons of the extensor carpi ulnaris and extensor digiti minimi have separate compartments and sheaths on the medial side of the wrist.
- The tendons of the abductor pollicis longus and extensor pollicis brevis muscles, the extensor carpi radialis longus and extensor carpi radialis brevis muscles, and the extensor pollicis longus muscle pass through three compartments on the lateral surface of the wrist.

In the clinic

Carpal tunnel syndrome

Carpal tunnel syndrome is an entrapment syndrome caused by pressure on the median nerve within the carpal tunnel. The etiology of this condition is often obscure, though in some instances the nerve injury may be a direct effect of increased pressure on the median nerve caused by overuse, swelling of the tendons and tendon sheaths (e.g., rheumatoid arthritis), and cysts arising from the carpal joints. Increased pressure in the carpal tunnel is thought to cause venous congestion that produces nerve edema and anoxic damage to the capillary endothelium of the median nerve itself.

Patients typically report pain and pins-and-needles sensations in the distribution of the median nerve. Weakness and loss of muscle bulk of the thenar muscles may also occur. Gently tapping over the median nerve (in the region of the flexor retinaculum) readily produces these symptoms (Tinel's sign).

Initial treatment is aimed at reducing the inflammation and removing any repetitive insults that produce the symptoms. If this does not lead to improvement, nerve conduction studies will be necessary to confirm nerve entrapment, which may require surgical decompression of the flexor retinaculum.

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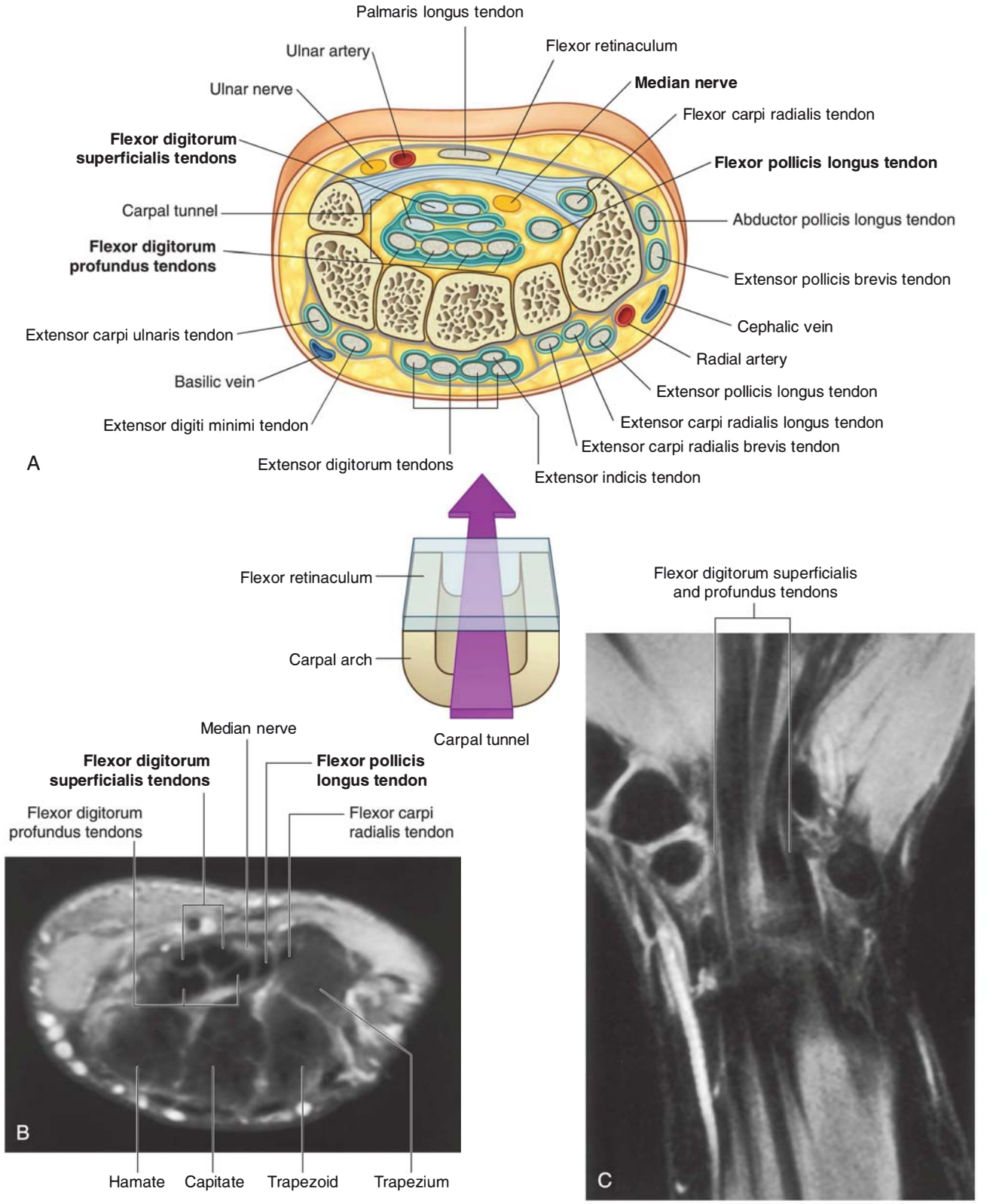


Fig. 7.95 Carpal tunnel. **A.** Structure and relations. **B.** Magnetic resonance image of a normal wrist in the axial plane. **C.** Magnetic resonance image of a normal wrist in the coronal plane.

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Palmar aponeurosis

The **palmar aponeurosis** is a triangular condensation of deep fascia that covers the palm and is anchored to the skin in distal regions (Fig. 7.96).

The apex of the triangle is continuous with the palmaris longus tendon, when present; otherwise, it is anchored to the flexor retinaculum. From this point, fibers radiate to extensions at the bases of the digits that project into each of the index, middle, ring, and little fingers and, to a lesser extent, the thumb.

Transverse fibers interconnect the more longitudinally arranged bundles that continue into the digits.

Vessels, nerves, and long flexor tendons lie deep to the palmar aponeurosis in the palm.

Palmaris brevis

The **palmaris brevis**, a small intrinsic muscle of the hand, is a quadrangular-shaped subcutaneous muscle that overlies the hypothenar muscles, ulnar artery, and superficial branch of the ulnar nerve at the medial side of the palm (Fig. 7.96). It originates from the palmar aponeurosis and flexor retinaculum and inserts into the dermis of the skin on the medial margin of the hand.

The palmaris brevis deepens the cup of the palm by pulling on skin over the hypothenar eminence and forming a distinct ridge. This may improve grip.

The palmaris brevis is innervated by the superficial branch of the ulnar nerve.

Anatomical snuffbox

The "anatomical snuffbox" is a term given to the triangular depression formed on the posterolateral side of the wrist and metacarpal I by the extensor tendons passing into the thumb (Fig. 7.97). Historically, ground tobacco (snuff) was placed in this depression before being inhaled into the nose. The base of the triangle is at the wrist and the apex is directed into the thumb. The impression is most apparent when the thumb is extended:

- The lateral border is formed by the tendons of the abductor pollicis longus and extensor pollicis brevis.

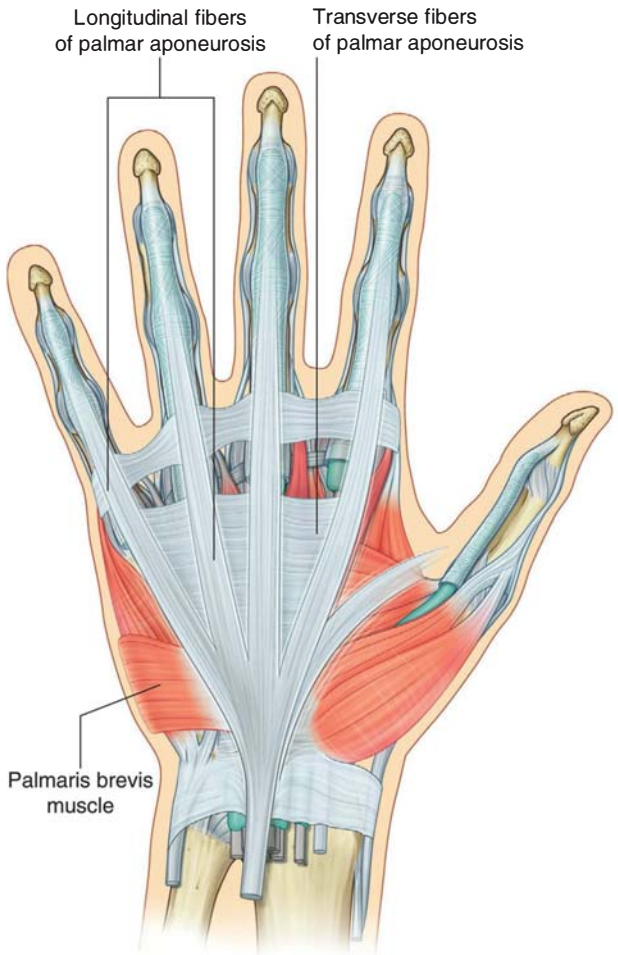


Fig. 7.96 Palmar aponeurosis.

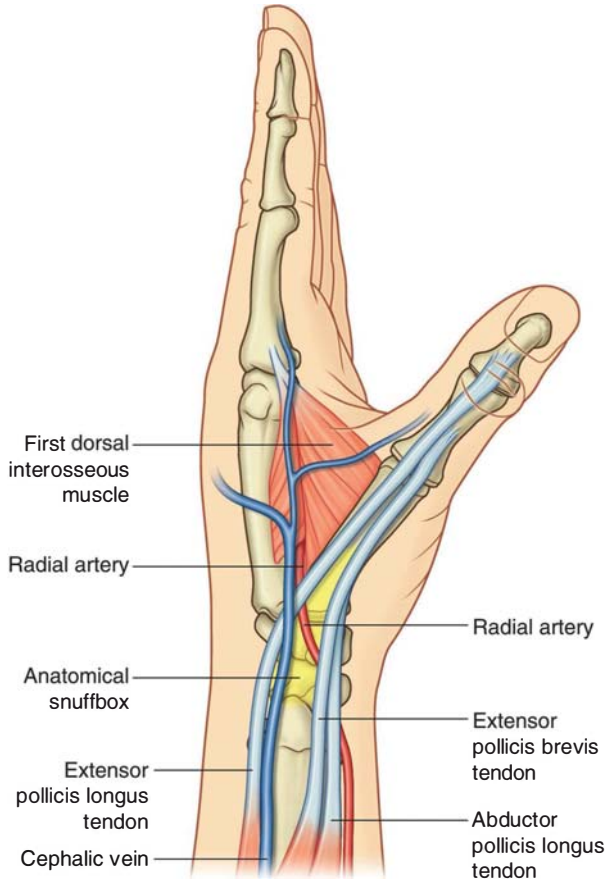


Fig. 7.97 Anatomical snuffbox.

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- The medial border is formed by the tendon of the extensor pollicis longus.
- The floor of the impression is formed by the scaphoid and trapezium, and the distal ends of the tendons of the extensor carpi radialis longus and extensor carpi radialis brevis.

The radial artery passes obliquely through the anatomical snuffbox, deep to the extensor tendons of the thumb and lies adjacent to the scaphoid and trapezium.

Terminal parts of the superficial branch of the radial nerve pass subcutaneously over the snuffbox as does the origin of the cephalic vein from the dorsal venous arch of the hand.

In the clinic

Snuffbox

The anatomical snuffbox is an important clinical region. When the hand is in ulnar deviation, the scaphoid becomes palpable within the snuffbox. This position enables the physician to palpate the bone to assess for a fracture. The pulse of the radial artery can also be felt in the snuffbox.

Fibrous digital sheaths

After exiting the carpal tunnel, the tendons of the flexor digitorum superficialis and profundus muscles cross the palm and enter fibrous sheaths on the palmar aspect of the digits (Fig. 7.98). These fibrous sheaths:

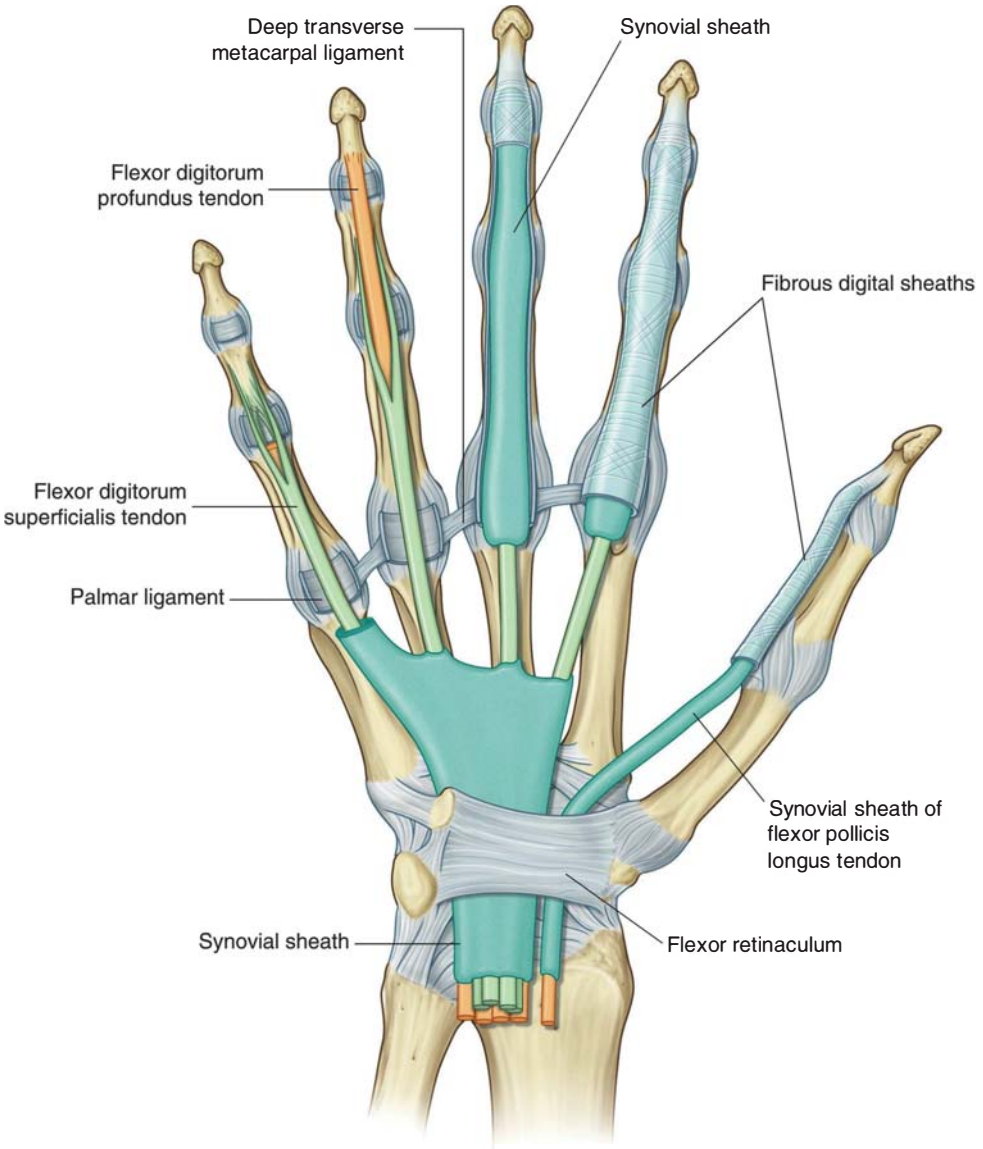


Fig. 7.98 Fibrous digital sheaths and synovial sheaths of the hand.

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- begin proximally, anterior to the metacarpophalangeal joints, and extend to the distal phalanges;
- are formed by fibrous arches and cruciate (cross-shaped) ligaments, which are attached posteriorly to the margins of the phalanges and to the palmar ligaments associated with the metacarpophalangeal and interphalangeal joints; and
- hold the tendons to the bony plane and prevent the tendons from bowing when the digits are flexed.

Within each tunnel, the tendons are surrounded by a synovial sheath. The synovial sheaths of the thumb and little finger are continuous with the sheaths associated with the tendons in the carpal tunnel (Fig. 7.98).

In the clinic

De Quervain's syndrome

De Quervain's syndrome is an inflammatory disorder that occurs within the first dorsal extensor compartment and involves the extensor pollicis brevis tendon and abductor pollicis longus tendon and their common tendon sheath. Patients typically present with significant wrist pain preventing appropriate flexion/extension and abduction of the thumb. The cause of this disorder is often overuse. For example, the syndrome is common in young mothers who are constantly lifting young children. Other causes include inflammatory disorders such as rheumatoid arthritis.

In the clinic

Tenosynovitis

Tenosynovitis is inflammation of a tendon and its sheath. The condition may be caused by overuse; however, it can also be associated with other disorders such as rheumatoid arthritis and connective tissue pathologies. If the inflammation becomes severe and ensuing fibrosis occurs, the tendon will not run smoothly within the tendon sheath, and typically within the fingers the tendon may stick or require excess force to fully extend and flex, producing a "triggering" phenomenon.

In the clinic

Trigger finger

Trigger finger is a common disorder of late childhood and adulthood and is typically characterized by catching or snapping and occasionally locking of the flexor tendon(s) in the hand. Trigger finger can be associated with significant dysfunction and pain. The triggering is usually related to fibrosis and tightening of the flexor tendon sheath at the level of the metacarpophalangeal joint.

Extensor hoods

The tendons of the extensor digitorum and extensor pollicis longus muscles pass onto the dorsal aspect of the digits and expand over the proximal phalanges to form complex "**extensor hoods**" or "**dorsal digital expansions**" (Fig. 7.99A). The tendons of the extensor digiti minimi, extensor indicis, and extensor pollicis brevis muscles join these hoods.

Each extensor hood is triangular, with:

- the apex attached to the distal phalanx,
- the central region attached to the middle phalanx (index, middle, ring, and little fingers) or proximal phalanx (thumb), and
- each corner of the base wrapped around the sides of the metacarpophalangeal joint—in the index, middle, ring, and little fingers, the corners of the hoods attach mainly to the deep transverse metacarpal ligaments; in the thumb, the hood is attached on each side to muscles.

In addition to other attachments, many of the intrinsic muscles of the hand insert into the free margin of the hood on each side. By inserting into the extensor hood, these intrinsic muscles are responsible for complex delicate movements of the digits that could not be accomplished with the long flexor and extensor tendons alone.

In the index, middle, ring, and little fingers, the lumbrical, interossei, and abductor digiti minimi muscles attach to the extensor hoods. In the thumb, the adductor pollicis and abductor pollicis brevis muscles insert into and anchor the extensor hood.

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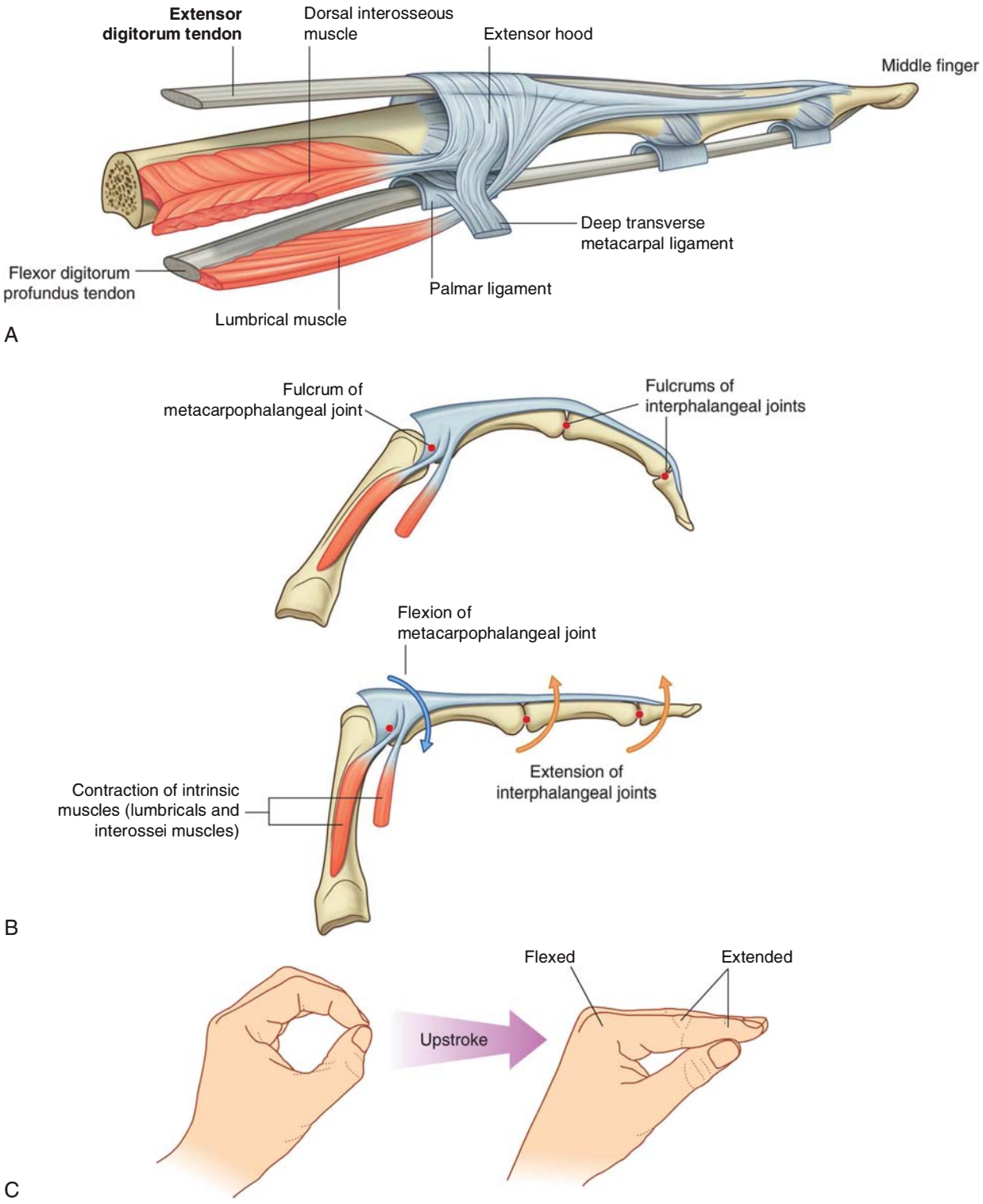


Fig. 7.99 Extensor hood. **A** and **B.** Middle finger, left hand. **C.** Function of extensor hoods and intrinsic muscles.

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Because force from the small intrinsic muscles of the hand is applied to the extensor hood distal to the fulcrum of the metacarpophalangeal joints, the muscles flex these joints (Fig. 7.99B). Simultaneously, the force is transferred dorsally through the hood to extend the interphalangeal joints. This ability to flex the metacarpophalangeal joints, while at the same time extending the interphalangeal joints, is entirely due to the intrinsic muscles of the hand working through the extensor hoods. This type of precision movement is used in the upstroke when writing a *t* (Fig. 7.99C).

Muscles

The intrinsic muscles of the hand are the palmaris brevis (described on p. 800; see Fig. 7.96), interossei, adductor pollicis, thenar, hypothenar, and lumbrical muscles (Figs. 7.100 to 7.104). Unlike the extrinsic muscles that originate in the forearm, insert in the hand, and function in forcefully gripping ("power grip") with the hand, the intrinsic muscles occur entirely in the hand and mainly execute precision movements ("precision grip") with the fingers and thumb.

All of the intrinsic muscles of the hand are innervated by the deep branch of the ulnar nerve except for the three thenar and two lateral lumbrical muscles, which are innervated by the median nerve. The intrinsic muscles are predominantly innervated by spinal cord segment T1 with a contribution from C8.

The interossei are muscles between and attached to the metacarpals (Figs. 7.100 and 7.101). They insert into the proximal phalanx of each digit and into the extensor hood and are divided into two groups, the dorsal interossei and the palmar interossei. All of the interossei are innervated by the deep branch of the ulnar nerve. Collectively, the interossei abduct and adduct the digits and contribute to the complex flexion and extension movements generated by the extensor hoods.

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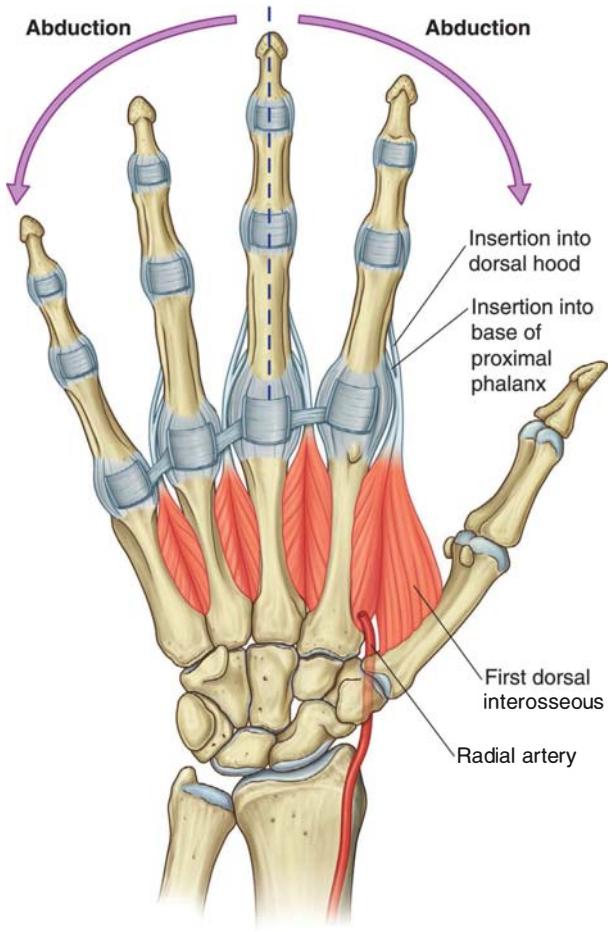


Fig. 7.100 Dorsal interossei (palmar view).

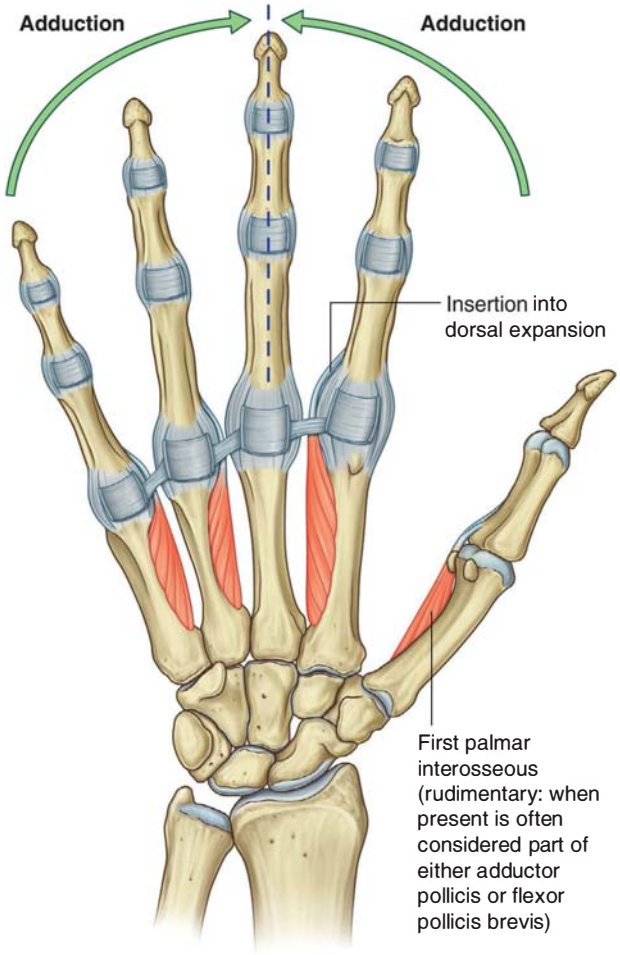


Fig. 7.101 Palmar interossei (palmar view).

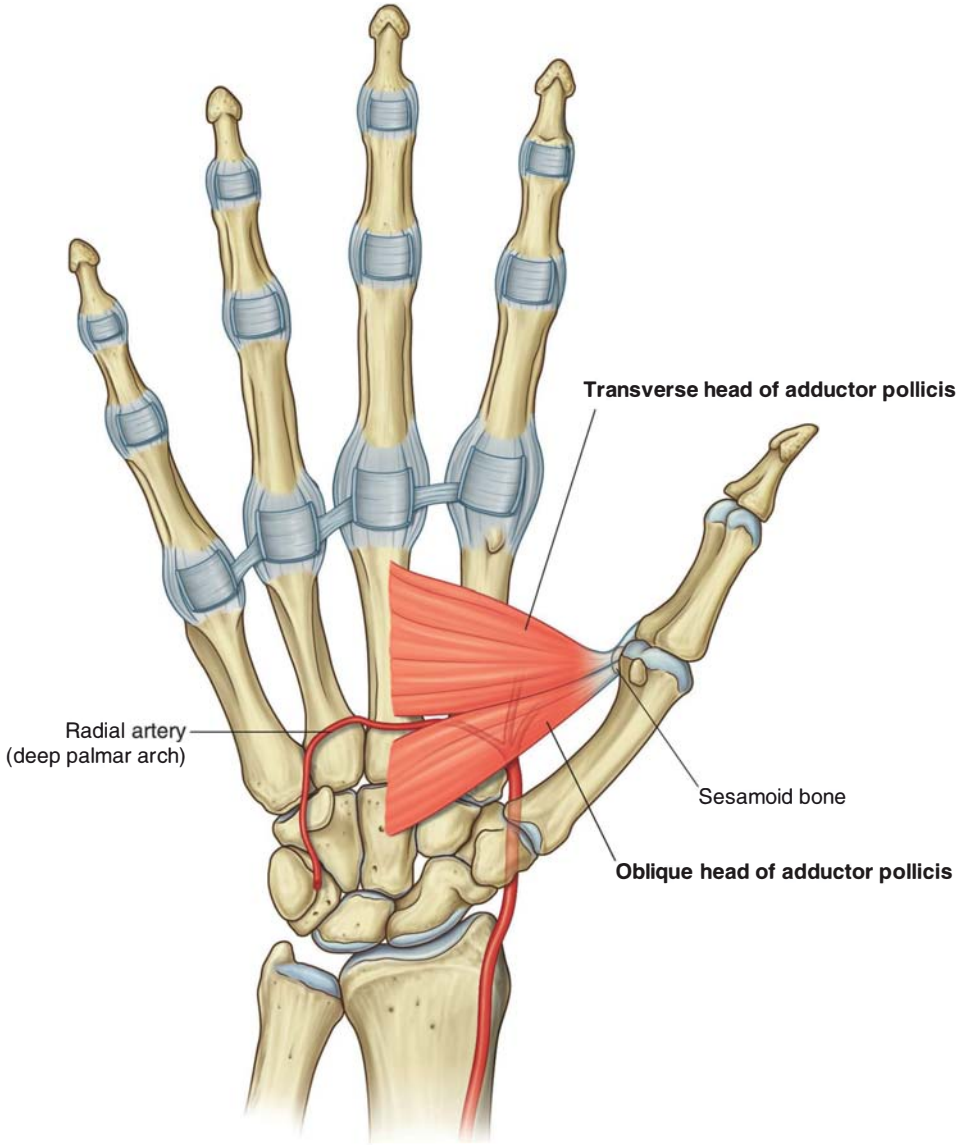


Fig. 7.102 Adductor pollicis.

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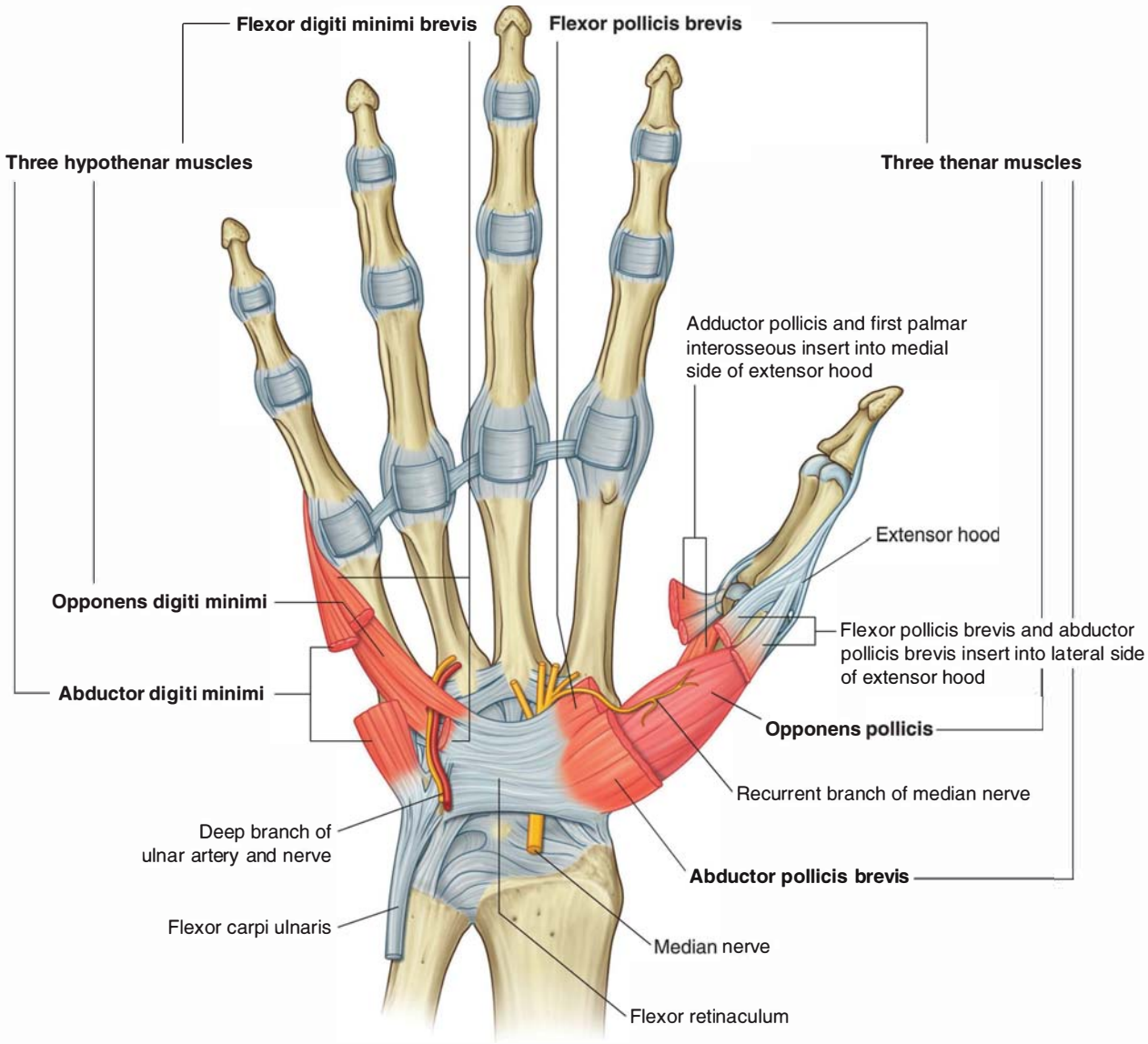


Fig. 7.103 Thenar and hypothenar muscles.

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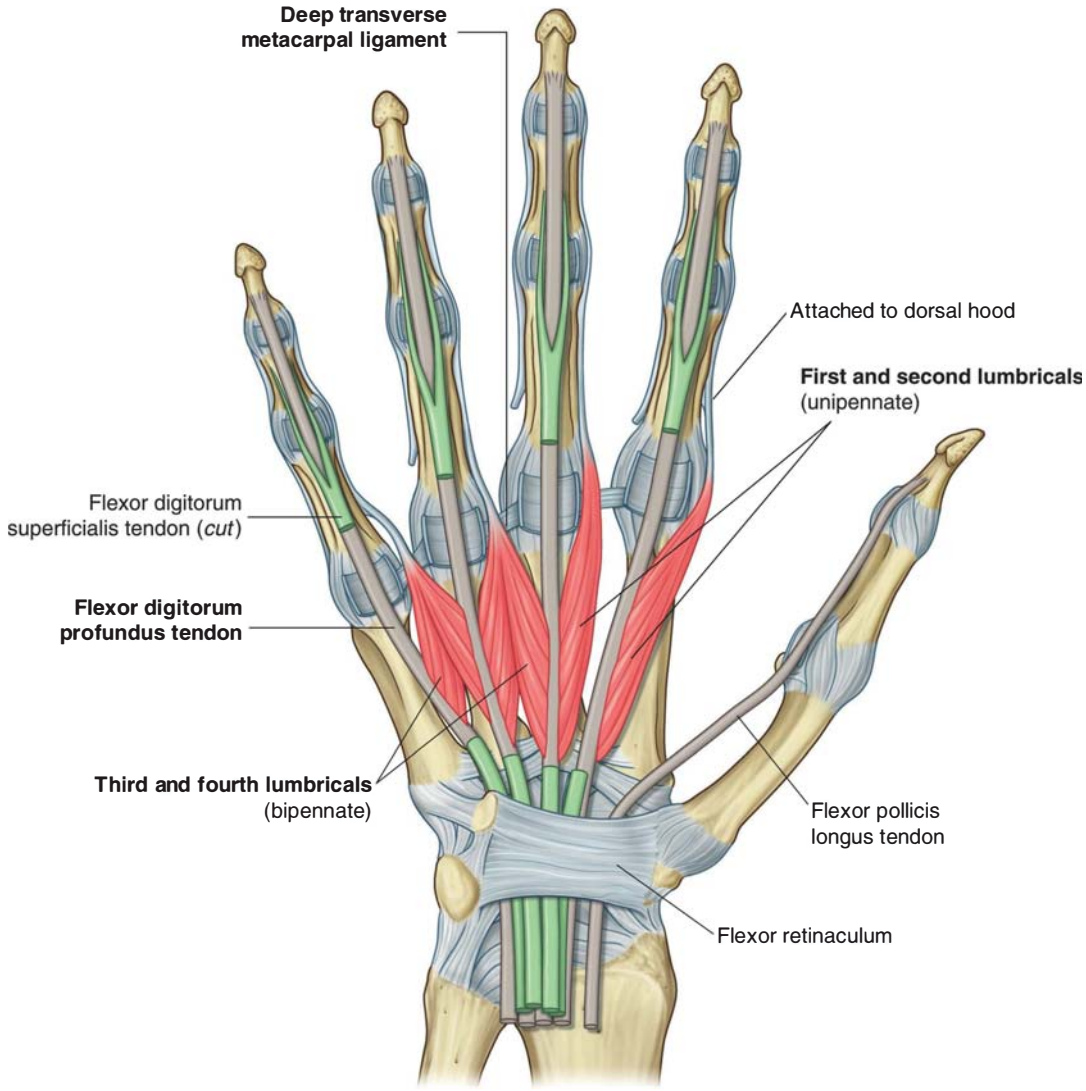


Fig. 7.104 Lumbrical muscles.

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Dorsal interossei

Dorsal interossei are the most dorsally situated of all of the intrinsic muscles and can be palpated through the skin on the dorsal aspect of the hand (Fig. 7.100). There are four bipennate dorsal interosseous muscles between, and attached to, the shafts of adjacent metacarpal bones (Fig. 7.100). Each muscle inserts both into the base of the proximal phalanx and into the extensor hood of its related digit.

The tendons of the dorsal interossei pass dorsal to the deep transverse metacarpal ligaments:

- The first dorsal interosseous muscle is the largest and inserts into the lateral side of the index finger.
- The second and third dorsal interossei insert into the lateral and medial sides, respectively, of the middle finger.
- The fourth dorsal interosseous muscle inserts into the medial side of the ring finger.

In addition to generating flexion and extension movements of the fingers through their attachments to the extensor hoods, the dorsal interossei are the major abductors of the index, middle, and ring fingers, at the metacarpophalangeal joints (Table 7.15).

The middle finger can abduct medially and laterally with respect to the long axis of the middle finger and consequently has a dorsal interosseous muscle on each side. The thumb and little finger have their own abductors in the

Table 7.15 Intrinsic muscles of the hand (spinal segments indicated in bold are the major segments innervating the muscle)

Muscle	Origin	Insertion	Innervation	Function
Palmaris brevis	Palmar aponeurosis and flexor retinaculum	Dermis of skin on the medial margin of the hand	Superficial branch of the ulnar nerve (C8, **T1**)	Improves grip
Dorsal interossei (four muscles)	Adjacent sides of metacarpals	Extensor hood and base of proximal phalanges of index, middle, and ring fingers	Deep branch of ulnar nerve (C8, **T1**)	Abduction of index, middle, and ring fingers at the metacarpophalangeal joints
Palmar interossei (three or four muscles)	Sides of metacarpals	Extensor hoods of the thumb, index, ring, and little fingers and the proximal phalanx of thumb	Deep branch of ulnar nerve (C8, **T1**)	Adduction of the thumb, index, ring, and little fingers at the metacarpophalangeal joints
Adductor pollicis	Transverse head—metacarpal III; oblique head—capitate and bases of metacarpals II and III	Base of proximal phalanx and extensor hood of thumb	Deep branch of ulnar nerve (C8, **T1**)	Adducts thumb
Lumbricals (four muscles)	Tendons of flexor digitorum profundus	Extensor hoods of index, ring, middle, and little fingers	Medial two by the deep branch of the ulnar nerve; lateral two by digital branches of the median nerve	Flex metacarpophalangeal joints while extending interphalangeal joints
THENAR MUSCLES				
Opponens pollicis	Tubercle of trapezium and flexor retinaculum	Lateral margin and adjacent palmar surface of metacarpal I	Recurrent branch of median nerve (C8, **T1**)	Medially rotates thumb
Abductor pollicis brevis	Tubercles of scaphoid and trapezium and adjacent flexor retinaculum	Proximal phalanx and extensor hood of thumb	Recurrent branch of median nerve (C8, **T1**)	Abducts thumb at metacarpophalangeal joint
Flexor pollicis brevis	Tubercle of the trapezium and flexor retinaculum	Proximal phalanx of the thumb	Recurrent branch of median nerve (C8, **T1**)	Flexes thumb at metacarpophalangeal joint
HYPOTHENAR MUSCLES				
Opponens digiti minimi	Hook of hamate and flexor retinaculum	Medial aspect of metacarpal V	Deep branch of ulnar nerve (C8, **T1**)	Laterally rotates metacarpal V
Abductor digiti minimi	Pisiform, the pisohamate ligament, and tendon of flexor carpi ulnaris	Proximal phalanx of little finger	Deep branch of ulnar nerve (C8, **T1**)	Abducts little finger at metacarpophalangeal joint
Flexor digiti minimi brevis	Hook of the hamate and flexor retinaculum	Proximal phalanx of little finger	Deep branch of ulnar nerve (C8, **T1**)	Flexes little finger at metacarpophalangeal joint

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thenar and hypothenar muscle groups, respectively, and therefore do not have dorsal interossei.

The radial artery passes between the two heads of the first dorsal interosseous muscle as it passes from the anatomical snuffbox on the posterolateral side of the wrist into the deep aspect of the palm.

Palmar interossei

The three (or four) **palmar interossei** are anterior to the dorsal interossei, and are unipennate muscles originating from the metacarpals of the digits with which each is associated (Fig. 7.101).

The first palmar interosseous muscle is rudimentary and often considered part of either the adductor pollicis or the flexor pollicis brevis. When present, it originates from the medial side of the palmar surface of metacarpal I and inserts into both the base of the proximal phalanx of the thumb and into the extensor hood. A sesamoid bone often occurs in the tendon attached to the base of the phalanx.

The second palmar interosseous muscle originates from the medial surface of metacarpal II and inserts into the medial side of the extensor hood of the index finger.

The third and fourth palmar interossei originate from the lateral surfaces of metacarpals IV and V and insert into the lateral sides of the respective extensor hoods.

Like the tendons of the dorsal interossei, the tendons of the palmar interossei pass dorsal to the deep transverse metacarpal ligaments.

The palmar interossei adduct the thumb, index, ring, and little fingers with respect to a long axis through the middle finger. The movements occur at the metacarpophalangeal joints. Because the muscles insert into the extensor hoods, they also produce complex flexion and extension movements of the digits (Table 7.15).

Adductor pollicis

The **adductor pollicis** is a large triangular muscle anterior to the plane of the interossei that crosses the palm (Fig. 7.102). It originates as two heads:

- a **transverse head** from the anterior aspect of the shaft of metacarpal III, and
- an **oblique head**, from the capitate and adjacent bases of metacarpals II and III.

The two heads converge laterally to form a tendon, which often contains a sesamoid bone, that inserts into both the medial side of the base of the proximal phalanx of the thumb and into the extensor hood.

The radial artery passes anteriorly and medially between the two heads of the muscle to enter the deep plane of the palm and form the deep palmar arch.

The adductor pollicis is a powerful adductor of the thumb and opposes the thumb to the rest of the digits in gripping (Table 7.15).

Thenar muscles

The three thenar muscles (the opponens pollicis, flexor pollicis brevis, and abductor pollicis brevis muscles) are associated with opposition of the thumb to the fingers and with delicate movements of the thumb (Fig. 7.103) and are responsible for the prominent swelling (**thenar eminence**) on the lateral side of the palm at the base of the thumb.

The thenar muscles are innervated by the recurrent branch of the median nerve.

Opponens pollicis

The **opponens pollicis** muscle is the largest of the thenar muscles and lies deep to the other two (Fig. 7.103). Originating from the tubercle of the trapezium and the adjacent flexor retinaculum, it inserts along the entire length of the lateral margin and adjacent lateral palmar surface of metacarpal I.

The opponens pollicis rotates and flexes metacarpal I on the trapezium, so bringing the pad of the thumb into a position facing the pads of the fingers (Table 7.15).

Abductor pollicis brevis

The **abductor pollicis brevis** muscle overlies the opponens pollicis and is proximal to the flexor pollicis brevis muscle (Fig. 7.103). It originates from the tubercles of the scaphoid and trapezium and from the adjacent flexor retinaculum, and inserts into the lateral side of the base of the proximal phalanx of the thumb and into the extensor hood.

The abductor pollicis brevis abducts the thumb, principally at the metacarpophalangeal joint. Its action is most apparent when the thumb is maximally abducted and the proximal phalanx is moved out of line with the long axis of the metacarpal bone (Table 7.15).

Flexor pollicis brevis

The **flexor pollicis brevis** muscle is distal to the abductor pollicis brevis (Fig. 7.103). It originates mainly from the tubercle of the trapezium and adjacent flexor retinaculum, but it may also have deeper attachments to other carpal bones and associated ligaments. It inserts into the lateral side of the base of the proximal phalanx of the thumb. The tendon often contains a sesamoid bone.

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The flexor pollicis brevis flexes the metacarpophalangeal joint of the thumb (Table 7.15).

Hypothenar muscles

The hypothenar muscles (the opponens digiti minimi, abductor digiti minimi, and flexor digiti minimi brevis) contribute to the swelling (**hypothenar eminence**) on the medial side of the palm at the base of the little finger (Fig. 7.103). The hypothenar muscles are similar to the thenar muscles in name and in organization.

Unlike the thenar muscles, the hypothenar muscles are innervated by the deep branch of the ulnar nerve and not by the recurrent branch of the median nerve.

Opponens digiti minimi

The **opponens digiti minimi** muscle lies deep to the other two hypothenar muscles (Fig. 7.103). It originates from the hook of the hamate and from the adjacent flexor retinaculum and it inserts into the medial margin and palmar surface of metacarpal V. Its base is penetrated by the deep branches of the ulnar nerve and ulnar artery.

The opponens digiti minimi rotates metacarpal V toward the palm; however, because of the simple shape of the carpometacarpal joint and the presence of a deep transverse metacarpal ligament, which attaches the head of metacarpal V to that of the ring finger, the movement is much less dramatic than that of the thumb (Table 7.15).

Abductor digiti minimi

The **abductor digiti minimi** muscle overlies the opponens digiti minimi (Fig. 7.103). It originates from the pisiform bone, the pisohamate ligament, and the tendon of the flexor carpi ulnaris, and inserts into the medial side of the base of the proximal phalanx of the little finger and into the extensor hood.

The abductor digiti minimi is the principal abductor of the little finger (Table 7.15).

Flexor digiti minimi brevis

The **flexor digiti minimi brevis** muscle is lateral to the abductor digiti minimi (Fig. 7.103). It originates from the hook of the hamate bone and the adjacent flexor retinaculum and inserts with the abductor digiti minimi muscle into the medial side of the base of the proximal phalanx of the little finger.

The flexor digiti minimi brevis flexes the metacarpophalangeal joint.

Lumbrical muscles

There are four lumbrical (worm-like) muscles, each of which is associated with one of the fingers. The muscles originate from the tendons of the flexor digitorum profundus in the palm:

- The medial two lumbricals are bipennate and originate from the flexor digitorum profundus tendons associated with the middle and ring fingers and the ring and little fingers, respectively.
- The lateral two lumbricals are unipennate muscles, originating from the flexor digitorum profundus tendons associated with the index and middle fingers, respectively.

The lumbricals pass dorsally around the lateral side of each finger, and insert into the extensor hood (Fig. 7.104). The tendons of the muscles are anterior to the deep transverse metacarpal ligaments.

The lumbricals are unique because they link flexor tendons with extensor tendons. Through their insertion into the extensor hoods, they participate in flexing the metacarpophalangeal joints and extending the interphalangeal joints.

The medial two lumbricals are innervated by the deep branch of the ulnar nerve; the lateral two lumbricals are innervated by digital branches of the median nerve (Table 7.15).

Arteries and veins

The blood supply to the hand is by the radial and ulnar arteries, which form two interconnected vascular arches (superficial and deep) in the palm (Fig. 7.105). Vessels to the digits, muscles, and joints originate from the two arches and the parent arteries:

- The radial artery contributes substantially to the supply of the thumb and the lateral side of the index finger.

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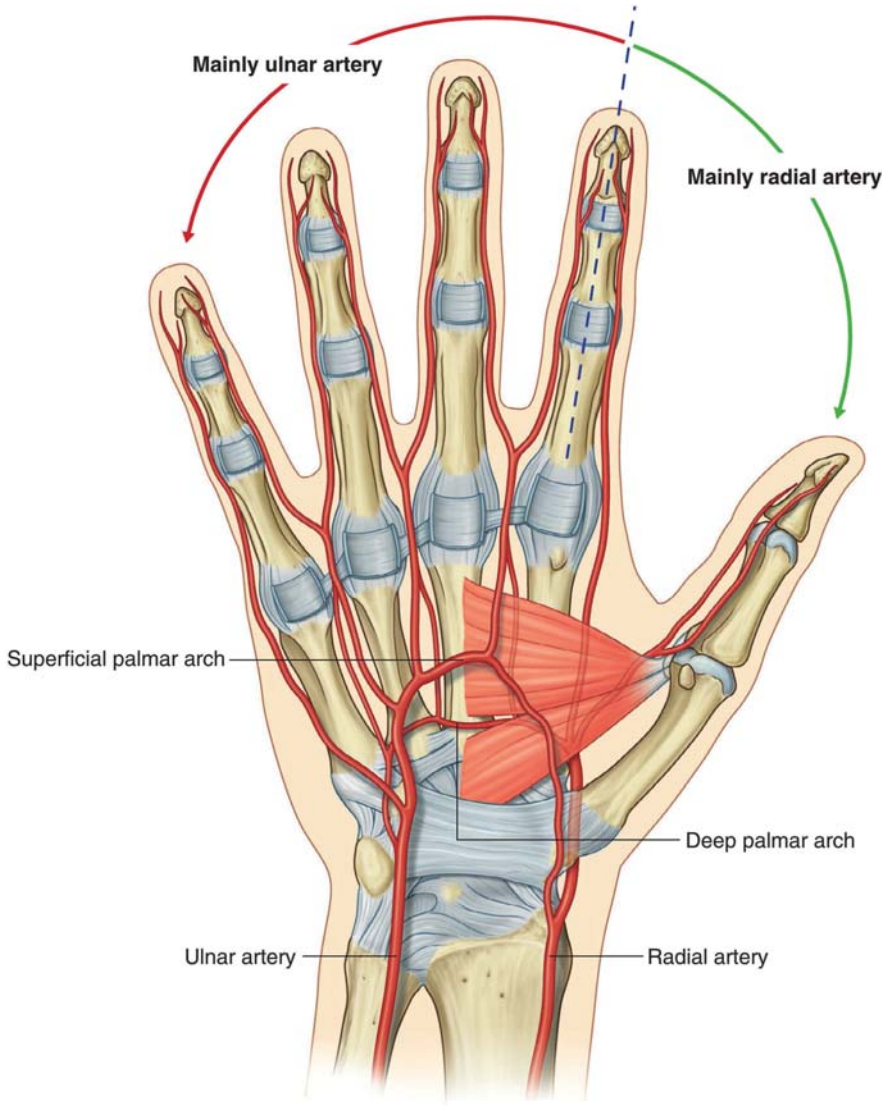


Fig. 7.105 Arterial supply of the hand.

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- The remaining digits and the medial side of the index finger are supplied mainly by the ulnar artery.

Ulnar artery and superficial palmar arch

The **ulnar artery** and ulnar nerve enter the hand on the medial side of the wrist (Fig. 7.106). The vessel lies between the palmaris brevis and the flexor retinaculum and is lateral to the ulnar nerve and the pisiform bone. Distally, the ulnar artery is medial to the hook of the hamate bone and then swings laterally across the palm, forming the **superficial palmar arch**, which is superficial to the long flexor tendons of the digits and just deep to the palmar aponeurosis. On the lateral side of the palm, the arch communicates with a palmar branch of the radial artery.

One branch of the ulnar artery in the hand is the **deep palmar branch** (Figs. 7.105 and 7.106), which arises from the medial aspect of the ulnar artery, just distal to the pisiform, and penetrates the origin of the hypothenar muscles. It curves medially around the hook of the hamate to access the deep plane of the palm and to anastomose with the deep palmar arch derived from the radial artery.

Branches from the superficial palmar arch include:

- a palmar digital artery to the medial side of the little finger, and
- three large, **common palmar digital arteries**, which ultimately provide the principal blood supply to the lateral side of the little finger, both sides of the ring and middle fingers, and the medial side of the index finger (Fig. 7.106); they are joined by palmar metacarpal arteries from the deep palmar arch before bifurcating into the **proper palmar digital arteries**, which enter the fingers.

Radial artery and deep palmar arch

The **radial artery** curves around the lateral side of the wrist and passes over the floor of the anatomical snuffbox and into the deep plane of the palm by penetrating anteriorly through the back of the hand (Figs. 7.105 and 7.107). It passes between the two heads of the first dorsal interosseous muscle and then between the two heads of the

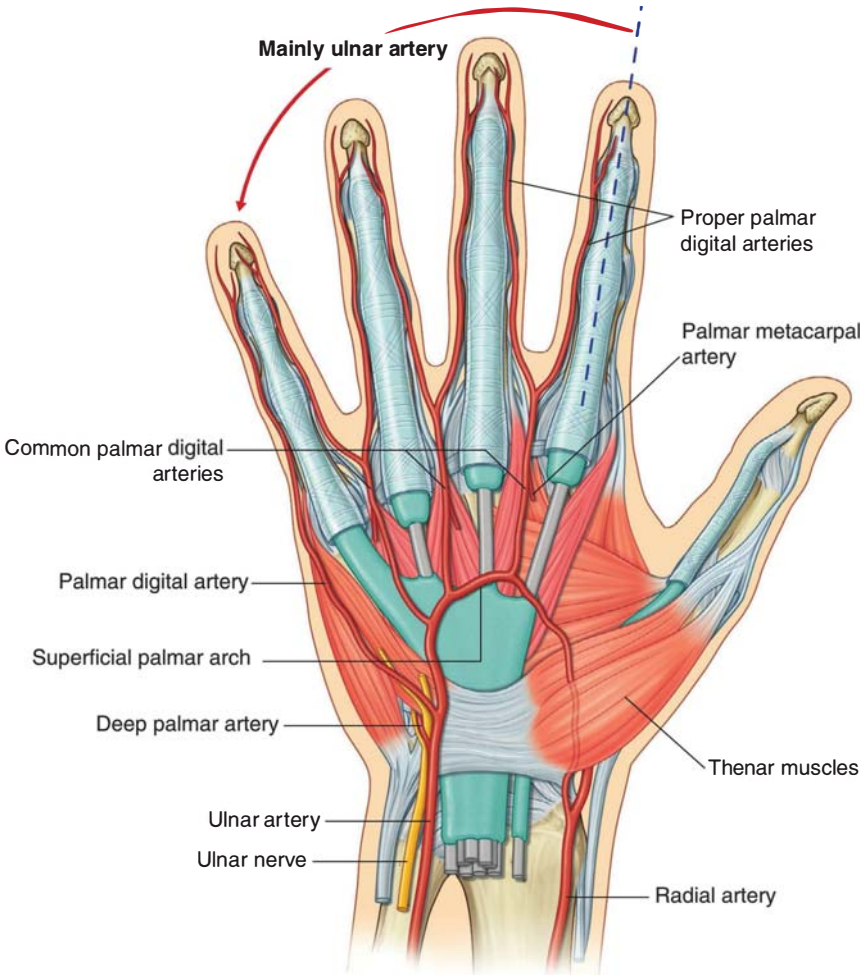


Fig. 7.106 Superficial palmar arch.

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adductor pollicis to access the deep plane of the palm and form the deep palmar arch.

The **deep palmar arch** passes medially through the palm between the metacarpal bones and the long flexor tendons of the digits. On the medial side of the palm, it communicates with the deep palmar branch of the ulnar artery (Figs. 7.105 and 7.107).

Before penetrating the back of the hand, the radial artery gives rise to two vessels:

- a **dorsal carpal branch**, which passes medially as the **dorsal carpal arch**, across the wrist and gives rise to **three dorsal metacarpal arteries**, which subsequently divide to become small dorsal digital arteries, which enter the fingers; and
- the **first dorsal metacarpal artery**, which supplies adjacent sides of the index finger and thumb.

Two vessels, the **princeps pollicis artery** and the **radialis indicis artery**, arise from the radial artery in the plane between the first dorsal interosseous and adductor pollicis. The princeps pollicis artery is the major blood supply to the thumb, and the radialis indicis artery supplies the lateral side of the index finger.

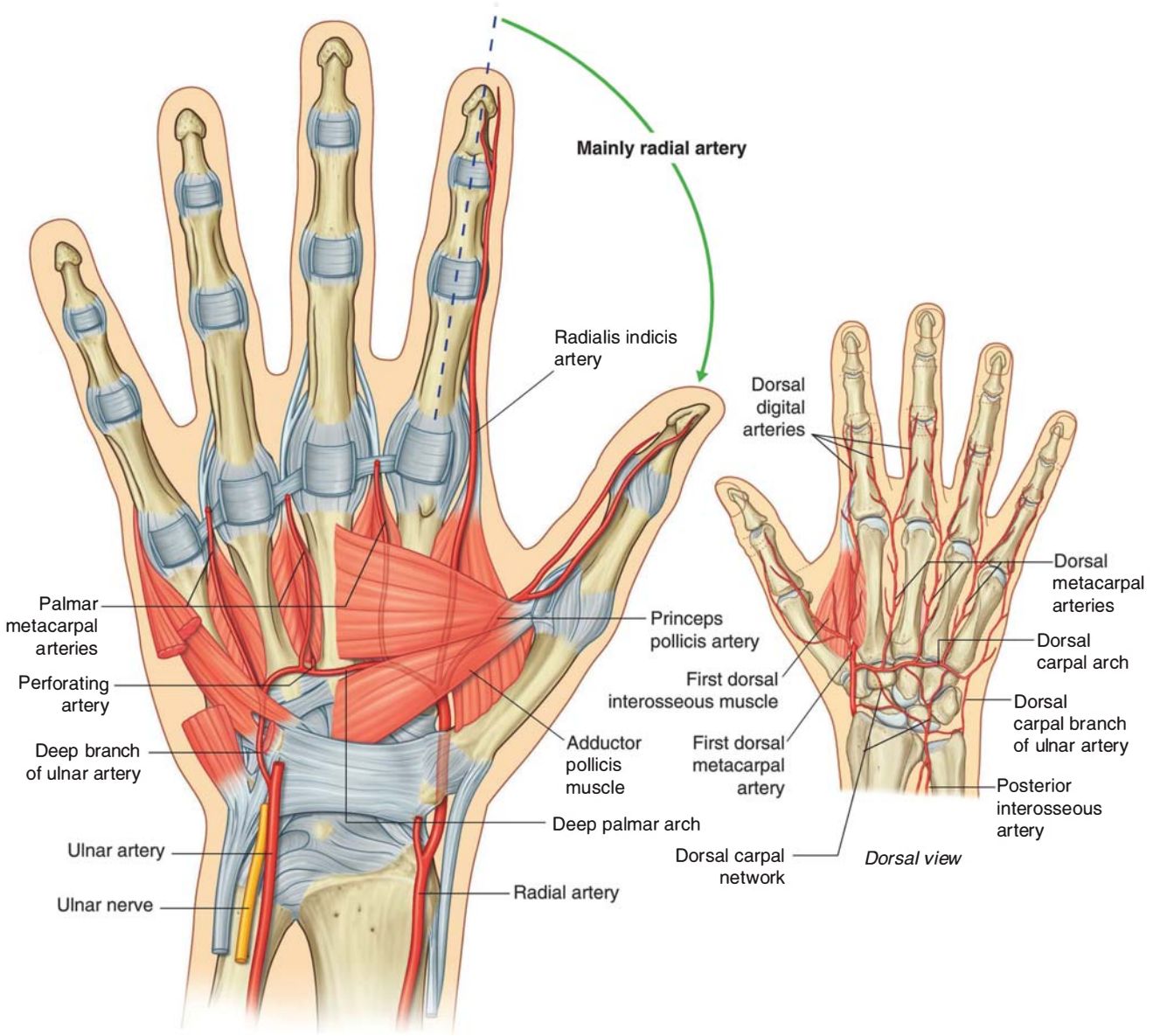


Fig. 7.107 Deep palmar arch.

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The deep palmar arch gives rise to:

- three **palmar metacarpal arteries**, which join the common palmar digital arteries from the superficial palmar arch; and
- three **perforating branches**, which pass posteriorly between the heads of origin of the dorsal interossei to anastomose with the dorsal metacarpal arteries from the dorsal carpal arch.

In the clinic

Allen's test

To test for adequate anastomoses between the radial and ulnar arteries, compress both the radial and ulnar arteries at the wrist, then release pressure from one or the other, and determine the filling pattern of the hand. If there is little connection between the deep and superficial palmar arteries, only the thumb and lateral side of the index finger will fill with blood (become red) when pressure on the radial artery alone is released.

Veins

As generally found in the upper limb, the hand contains interconnected networks of deep and superficial veins. The deep veins follow the arteries; the superficial veins drain into a dorsal venous network on the back of the hand over the metacarpal bones (Fig. 7.108).

The cephalic vein originates from the lateral side of the dorsal venous network and passes over the anatomical snuffbox into the forearm.

The basilic vein originates from the medial side of the dorsal venous network and passes into the dorsomedial aspect of the forearm.

In the clinic

Venipuncture

In many patients, venous access is necessary for obtaining blood for laboratory testing and administering fluid and intravenous drugs. The ideal sites for venous access are typically in the cubital fossa and in the cephalic vein adjacent to the anatomical snuffbox. The veins are simply distended by use of a tourniquet. A tourniquet should be applied enough to allow the veins to become prominent. For straightforward blood tests the antecubital vein is usually the preferred site, and although it may not always be visible, it is easily palpated. The cephalic vein is generally the preferred site for a short-term intravenous cannula.

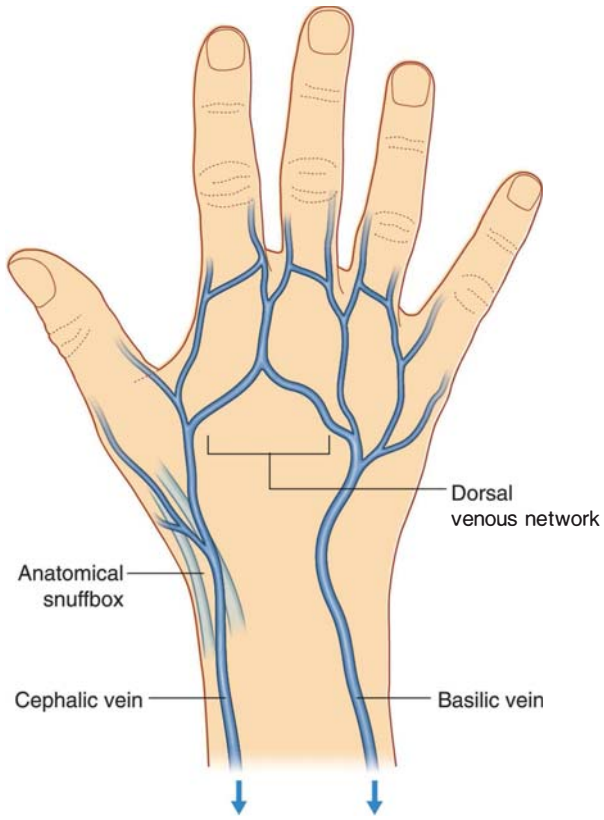


Fig. 7.108 Dorsal venous arch of the hand.

Nerves

The hand is supplied by the ulnar, median, and radial nerves (Figs. 7.109 to 7.111). All three nerves contribute to cutaneous or general sensory innervation. The ulnar nerve innervates all intrinsic muscles of the hand except for the three thenar muscles and the two lateral lumbricals, which are innervated by the median nerve. The radial nerve only innervates skin on the dorsolateral side of the hand.

Ulnar nerve

The ulnar nerve enters the hand lateral to the pisiform and posteromedially to the ulnar artery (Fig. 7.109). Immediately distal to the pisiform, it divides into a deep branch, which is mainly motor, and a superficial branch, which is mainly sensory.

The **deep branch** of the ulnar nerve passes with the deep branch of the ulnar artery (Fig. 7.109). It penetrates and supplies the hypothenar muscles to reach the deep aspect of the palm, arches laterally across the palm, deep to the long flexors of the digits, and supplies the interossei, the adductor pollicis, and the two medial lumbricals. In addition, the deep branch of the ulnar nerve contributes small articular branches to the wrist joint.

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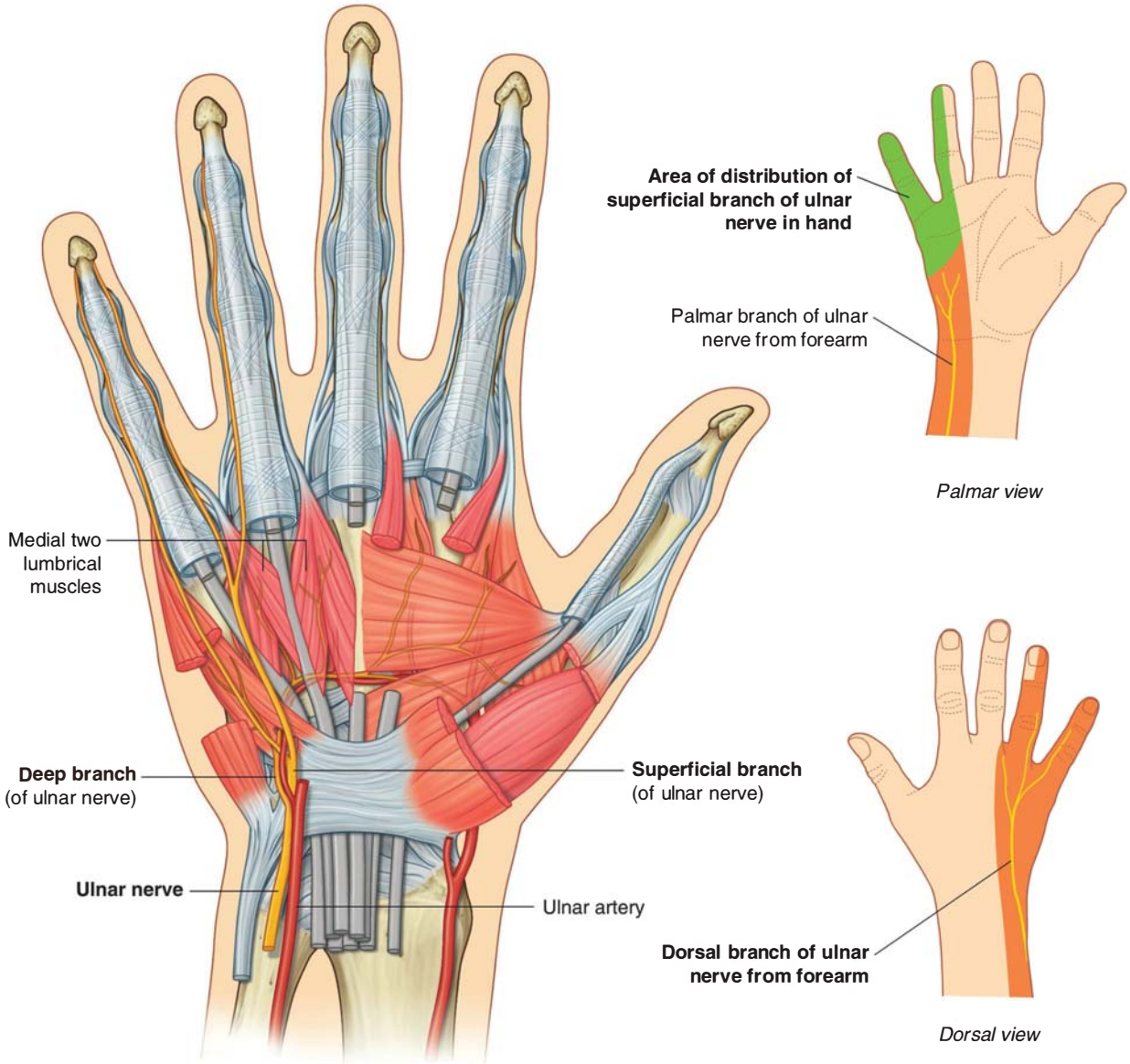


Fig. 7.109 Ulnar nerve in the hand.

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In the clinic

Ulnar nerve injury

The ulnar nerve is most commonly injured at two sites: the elbow and the wrist.

- At the elbow, the nerve lies posterior to the medial epicondyle.
- At the wrist, the ulnar nerve passes superficial to the flexor retinaculum and lies lateral to the pisiform bone.

Ulnar nerve lesions are characterized by "clawing" of the hand, in which the metacarpophalangeal joints of the fingers are hyperextended and the interphalangeal joints are flexed because the function of most of the intrinsic muscles of the hand is lost (Fig. 7.110).

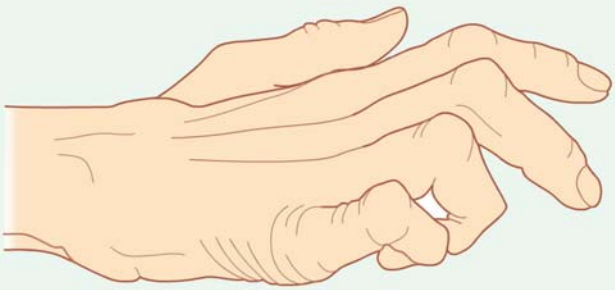


Fig. 7.110 Typical appearance of a "clawed hand" due to a lesion of the ulnar nerve.

Clawing is most pronounced in the medial fingers because the function of all intrinsic muscles of these digits is lost while in the lateral two digits, the lumbricals are innervated by the median nerve. Function of the adductor pollicis muscle is also lost.

In lesions of the ulnar nerve at the elbow, function of the flexor carpi ulnaris muscle and flexor digitorum profundus to the medial two digits is lost as well. Clawing of the hand, particularly of the little and ring fingers, is worse with lesions of the ulnar nerve at the wrist than at the elbow because interruption of the nerve at the elbow paralyzes the ulnar half of the flexor digitorum profundus, which leads to lack of flexion at the distal interphalangeal joints in these fingers.

Ulnar nerve lesions at the elbow and wrist result in impaired sensory innervation on the palmar aspect of the medial one and one-half digits.

Damage to the ulnar nerve at the wrist or at a site proximal to the wrist can be distinguished by evaluating the status of function of the **dorsal branch** (cutaneous) of the ulnar nerve, which originates in distal regions of the forearm. This branch innervates skin over the dorsal surface of the hand on the medial side.

As the deep branch of the ulnar nerve passes across the palm, it lies in a fibro-osseous tunnel (Guyon's canal) between the hook of the hamate and the flexor tendons. Occasionally, small outpouchings of synovial membrane (ganglia) from the joints of the carpus compress the nerve within this canal, producing sensory and motor symptoms.

The superficial branch of the ulnar nerve innervates the palmaris brevis muscle and continues across the palm to supply skin on the palmar surface of the little finger and the medial half of the ring finger (Fig. 7.109).

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Median nerve

The median nerve is the most important sensory nerve in the hand because it innervates skin on the thumb, index and middle fingers, and lateral side of the ring finger (Fig. 7.111). The nervous system, using touch, gathers information about the environment from this area, particularly from the skin on the thumb and index finger. In addition, sensory information from the lateral three and one-half digits enables the fingers to be positioned with the appropriate amount of force when using precision grip.

The median nerve also innervates the thenar muscles that are responsible for opposition of the thumb to the other digits.

The median nerve enters the hand by passing through the carpal tunnel and divides into a recurrent branch and palmar digital branches (Fig. 7.111).

The **recurrent branch** of the median nerve innervates the three thenar muscles. Originating from the lateral side of the median nerve near the distal margin of the flexor retinaculum, it curves around the margin of the retinaculum and passes proximally over the flexor pollicis brevis muscle. The recurrent branch then passes between the flexor pollicis brevis and abductor pollicis brevis to end in the opponens pollicis.

The **palmar digital nerves** cross the palm deep to the palmar aponeurosis and the superficial palmar arch and enter the digits. They innervate skin on the palmar surfaces of the lateral three and one-half digits and cutaneous regions over the dorsal aspects of the distal phalanges (nail beds) of the same digits. In addition to skin, the digital nerves supply the lateral two lumbrical muscles.

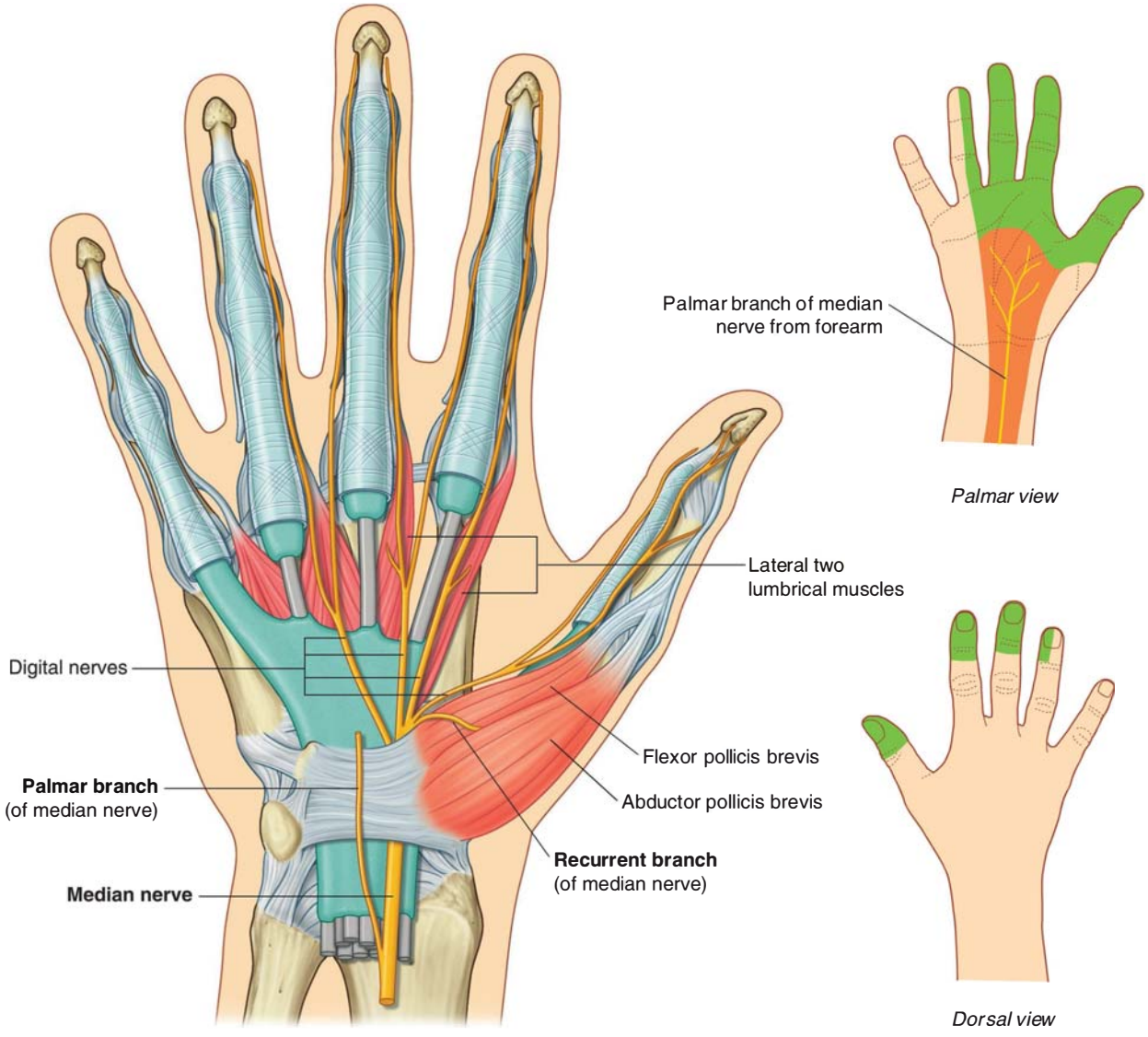


Fig. 7.111 Median nerve in the hand.

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Superficial branch of the radial nerve

The only part of the radial nerve that enters the hand is the superficial branch (Fig. 7.112). It enters the hand by passing over the anatomical snuffbox on the dorsolateral side of the wrist. Terminal branches of the nerve can be palpated or "rolled" against the tendon of the extensor pollicis longus as they cross the anatomical snuffbox.

The superficial branch of the radial nerve innervates skin over the dorsolateral aspect of the palm and the dorsal aspects of the lateral three and one-half digits distally to approximately the terminal interphalangeal joints.

In the clinic

Radial nerve injury

Around the elbow joint the radial nerve divides into its two terminal branches—the superficial branch and the deep branch.

The most common radial nerve injury is damage to the nerve in the radial groove of the humerus, which produces a global paralysis of the muscles of the posterior compartment, resulting in wrist drop. Radial nerve damage can result from fracture of the shaft of the humerus as the radial nerve spirals around in the radial groove. The typical injury produces reduction of sensation in the cutaneous distribution, predominantly over the posterior aspect of the hand. Severing the posterior interosseous nerve (continuation of deep branch of radial nerve) may paralyze the muscles of the posterior compartment of the forearm, but the nerve supply is variable. Typically, the patient may not be able to extend the fingers.

The distal branches of the superficial branch of the radial nerve can be readily palpated as "cords" passing over the tendon of the extensor pollicis longus in the anatomical snuffbox. Damage to these branches is of little consequence because they supply only a small area of skin.

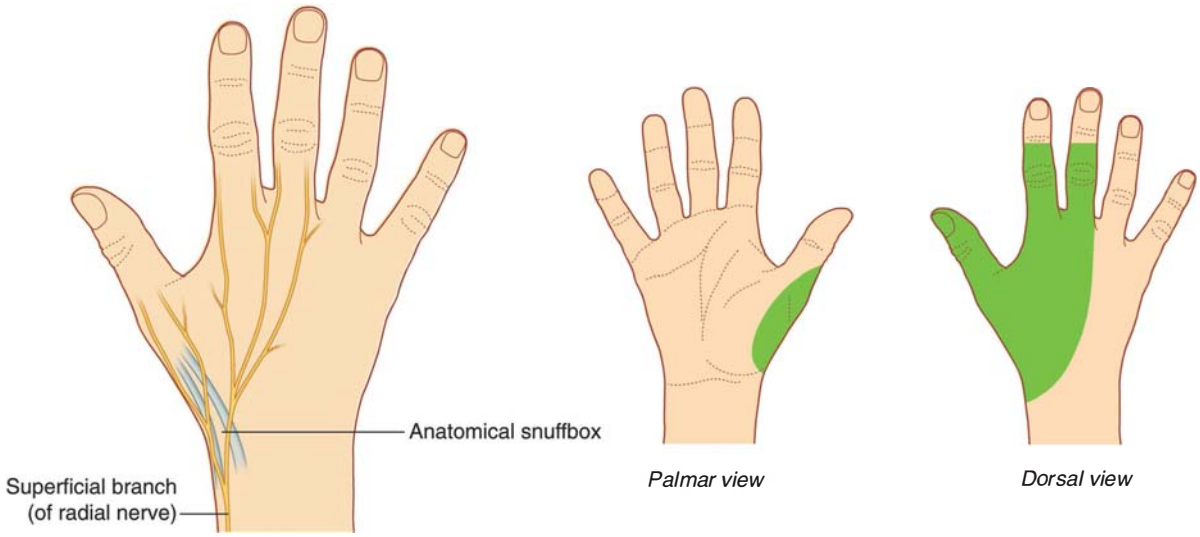


Fig. 7.112 Radial nerve in the hand.

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Surface anatomy

Upper limb surface anatomy

Tendons, muscles, and bony landmarks in the upper limb are used to locate major arteries, veins, and nerves. Asking patients to maneuver their upper limbs in specific ways is essential for performing neurological examinations.

- Tendons are used to test reflexes associated with specific spinal cord segments.
- Vessels are used clinically as points of entry into the vascular system (for collecting blood and administering pharmaceuticals and nutrients), and for taking blood pressure and pulses.
- Nerves can become entrapped or be damaged in regions where they are related to bone or pass through confined spaces.

Bony landmarks and muscles of the posterior scapular region

The medial border, inferior angle, and part of the lateral border of the scapula can be palpated on a patient, as can the spine and acromion. The superior border and angle of the scapula are deep to soft tissue and are not readily palpable. The supraspinatus and infraspinatus muscles can be palpated above and below the spine, respectively (Fig. 7.113).

The trapezius muscle is responsible for the smooth contour of the lateral side of the neck and over the superior aspect of the shoulder.

The deltoid muscles form the muscular eminence inferior to the acromion and around the glenohumeral joint. The axillary nerve passes posteriorly around the surgical neck of the humerus deep to the deltoid muscle.

The latissimus dorsi muscle forms much of the muscle mass underlying the posterior axillary skin fold extending obliquely upward from the trunk to the arm. The teres major muscle passes from the inferior angle of the scapula to the upper humerus and contributes to this posterior axillary skin fold laterally.

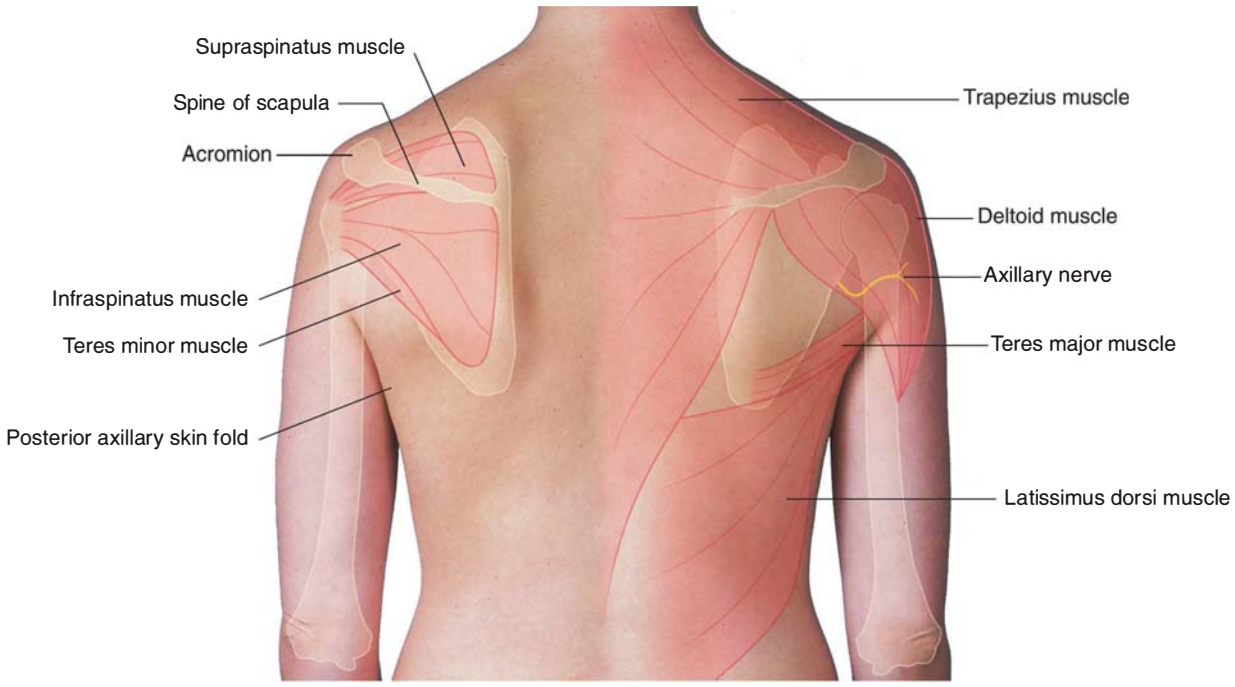


Fig. 7.113 Bony landmarks and muscles of the posterior scapular region. Posterior view of shoulder and back.

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Visualizing the axilla and locating contents and related structures

The axillary inlet and outlet and walls of the axilla can be established using skin folds and palpable bony landmarks (Fig. 7.114).

- The anterior margin of the axillary inlet is the clavicle, which can be palpated along its entire length. The lateral limit of the axillary inlet is approximated by the tip of the coracoid process, which is palpable immediately below the lateral third of the clavicle and deep to the medial margin of the deltoid muscle.
- The inferior margin of the anterior axillary wall is the anterior axillary skin fold, which overlies the lower margin of the pectoralis major muscle.
- The inferior margin of the posterior axillary wall is the posterior axillary skin fold, which overlies the margins of the teres major muscle laterally and latissimus dorsi muscle medially.
- The medial wall of the axilla is the upper part of the serratus anterior muscle overlying the thoracic wall.

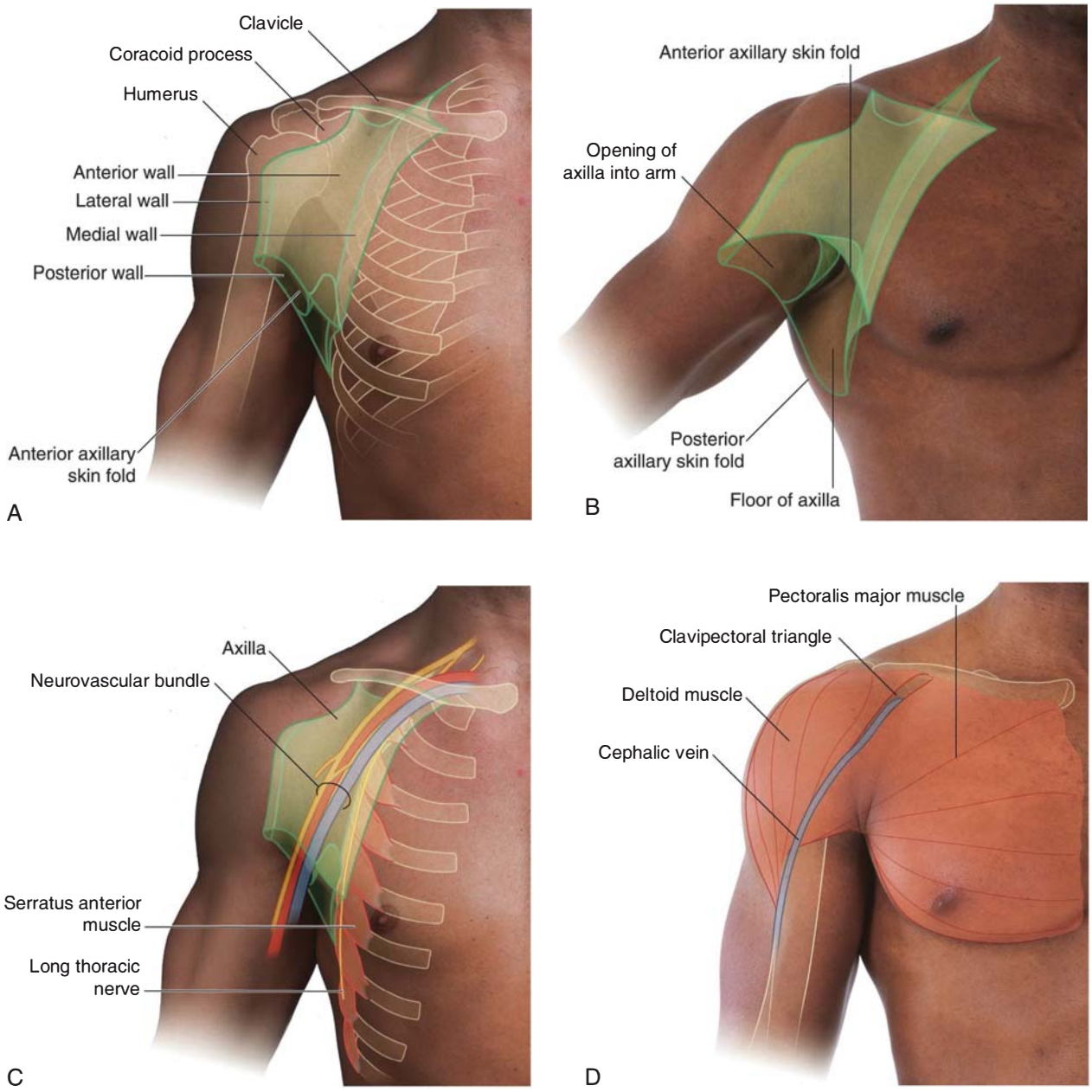


Fig. 7.114 Visualizing the axilla and locating its contents and related structures. **A.** Anterior shoulder showing folds and walls of the axilla. **B.** Anterior shoulder showing outlet and floor of the axilla. **C.** Anterior view showing the axillary neurovascular bundle and long thoracic nerve. **D.** Anterior view of the shoulder showing the clavipectoral triangle with the cephalic vein.

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The long thoracic nerve passes vertically out of the axilla and down the lateral surface of the serratus anterior muscle in a position just anterior to the posterior axillary skin fold.

- The lateral boundary of the axilla is the humerus.
- The floor of the axilla is the dome of skin between the posterior and anterior axillary skin folds.

Major vessels, nerves, and lymphatics travel between the upper limb and the trunk by passing through the axilla.

The axillary artery, axillary vein, and components of the brachial plexus pass through the axilla and into the arm by traveling lateral to the dome of skin that forms the floor. This neurovascular bundle can be palpated by placing a hand into this dome of skin and pressing laterally against the humerus.

The cephalic vein travels in superficial fascia in the cleft between the deltoid muscle and the pectoralis major muscle and penetrates deep fascia in the clavipectoral triangle to join with the axillary vein.

Locating the brachial artery in the arm

The brachial artery is on the medial side of the arm in the cleft between the biceps brachii and triceps brachii muscles (Fig. 7.115). The median nerve courses with the brachial artery, whereas the ulnar nerve deviates posteriorly from the vessel in distal regions.

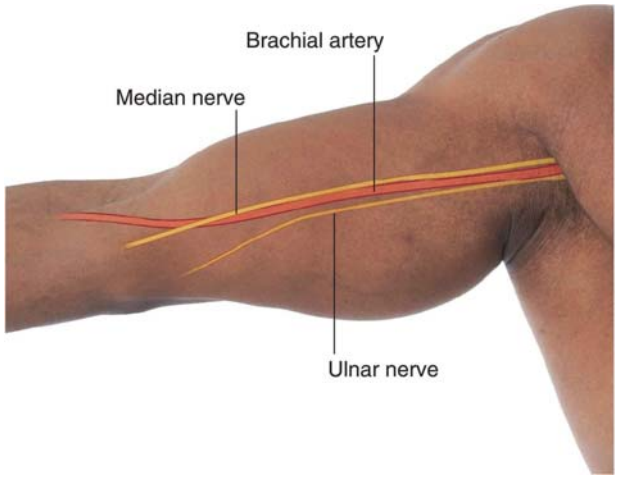


Fig. 7.115 Locating the brachial artery in the arm (medial view of arm with brachial artery, median nerve, and ulnar nerve).

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The triceps brachii tendon and position of the radial nerve

The triceps brachii muscle forms the soft tissue mass posterior to the humerus, and the tendon inserts onto the olecranon of the ulna, which is readily palpable and forms the bony protuberance at the "tip" of the elbow (Fig. 7.116).

The brachioradialis muscle is also visible as a muscular bulge on the lateral aspect of the arm. It is particularly prominent when the forearm is half pronated, flexed at the elbow against resistance, and viewed anteriorly.

The radial nerve in the distal arm emerges from behind the humerus to lie deep to the brachioradialis muscle.

Cubital fossa (anterior view)

The cubital fossa lies anterior to the elbow joint and contains the biceps brachii tendon, the brachial artery, and the median nerve (Fig. 7.117).

The base of the cubital fossa is an imaginary line between the readily palpable medial and lateral epicondyles of the humerus. The lateral and medial borders are formed by the brachioradialis and pronator teres muscles, respectively. The margin of the brachioradialis can be found by asking a subject to flex the semipronated forearm against resistance. The margin of the pronator teres can be estimated by an oblique line extending between the medial epicondyle and the midpoint along the length of the lateral surface of the forearm. The approximate apex of the cubital fossa is where this line meets the margin of the brachioradialis muscle.

Contents of the cubital fossa, from lateral to medial, are the tendon of the biceps brachii, the brachial artery, and the median nerve. The tendon of the biceps brachii is easily palpable. Often the cephalic, basilic, and median cubital veins are visible in the subcutaneous fascia overlying the cubital fossa.

The ulnar nerve passes behind the medial epicondyle of the humerus and can be "rolled" here against the bone.

The radial nerve travels into the forearm deep to the margin of the brachioradialis muscle anterior to the elbow joint.

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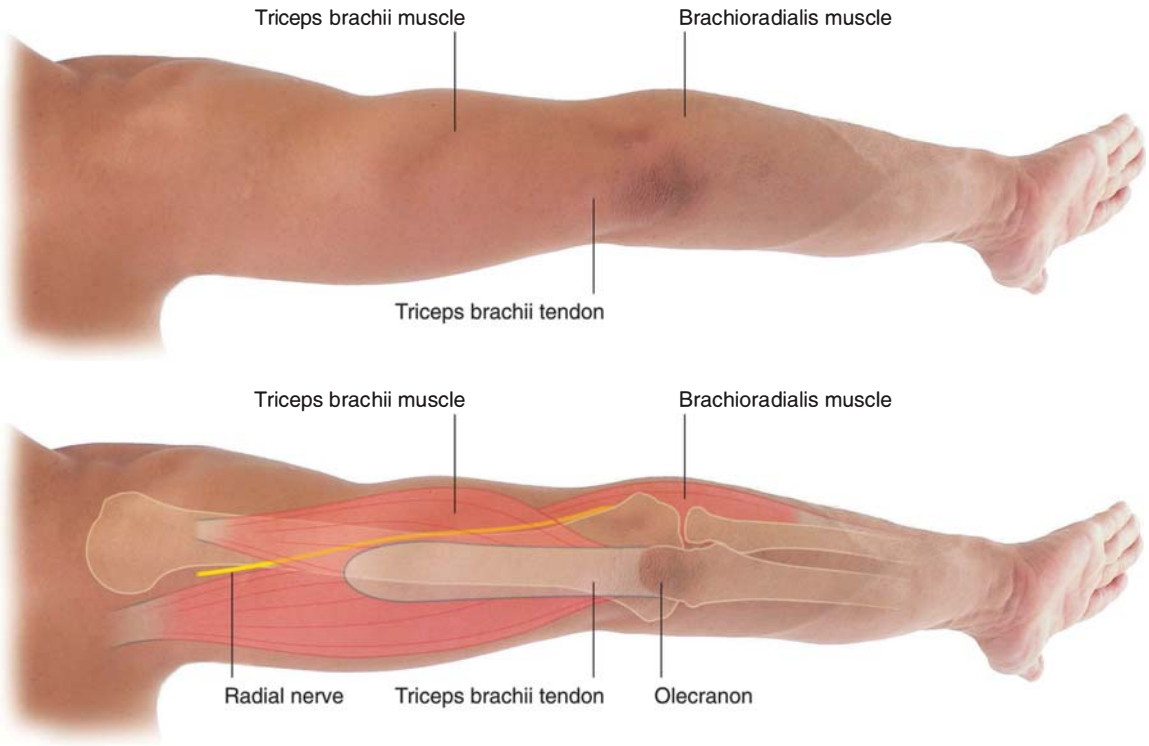


Fig. 7.116 Triceps brachii tendon and position of the radial nerve (posterior view of arm).

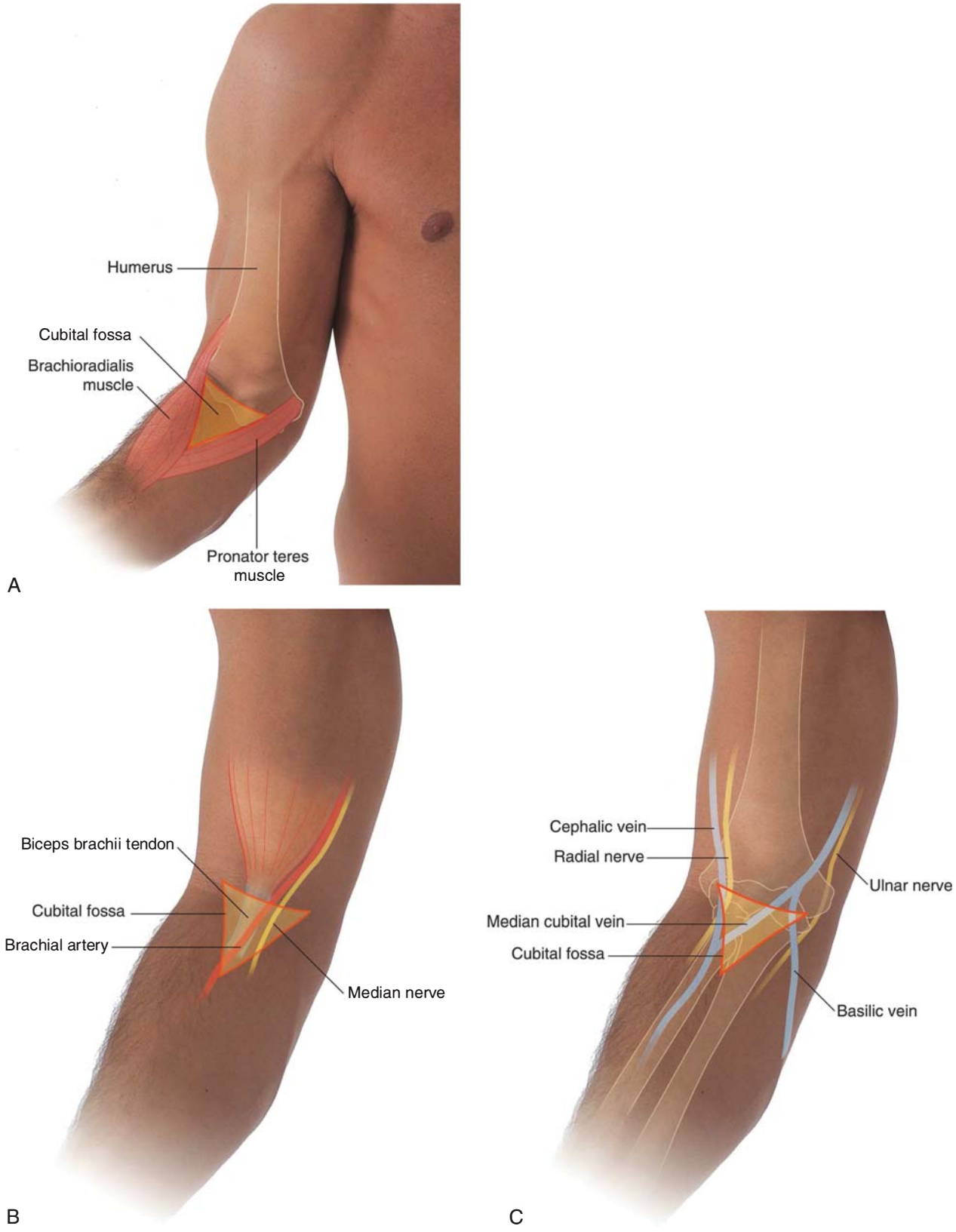


Fig. 7.117 Cubital fossa (anterior view). **A.** Anterior view. **B.** Boundaries and contents. **C.** Showing radial and ulnar nerves, and veins.

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Identifying tendons and locating major vessels and nerves in the distal forearm

Tendons that pass from the forearm into the hand are readily visible in the distal forearm and can be used as landmarks to locate major vessels and nerves.

In the anterior aspect of the distal forearm, the tendons of the flexor carpi radialis, flexor carpi ulnaris, and palmaris longus muscles can be easily located either by palpating or by asking a patient to flex the wrist against resistance.

- The tendon of flexor carpi radialis is located approximately at the junction between the lateral and middle thirds of an imaginary line drawn transversely across the distal forearm. The radial artery is immediately lateral to this tendon and this site is used for taking a radial pulse (Fig. 7.118A).
- The tendon of the flexor carpi ulnaris is easily palpated along the medial margin of the forearm and inserts on the pisiform, which can also be palpated by following the tendon to the base of the hypothenar eminence of the hand. The ulnar artery and ulnar nerve travel

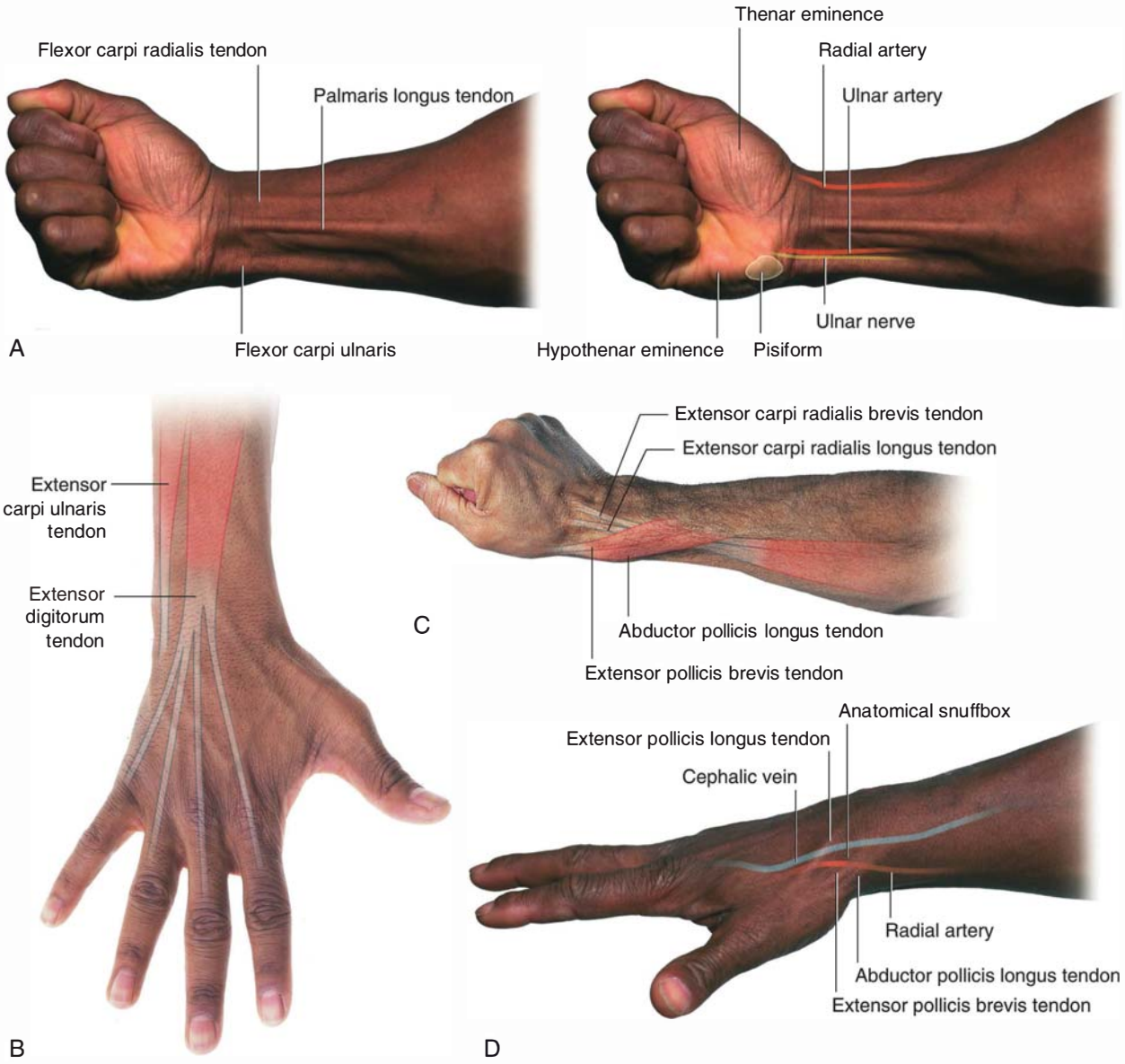


Fig. 7.118 Identifying tendons and locating major vessels and nerves in the distal forearm. **A.** Anterior distal forearm and wrist. **B.** Posterior distal forearm and wrist. **C.** Lateral view of posterior wrist and forearm. **D.** Anatomical snuffbox.

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through the distal forearm and into the hand under the lateral lip of the flexor carpi ulnaris tendon and lateral to the pisiform.

- The palmaris longus tendon may be absent, but when present, lies medial to the flexor carpi radialis tendon and is particularly prominent when the wrist is flexed against resistance. The median nerve is also medial to the flexor carpi radialis tendon and lies under the palmaris longus tendon.
- The long tendons of the digits of the hand are deep to the median nerve and between the long flexors of the wrist. Their position can be visualized by rapidly and repeatedly flexing and extending the fingers from medial to lateral.
- In the posterior distal forearm and wrist, the tendons of the extensor digitorum (Fig. 7.118B) are in the midline and radiate into the index, middle, ring, and little fingers from the wrist.
- The distal ends of the tendons of the extensor carpi radialis longus and brevis muscles are on the lateral side of the wrist (Fig. 7.118C) and can be accentuated by making a tight fist and extending the wrist against resistance.
- The tendon of the extensor carpi ulnaris can be felt on the far medial side of the wrist between the distal end of the ulna and the wrist.
- Hyperextension and abduction of the thumb reveals the anatomical snuffbox (Fig. 7.118D). The medial margin of this triangular area is the tendon of the extensor pollicis longus, which swings around the dorsal tubercle of the radius and then travels into the thumb. The lateral margin is formed by the tendons of the extensor pollicis brevis and abductor pollicis longus. The radial artery passes through the anatomical snuffbox when traveling laterally around the wrist to reach the back of the hand and penetrate the base of the first dorsal interosseous muscle to access the deep aspect of the palm of the hand. The pulse of the radial artery can be felt in the floor of the anatomical snuffbox in the relaxed wrist. The cephalic vein crosses the roof of the anatomical snuffbox, and cutaneous branches of the radial nerve can be felt by moving a finger back and forth along the tendon of the extensor pollicis longus muscle.

Normal appearance of the hand

In the resting position, the palm and digits of the hand have a characteristic appearance. The fingers form a flexed arcade, with the little finger flexed the most and the index finger flexed the least (Fig. 7.119A). The pad of the thumb is positioned at a 90° angle to the pads of the fingers.

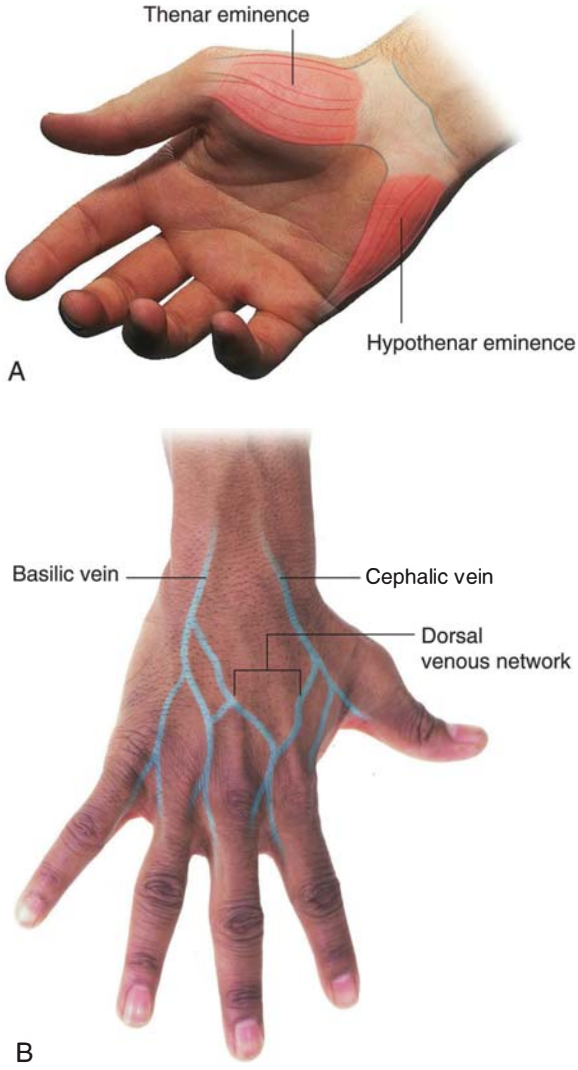


Fig. 7.119 Normal appearances of the hand. **A.** Palmar view with the thenar and hypothenar eminences and finger arcade. **B.** Dorsal view with dorsal venous network.

A thenar eminence occurs at the base of the thumb and is formed by the underlying thenar muscles. A similar hypothenar eminence occurs along the medial margin of the palm at the base of the little finger. The appearance of the thenar and hypothenar eminences, and the positions of the fingers change when the ulnar and median nerves are compromised.

Major superficial veins of the upper limb begin in the hand from a dorsal venous network (Fig. 7.119B), which overlies the metacarpals. The basilic vein originates from the medial side of the network and the cephalic vein originates from the lateral side.

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Position of the flexor retinaculum and the recurrent branch of the median nerve

The proximal margin of the flexor retinaculum can be determined using two bony landmarks.

- The pisiform bone is readily palpable at the distal end of the flexor carpi ulnaris tendon.
- The tubercle of the scaphoid can be palpated at the distal end of the flexor carpi radialis tendon as it enters the wrist (Fig. 7.120).

An imaginary line between these two points marks the proximal margin of the flexor retinaculum. The distal margin of the flexor retinaculum is approximately deep to the point where the anterior margin of the thenar eminence meets the hypothenar eminence near the base of the palm.

The recurrent branch of the median nerve lies deep to the skin and deep fascia overlying the anterior margin of the thenar eminence near the midline of the palm.

Motor function of the median and ulnar nerves in the hand

The ability to flex the metacarpophalangeal joints while at the same time extending the interphalangeal joints of the fingers is entirely dependent on the intrinsic muscles of the hand (Fig. 7.121A). These muscles are mainly innervated by the deep branch of the ulnar nerve, which carries fibers from spinal cord level (C8)T1.

Adducting the fingers to grasp an object placed between them is caused by the palmar interossei muscles, which are innervated by the deep branch of the ulnar nerve carrying fibers from spinal cord level (C8)T1.

The ability to grasp an object between the pad of the thumb and the pad of one of the fingers depends on normal

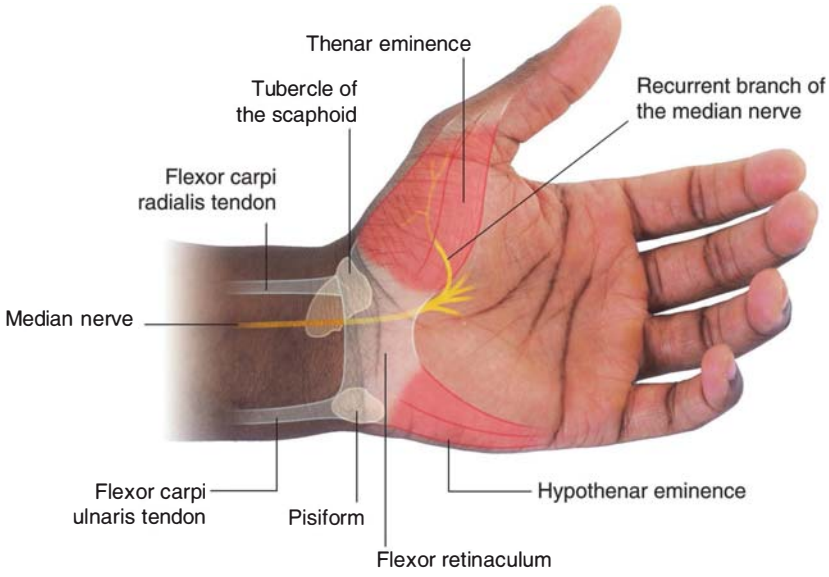


Fig. 7.120 Anterior hand to show the position of the flexor retinaculum and recurrent branch of the median nerve.

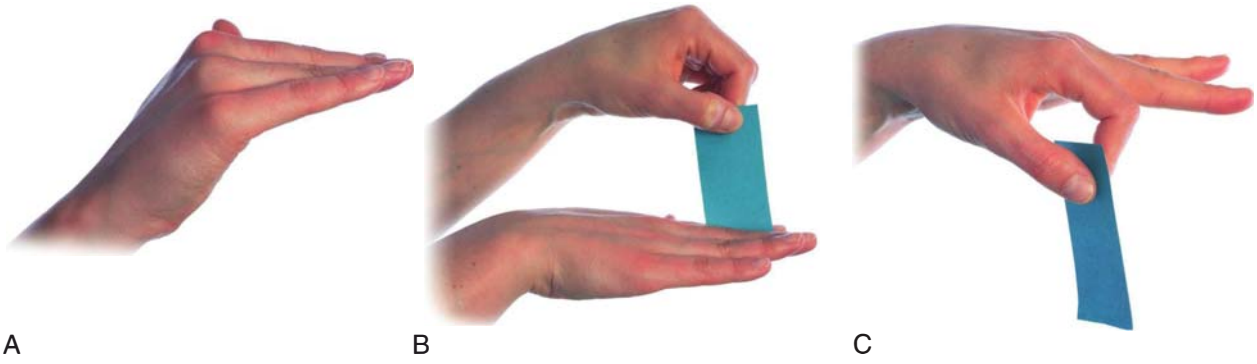


Fig. 7.121 Motor function of the ulnar and median nerves in the hand. **A.** Flexing the metacarpophalangeal joints and extending the interphalangeal joints: the "ta-ta" position. **B.** Grasping an object between the fingers. **C.** Grasping an object between the pad of the thumb and pad of the index finger.

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functioning of the thenar muscles, which are innervated by the recurrent branch of the median nerve carrying fibers from spinal cord level C8(T1).

Visualizing the positions of the superficial and deep palmar arches

The positions of the superficial and deep palmar arches in the hand can be visualized using bony landmarks, muscle eminences, and skin creases (Fig. 7.122).

- The superficial palmar arch begins as a continuation of the ulnar artery, which lies lateral to the pisiform bone at the wrist. The arch curves laterally across the palm anterior to the long flexor tendons in the hand. The arch reaches as high as the proximal transverse skin crease of the palm and terminates laterally by joining a vessel of variable size, which crosses the thenar eminence from the radial artery in the distal forearm.
- The deep palmar arch originates on the lateral side of the palm deep to the long flexor tendons and between the proximal ends of metacarpals I and II. It arches medially across the palm and terminates by joining the deep branch of the ulnar artery, which passes through the base of the hypothenar muscles and between the pisiform and hook of the hamate. The deep palmar arch is more proximal in the hand than the superficial palmar arch and lies approximately one-half of the distance between the distal wrist crease and the proximal transverse skin crease of the palm.

Pulse points

Peripheral pulses can be felt at six locations in the upper limb (Fig. 7.123).

- Axillary pulse: axillary artery in the axilla lateral to the apex of the dome of skin covering the floor of the axilla.
- Brachial pulse in midarm: brachial artery on the medial side of the arm in the cleft between the biceps brachii and triceps brachii muscles. This is the position where a blood pressure cuff is placed.
- Brachial pulse in the cubital fossa: brachial artery medial to the tendon of the biceps brachii muscle. This is the position where a stethoscope is placed to hear the pulse of the vessel when taking a blood pressure reading.
- Radial pulse in the distal forearm: radial artery immediately lateral to the tendon of the flexor carpi radialis muscle. This is the most common site for "taking a pulse."
- Ulnar pulse in the distal forearm: ulnar artery immediately under the lateral margin of the flexor carpi ulnaris tendon and proximal to the pisiform.
- Radial pulse in the anatomical snuffbox: radial artery as it crosses the lateral side of the wrist between the tendon of the extensor pollicis longus muscle and the tendons of the extensor pollicis brevis and abductor pollicis longus muscles.

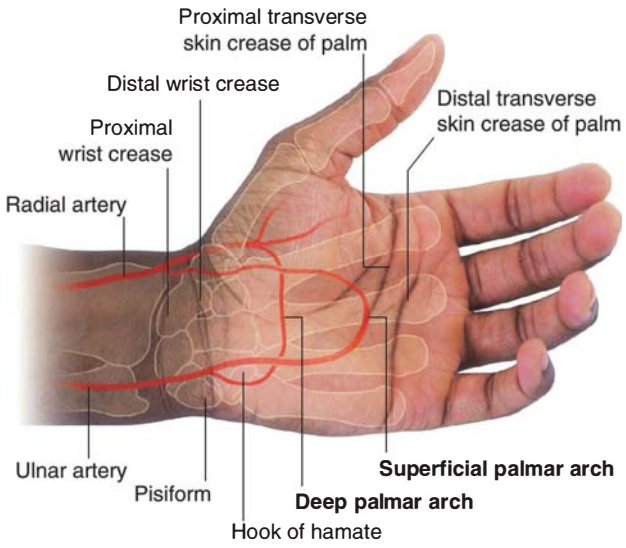


Fig. 7.122 Visualizing the positions of the superficial and deep palmar arches. The proximal transverse skin crease of the palm and distal wrist crease are labeled and the superficial and deep palmar arches shown in overlay. This also shows the position of the pisiform and the hook of the hamate.

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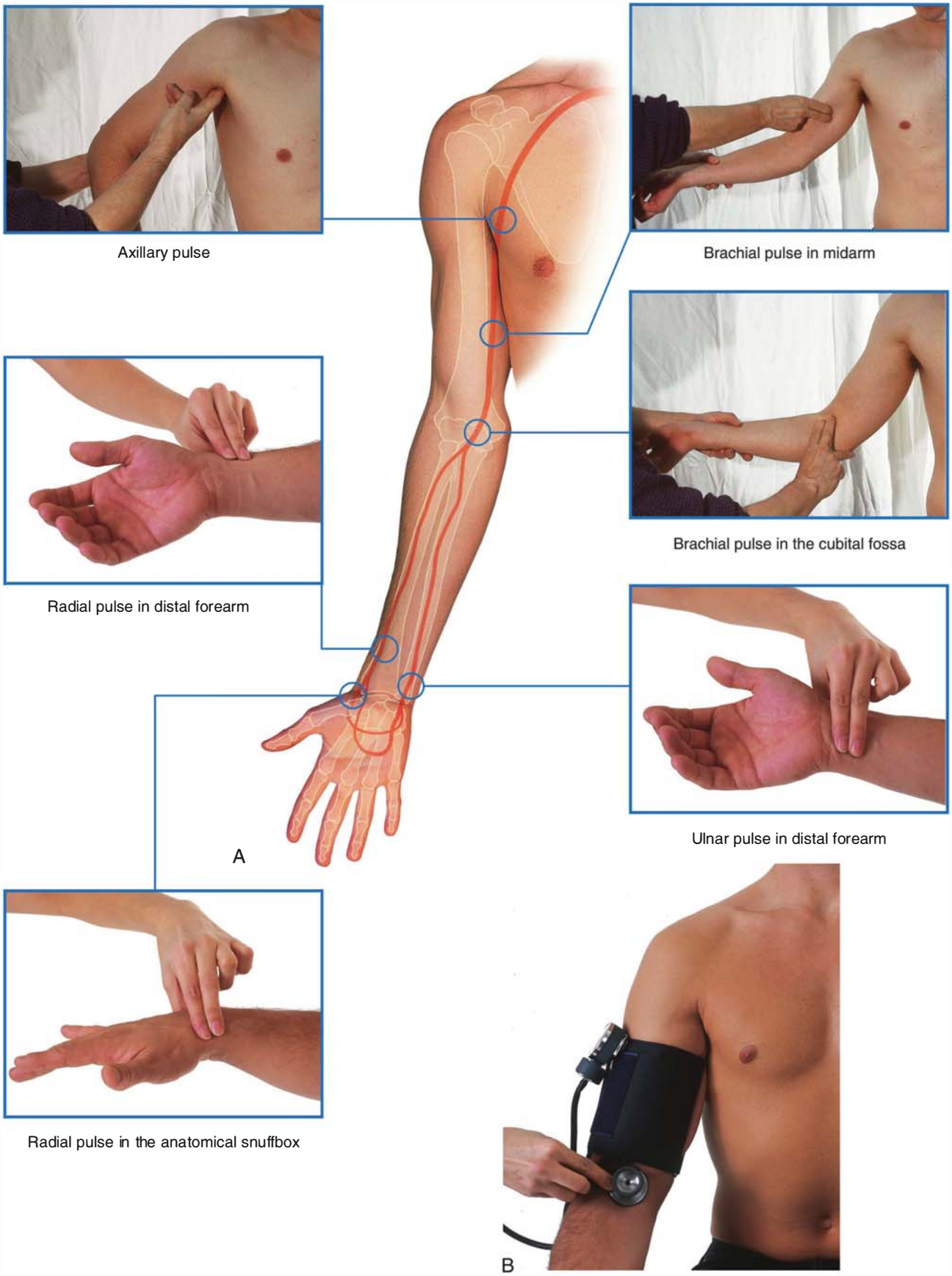


Fig. 7.123 Where to take peripheral artery pulses in the upper limb. **A.** Pulse points. **B.** Placement of blood pressure cuff and stethoscope.

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Clinical cases

Case 1

SHOULDER PROBLEM AFTER FALLING ON AN OUTSTRETCHED HAND

A 45-year-old man came to his physician complaining of pain and weakness in his right shoulder. The pain began after a fall on his outstretched hand approximately 6 months previously. The patient recalled having some minor shoulder tenderness but no other specific symptoms. He was otherwise fit and well.

On examination of the shoulder, there was marked wasting of the muscles in the supraspinous and infraspinous fossae. The patient found initiation of abduction difficult and there was a weakness of lateral rotation of the humerus.

The wasted muscles were the supraspinatus and infraspinatus. The cause of the muscle wasting was disuse.

Muscle atrophy (wasting) occurs through a variety of disorders. Disuse atrophy is one of the most common causes. Examples of disuse atrophy include the loss of muscle bulk after fracture immobilization in a plaster cast.

The opposite effect can also be demonstrated—when muscles are overused they become bulkier (hypertrophy).

The supraspinatus and infraspinatus muscles are supplied by the suprascapular nerve (C5, C6), which originates from the superior trunk of the brachial plexus. Given that only these muscles were involved, it is highly likely that the muscle atrophy is caused by denervation. Denervation may result from a direct nerve transection, nerve compression, or a pharmacological effect on the nerve.

The typical site for compression of the suprascapular nerve is the suprascapular notch (foramen) on the superior margin of the scapula.

The patient's apparently minor injury damaged the fibrocartilaginous glenoid labrum, which allowed a cyst to form and pass along the superior border of the scapula to the suprascapular notch (foramen), where the cyst compressed the suprascapular nerve.

Surgical excision of the damaged glenoid labrum and removal of the cyst improved the patient's symptoms.

Case 2

WINGED SCAPULA

A 57-year-old woman underwent a right mastectomy for a breast cancer. The surgical note reported that all of the breast tissue had been removed, including the axillary process. In addition, the surgeon had dissected all lymph nodes within the axilla with their surrounding fat. The patient made an uneventful recovery.

At the first follow-up appointment, the patient's husband told the surgeon that she had now developed a bony "spike" on her back. The surgeon was intrigued and asked the patient to reveal this spike. At examination, the spike was the inferior angle of the scapula, which appeared to be sticking out posteriorly ("winged"). Raising the arms accentuated this structure.

The medial border of the scapula was accentuated and it was noted that there was some loss of bulk of the serratus anterior muscle, which attaches to the tip of the scapula.

The nerve to this muscle was damaged.

During the surgery on the axilla, the long thoracic nerve was damaged as it passed down the lateral thoracic wall on the external surface of the serratus anterior, just deep to the skin and subcutaneous fascia.

Because the nerve was transected, it is unlikely that the patient will improve, but she was happy that she had an adequate explanation for the spike.

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Case 3

BRACHIAL PLEXUS NERVE BLOCK

A surgeon wished to carry out a complex procedure on a patient's wrist, and asked the anesthesiologist whether the whole arm could be numbed while the patient was awake. Within 20 minutes the anesthesiologist had carried out the procedure after injecting 10 mL of local anesthetic into the axilla. The surgeon went ahead with the operation and the patient did not feel a thing.

The anesthetic was injected into the axillary sheath.

It would be almost impossible to anesthetize the wrist in the forearm because local anesthetic would have to be placed accurately around the ulnar, median, and radial nerves. Furthermore, all of the cutaneous branches of the forearm would also have to be anesthetized individually, which would take a considerable amount of time and probably produce subtotal anesthesia.

The nerves of the upper limb originate from the brachial plexus, which surrounds the axillary artery within the axilla. Importantly, the axillary artery, axillary vein, and brachial plexus lie within the sleeve-like covering of fascia termed the axillary sheath. By injecting the anesthetic into the space enclosed by the axillary sheath, all of the nerves of the brachial plexus were paralyzed.

It is possible with a patient's arm abducted and externally rotated (palm behind the head) to easily palpate the axillary artery and therefore locate the position of the axillary sheath. Once the axillary artery has been identified, a small needle can be placed beside the vessel and local anesthetic can be injected on both sides of it. The local anesthetic tracks along the axillary sheath in this region. The brachial plexus surrounding the axillary artery is therefore completely anesthetized and an effective local anesthetic "block" is achieved.

"Could there be any complications?" asks the patient.

Potential complications are a direct needle spike of the branches of the brachial plexus, damage to the axillary artery, and inadvertent arterial injection of the local anesthetic. Fortunately, these are rare in skilled hands.

Case 4

COMPLICATION OF A FRACTURED FIRST RIB

A 25-year-old woman was involved in a motor vehicle accident and thrown from her motorcycle. When she was admitted to the emergency room, she was unconscious. A series of tests and investigations were performed, one of which included chest radiography. The attending physician noted a complex fracture of the first rib on the left.

Many important structures that supply the upper limb pass over rib I.

It is important to test the nerves that supply the arm and hand, although this is extremely difficult to do in an unconscious patient. However, some muscle reflexes can be determined using a tendon hammer. Also, it may be possible to test for pain reflexes in patients with altered consciousness levels. Palpation of the axillary artery, brachial artery, radial artery, and ulnar artery pulses is necessary because a fracture of the first rib can sever and denude the subclavian artery, which passes over it.

A chest drain was immediately inserted because the lung had collapsed. The fractured first rib had damaged the visceral and parietal pleurae, allowing air from a torn lung to escape into the pleural cavity. The lung collapsed, and the pleural cavity filled with air, which impaired lung function.

A tube was inserted between the ribs, and the air was sucked out to re-inflate the lung.

The first rib is a deep structure at the base of the neck. It is not uncommon for ribs to be broken after minor injuries, including sports injuries. However, rib I, which lies at the base of the neck, is surrounded by muscles and soft tissues that provide it with considerable protection. Therefore a patient with a fracture of the first rib has undoubtedly been subjected to a considerable force, which usually occurs in a deceleration injury.
Other injuries should always be sought and the patient should be managed with a high level of concern for deep neck and mediastinal injuries.

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Case 5

MEDIAN NERVE COMPRESSION

A 35-year-old woman comes to her physician complaining of tingling and numbness in the fingertips of the first, second, and third digits (thumb, index, and middle fingers). The symptoms were provoked by arm extension. Local anesthesia was also present around the base of the thenar eminence.

The problem was diagnosed as median nerve compression.

The median nerve is formed from the lateral and medial cords of the brachial plexus anterior to the axillary artery and passes into the arm anterior to the brachial artery. At the level of the elbow joint it sits medial to the brachial artery, both of which are medial to the tendon of the biceps. In the forearm the nerve courses through the anterior compartment and passes deep to the flexor retinaculum. It innervates most of the muscles of the forearm, the thenar muscles, the two lateral lumbricals, and the skin over the palmar surface of the lateral three and one-half digits and over the lateral side of the palm and the middle of the wrist.

In this patient, the median nerve initially was believed to be trapped below the flexor retinaculum (carpal tunnel syndrome).

Carpal tunnel syndrome is a common problem in young to middle-aged patients. Typically the nerve becomes compressed within the carpal tunnel. This syndrome may be associated with a number of medical conditions, such as thyroid disease and pregnancy. Occasionally a small ganglion or a tumor situated within the carpal tunnel can also compress the nerve. Other possibilities include tenosynovitis in patients with rheumatoid arthritis.

Nerve conduction studies were performed to confirm the clinical findings. Nerve conduction studies are a series of tests that send small electrical impulses along the length of a variety of nerves in order to measure the speed at which the nerve conducts these pulses. The speed of the nerve pulse can be measured and is referred to as the latency. In our patient it was noted that the nerve had normal latency to the elbow joint; however, below the elbow joint there was increased latency.

The nerve conduction studies indicated that the compression site was at the elbow joint.

The clinical findings are not consistent with carpal tunnel syndrome. The clinician should have been alerted to this problem given that the patient experienced numbness over the thenar eminence of the hand. This clue indicates an understanding of the anatomy. Compression of the nerve within the carpal tunnel does not produce this numbness, because the small cutaneous branch that supplies this region is proximal to the flexor retinaculum.

The nerve compromise was caused by the ligament of Struthers, which is an embryological remnant of the coracobrachialis muscle. It is an extremely rare finding. Occasionally it may ossify and cross the nerve, artery, and vein to produce compression in arm extension. Although this is very rare and unusual, it illustrates the complex course of the median nerve.

Case 6

IMMOBILIZING THE EXTENSOR DIGITORUM MUSCLE

After a hard day's studying, two medical students decided to meet for coffee. The more senior student said to the freshman that he would bet him $50 that he could not lift a matchbook with a finger. The freshman placed $50 on the table and the bet was on. The senior medical student told the freshman to make a fist and place it in a palm-downward position, so that the middle phalanges of the fingers were in direct contact with the bar counter. He was then told to extend his middle finger so that it stuck forward while maintaining the middle phalanges of the index finger, the ring finger, and the little finger on the bar surface.

A matchbook was placed on top of the freshman's middle fingernail and he was told to flip it. He couldn't. He lost the $50.

Extension of the index, middle, ring, and little fingers is performed by the extensor digitorum muscle.

Placing the fist in a palm-down position on the table and pressing the middle phalanges onto the table effectively immobilizes the action of the extensor digitorum. The freshman was therefore unable to elevate his middle finger (which was sticking out). It is important to remember that if this same procedure is carried out leaving the index or little finger free to move, they do. This is because these two digits are extended not only by the extensor digitorum muscle but also by the extensor indicis muscle (index finger) and extensor digiti minimi muscle (little finger).

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Case 7

TORN SUPRASPINATUS TENDON

A 70-year-old woman came to an orthopedic surgeon with right shoulder pain and failure to initiate abduction of the shoulder. Further examination revealed loss of muscle bulk in the supraspinous fossa. The supraspinatus muscle was damaged.

Abduction of the humerus at the glenohumeral joint is initiated by the supraspinatus muscle. After the shoulder has been abducted to 10°–15°, the deltoid muscle continues the movement. The patient was able to abduct her arm by lowering and tilting the glenohumeral joint inferiorly to enable the deltoid to obtain its mechanical advantage.

The loss of muscle bulk in the supraspinous fossa suggested muscle atrophy.

Muscle atrophy occurs when a muscle is not used. The orthopedic surgeon thought that there was a tear of the supraspinatus tendon beneath the acromion. If this was so, the muscle would atrophy.

The diagnosis was confirmed by ultrasound scan.

The patient was seated on a stool and her right shoulder was uncovered. The patient's hand was placed over her right buttock, a position that acts to externally rotate and extend the shoulder, exposing the supraspinatus tendon for ultrasound scan examination. The ultrasound revealed a completely torn tendon with fluid in the subacromial subdeltoid bursa (Fig. 7.124). The patient underwent a surgical repair and made a good recovery.

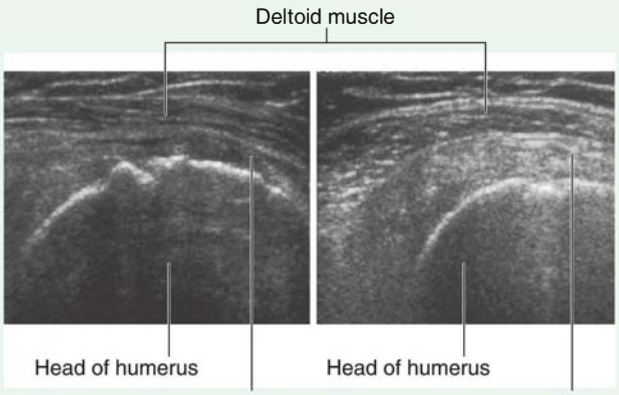


Fig. 7.124 Ultrasound showing a completely torn supraspinatus tendon with fluid in the subacromial subdeltoid bursa.

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Case 8

HOW TO EXAMINE THE HAND

A resident was asked to carry out a clinical assessment of a patient's hand. He examined the following:

Musculoskeletal system

The musculoskeletal system includes the bones, joints, muscles, and tendons. The resident looked for abnormalities and muscle wasting. Knowing which areas are wasted identifies the nerve that supplies them. She palpated the individual bones and palpated the scaphoid with the wrist in ulnar deviation. She examined the movement of joints because they may be restricted by joint disease or inability of muscular contraction.

Circulation

Palpation of both radial and ulnar pulses is necessary. The resident looked for capillary return to assess how well the hand was perfused.

Examination of the nerves

The three main nerves to the hand should be tested.

Median nerve

The median nerve innervates the skin on the palmar aspect of the lateral three and one-half digits, the dorsal aspect of the distal phalanx, half of the middle phalanges of the same fingers, and a variable amount on the radial side of the palm of the hand. Median nerve damage results in wasting of the thenar eminence, absence of abduction of the thumb, and absence of opposition of the thumb.

Ulnar nerve

The ulnar nerve innervates the skin of the anterior and posterior surfaces of the little finger and the ulnar side of the ring finger, the skin over the hypothenar eminence, and a similar strip of skin posteriorly. Sometimes the ulnar nerve innervates all the skin of the ring finger and the ulnar side of the middle finger.

An ulnar nerve palsy results in wasting of the hypothenar eminence, absent flexion of the distal interphalangeal joints of the little and ring fingers, and absent abduction and adduction of the fingers. Adduction of the thumb also is affected.

Radial nerve

The radial nerve innervates a small area of skin over the lateral aspect of metacarpal I and the back of the first web space.

The radial nerve also produces extension of the wrist and extension of the metacarpophalangeal and interphalangeal joints and of the digits.

A very simple examination would include tests for the median nerve by opposition of the thumb, for the ulnar nerve by abduction and adduction of the digits, and for the radial nerve by extension of the wrist and fingers and feeling on the back of the first web space.

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Case 9

SHOULDER JOINT PROBLEM

A 35-year-old baseball pitcher came to the clinic with a history of a recurrent dislocation of the shoulder (Fig. 7.125). An MRI scan was performed to assess the shoulder joint prior to any treatment.

The MRI demonstrates the anatomical structures in multiple planes, allowing the physician to obtain an overview of the shoulder and to assess any intraarticular or extraarticular structures that may have been damaged and require surgical repair.

The MRI demonstrated a divot in the posterosuperior aspect of the humeral head and a small fragment of bone and glenoid labrum that had become separated in the anteroinferior aspect of the glenoid cavity.

Shoulder dislocation is not an uncommon problem and may occur as a "once-off" or with repetitive injury may be recurrent. Recurrent dislocations may be bilateral and symmetrical (a memory aid is "torn loose or born loose").

The MRI findings are typical for an anteroinferior dislocation, which is the most common type; moreover the MRI demonstrates the injuries that occur within the joint at the time of dislocation. These injuries include the abutment of the posterosuperior aspect of the humeral head on the anteroinferior aspect of the glenoid cavity. This type of injury, when recurrent, may avulse a small fragment of the glenoid labrum, and in some cases this may attach to a small fragment of bone (the Bankart lesion). When the shoulder is relocated, the integrity of the capsular attachment anteroinferiorly has been disrupted, potentially making the shoulder somewhat prone to further dislocation.

An arthroscopic repair was performed.

Arthroscopy of the shoulder is an established method for assessing the shoulder joint. Portals of entry are anterior and posterior and small holes in the capsule are made percutaneously. The shoulder joint is filled with saline, which distends it, allowing the arthroscope to move around the joint and inspect the joint surfaces, including the labrum. The labrum and its bony fragment were reattached and sutured using anchor sutures (somewhat similar to staples). The anterior aspect of the capsule was also tightened.

The patient made an uneventful recovery.

After the procedure the arm was held in internal rotation and remained adducted. Gentle exercise and physiotherapy were performed and the patient returned to playing baseball.



Fig. 7.125 The radiograph, anteroposterior view, demonstrates an anteroinferior dislocation of the humeral head at the glenohumeral joint.

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8

Head and Neck

ADDITIONAL LEARNING RESOURCES for Chapter 8, *Head and Neck*, on STUDENT CONSULT (www.studentconsult.com):

- Image Library—illustrations of head and neck anatomy, Chapter 8
- Self-Assessment—National Board style multiple-choice questions, Chapter 8
- Short Questions—these are questions requiring short responses, Chapter 8
- Interactive Surface Anatomy—interactive surface animations, Chapter 8
- Medical Clinical Case Studies, Chapter 8
  - Pancoast's tumor
  - Subclavian steal syndrome
  - Branchial cyst
  - Epiglottitis
  - Otosclerosis
  - Scalp laceration
  - Scaphocephaly
  - Temporal arteritis
  - Tonsillitis
- Clinical Cases, Chapter 8
  - Multinodular goiter
  - Parotid duct calculus
  - Extradural hematoma
  - Stenosis of the internal carotid artery
  - Posterior communicating artery aneurysm
  - Recurrent epistaxis
  - Complication of orbital fracture
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Conceptual overview

GENERAL DESCRIPTION

The head and neck are anatomically complex areas of the body.

Head

Major compartments

The head is composed of a series of compartments, which are formed by bone and soft tissues. They are:

- the cranial cavity,
- two ears,
- two orbits,
- two nasal cavities, and
- an oral cavity (Fig. 8.1).

The **cranial cavity** is the largest compartment and contains the brain and associated membranes (meninges).

Most of the ear apparatus on each side is contained within one of the bones forming the floor of the cranial cavity. The external parts of the ears extend laterally from these regions.

The two **orbits** contain the eyes. They are cone-shaped chambers immediately inferior to the anterior aspect of the cranial cavity, and the apex of each cone is directed posteromedially. The walls of the orbits are bone, whereas the base of each conical chamber can be opened and closed by the eyelids.

The **nasal cavities** are the upper parts of the respiratory tract and are between the orbits. They have walls, floors, and ceilings, which are predominantly composed of bone and cartilage. The anterior openings to the nasal cavities are **nares** (**nostrils**), and the posterior openings are **choanae** (**posterior nasal apertures**).

Continuous with the nasal cavities are air-filled extensions (**paranasal sinuses**), which project laterally, superiorly, and posteriorly into surrounding bones. The largest, the **maxillary sinuses**, are inferior to the orbits.

The **oral cavity** is inferior to the nasal cavities, and separated from them by the **hard** and **soft palates**. The floor of the oral cavity is formed entirely of soft tissues.

The anterior opening to the oral cavity is the **oral fissure** (mouth), and the posterior opening is the **oropharyngeal isthmus**. Unlike the nares and choanae, which are continuously open, both the oral fissure and oropharyngeal isthmus can be opened and closed by surrounding soft tissues.

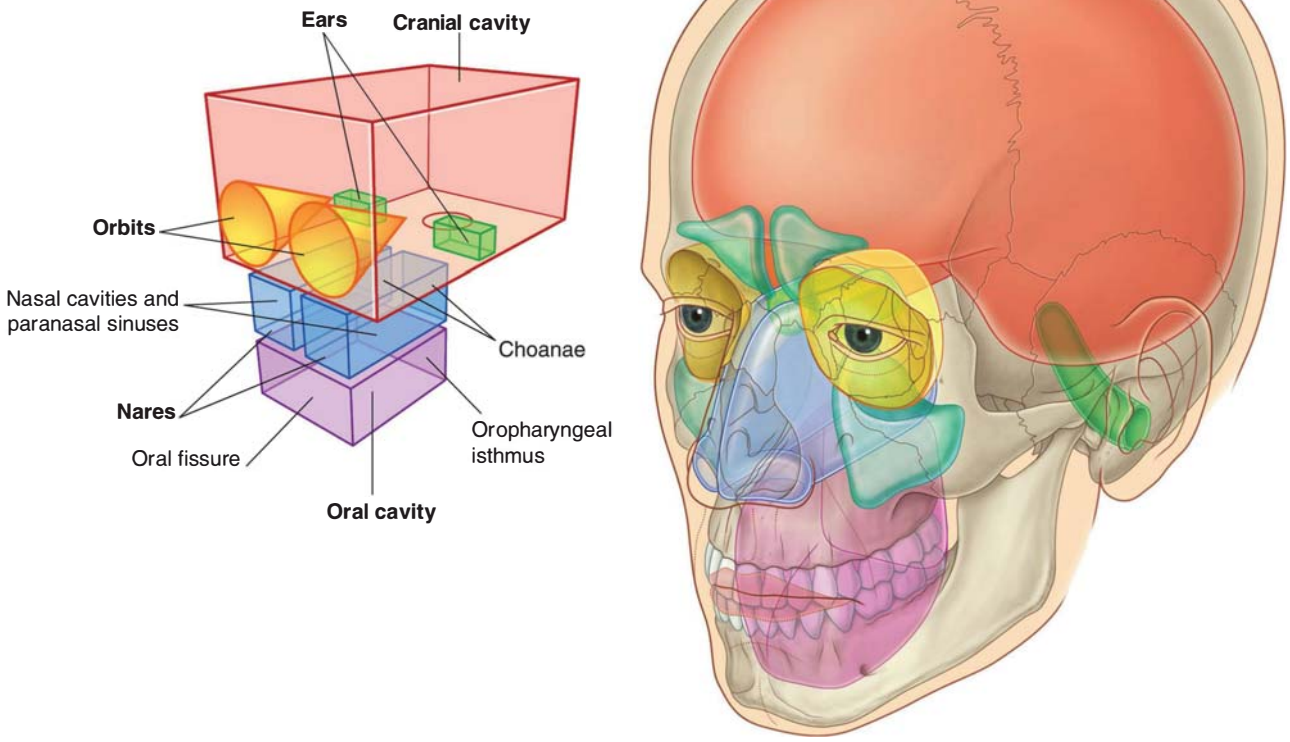


Fig. 8.1 Major compartments of the head and neck.

Other anatomically defined regions

In addition to the major compartments of the head, two other anatomically defined regions (infratemporal fossa and pterygopalatine fossa) of the head on each side are areas of transition from one compartment of the head to another (Fig. 8.2). The face and scalp also are anatomically defined areas of the head and are related to external surfaces.

The **infratemporal fossa** is an area between the posterior aspect (ramus) of the mandible and a flat region of bone (lateral plate of the pterygoid process) just posterior to the upper jaw (maxilla). This fossa, bounded by bone and soft tissues, is a conduit for one of the major cranial nerves—the mandibular nerve (the mandibular division of the trigeminal nerve [V_3]), which passes between the cranial and oral cavities.

The **pterygopalatine fossa** on each side is just posterior to the upper jaw. This small fossa communicates with the cranial cavity, the infratemporal fossa, the orbit, the nasal cavity, and the oral cavity. A major structure passing through the pterygopalatine fossa is the maxillary nerve (the maxillary division of the trigeminal nerve [V_2]).

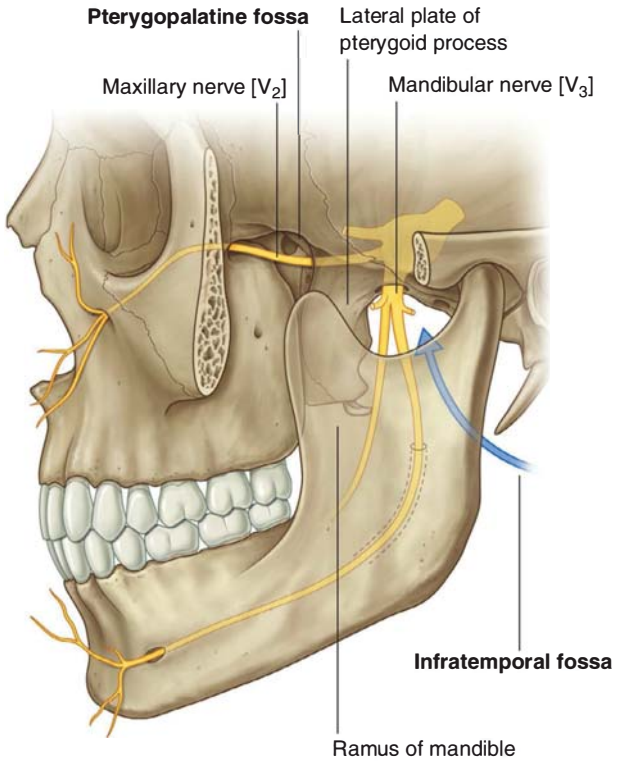


Fig. 8.2 Areas of transition from one compartment of the head to another.

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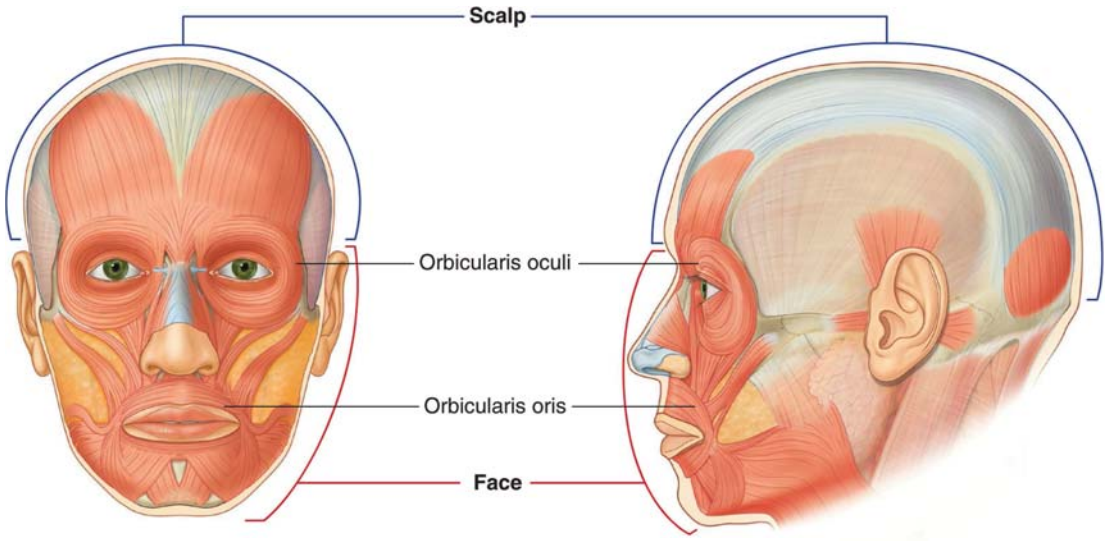


Fig. 8.3 Muscles of the face.

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The **face** is the anterior aspect of the head and contains a unique group of muscles that move the skin relative to underlying bone and control the anterior openings to the orbits and oral cavity (Fig. 8.3).

The **scalp** covers the superior, posterior, and lateral regions of the head (Fig. 8.3).

Neck

The **neck** extends from the head above to the shoulders and thorax below (Fig. 8.4). Its superior boundary is along the inferior margins of the mandible and bone features on the posterior aspect of the skull. The posterior neck is higher than the anterior neck to connect cervical viscera with the posterior openings of the nasal and oral cavities.

The inferior boundary of the neck extends from the top of the sternum, along the clavicle, and onto the adjacent acromion, a bony projection of the scapula. Posteriorly, the inferior limit of the neck is less well defined, but can be approximated by a line between the acromion and the spinous process of vertebra CVII, which is prominent and easily palpable. The inferior border of the neck encloses the **base of the neck**.

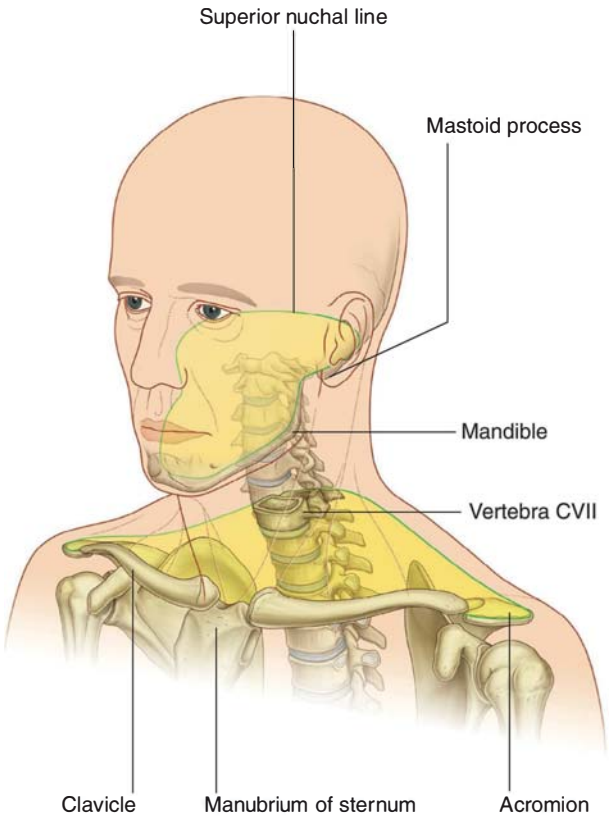


Fig. 8.4 Boundaries of the neck.

Compartments

The neck has four major compartments (Fig. 8.5), which are enclosed by an outer musculofascial collar:

- The vertebral compartment contains the cervical vertebrae and associated postural muscles.
- The visceral compartment contains important glands (thyroid, parathyroid, and thymus), and parts of the respiratory and digestive tracts that pass between the head and thorax.
- The two vascular compartments, one on each side, contain the major blood vessels and the vagus nerve.

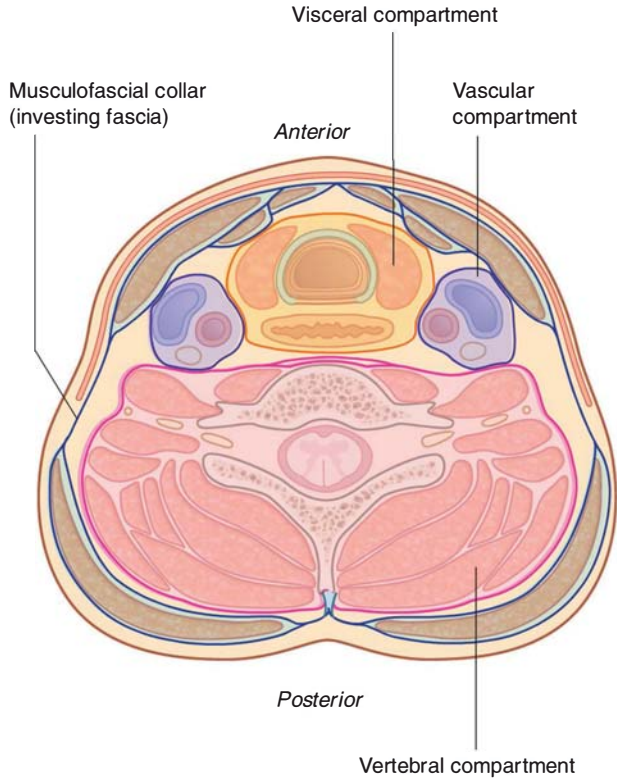


Fig. 8.5 Major compartments of the neck.

Larynx and pharynx

The neck contains two specialized structures associated with the digestive and respiratory tracts—the larynx and pharynx.

The **larynx** (Fig. 8.6) is the upper part of the lower airway and is attached below to the top of the trachea and above, by a flexible membrane, to the hyoid bone, which in turn is attached to the floor of the oral cavity. A number of cartilages form a supportive framework for the larynx, which has a hollow central channel. The dimensions of

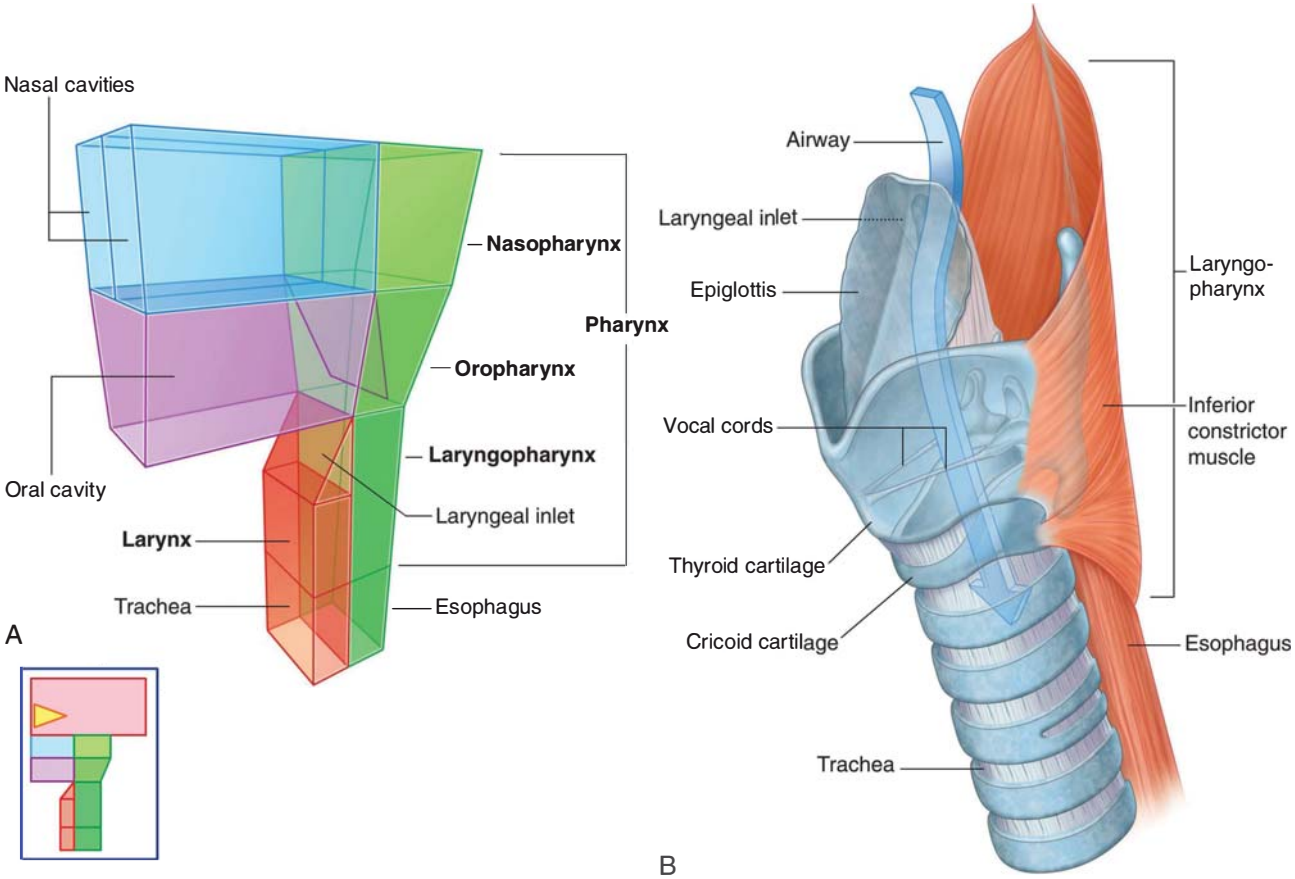


Fig. 8.6 Specialized structures of the neck. **A.** Conceptual view. **B.** Anatomical view.

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this central channel can be adjusted by soft tissue structures associated with the laryngeal wall. The most important of these are two lateral vocal folds, which project toward each other from adjacent sides of the laryngeal cavity. The upper opening of the larynx (**laryngeal inlet**) is tilted posteriorly, and is continuous with the pharynx.

The **pharynx** (Fig. 8.6) is a chamber in the shape of a half-cylinder with walls formed by muscles and fascia. Above, the walls are attached to the base of the skull, and below to the margins of the esophagus. On each side, the walls are attached to the lateral margins of the nasal cavities, the oral cavity, and the larynx. The two nasal cavities, the oral cavity, and the larynx therefore open into the anterior aspect of the pharynx, and the esophagus opens inferiorly.

The part of the pharynx posterior to the nasal cavities is the **nasopharynx**. Those parts posterior to the oral cavity and larynx are the **oropharynx** and **laryngopharynx**, respectively.

FUNCTIONS

Protection

The head houses and protects the brain and all the receptor systems associated with the special senses—the nasal cavities associated with smell, the orbits with vision, the ears with hearing and balance, and the oral cavity with taste.

Contains upper parts of respiratory and digestive tracts

The head contains the upper parts of the respiratory and digestive systems—the nasal and oral cavities—which have structural features for modifying the air or food passing into each system.

Communication

The head and neck are involved in communication. Sounds produced by the larynx are modified in the pharynx and oral cavity to produce speech. In addition, the muscles of facial expression adjust the contours of the face to relay nonverbal signals.

Positioning the head

The neck supports and positions the head. Importantly, it enables an individual to position sensory systems in the head relative to environmental cues without moving the entire body.

Connects the upper and lower respiratory and digestive tracts

The neck contains specialized structures (pharynx and larynx) that connect the upper parts of the digestive and respiratory tracts (nasal and oral cavities) in the head, with the esophagus and trachea, which begin relatively low in the neck and pass into the thorax.

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COMPONENT PARTS

Skull

The many bones of the head collectively form the skull (Fig. 8.7A). Most of these bones are interconnected by **sutures**, which are immovable fibrous joints (Fig. 8.7B).

In the fetus and newborn, large membranous and unossified gaps (**fontanelles**) between the bones of the skull, particularly between the large flat bones that cover the top of the cranial cavity (Fig. 8.7C), allow:

- the head to deform during its passage through the birth canal, and
- postnatal growth.

Most of the fontanelles close during the first year of life. Full ossification of the thin connective tissue ligaments separating the bones at the suture lines begins in the late twenties, and is normally completed in the fifth decade of life.

There are only three pairs of synovial joints on each side in the head. The largest are the temporomandibular joints between the lower jaw (mandible) and the temporal bone. The other two synovial joints are between the three tiny bones in the middle ear, the malleus, incus, and stapes.

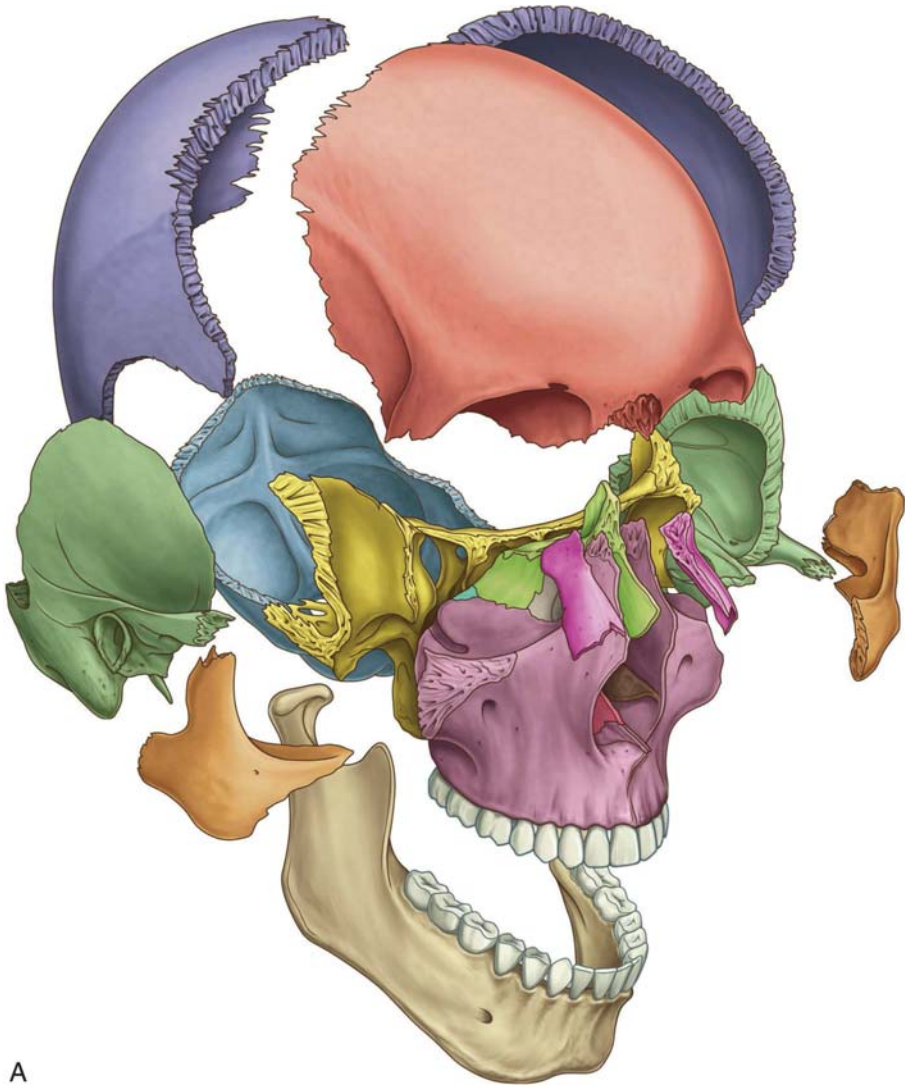


Fig. 8.7 Skull. **A.** Bones.

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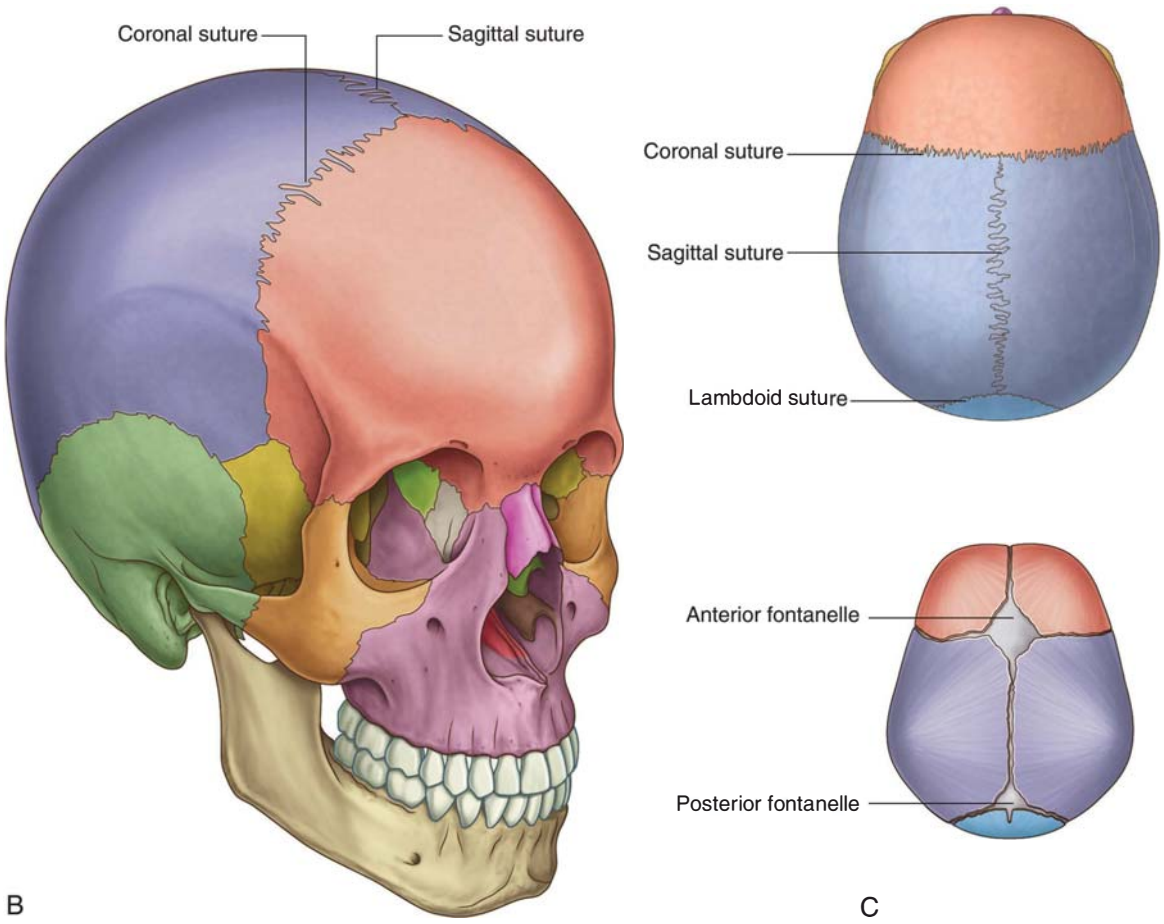


Fig. 8.7, cont'd Skull. **B.** Sutures. **C.** Fontanelles and lambdoid suture.

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Cervical vertebrae

The seven cervical vertebrae form the bony framework of the neck.

Cervical vertebrae (Fig. 8.8A) are characterized by:

- small bodies,
- bifid spinous processes, and
- transverse processes that contain a foramen (**foramen transversarium**).

Together the foramina transversaria form a longitudinal passage on each side of the cervical vertebral column for blood vessels (vertebral artery and veins) passing between the base of the neck and the cranial cavity.

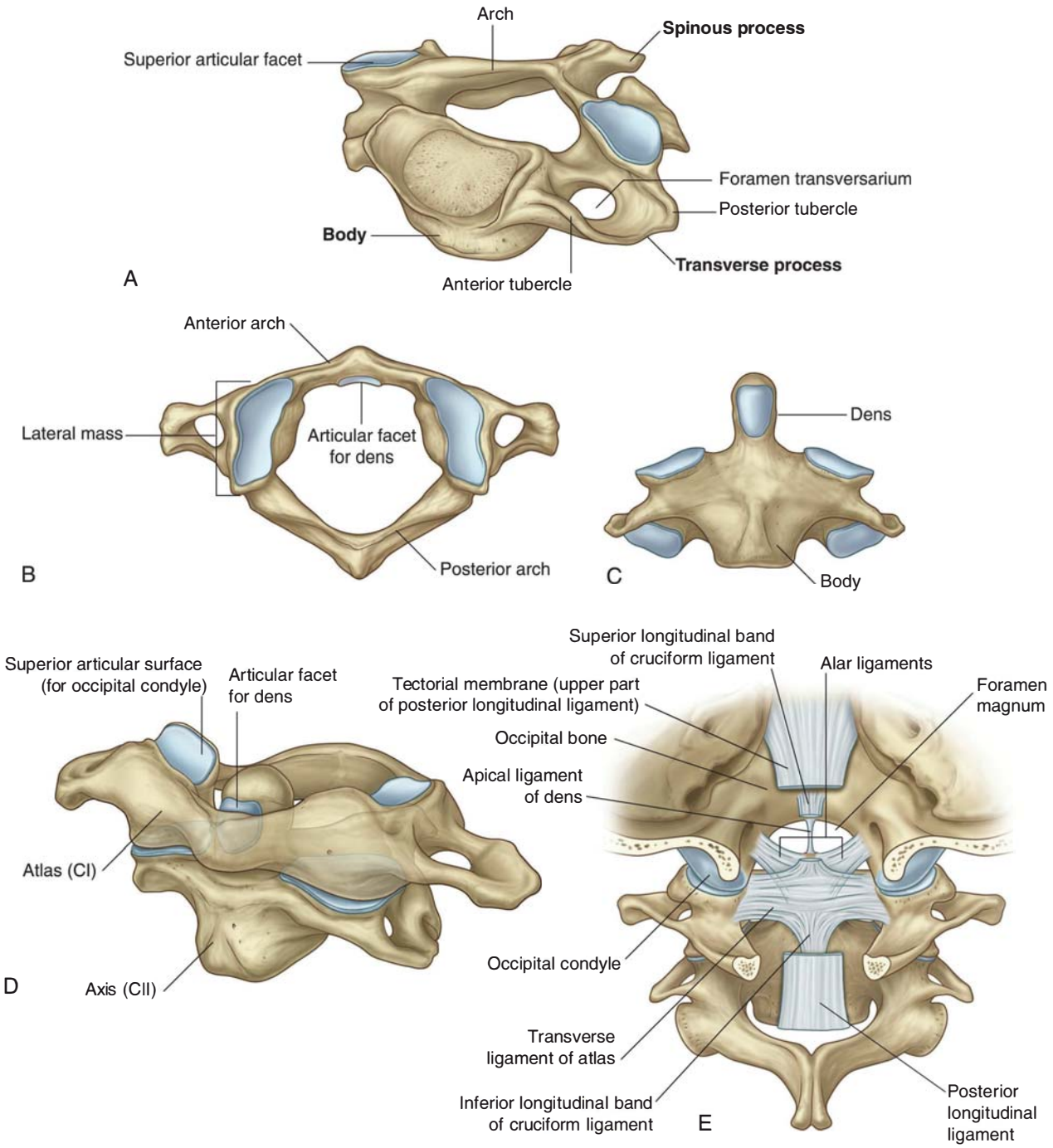


Fig. 8.8 Cervical vertebrae. **A.** Typical features. **B.** Atlas—vertebra CI (superior view). **C.** Axis—vertebra CII (anterior view). **D.** Atlas and axis (anterolateral view). **E.** Atlanto-occipital joint (posterior view).

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The typical transverse process of a cervical vertebra also has **anterior** and **posterior tubercles** for muscle attachment. The anterior tubercles are derived from the same embryological elements that give rise to ribs in the thoracic region. Occasionally, cervical ribs develop from these elements, particularly in association with the lower cervical vertebrae.

The upper two cervical vertebrae (CI and CII) are modified for moving the head (Fig. 8.8B–E; see also Chapter 2).

Hyoid bone

The hyoid bone is a small U-shaped bone (Fig. 8.9A) oriented in the horizontal plane just superior to the larynx, where it can be palpated and moved from side to side.

- The **body of the hyoid bone** is anterior and forms the base of the U.
- The two arms of the U (**greater horns**) project posteriorly from the lateral ends of the body.

The hyoid bone does not articulate directly with any other skeletal elements in the head and neck.

The hyoid bone is a highly movable and strong bony anchor for a number of muscles and soft tissue structures in the head and neck. Significantly, it is at the interface between three dynamic compartments:

- Superiorly, it is attached to the floor of the oral cavity.
- Inferiorly, it is attached to the larynx.
- Posteriorly, it is attached to the pharynx (Fig. 8.9B).

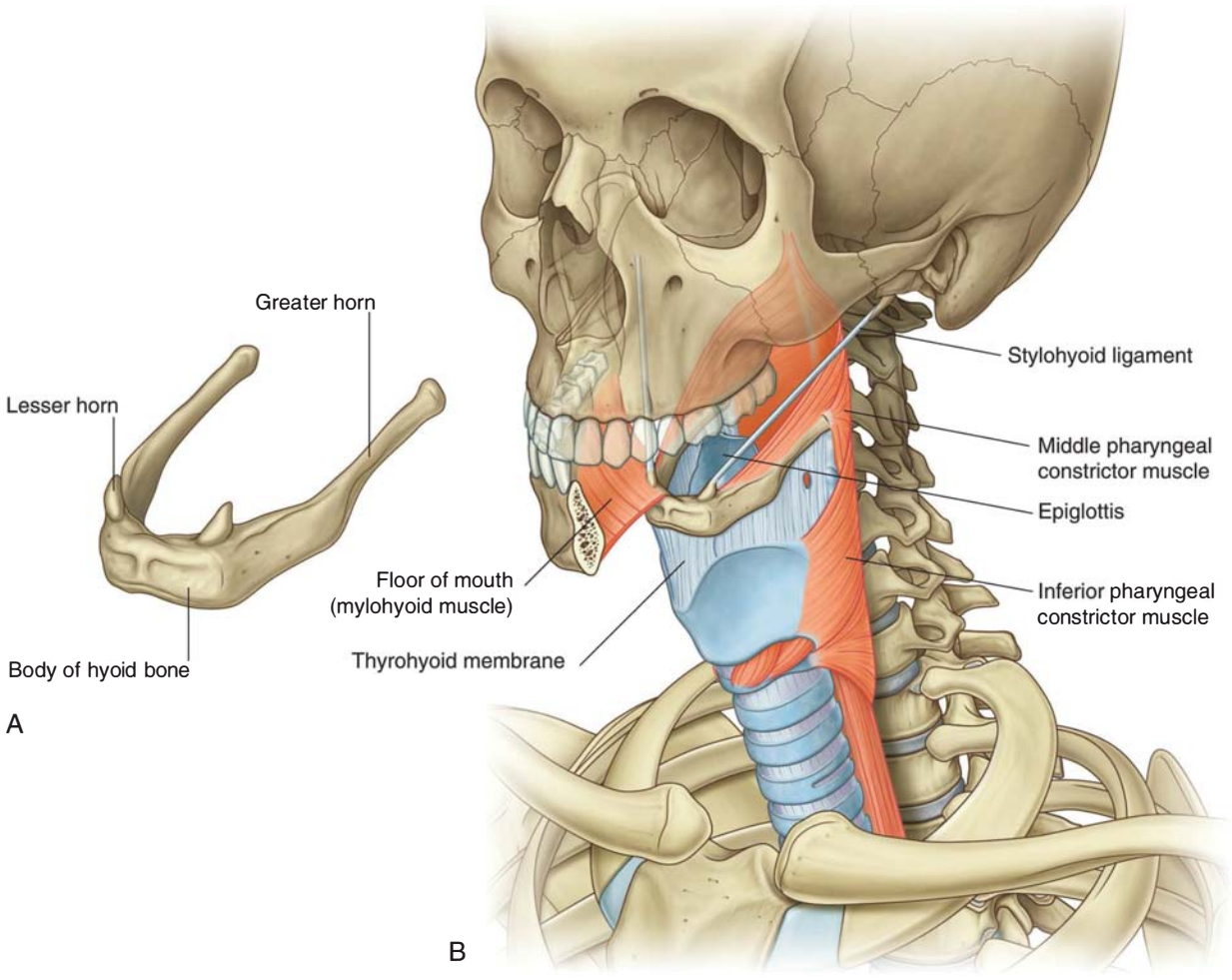


Fig. 8.9 Hyoid. **A.** Bone. **B.** Attachments.

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Soft palate

The soft palate is a soft tissue flap-like structure "hinged" to the back of the hard palate (Fig. 8.10A) with a free posterior margin. It can be elevated and depressed by muscles (Fig. 8.10B).

The soft palate and associated structures can be clearly seen through an open mouth.

Muscles

The skeletal muscles of the head and neck can be grouped on the basis of function, innervation, and embryological derivation.

In the head

The muscle groups in the head include:

- the extra-ocular muscles (move the eyeball and open the upper eyelid),
- muscles of the middle ear (adjust the movement of the middle ear bones),
- muscles of facial expression (move the face),
- muscles of mastication (move the jaw—temporomandibular joint),
- muscles of the soft palate (elevate and depress the palate), and
- muscles of the tongue (move and change the contour of the tongue).

In the neck

In the neck, major muscle groups include:

- muscles of the pharynx (constrict and elevate the pharynx),
- muscles of the larynx (adjust the dimensions of the air pathway),
- strap muscles (position the larynx and hyoid bone in the neck),
- muscles of the outer cervical collar (move the head and upper limb), and
- postural muscles in the muscular compartment of the neck (position the neck and head).

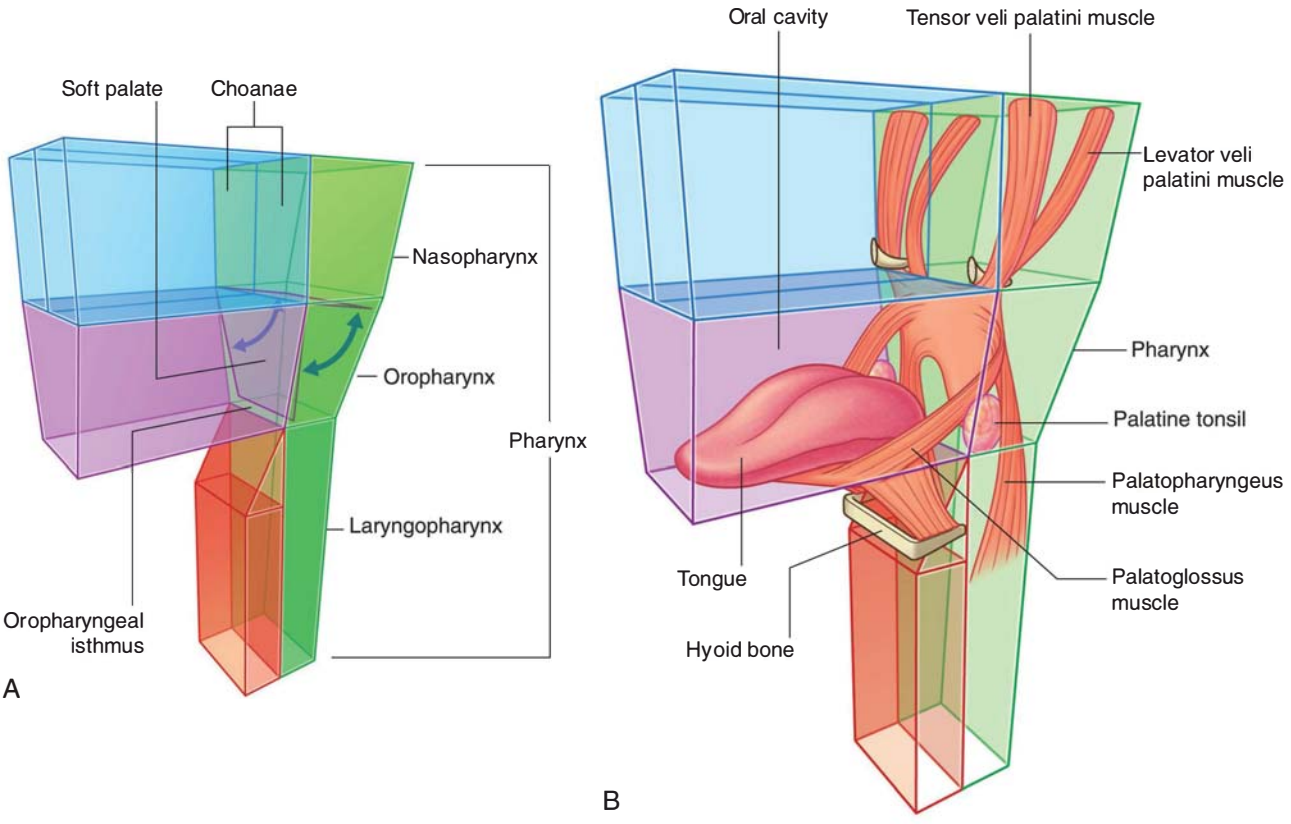


Fig. 8.10 Soft palate. **A.** Position. **B.** Muscles.

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RELATIONSHIP TO OTHER REGIONS

Thorax

The **superior thoracic aperture** (**thoracic inlet**) opens directly into the base of the neck (Fig. 8.11). Structures passing between the head and thorax pass up and down through the superior thoracic aperture and the visceral compartment of the neck. At the base of the neck, the trachea is immediately anterior to the esophagus, which is directly anterior to the vertebral column. There are major veins, arteries, and nerves anterior and lateral to the trachea.

Upper limbs

There is an axillary inlet (gateway to the upper limb) on each side of the superior thoracic aperture at the base of the neck (Fig. 8.11):

- Structures such as blood vessels pass over rib I when passing between the axillary inlet and thorax.
- Cervical components of the brachial plexus pass directly from the neck through the axillary inlets to enter the upper limb.

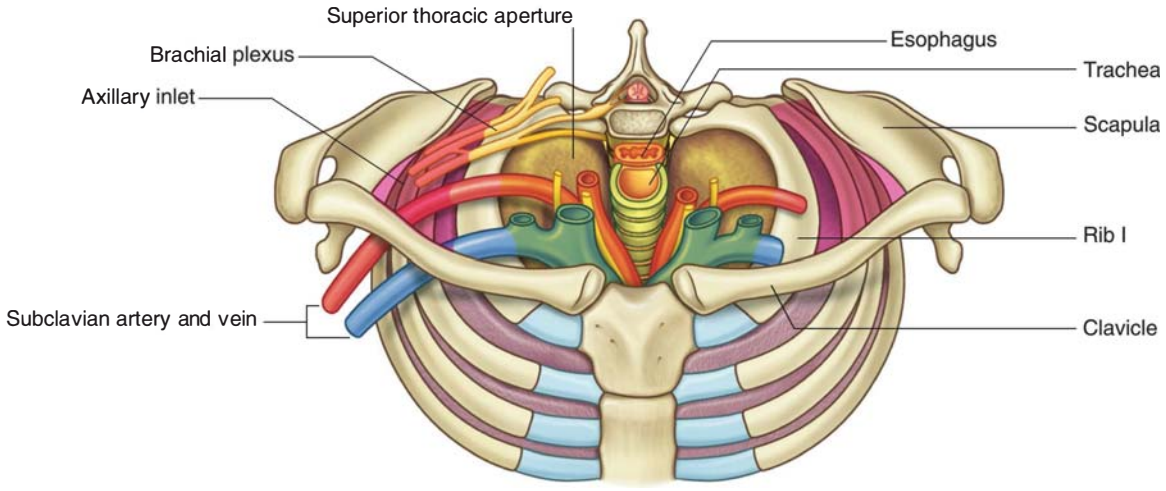


Fig. 8.11 Superior thoracic aperture and axillary inlets.

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KEY FEATURES

Vertebral levels CIII/IV and CV/VI

In the neck, the two important vertebral levels (Fig. 8.12) are:

- between CIII and CIV, at approximately the superior border of the thyroid cartilage of the larynx (which can be palpated) and where the major artery on each side of the neck (the **common carotid artery**) bifurcates into internal and external carotid arteries; and
- between CV and CVI, which marks the lower limit of the pharynx and larynx, and the superior limit of the trachea and esophagus—the indentation between the cricoid cartilage of the larynx and the first tracheal ring can be palpated.

The internal carotid artery has no branches in the neck and ascends into the skull to supply much of the brain. It also supplies the eye and orbit. Other regions of the head and neck are supplied by branches of the external carotid artery.

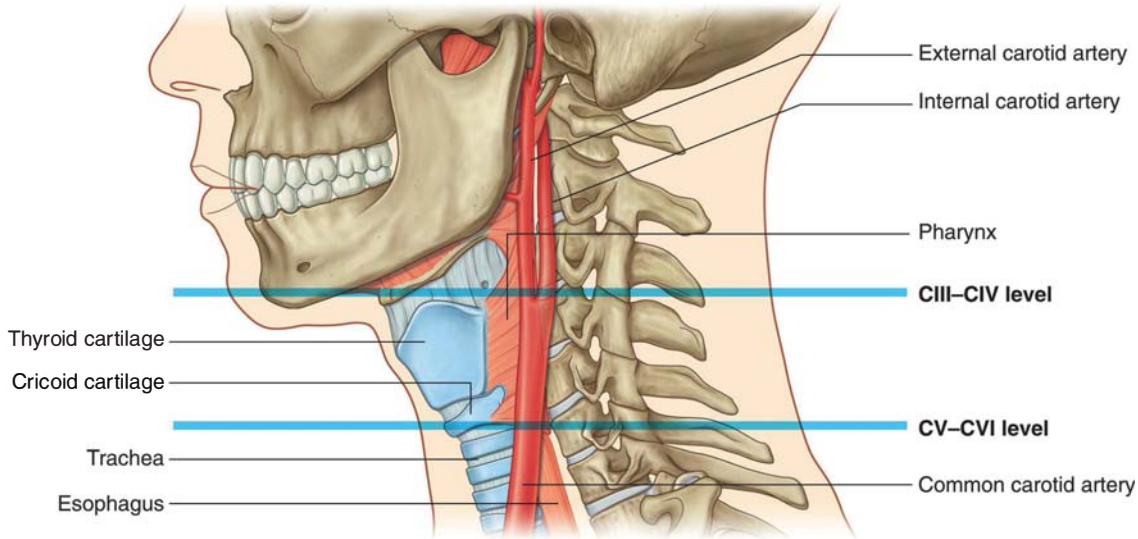


Fig. 8.12 Important vertebral levels—CIII/CIV and CV/CVI.

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Airway in the neck

The larynx (Fig. 8.13) and the trachea are anterior to the digestive tract in the neck, and can be accessed directly when upper parts of the system are blocked. A **cricothyrotomy** makes use of the easiest route of access through the **cricothyroid ligament** (cricovocal membrane, cricothyroid membrane) between the cricoid and thyroid cartilages of the larynx. The ligament can be palpated in the midline, and usually there are only small blood vessels, connective tissue, and skin (though occasionally, a small lobe of the thyroid gland—pyramidal lobe) overlying it. At a lower level, the airway can be accessed surgically through the anterior wall of the trachea by **tracheostomy**. This route of entry is complicated because large veins and part of the thyroid gland overlie this region.

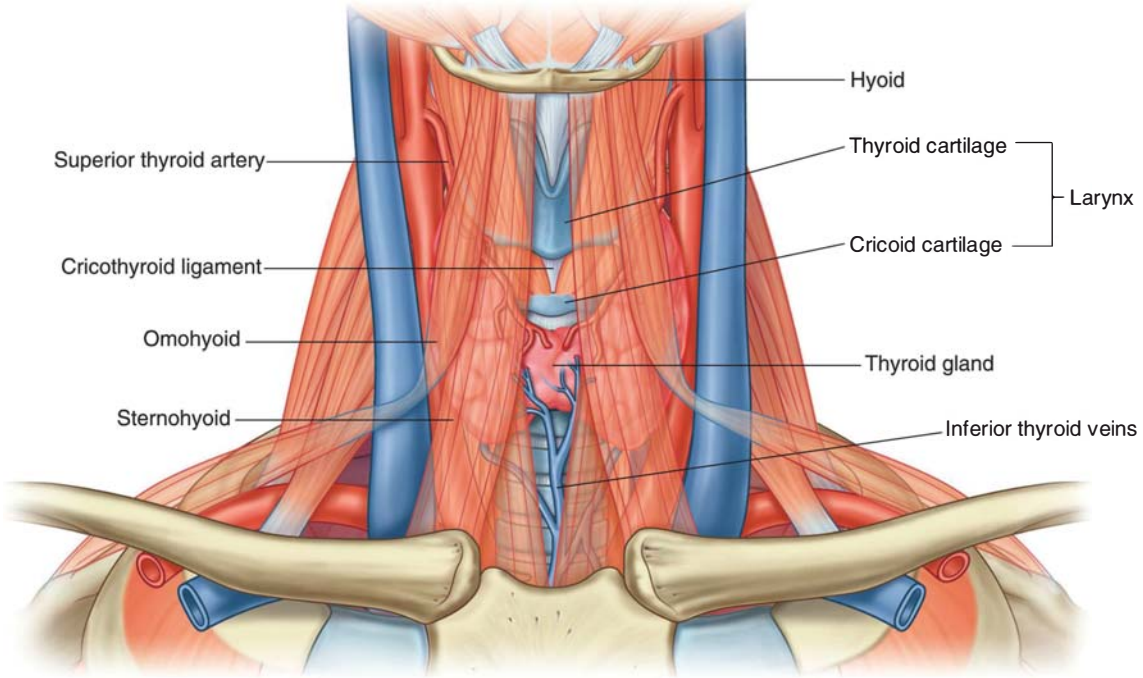


Fig. 8.13 Larynx and associated structures in the neck.

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Cranial nerves

There are twelve pairs of cranial nerves and their defining feature is that they exit the cranial cavity through foramina or fissures.

All cranial nerves innervate structures in the head or neck. In addition, the **vagus nerve [X]** descends through the neck and into the thorax and abdomen where it innervates viscera.

Parasympathetic fibers in the head are carried out of the brain as part of four cranial nerves—the oculomotor nerve [III], the facial nerve [VII], the glossopharyngeal nerve [IX], and the vagus nerve [X] (Fig. 8.14). Parasympathetic fibers in the oculomotor nerve [III], the facial nerve [VII], and the glossopharyngeal nerve [IX] destined for target tissues in the head leave these nerves, and are distributed with branches of the trigeminal nerve [V].

The vagus nerve [X] leaves the head and neck to deliver parasympathetic fibers to the thoracic and abdominal viscera.

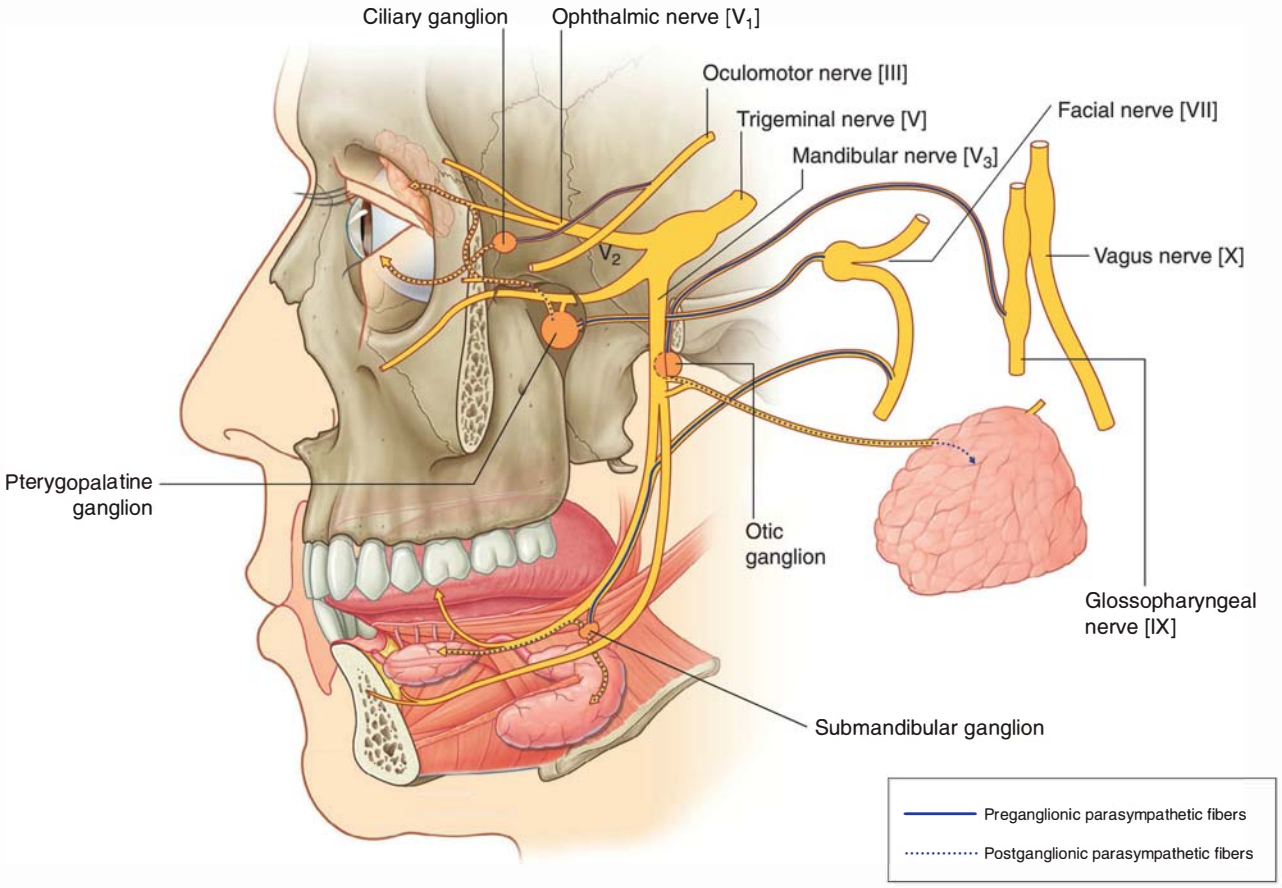


Fig. 8.14 Cranial nerves and parasympathetic innervation.

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Cervical nerves

There are eight cervical nerves (C1 to C8):

- C1 to C7 emerge from the vertebral canal above their respective vertebrae.
- C8 emerges between vertebrae CVII and TI (Fig. 8.15A).

The anterior rami of C1 to C4 form the **cervical plexus**. The major branches from this plexus supply the strap muscles, the diaphragm (phrenic nerve), skin on the anterior and lateral parts of the neck, skin on the upper anterior thoracic wall, and skin on the inferior parts of the head (Fig. 8.15B).

The anterior rami of C5 to C8, together with a large component of the anterior ramus of T1, form the **brachial plexus**, which innervates the upper limb.

Functional separation of the digestive and respiratory passages

The pharynx is a common chamber for the digestive and respiratory tracts. Consequently, breathing can take place through the mouth as well as through the nose, and

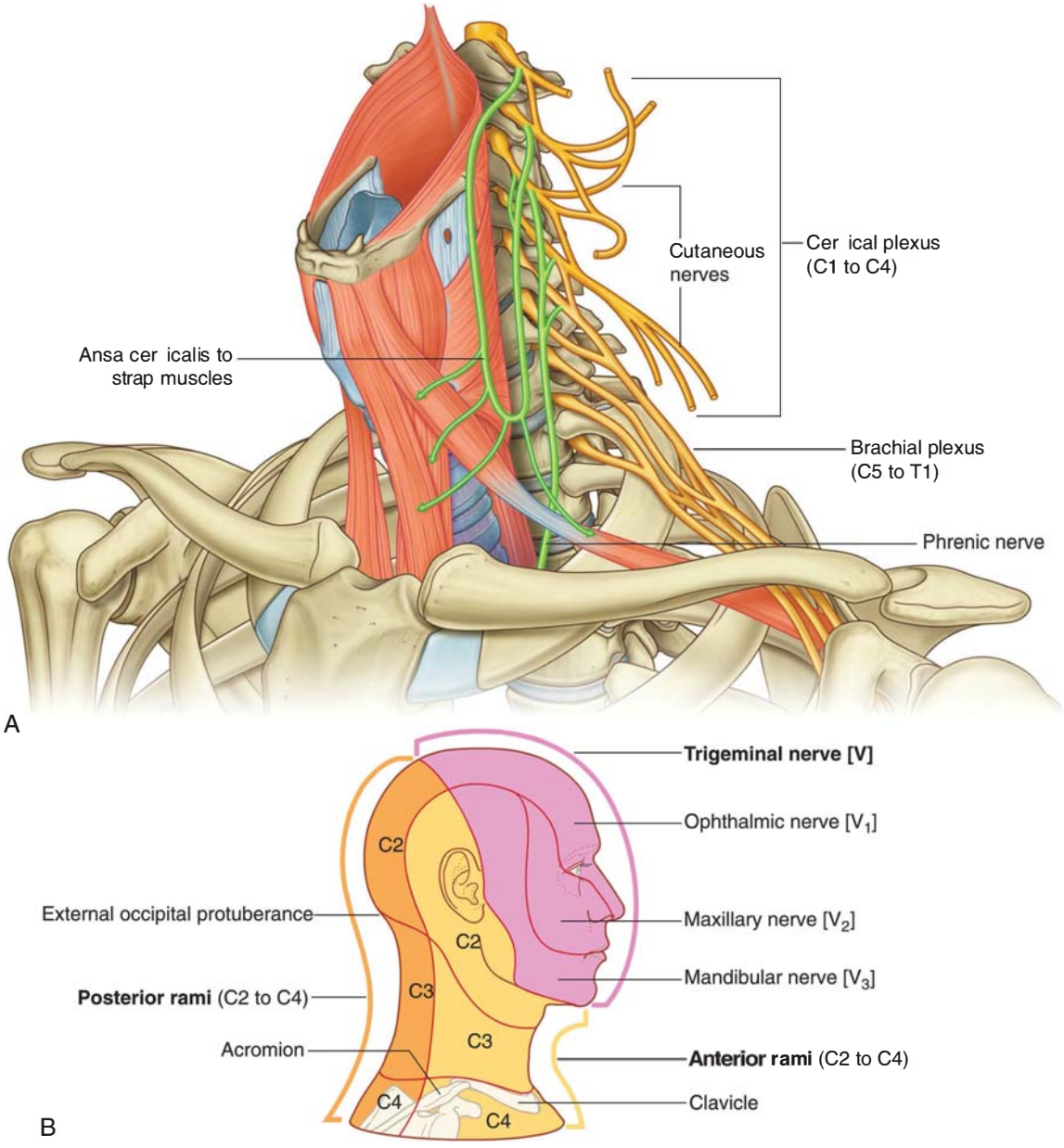


Fig. 8.15 Cervical nerves. **A.** Structure. **B.** Dermatomes.

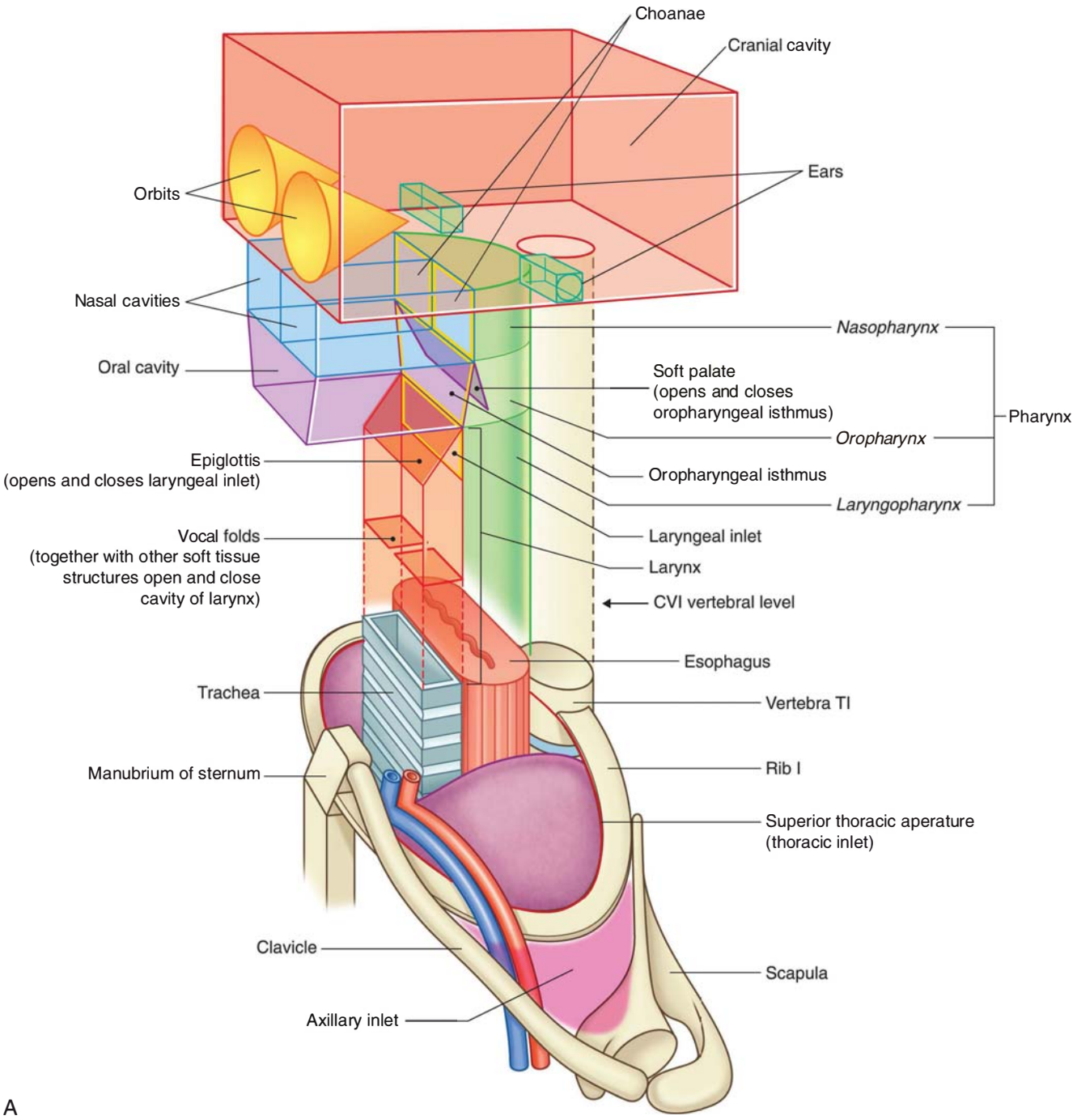
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material from the oral cavity can potentially enter either the esophagus or the larynx. Importantly:

- The lower airway can be accessed through the oral cavity by intubation.
- The digestive tract (esophagus) can be accessed through the nasal cavity by feeding tubes.

Normally, the soft palate, epiglottis, and soft tissue structures within the larynx act as valves to prevent food and liquid from entering lower parts of the respiratory tract (Fig. 8.16A).

During normal breathing, the airway is open and air passes freely through the nasal cavities (or oral cavity), pharynx, larynx, and trachea (Fig. 8.16A). The lumen of



A

Fig. 8.16 Larynx, soft palate, epiglottis, and oropharyngeal isthmus. **A.** Overall design.

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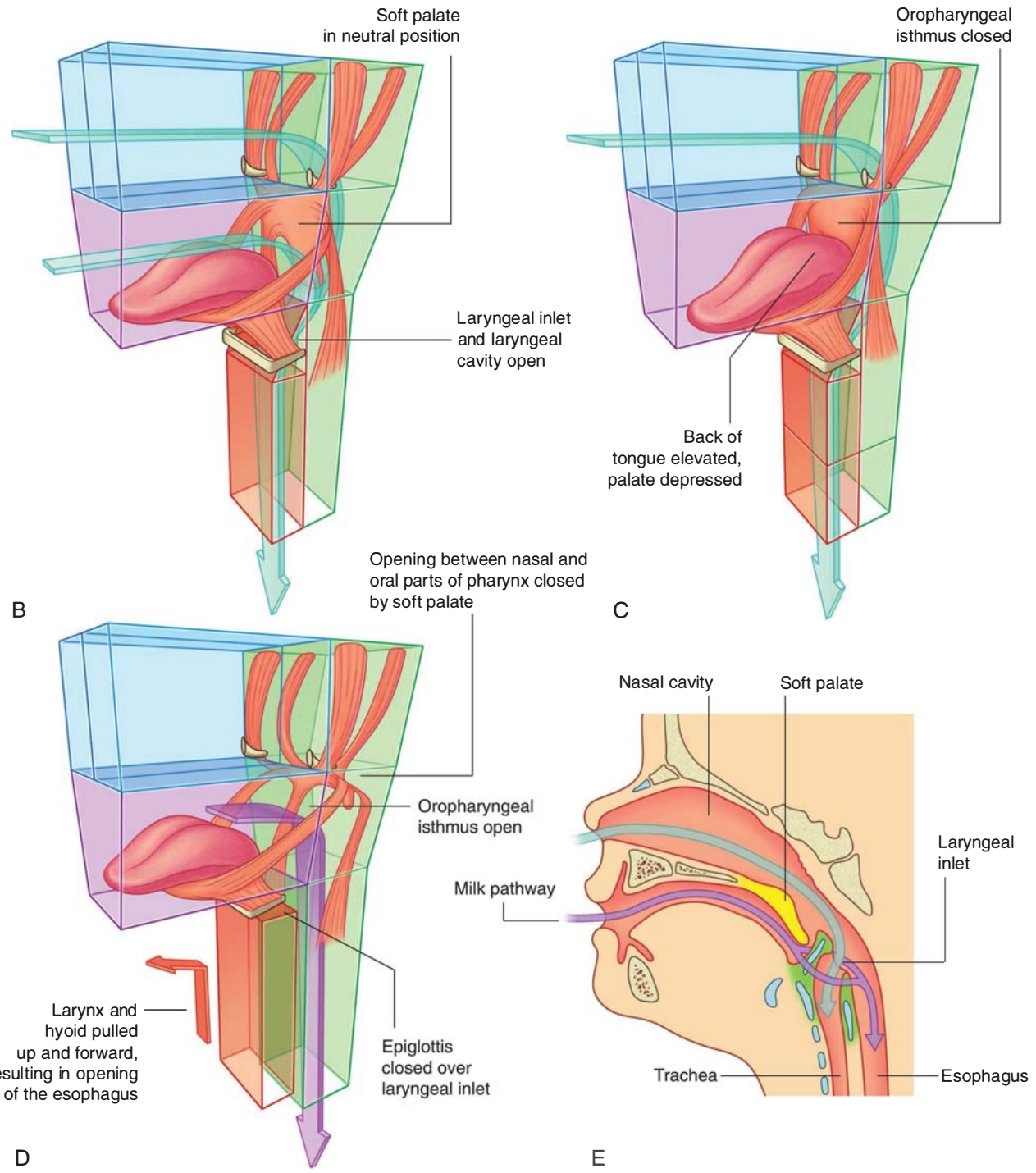


Fig. 8.16, cont'd **B.** Normal breathing. **C.** Breathing with food or liquid in the oral cavity. **D.** Swallowing. **E.** In a newborn child.

the esophagus is normally closed because, unlike the airway, it has no skeletal support structures to hold it open.

When the oral cavity is full of liquid or food, the soft palate is swung down (depressed) to close the oropharyngeal isthmus, thereby allowing manipulation of food and fluid in the oral cavity while breathing (Fig. 8.16C).

When swallowing, the soft palate and parts of the larynx act as valves to ensure proper movement of food from the oral cavity into the esophagus (Fig. 8.16D).

The soft palate elevates to open the oropharyngeal isthmus while at the same time sealing off the nasal part of the pharynx from the oral part. This prevents food and

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fluid from moving upward into the nasopharynx and nasal cavities.

The epiglottis of the larynx closes the laryngeal inlet and much of the laryngeal cavity becomes occluded by opposition of the vocal folds and soft tissue folds superior to them. In addition, the larynx is pulled up and forward to facilitate the moving of food and fluid over and around the closed larynx and into the esophagus.

In newborns, the larynx is high in the neck and the epiglottis is above the level of the soft palate (Fig. 8.16E). Babies can therefore suckle and breathe at the same time. Liquid flows around the larynx without any danger of entering the airway. During the second year of life, the larynx descends into the low cervical position characteristic of adults.

Triangles of the neck

The two muscles (trapezius and sternocleidomastoid) that form part of the outer cervical collar divide the neck into anterior and posterior triangles on each side (Fig. 8.17).

The boundaries of each anterior triangle are:

- the median vertical line of the neck,
- the inferior margin of the mandible, and
- the anterior margin of the sternocleidomastoid muscle.

The posterior triangle is bounded by:

- the middle one-third of the clavicle,
- the anterior margin of the trapezius, and
- the posterior margin of the sternocleidomastoid.

Major structures that pass between the head and thorax can be accessed through the anterior triangle.

The posterior triangle in part lies over the axillary inlet, and is associated with structures (nerves and vessels) that pass into and out of the upper limb.

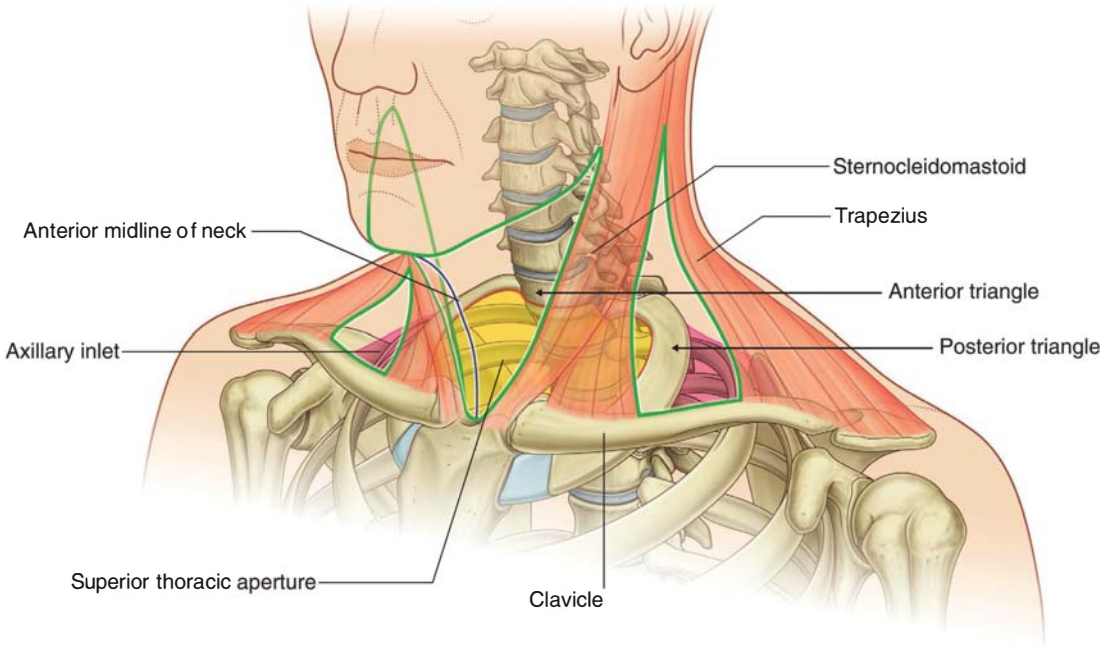


Fig. 8.17 Anterior and posterior triangles of neck.

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Regional anatomy

SKULL

The skull has 22 bones, excluding the ossicles of the ear. Except for the mandible, which forms the lower jaw, the bones of the skull are attached to each other by sutures, are immobile, and form the **cranium**.

The cranium can be subdivided into:

- an upper domed part (the **calvaria**), which covers the cranial cavity containing the brain,
- a base that consists of the floor of the cranial cavity, and
- a lower anterior part—the **facial skeleton** (**viscerocranium**).

The bones forming the calvaria are mainly the paired temporal and parietal bones, and parts of the unpaired frontal, sphenoid, and occipital bones.

The bones forming the base of the cranium are mainly parts of the sphenoid, temporal, and occipital bones.

The bones forming the facial skeleton are the paired nasal bones, palatine bones, lacrimal bones, zygomatic bones, maxillae and inferior nasal conchae and the unpaired vomer.

The mandible is not part of the cranium nor part of the facial skeleton.

Anterior view

The anterior view of the skull includes the **forehead** superiorly, and, inferiorly, the orbits, the **nasal region**, the part of the face between the orbit and the upper jaw, the upper jaw, and the lower jaw (Fig. 8.18).

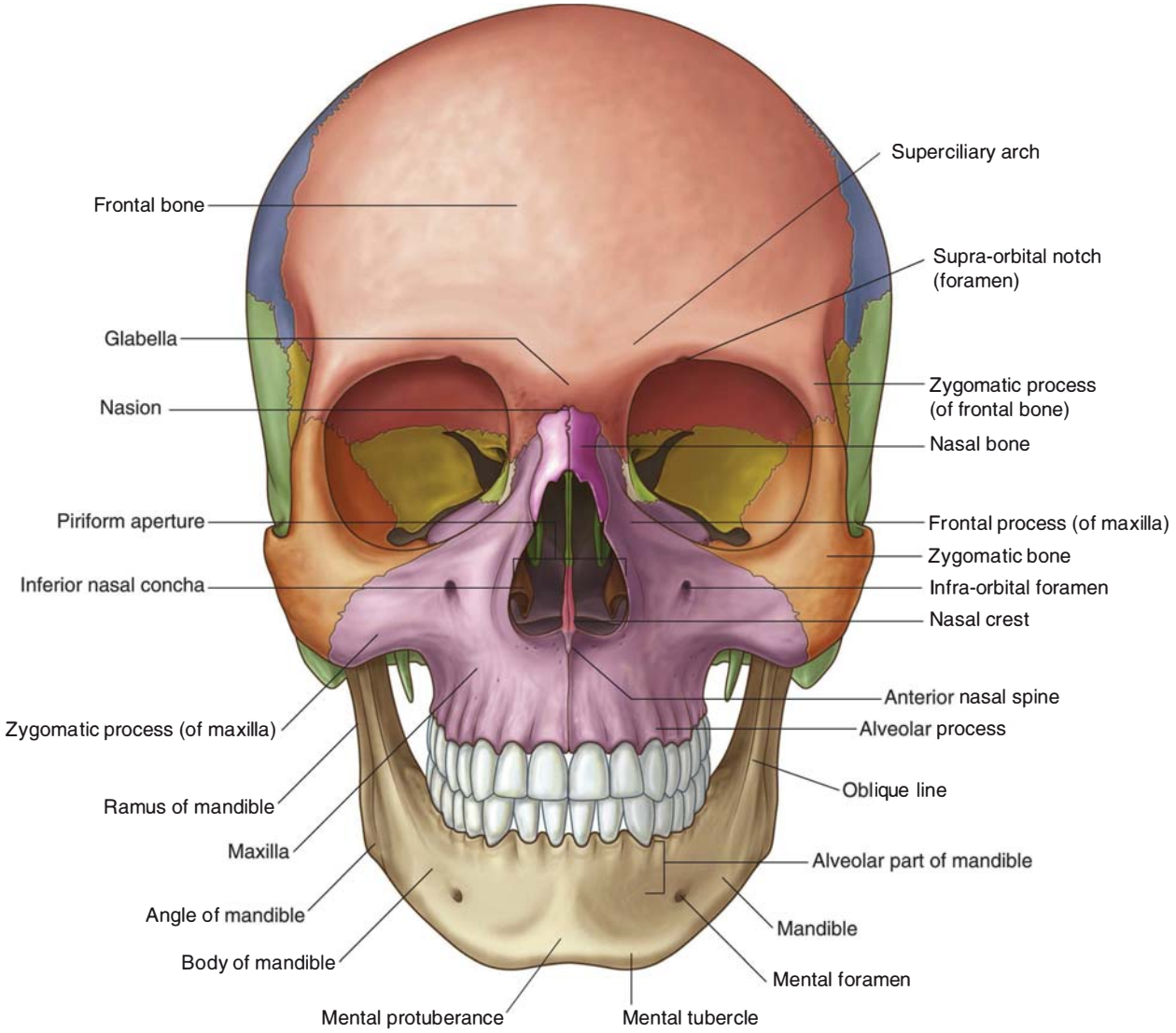


Fig. 8.18 Anterior view of the skull.

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Frontal bone

The forehead consists of the **frontal bone**, which also forms the superior part of the rim of each orbit (Fig. 8.18).

Just superior to the rim of the orbit on each side are the raised **superciliary arches**. These are more pronounced in men than in women. Between these arches is a small depression (the **glabella**).

Clearly visible in the medial part of the superior rim of each orbit is the **supra-orbital foramen** (**supra-orbital notch**; Table 8.1).

Medially, the frontal bone projects inferiorly forming a part of the medial rim of the orbit.

Laterally, the **zygomatic process** of the frontal bone projects inferiorly forming the upper lateral rim of the orbit. This process articulates with the **frontal process** of the zygomatic bone.

Table 8.1 External foramina of the skull

Foramen	Structures passing through foramen
ANTERIOR VIEW	
Supra-orbital foramen	Supra-orbital nerve and vessels
Infra-orbital foramen	Infra-orbital nerve and vessels
Mental foramen	Mental nerve and vessels
LATERAL VIEW	
Zygomaticofacial foramen	Zygomaticofacial nerve
SUPERIOR VIEW	
Parietal foramen	Emissary veins
INFERIOR VIEW	
Incisive foramen	Nasopalatine nerve; sphenopalatine vessels
Greater palatine foramen	Greater palatine nerve and vessels
Lesser palatine foramen	Lesser palatine nerves and vessels
Pterygoid canal	Pterygoid nerve and vessels
Foramen ovale	Mandibular nerve [V_3]; lesser petrosal nerve
Foramen spinosum	Middle meningeal artery
Foramen lacerum	Filled with cartilage
Carotid canal	Internal carotid artery and nerve plexus
Foramen magnum	Continuation of brain and spinal cord; vertebral arteries and nerve plexuses; anterior spinal artery; posterior spinal arteries; roots of accessory nerve [XI]; meninges
Condylar canal	Emissary veins
Hypoglossal canal	Hypoglossal nerve [XII] and vessels
Jugular foramen	Internal jugular vein; inferior petrosal sinus; glossopharyngeal nerve [IX]; vagus nerve [X]; accessory nerve [XI]
Stylomastoid foramen	Facial nerve [VII]

Zygomatic and nasal bones

The lower lateral rim of the orbit, as well as the lateral part of the inferior rim of the orbit is formed by the **zygomatic bone** (the cheekbone).

Superiorly, in the nasal region the paired nasal bones articulate with each other in the midline, and with the frontal bone superiorly. The center of the **frontonasal suture** formed by the articulation of the nasal bones and the frontal bone is the **nasion**.

Laterally, each nasal bone articulates with the **frontal process** of each maxilla.

Inferiorly, the **piriform aperture** is the large opening in the nasal region and the anterior opening of the nasal cavity. It is bounded superiorly by the nasal bones and laterally and inferiorly by each maxilla.

Visible through the piriform aperture are the fused **nasal crests**, forming the lower part of the bony **nasal septum** and ending anteriorly as the **anterior nasal spine**, and the paired **inferior nasal conchae**.

Maxillae

The part of the face between the orbit and the upper teeth and each upper jaw is formed by the paired maxillae.

Superiorly, each maxilla contributes to the inferior and medial rims of the orbit.

Laterally, the **zygomatic process** of each maxilla articulates with the zygomatic bone and medially, the frontal process of each maxilla articulates with the frontal bone.

Inferiorly, the part of each maxilla, lateral to the opening of the nasal cavity, is the **body of the maxilla**.

On the anterior surface of the body of the maxilla, just below the inferior rim of the orbit, is the **infra-orbital foramen** (Table 8.1).

Inferiorly, each maxilla ends as the **alveolar process**, which contains the teeth and forms the upper jaw.

Mandible

The lower jaw (mandible) is the most inferior structure in the anterior view of the skull. It consists of the **body of the mandible** anteriorly and the **ramus of the mandible** posteriorly. These meet posteriorly at the **angle of the mandible**. All these parts of the mandible are visible, to some extent, in the anterior view.

The body of the mandible is arbitrarily divided into two parts:

- The lower part is the **base of the mandible**.
- The upper part is the **alveolar part of the mandible**.

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The alveolar part of the mandible contains the teeth and is resorbed when the teeth are removed. The base of the mandible has a midline swelling (the **mental protuberance**) on its anterior surface where the two sides of the mandible come together. Just lateral to the mental protuberance, on either side, are slightly more pronounced bumps (**mental tubercles**).

Laterally, a **mental foramen** (Table 8.1) is visible halfway between the upper border of the alveolar part of the mandible and the lower border of the base of the mandible. Continuing past this foramen is a ridge (the **oblique line**) passing from the front of the ramus onto the body of the mandible. The oblique line is a point of attachment for muscles that depress the lower lip.

Lateral view

The lateral view of the skull consists of the lateral wall of the cranium, which includes lateral portions of the calvaria and the facial skeleton, and half of the lower jaw (Fig. 8.19):

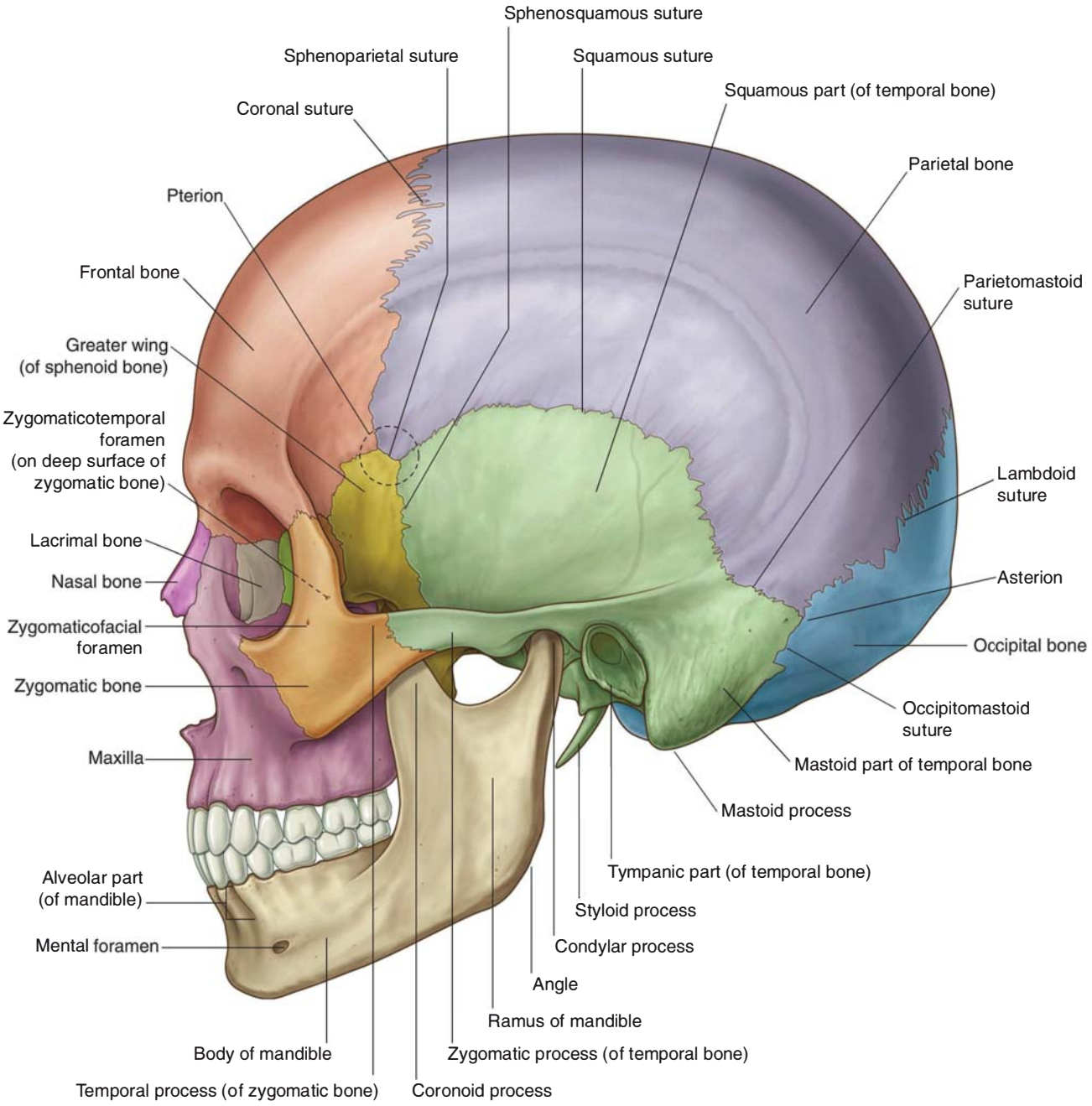


Fig. 8.19 Lateral view of the skull.

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- Bones forming the lateral portion of the calvaria include the frontal, parietal, occipital, sphenoid, and temporal bones.
- Bones forming the visible part of the facial skeleton include the nasal, maxilla, and zygomatic bones.
- The mandible forms the visible part of the lower jaw.

Lateral portion of the calvaria

The lateral portion of the calvaria begins anteriorly with the frontal bone. In upper regions, the frontal bone articulates with the parietal bone at the **coronal suture**. The parietal bone then articulates with the occipital bone at the **lambdoid suture**.

In lower parts of the lateral portion of the calvaria, the frontal bone articulates with the **greater wing of the sphenoid bone** (Fig. 8.19), which then articulates with the parietal bone at the **sphenoparietal suture**, and with the anterior edge of the temporal bone at the **sphenosquamous suture**.

The junction where the frontal, parietal, sphenoid, and temporal bones are in close proximity is the **pterion**. The clinical consequences of a skull fracture in this area can be very serious. The bone in this area is particularly thin and overlies the anterior division of the middle meningeal artery, which can be torn by a skull fracture in this area, resulting in an extradural hematoma.

The final articulation across the lower part of the lateral portion of the calvaria is between the temporal bone and the occipital bone at the **occipitomastoid suture**.

Temporal bone

A major contributor to the lower portion of the lateral wall of the cranium is the temporal bone (Fig. 8.19), which consists of several parts:

- The **squamous part** has the appearance of a large flat plate, forms the anterior and superior parts of the temporal bone, contributes to the lateral wall of the cranium, and articulates anteriorly with the greater wing of the sphenoid bone at the sphenosquamous suture, and with the parietal bone superiorly at the squamous suture.
- The **zygomatic process** is an anterior bony projection from the lower surface of the squamous part of the temporal bone that initially projects laterally and then curves anteriorly to articulate with the temporal process of the zygomatic bone to form the **zygomatic arch**.
- Immediately below the origin of the zygomatic process from the squamous part of the temporal bone is the **tympanic part** of the temporal bone, and clearly visible on the surface of this part is the **external acoustic opening** leading to the **external acoustic meatus** (ear canal).
- The petromastoid part, which is usually separated into a **petrous part** and a **mastoid part** for descriptive purposes.

The mastoid part is the most posterior part of the temporal bone, and is the only part of the petromastoid part of the temporal bone seen on a lateral view of the skull. It is continuous with the squamous part of the temporal bone anteriorly, and articulates with the parietal bone superiorly at the **parietomastoid suture**, and with the occipital bone posteriorly at the occipitomastoid suture. These two sutures are continuous with each other, and the parietomastoid suture is continuous with the squamous suture.

Inferiorly, a large bony prominence (the **mastoid process**) projects from the inferior border of the mastoid part of the temporal bone. This is a point of attachment for several muscles.

Medial to the mastoid process, the **styloid process** projects from the lower border of the temporal bone.

Visible part of the facial skeleton

The bones of the viscerocranium visible in a lateral view of the skull include the nasal, maxilla, and zygomatic bones (Fig. 8.19) as follows:

- A nasal bone anteriorly.
- The maxilla with its alveolar process containing teeth forming the upper jaw; anteriorly, it articulates with the nasal bone; superiorly, it contributes to the formation of the inferior and medial borders of the orbit; medially, its frontal process articulates with the frontal bone; laterally, its zygomatic process articulates with the zygomatic bone.
- The zygomatic bone, an irregularly shaped bone with a rounded lateral surface that forms the prominence of the cheek, is a visual centerpiece in this view— medially, it assists in the formation of the inferior rim of the orbit through its articulation with the zygomatic process of the maxilla; superiorly, its frontal process articulates with the zygomatic process of the frontal bone assisting in the formation of the lateral rim of the orbit; laterally, seen prominently in this view of the skull, the horizontal temporal process of the zygomatic bone projects backward to articulate with the zygomatic process of the temporal bone and so form the zygomatic arch.

Usually a small foramen (the **zygomaticofacial foramen**; Table 8.1) is visible on the lateral surface of the zygomatic bone. A **zygomaticotemporal foramen** is present on the medial deep surface of the bone.

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Mandible

The final bony structure visible in a lateral view of the skull is the mandible. Inferiorly in the anterior part of this view, it consists of the anterior body of the mandible, a posterior ramus of the mandible, and the angle of the mandible where the inferior margin of the mandible meets the posterior margin of the ramus (Fig. 8.19).

The teeth are in the alveolar part of the body of the mandible and the mental protuberance is visible in this view.

The mental foramen is on the lateral surface of the body, and on the superior part of the ramus **condylar** and **coronoid processes** extend upward.

The condylar process is involved in articulation of the mandible with the temporal bone, and the coronoid process is the point of attachment for the temporalis muscle.

Posterior view

The occipital, parietal, and temporal bones are seen in the posterior view of the skull.

Occipital bone

Centrally the flat or **squamous part of the occipital bone** is the main structure in this view of the skull (Fig. 8.20). It articulates superiorly with the paired parietal bones at the lambdoid suture and laterally with each temporal bone at the occipitomastoid sutures. Along the lambdoid suture small islands of bone (**sutural bones** or wormian bones) may be observed.

Several bony landmarks are visible on the occipital bone. There is a midline projection (the **external occipital protuberance**) with curved lines extending laterally from

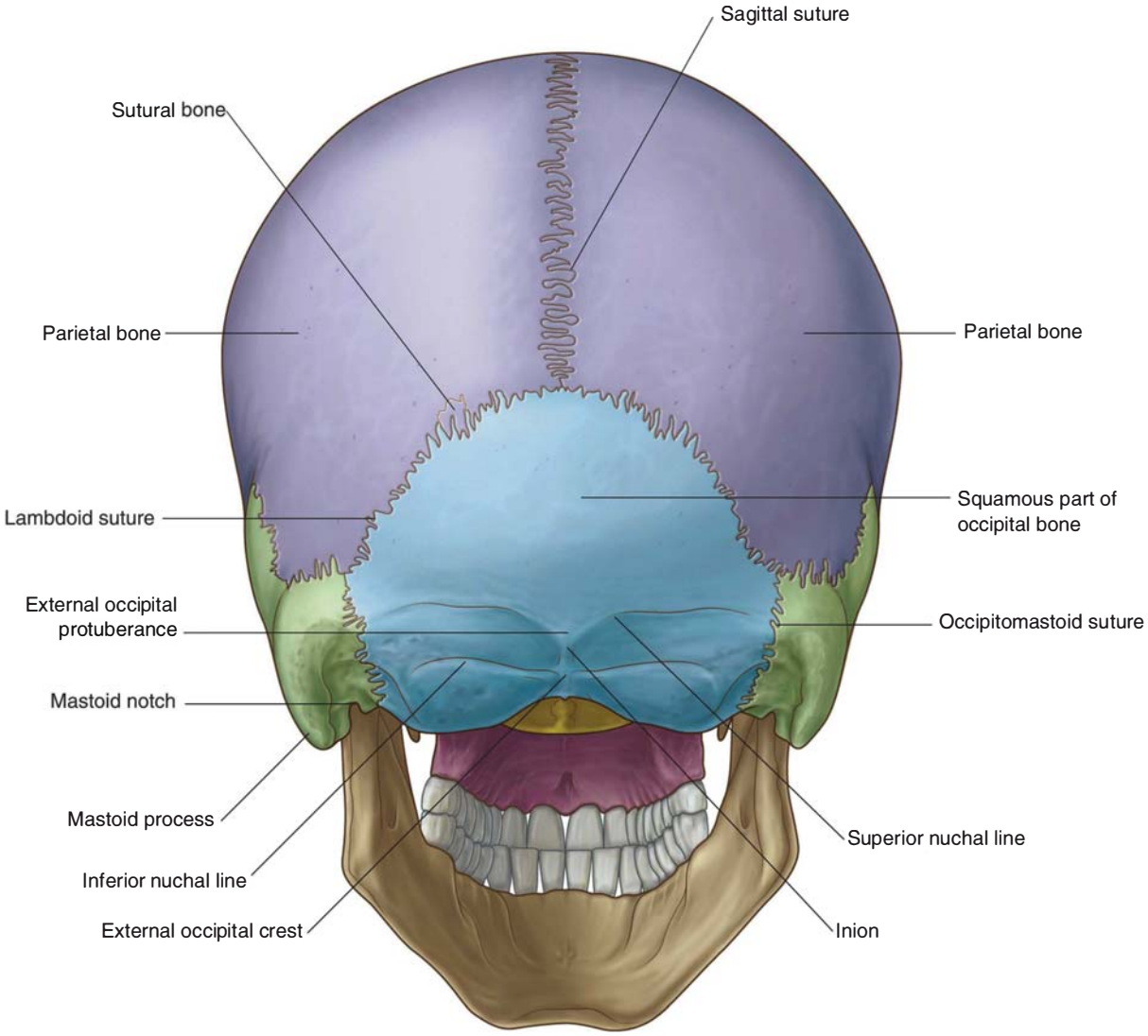


Fig. 8.20 Posterior view of the skull.

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it (**superior nuchal lines**). The most prominent point of the external occipital protuberance is the **inion**. About 1 inch (2.5 cm) below the superior nuchal lines two additional lines (the **inferior nuchal lines**) curve laterally. Extending downward from the external occipital protuberance is the **external occipital crest**.

Temporal bones

Laterally, the temporal bones are visible in the posterior view of the skull, with the mastoid processes being the prominent feature (Fig. 8.20). On the inferomedial border of each mastoid process is a notch (the **mastoid notch**), which is a point of attachment for the posterior belly of the digastric muscle.

Superior view

The frontal bone, parietal bones, and occipital bone are seen in a superior view of the skull (Fig. 8.21). These bones make up the superior part of the calvaria or the **calva** (skullcap).

In an anterior to posterior direction:

- The unpaired frontal bone articulates with the paired parietal bones at the coronal suture.
- The two parietal bones articulate with each other in the midline at the sagittal suture.
- The parietal bones articulate with the unpaired occipital bone at the lambdoid suture.

The junction of the sagittal and coronal sutures is the **bregma**, and the junction of the sagittal and lambdoid sutures is the **lambda**.

The only foramina visible in this view of the skull may be the paired parietal foramina, posteriorly, one on each parietal bone just lateral to the sagittal suture (Fig. 8.21).

The bones making up the calvaria (Fig. 8.22) are unique in their structure, consisting of dense internal and external tables of compact bone separated by a layer of spongy bone (the **diploë**).

Inferior view

The base of the skull is seen in the inferior view and extends anteriorly from the middle incisor teeth posteriorly to the superior nuchal lines and laterally to the mastoid processes and zygomatic arches (Fig. 8.23).

For descriptive purposes the base of the skull is often divided into:

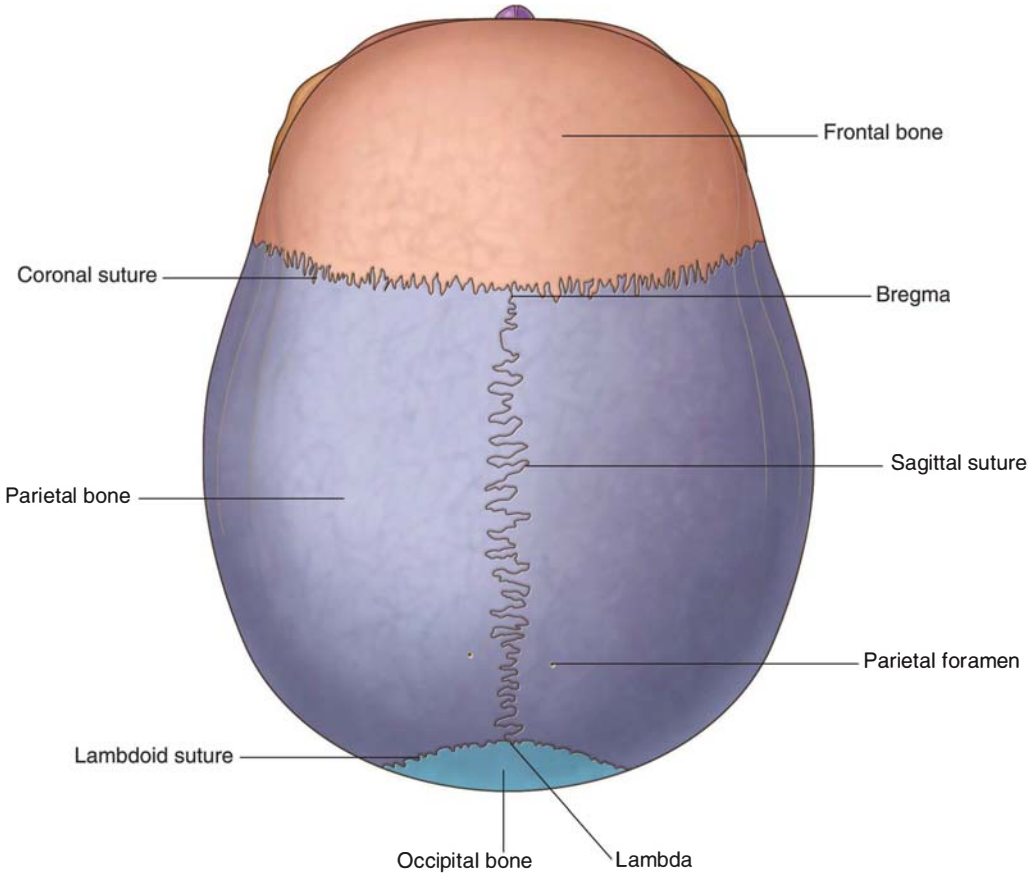


Fig. 8.21 Superior view of the skull.

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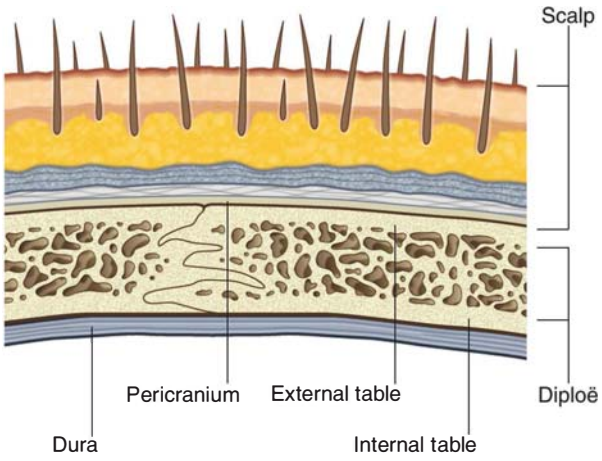


Fig. 8.22 Calvaria.

- an anterior part, which includes the teeth and the hard palate,
- a middle part, which extends from behind the hard palate to the anterior margin of the foramen magnum, and
- a posterior part, which extends from the anterior edge of the foramen magnum to the superior nuchal lines.

Anterior part

The main features of the anterior part of the base of the skull are the teeth and the hard palate.

The teeth project from the **alveolar processes** of the two maxillae. These processes are together arranged in a U-shaped alveolar arch that borders the hard palate on three sides (Fig. 8.23).

The **hard palate** is composed of the **palatine processes** of each maxilla anteriorly and the **horizontal plates** of each **palatine bone** posteriorly.

The paired palatine processes of each maxilla meet in the midline at the **intermaxillary suture**, the paired maxillae and the paired palatine bones meet at the **palatomaxillary suture**, and the paired horizontal plates of each palatine bone meet in the midline at the **interpalatine suture**.

Several additional features are also visible when the hard palate is examined:

- the **incisive fossa** in the anterior midline immediately posterior to the teeth, the walls of which contain **incisive foramina** (the openings of the **incisive canals**, which are passageways between the hard palate and nasal cavity);
- the **greater palatine foramina** near the posterolateral border of the hard palate on each side, which lead to **greater palatine canals**;
- just posterior to the greater palatine foramina, the **lesser palatine foramina** in the **pyramidal process** of each palatine bone, which lead to **lesser palatine canals**;
- a midline pointed projection (the **posterior nasal spine**) in the free posterior border of the hard palate.

Middle part

The middle part of the base of the skull is complex:

- Forming the anterior half are the vomer and sphenoid bones.
- Forming the posterior half are the occipital and paired temporal bones.

Anterior half

Vomer

Anteriorly, the small vomer is in the midline, resting on the sphenoid bone (Fig. 8.23). It contributes to the formation of the bony nasal septum separating the two choanae.

Sphenoid

Most of the anterior part of the middle part of the base of the skull consists of the sphenoid bone.

The sphenoid bone is made up of a centrally placed **body**, paired **greater and lesser wings** projecting laterally from the body, and two downward projecting **pterygoid processes** immediately lateral to each choana.

Three parts of the sphenoid bone, the body, greater wings, and pterygoid processes, are seen in the inferior view of the skull (Fig. 8.23). The lesser wing of the sphenoid is not seen in the inferior view.

Body

The body of the sphenoid is a centrally placed cube of bone containing two large air sinuses separated by a septum.

It articulates anteriorly with the vomer, ethmoid, and palatine bones, posterolaterally with the temporal bones, and posteriorly with the occipital bone.

Pterygoid processes

Extending downward from the junction of the body and the greater wings are the pterygoid processes (Fig. 8.23). Each of these processes consists of a narrow **medial plate** and broader **lateral plate** separated by the **pterygoid fossa**.

Each medial plate of the pterygoid process ends inferiorly with a hook-like projection, the **pterygoid hamulus**, and divides superiorly to form the small, shallow **scaphoid fossa**.

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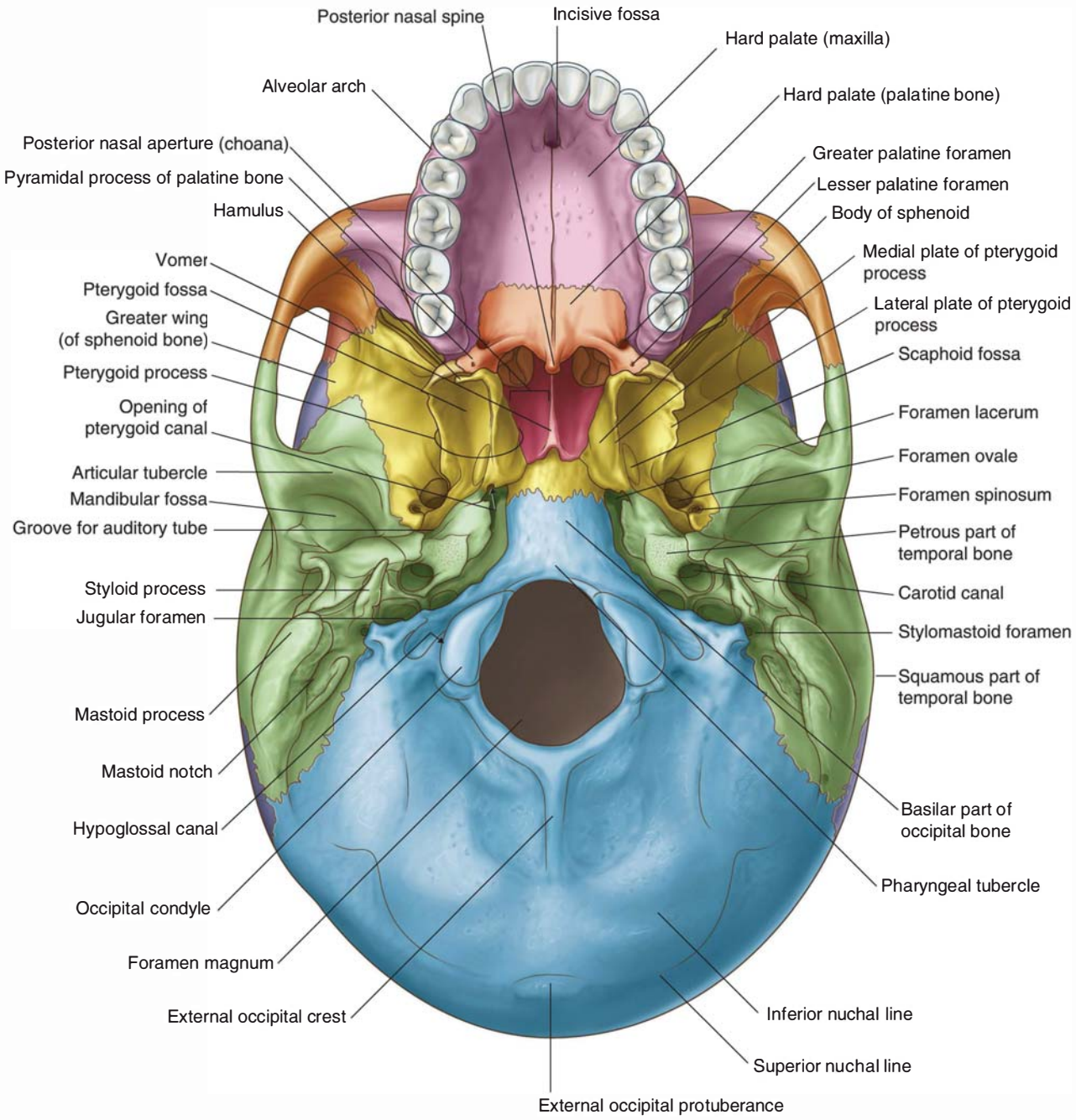


Fig. 8.23 Inferior view of the skull.

Just superior to the scaphoid fossa, at the root of the medial plate of the pterygoid process is the opening of the **pterygoid canal**, which passes forward from near the anterior margin of the foramen lacerum.

Greater wing

Lateral to the lateral plate of the pterygoid process is the greater wing of the sphenoid (Fig. 8.23), which not only forms a part of the base of the skull but also continues laterally to form part of the lateral wall of the skull. It articulates laterally and posteriorly with parts of the temporal bone.

Important features visible on the surface of the greater wing in an inferior view of the skull are the foramen ovale and the foramen spinosum on the posterolateral border

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extending outward from the upper end of the lateral plate of the pterygoid process.

Posterior half

In the posterior half of the middle part of the base of the skull are the occipital bone and the paired temporal bones (Fig. 8.23).

Occipital bone

The occipital bone, or more specifically its **basilar part**, is in the midline immediately posterior to the body of the sphenoid. It extends posteriorly to the **foramen magnum** and is bounded laterally by the temporal bones.

Prominent on the basilar part of the occipital bone is the **pharyngeal tubercle**, a bony protuberance for the attachment of parts of the pharynx to the base of the skull (Fig. 8.23).

Temporal bone

Immediately lateral to the basilar part of the occipital bone is the petrous part of the petromastoid part of each temporal bone.

Wedge-shaped in its appearance, with its **apex** anteromedial, the petrous part of the temporal bone is between the greater wing of the sphenoid anteriorly and the basilar part of the occipital bone posteriorly. The apex forms one of the boundaries of the **foramen lacerum**, an irregular opening filled in life with cartilage (Fig. 8.23).

The other boundaries of the foramen lacerum are the basilar part of the occipital bone medially and the body of the sphenoid anteriorly.

Posterolateral from the foramen lacerum along the petrous part of the temporal bone is the large circular opening for the **carotid canal**.

Between the petrous part of the temporal bone and the greater wing of the sphenoid is a groove for the cartilaginous part of the **pharyngotympanic tube** (**auditory tube**). This groove continues posterolaterally into a bony canal in the petrous part of the temporal bone for the pharyngotympanic tube.

Just lateral to the greater wing of the sphenoid is the squamous part of the temporal bone, which participates in the temporomandibular joint. It contains the **mandibular fossa**, which is a concavity where the head of the mandible articulates with the base of the skull. An important feature of this articulation is the prominent **articular tubercle**, which is the downward projection of the anterior border of the mandibular fossa (Fig. 8.23).

Posterior part

The posterior part of the base of the skull extends from the anterior edge of the foramen magnum posteriorly to the superior nuchal lines (Fig. 8.23). It consists of parts of the occipital bone centrally and the temporal bones laterally.

Occipital bone

The occipital bone is the major bony element of this part of the base of the skull (Fig. 8.23). It has four parts organized around the foramen magnum, which is a prominent feature of this part of the base of the skull and through which the brain and spinal cord are continuous.

The parts of the occipital bone are the squamous part, which is posterior to the foramen magnum, the **lateral parts**, which are lateral to the foramen magnum, and the **basilar part**, which is anterior to the foramen magnum (Fig. 8.23).

The squamous and lateral parts are components of the posterior part of the base of the skull.

The most visible feature of the squamous part of the occipital bone when examining the inferior view of the skull is a ridge of bone (the external occipital crest), which extends downward from the external occipital protuberance toward the foramen magnum. The inferior nuchal lines arc laterally from the midpoint of the crest.

Immediately lateral to the foramen magnum are the lateral parts of the occipital bones, which contain numerous important structural features.

On each anterolateral border of the foramen magnum are the rounded **occipital condyles** (Fig. 8.23). These paired structures articulate with the atlas (vertebra CI). Posterior to each condyle is a depression (the **condylar fossa**) containing a **condylar canal**, and anterior and superior to each condyle is the large **hypoglossal canal**. Lateral to each hypoglossal canal is a large, irregular **jugular foramen** formed by opposition of the **jugular notch** of the occipital bone and **jugular notch** of the temporal bone.

Temporal bone

Laterally in the posterior part of the base of the skull is the temporal bone. The parts of the temporal bone seen in this location are the mastoid part of the petromastoid part and the styloid process (Fig. 8.23).

The lateral edge of the mastoid part is identified by the large cone-shaped mastoid process projecting from its inferior surface. This prominent bony structure is the point of attachment for several muscles. On the medial aspect of the mastoid process is the deep mastoid notch, which is also an attachment point for a muscle.

Anteromedial to the mastoid process is the needle-shaped styloid process projecting from the lower border of the temporal bone. The styloid process is also a point of attachment for numerous muscles and ligaments.

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Finally, between the styloid process and the mastoid process is the stylomastoid foramen.

CRANIAL CAVITY

The cranial cavity is the space within the cranium that contains the brain, meninges, proximal parts of the cranial nerves, blood vessels, and cranial venous sinuses.

Roof

The calvaria is the dome-shaped roof that protects the superior aspect of the brain. It consists mainly of the frontal bone anteriorly, the paired parietal bones in the middle, and the occipital bone posteriorly (Fig. 8.24).

Sutures visible internally include:

- the coronal suture, between the frontal and parietal bones,
- the sagittal suture, between the paired parietal bones, and
- the lambdoid suture, between the parietal and occipital bones.

Visible junctions of these sutures are the bregma, where the coronal and sagittal sutures meet, and the lambda, where the lambdoid and sagittal sutures meet.

Other markings on the internal surface of the calva include bony ridges and numerous grooves and pits.

From anterior to posterior, features seen on the bony roof of the cranial cavity are:

- a midline ridge of bone extending from the surface of the frontal bone (the **frontal crest**), which is a point of attachment for the **falx cerebri** (a specialization of the dura mater that partially separates the two cerebral hemispheres);

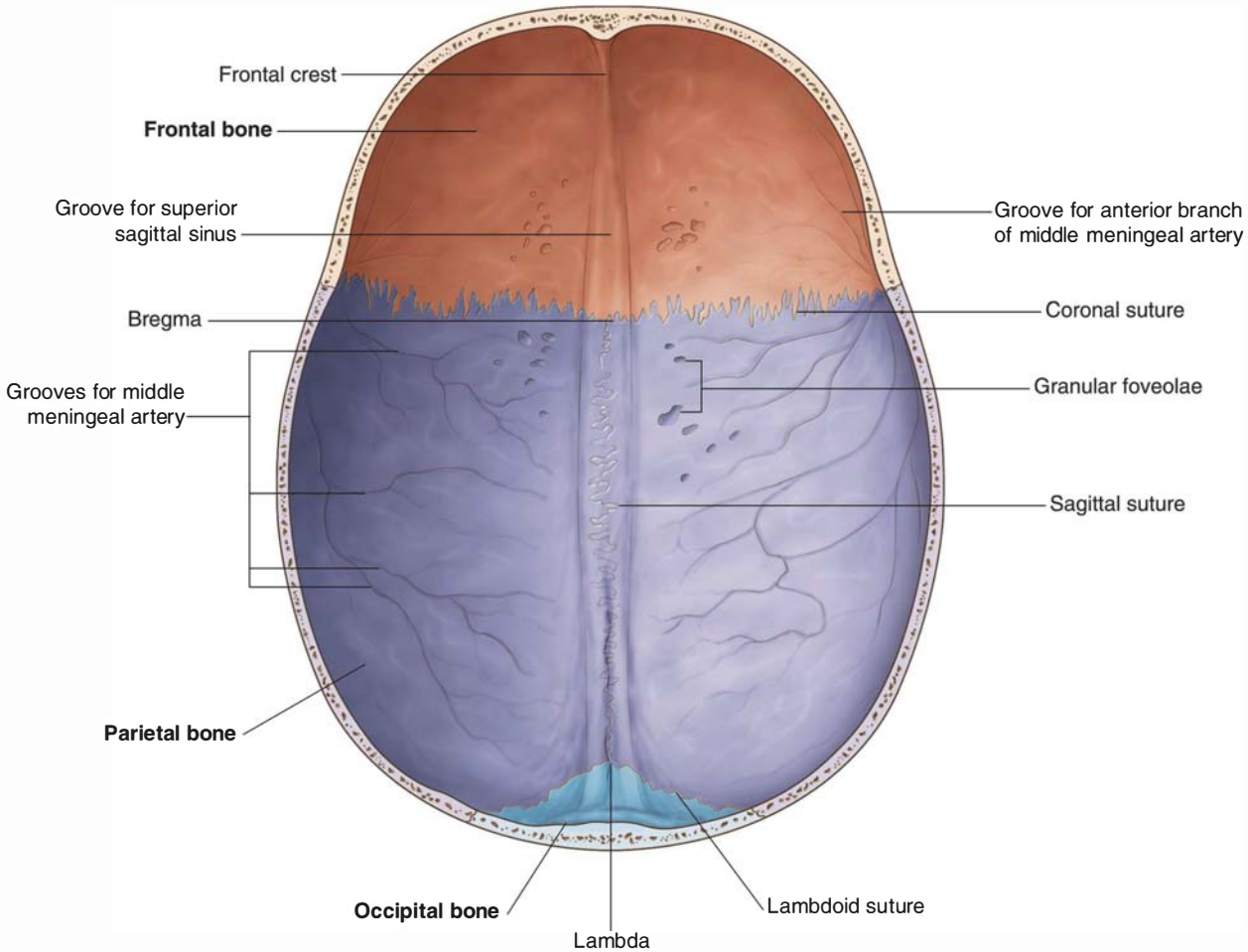


Fig. 8.24 Roof of the cranial cavity.

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- at the superior point of the termination of the frontal crest the beginning of the **groove for the superior sagittal sinus**, which widens and deepens posteriorly and marks the position of the superior sagittal sinus (an intradural venous structure);
- on either side of the groove for the superior sagittal sinus throughout its course, a small number of depressions and pits (the **granular foveolae**), which mark the location of arachnoid granulations (prominent structures readily identifiable when a brain with its meningeal coverings is examined; the arachnoid granulations are involved in the reabsorption of cerebrospinal fluid); and
- on the lateral aspects of the roof of the cranial cavity, smaller grooves created by various meningeal vessels.

Floor

The floor of the cranial cavity is divided into anterior, middle, and posterior cranial fossae.

Anterior cranial fossa

Parts of the frontal, ethmoid, and sphenoid bones form the anterior cranial fossa (Fig. 8.25). Its floor is composed of:

- frontal bone in the anterior and lateral direction,
- ethmoid bone in the midline, and
- two parts of the sphenoid bone posteriorly, the body (midline) and the lesser wings (laterally).

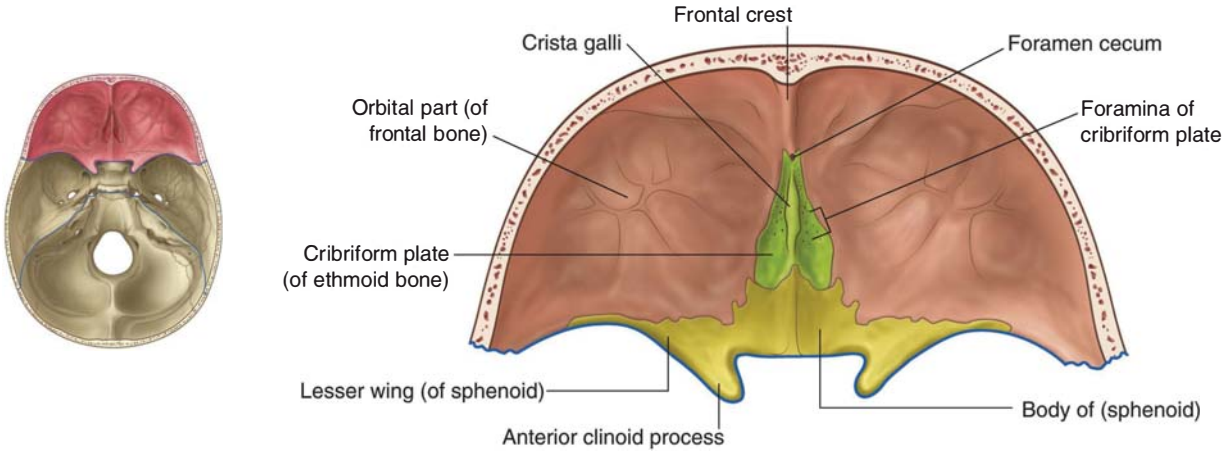


Fig. 8.25 Anterior cranial fossa.

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The anterior cranial fossa is above the nasal cavity and the orbits, and it is filled by the frontal lobes of the cerebral hemispheres.

Anteriorly, a small wedge-shaped midline crest of bone (the frontal crest) projects from the frontal bone. This is a point of attachment for the falx cerebri. Immediately posterior to the frontal crest is the **foramen cecum** (Table 8.2). This foramen between the frontal and ethmoid bones may transmit emissary veins connecting the nasal cavity with the superior sagittal sinus.

Posterior to the frontal crest is a prominent wedge of bone projecting superiorly from the **ethmoid** (the **crista galli**). This is another point of attachment for the falx cerebri, which is the vertical extension of dura mater partially separating the two cerebral hemispheres.

Lateral to the crista galli is the **cribriform plate** of the ethmoid bone (Fig. 8.25). This is a sieve-like structure, which allows small olfactory nerve fibers to pass through its foramina from the nasal mucosa to the olfactory bulb. The olfactory nerves are commonly referred to collectively as the olfactory nerve [I].

On each side of the ethmoid, the floor of the anterior cranial fossa is formed by relatively thin plates of frontal bone (the **orbital part** of the frontal bone), which also forms the roof of the orbit below. Posterior to both the frontal and ethmoid bones, the rest of the floor of the anterior cranial fossa is formed by the body and lesser wings of the sphenoid. In the midline, the body extends anteriorly between the orbital parts of the frontal bone to reach the ethmoid bone and posteriorly it extends into the middle cranial fossa.

The boundary between the anterior and middle cranial fossae in the midline is the anterior edge of the chiasmatic sulcus, a smooth groove stretching between the optic canals across the body of the sphenoid.

Lesser wings of the sphenoid

The two lesser wings of the sphenoid project laterally from the body of the sphenoid and form a distinct boundary between the lateral parts of the anterior and middle cranial fossae.

Overhanging the anterior part of the middle cranial fossae, each lesser wing ends laterally as a sharp point at the junction of the frontal bone and the greater wing of the sphenoid near the upper lateral edge of the superior orbital fissure that is formed between the greater and lesser wings.

Medially each lesser wing widens, curves posteriorly, and ends as a rounded **anterior clinoid process** (Fig. 8.25). These processes serve as the anterior point of

Table 8.2 Internal foramina of the skull

Foramen	Structures passing through foramen
ANTERIOR CRANIAL FOSSA	
Foramen cecum	Emissary veins to nasal cavity
Olfactory foramen in cribriform plate	Olfactory nerves [I]
MIDDLE CRANIAL FOSSA	
Optic canal	Optic nerve [II]; ophthalmic artery
Superior orbital fissure	Oculomotor nerve [III]; trochlear nerve [IV]; ophthalmic division of the trigeminal nerve [V_1]; abducent nerve [VI]; ophthalmic veins
Foramen rotundum	Maxillary division of the trigeminal nerve [V_2]
Foramen ovale	Mandibular division of the trigeminal nerve [V_3]; lesser petrosal nerve
Foramen spinosum	Middle meningeal artery
Hiatus for the greater petrosal nerve	Greater petrosal nerve
Hiatus for the lesser petrosal nerve	Lesser petrosal nerve
POSTERIOR CRANIAL FOSSA	
Foramen magnum	End of brainstem/beginning of spinal cord; vertebral arteries; spinal roots of the accessory nerve; meninges
Internal acoustic meatus	Facial nerve [VII]; vestibulocochlear nerve [VIII]; labyrinthine artery
Jugular foramen	Glossopharyngeal nerve [IX]; vagus nerve [X]; accessory nerve [XI]; inferior petrosal sinus, sigmoid sinus (forming internal jugular vein)
Hypoglossal canal	Hypoglossal nerve [XII]; meningeal branch of the ascending pharyngeal artery
Condylar canal	Emissary vein

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attachment for the **tentorium cerebelli**, which is a sheet of dura that separates the posterior part of the cerebral hemispheres from the cerebellum. Just anterior to each anterior clinoid process is a circular opening in the lesser wing of the sphenoid (the **optic canal**), through which the ophthalmic artery and optic nerve [II] pass as they exit the cranial cavity to enter the orbit. The optic canals are usually included in the middle cranial fossa.

Middle cranial fossa

The middle cranial fossa consists of parts of the sphenoid and temporal bones (Fig. 8.26).

The boundary between the anterior and middle cranial fossae in the midline is the anterior edge of the chiasmatic sulcus, which is a smooth groove stretching between the optic canals across the body of the sphenoid.

The posterior boundaries of the middle cranial fossa are formed by the anterior surface, as high as the superior border, of the petrous part of the petromastoid part of the temporal bone.

Sphenoid

The floor in the midline of the middle cranial fossa is elevated and formed by the body of the sphenoid. Lateral to this are large depressions formed on either side by the greater wing of the sphenoid and the squamous part of the temporal bone. These depressions contain the temporal lobes of the brain.

Sella turcica

Just posterior to the chiasmatic sulcus is the uniquely modified remainder of the body of the sphenoid (the **sella turcica**), which consists of a deep central area (the **hypophyseal fossa**) containing the pituitary gland with anterior and posterior vertical walls of bone (Fig. 8.26).

The anterior wall of the sella is vertical in position with its superior extent visible as a slight elevation (the **tuberculum sellae**) at the posterior edge of the chiasmatic sulcus.

Lateral projections from the corners of the tuberculum sellae (the **middle clinoid processes**) are sometimes evident.

The posterior wall of the sella turcica is the **dorsum sellae**, a large ridge of bone projecting upward and forward. At the top of this bony ridge the lateral edges contain rounded projections (the **posterior clinoid processes**), which are points of attachment, like the anterior clinoid processes, for the tentorium cerebelli.

Fissures and foramina

Lateral to each side of the body of the sphenoid, the floor of the middle cranial fossa is formed on either side by the greater wing of the sphenoid (Fig. 8.26).

A diagonal gap, the **superior orbital fissure**, separates the greater wing of the sphenoid from the lesser wing and is a major passageway between the middle cranial fossa and the orbit. Passing through the fissure are the

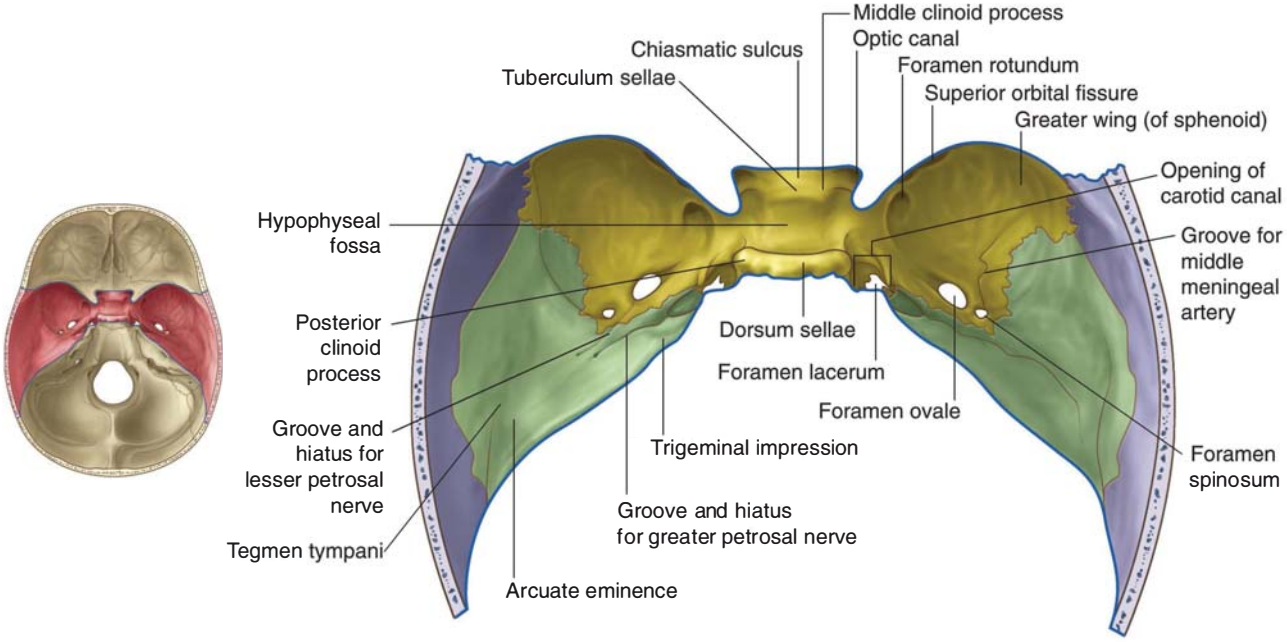


Fig. 8.26 Middle cranial fossa.

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oculomotor nerve [III], the trochlear nerve [IV], the ophthalmic nerve [V_1], the abducent nerve [VI], and ophthalmic veins.

Posterior to the medial end of the superior orbital fissure on the floor of the middle cranial fossa is a rounded foramen projecting in an anterior direction (the **foramen rotundum**), through which the maxillary nerve [V_2] passes from the middle cranial fossa to the pterygopalatine fossa.

Posterolateral to the **foramen rotundum** is a large oval opening (the **foramen ovale**), which allows structures to pass between the extracranial infratemporal fossa and the middle cranial fossa. The mandibular nerve [V_3], lesser petrosal nerve (carrying fibers from the tympanic plexus that originally came from the glossopharyngeal nerve [IX]) and, occasionally, a small vessel (the accessory middle meningeal artery), pass through this foramen.

Posterolateral from the foramen ovale is the small foramen spinosum (Fig. 8.26). This opening also connects the infratemporal fossa with the middle cranial fossa. The middle meningeal artery and its associated veins pass through this foramen and, once inside, the groove for the middle meningeal artery across the floor and lateral wall of the middle cranial fossa clearly marks their path.

Posteromedial to the foramen ovale is the **rounded intracranial opening** of the **carotid canal**. Directly inferior to this opening is an irregular foramen (the **foramen lacerum**) (Fig. 8.26). Clearly observed in the inferior view of the skull, the foramen lacerum is closed in life by a cartilaginous plug, and no structures pass through it completely.

Temporal bone

The posterior boundary of the middle cranial fossa is formed by the anterior surface of the petrous part of the petromastoid part of the temporal bone.

Medially, there is a slight depression (**trigeminal impression**) in the anterior surface of the petrous part of the temporal bone (Fig. 8.26), which marks the location of the sensory ganglion for the trigeminal nerve [V].

Lateral to the trigeminal impression and on the anterior surface of the petrous part of the temporal bone is a small linear groove that passes in a superolateral direction and ends in a foramen (the **groove** and **hiatus for the greater petrosal nerve**). The greater petrosal nerve is a branch of the facial nerve [VII].

Anterolateral to the groove for the greater petrosal nerve is a second, smaller **groove** and **hiatus for the lesser petrosal nerve**, a branch from the tympanic plexus carrying fibers that originally came from the glossopharyngeal nerve [IX] (Fig. 8.26).

Above and lateral to the small openings for the greater and lesser petrosal nerves, near the superior ridge of the petrous part of the temporal bone, is a rounded protrusion of bone (the **arcuate eminence**) produced by the underlying anterior semicircular canal of the inner ear.

Just anterior and lateral to the arcuate eminence the anterior surface of the petrous part of the temporal bone is slightly depressed. This region is the **tegmen tympani**, and marks the thin bony roof of the middle ear cavity.

Posterior cranial fossa

The posterior cranial fossa consists mostly of parts of the temporal and occipital bones, with small contributions from the sphenoid and parietal bones (Fig. 8.27). It is the largest and deepest of the three cranial fossae and contains the brainstem (midbrain, pons, and medulla) and the cerebellum.

Boundaries

The anterior boundaries of the posterior cranial fossa in the midline are the dorsum sellae and the **clivus** (Fig. 8.27). The clivus is a slope of bone that extends upward from the foramen magnum. It is formed by contributions from the body of the sphenoid and from the basilar part of the occipital bone.

Laterally the anterior boundaries of the posterior cranial fossa are the superior border of the petrous part of the petromastoid part of the temporal bone.

Posteriorly the squamous part of the occipital bone to the level of the transverse groove is the major boundary, while laterally the petromastoid part of the temporal bone and small parts of the occipital and parietal bones border the fossa.

Foramen magnum

Centrally, in the deepest part of the posterior cranial fossa, is the largest foramen in the skull, the foramen magnum. It is surrounded by the basilar part of the occipital bone anteriorly, the lateral parts of the occipital bone on either side, and the squamous part of the occipital bone posteriorly.

The spinal cord passes superiorly through the foramen magnum to continue as the brainstem.

Also passing through the foramen magnum are the vertebral arteries, the meninges, and the spinal roots of the accessory nerve [XI].

Grooves and foramina

The clivus slopes upward from the foramen magnum. Lateral to the clivus is a **groove for the inferior petrosal sinus** between the basilar part of the occipital bone and the petrous part of the petromastoid part of the temporal bone (Fig. 8.27).

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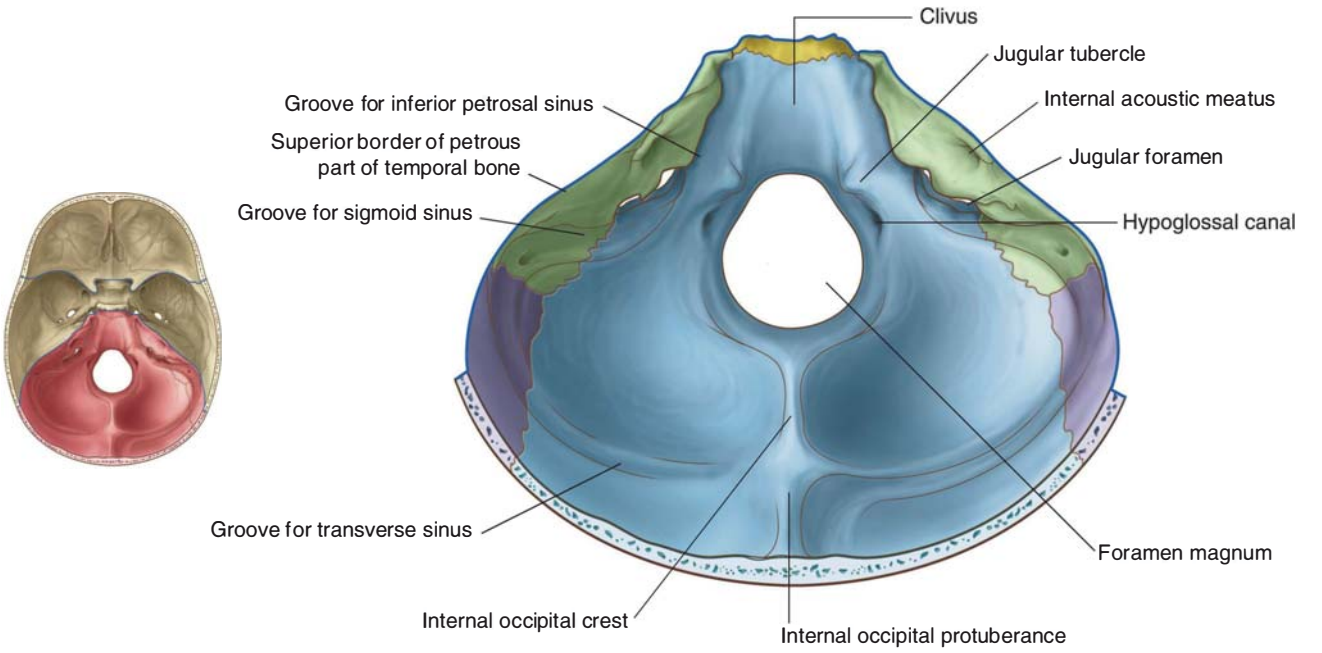


Fig. 8.27 Posterior cranial fossa.

Laterally, across the upper half of the posterior surface of the petrous part of the temporal bone, is an oval foramen (the **internal acoustic meatus**). The facial [VII] and vestibulocochlear [VIII] nerves, and the labyrinthine artery pass through it.

Inferior to the internal acoustic meatus the temporal bone is separated from the occipital bone by the large jugular foramen (Fig. 8.27). Leading to this foramen from the medial side is the groove for the inferior petrosal sinus, and from the lateral side the **groove for the sigmoid sinus**.

The sigmoid sinus passes into the jugular foramen, and is continuous with the internal jugular vein, while the inferior petrosal sinus empties into the internal jugular vein in the area of the jugular foramen.

Also passing through the jugular foramen are the glossopharyngeal nerve [IX], the vagus nerve [X], and the accessory nerve [XI].

Medial to the jugular foramen is a large rounded mound of the occipital bone (the **jugular tubercle**). Just inferior to this, and superior to the foramen magnum, is the **hypoglossal canal**, through which the hypoglossal nerve [XII] leaves the posterior cranial fossa, and a meningeal branch of the ascending pharyngeal artery enters the posterior cranial fossa.

Just posterolateral to the hypoglossal canal is the small **condylar canal** that, when present, transmits an emissary vein.

Squamous part of the occipital bone

The squamous part of the occipital bone has several prominent features (Fig. 8.27):

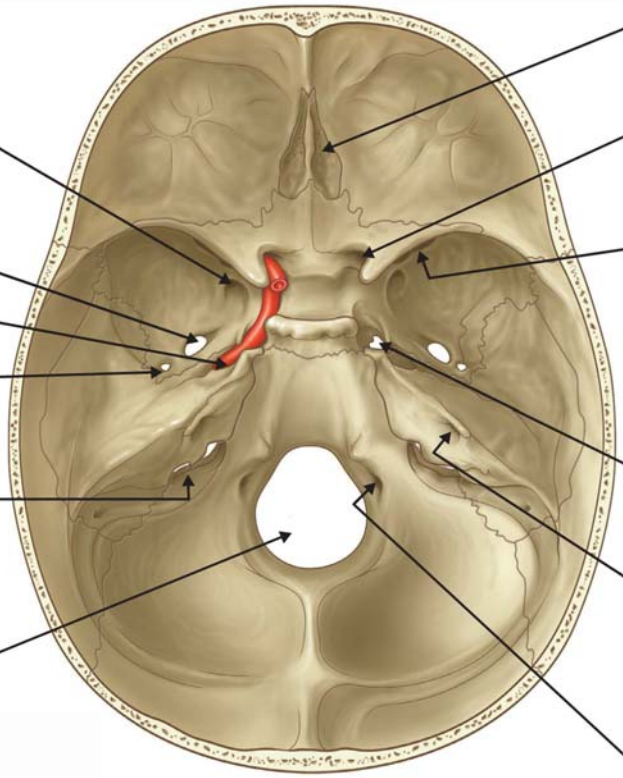
- Running upward in the midline from the foramen magnum is the internal occipital crest.
- On either side of the internal occipital crest, the floor of the posterior cranial fossa is concave to accommodate the cerebellar hemispheres.
- The internal occipital crest ends superiorly in a bony prominence (the internal occipital protuberance).
- Extending laterally from the internal occipital protuberance are grooves produced by the transverse sinuses, which continue laterally, eventually joining a groove for each sigmoid sinus—each of these grooves then turns inferiorly toward the jugular foramina.

The transverse and sigmoid sinuses are intradural venous sinuses.

Foramina and fissures through which major structures enter and leave the cranial cavity

Foramina and fissures through which major structures pass between the cranial cavity and other regions are summarized in Fig. 8.28.

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Foramen rotundum:
(*middle cranial fossa/ pterygopalatine fossa*)
- [V_2] Maxillary division of [V] (trigeminal nerve)

Foramen ovale:
(*middle cranial fossa/ infratemporal fossa*)
- [V_2] Mandibular division of [V] (trigeminal nerve)

Carotid canal:
(*middle cranial fossa/neck*)
- Internal carotid artery

Foramen spinosum:
(*middle cranial fossa/ infratemporal fossa*)
- Middle meningeal artery

Jugular foramen:
(*posterior cranial fossa/neck*)
- [IX] Glossopharyngeal nerve
- [X] Vagus nerve
- [XI] Accessory nerve
- Internal jugular vein

Foramen magnum:
(*posterior cranial fossa/neck*)
- Spinal cord
- Vertebral arteries
  - Roots of accessory nerve [XI] pass from upper region of spinal cord through the foramen magnum into the cranial cavity and then leave the cranial cavity through the jugular foramen

Cribiform plate:
(*anterior cranial fossa/nasal cavity*)
- [I] Olfactory nerves

Optic canal:
(*middle cranial fossa/orbit*)
- [II] Optic nerve
- Ophthalmic artery

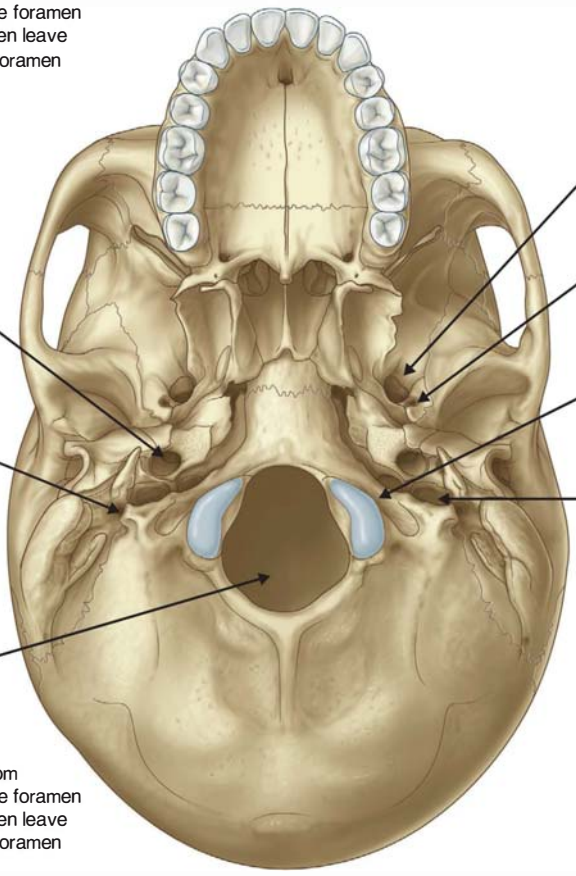
Superior orbital fissure:
(*middle cranial fossa/orbit*)
- [V_1] Ophthalmic division of [V] (trigeminal nerve)
- [III] Oculomotor nerve
- [IV] Trochlear nerve
- [VI] Abducent nerve
- Superior ophthalmic vein

Foramen lacerum
(filled with cartilage in life)

Internal acoustic meatus:
(*posterior cranial fossa/ear, and neck via stylomastoid foramen*)
- [VII] Facial nerve
- [VIII] Vestibulocochlear nerve
  - Labrynthine artery and vein

Hypoglossal canal:
(*posterior cranial fossa/neck*)
- [XII] Hypoglossal nerve

A



Carotid canal:
- Internal carotid artery

Stylomastoid foramen:
- [VII] Facial nerve

Foramen magnum:
- Spinal cord
- Vertebral arteries
  - Roots of accessory nerve [XI] pass from upper region of spinal cord through the foramen magnum into the cranial cavity and then leave the cranial cavity through the jugular foramen

Foramen ovale:
- [V_2] Mandibular division of [V] (trigeminal nerve)

Foramen spinosum:
- Middle meningeal artery

Hypoglossal canal:
- [XII] Hypoglossal nerve

Jugular foramen:
- [IX] Glossopharyngeal nerve
- [X] Vagus nerve
- [XI] Accessory nerve
- Internal jugular vein

B

Fig. 8.28 Summary of foramina and fissures through which major structures enter and leave the cranial cavity. **A.** Floor of cranial cavity. Also indicated are the regions between which each foramen or fissure communicates. **B.** Inferior aspect of cranium.

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In the clinic

Medical imaging of the head

Radiography

Until two decades ago the standard method of imaging the head was plain radiography. The radiographs are taken in three standard projections—the posteroanterior view, the lateral view, and the Towne's view (anteroposterior [AP] axial—head in anatomical position). Additional views are obtained to assess the foramina at the base of the skull and the facial bones. Currently, skull radiographs are used in cases of trauma, but such use is declining. Skull fractures are relatively easily detected (Fig. 8.29). The patient is assessed and treatment is based upon the underlying neurological or potential neurological complications.

Computed tomography

Since the development of the first computed tomography (CT) scanner, cerebral CT has been the "workhorse" of neuroradiological examination. It is ideally used for head injury because the brain and its coverings can be easily and quickly examined and blood is easily detected. By altering the mathematical algorithm of the data set the bones can also be demonstrated.

With intravenous contrast, CT angiography can be used to demonstrate the position and the size of an intracerebral aneurysm before endovascular treatment.

Magnetic resonance imaging

Magnetic resonance imaging (MRI) is unsurpassed by other imaging techniques in its ability for contrast resolution. The brain and its coverings, cerebrospinal fluid (CSF), and vertebral column can be easily and quickly examined. Newer imaging sequences permit CSF suppression to define periventricular lesions.

Magnetic resonance angiography has been extremely useful in determining the completeness of the intracranial vasculature (circle of Willis), which is necessary in some surgical conditions.

MRI is also a powerful tool in the assessment of carotid stenosis.

Ultrasonography

It is now possible to carry out intracranial Doppler studies, which enable a surgeon to detect whether a patient is experiencing cerebral embolization from a carotid plaque.

Extracranial ultrasound is extremely important in tumor staging and in assessing neck masses and the carotid bifurcation (Fig. 8.30).

Ultrasound is useful in children because they have an acoustic window through the fontanelles.



Fig. 8.29 Skull fracture seen on a skull radiograph (patient in supine position).

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In the clinic—cont'd

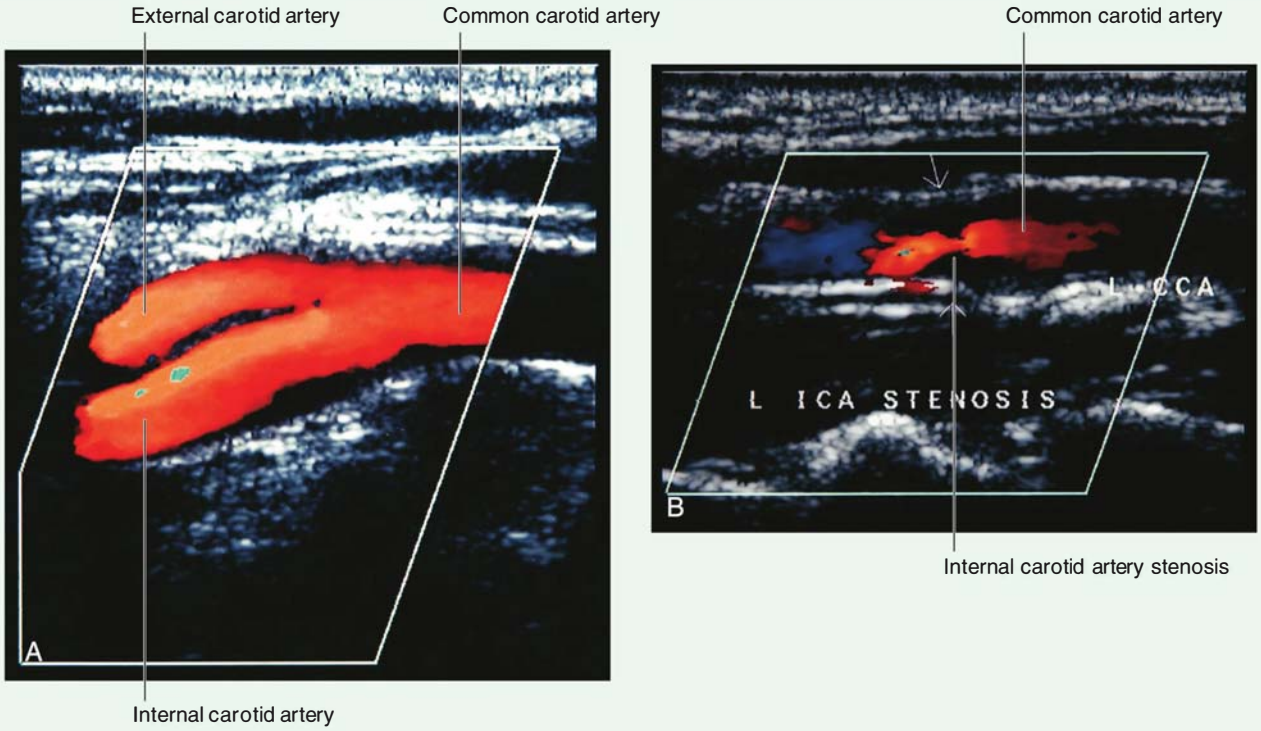


Fig. 8.30 Ultrasound scans. **A.** Normal carotid bifurcation. **B.** Internal carotid artery stenosis.

In the clinic

Fractures of the skull vault

The skull vault is a remarkably strong structure because it protects our most vital organ, the brain. The shape of the skull vault is of critical importance and its biomechanics prevent fracture. From a clinical standpoint skull fractures alert clinicians to the nature and force of an injury and potential complications. The fracture itself is usually of little consequence (unlike, say, a fracture of the tibia). Of key importance is the need to minimize the extent of primary brain injury and to treat potential secondary complications, rather than focusing on the skull fracture. Skull fractures that have particular significance include depressed skull fractures, compound fractures, and pterion fractures.

Depressed skull fractures

In a depressed skull fracture a bony fragment is depressed below the normal skull convexity. This may lead to secondary arterial and venous damage with hematoma formation. A primary brain injury can also result from this type of fracture.

Compound fractures

In a compound fracture there is a fracture of the bone together with a breach of the skin, which may allow an infection to enter. Typically these fractures are associated with scalp lacerations and can usually be treated with antibiotics.

Important complications of compound fractures include meningitis, which may be fatal.

A more subtle type of compound fracture involves fractures across the sinuses. These may not be appreciated on first inspection, but are an important potential cause of morbidity and should be considered in patients who develop intracranial infections secondary to trauma.

Pterion fractures

The pterion is an important clinical point on the lateral aspect of the skull. To find the precise point of the pterion, an imaginary line 1 inch (2.5 cm) above the zygomatic arch, and 1 inch (2.5 cm) posterior to the lateral orbital margin will approximate to this region. At the pterion the frontal, parietal, greater wing of the sphenoid, and temporal bones come together. Importantly, deep to this structure is the middle meningeal artery. An injury to this point of the skull is extremely serious because damage to this vessel may produce a significant extradural hematoma, which can be fatal.

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MENINGES

The brain, as well as the spinal cord, is surrounded by three layers of membranes (the **meninges**, Fig. 8.31A)—a tough, outer layer (the **dura mater**), a delicate, middle layer (the **arachnoid mater**), and an inner layer firmly attached to the surface of the brain (the **pia mater**).

The cranial meninges are continuous with, and similar to, the spinal meninges through the foramen magnum, with one important distinction—the cranial dura mater consists of two layers, and only one of these is continuous through the foramen magnum (Fig. 8.31B).

Cranial dura mater

The cranial dura mater is a thick, tough, outer covering of the brain. It consists of an outer periosteal layer and an inner meningeal layer (Fig. 8.31A):

- The outer **periosteal layer** is firmly attached to the skull, is the periosteum of the cranial cavity, contains the meningeal arteries, and is continuous with the periosteum on the outer surface of the skull at the foramen magnum and other intracranial foramina (Fig. 8.31B).
- The inner **meningeal layer** is in close contact with the arachnoid mater and is continuous with the spinal dura mater through the foramen magnum.

The two layers of dura separate from each other at numerous locations to form two unique types of structures (Fig. 8.31A):

- dural partitions, which project inward and incompletely separate parts of the brain, and
- intracranial venous structures.

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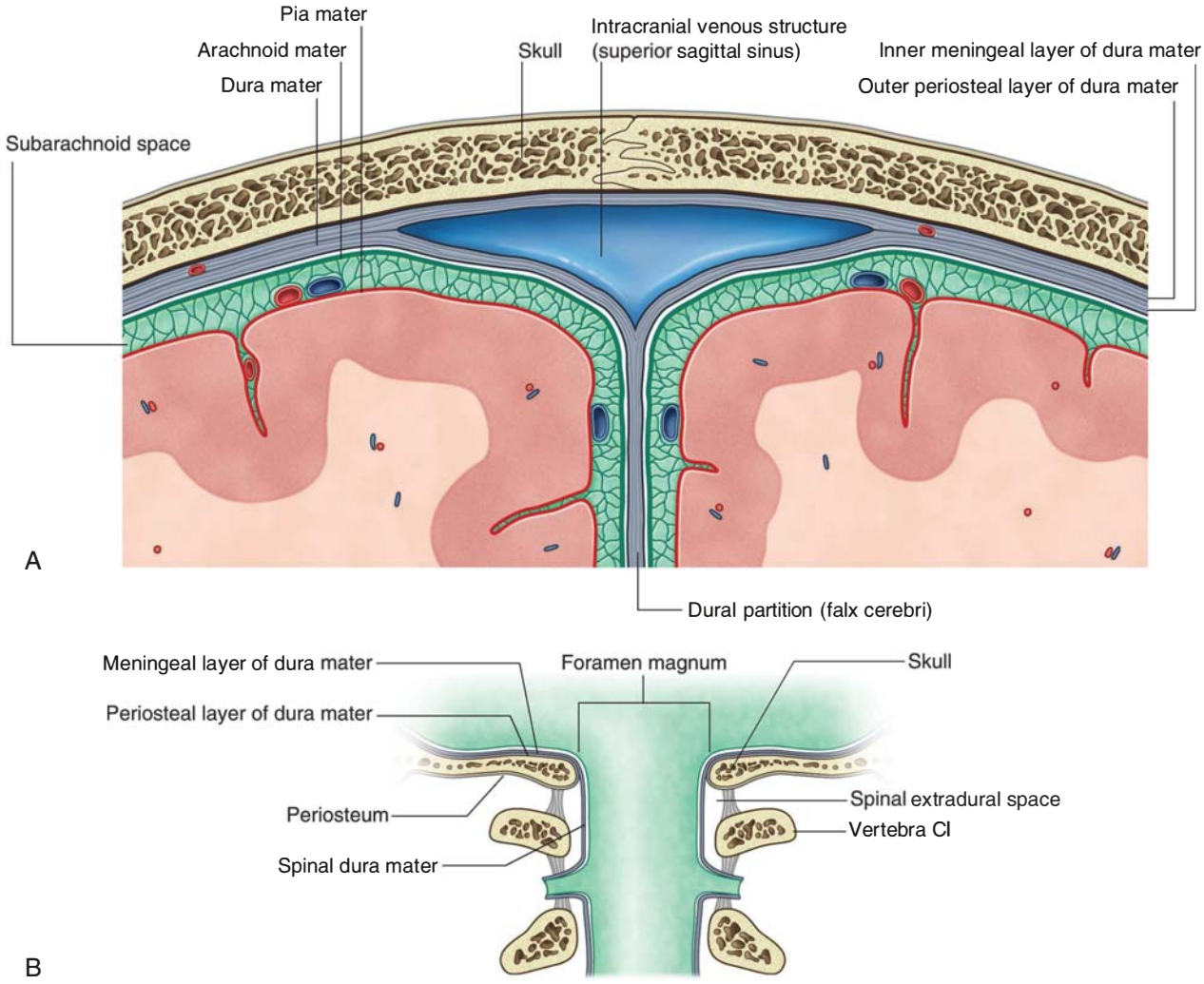


Fig. 8.31 Cranial meninges. **A.** Superior coronal view. **B.** Continuity with the spinal meninges.

Dural partitions

The dural partitions project into the cranial cavity and partially subdivide the cranial cavity. They include the falx cerebri, tentorium cerebelli, falx cerebelli, and diaphragma sellae.

Falx cerebri

The falx cerebri (Fig. 8.32) is a crescent-shaped downward projection of meningeal dura mater from the dura lining the calva that passes between the two cerebral hemispheres. It is attached anteriorly to the crista galli of the ethmoid bone and frontal crest of the frontal bone. Posteriorly it is attached to and blends with the tentorium cerebelli.

Tentorium cerebelli

The tentorium cerebelli (Fig. 8.32) is a horizontal projection of the meningeal dura mater that covers and separates the cerebellum in the posterior cranial fossa from the posterior parts of the cerebral hemispheres. It is attached posteriorly to the occipital bone along the grooves for the transverse sinuses. Laterally, it is attached to the superior border of the petrous part of the temporal bone, ending anteriorly at the anterior and posterior clinoid processes.

The anterior and medial borders of the tentorium cerebelli are free, forming an oval opening in the midline (the **tentorial notch**), through which the midbrain passes.

Falx cerebelli

The falx cerebelli (Fig. 8.32) is a small midline projection of meningeal dura mater in the posterior cranial fossa. It is attached posteriorly to the internal occipital crest of the occipital bone and superiorly to the tentorium cerebelli. Its anterior edge is free and is between the two cerebellar hemispheres.

Diaphragma sellae

The final dural projection is the diaphragma sellae (Fig. 8.32). This small horizontal shelf of meningeal dura mater covers the hypophyseal fossa in the sella turcica of the sphenoid bone. There is an opening in the center of the diaphragma sellae through which passes the **infundibulum**, connecting the pituitary gland with the base of the brain, and any accompanying blood vessels.

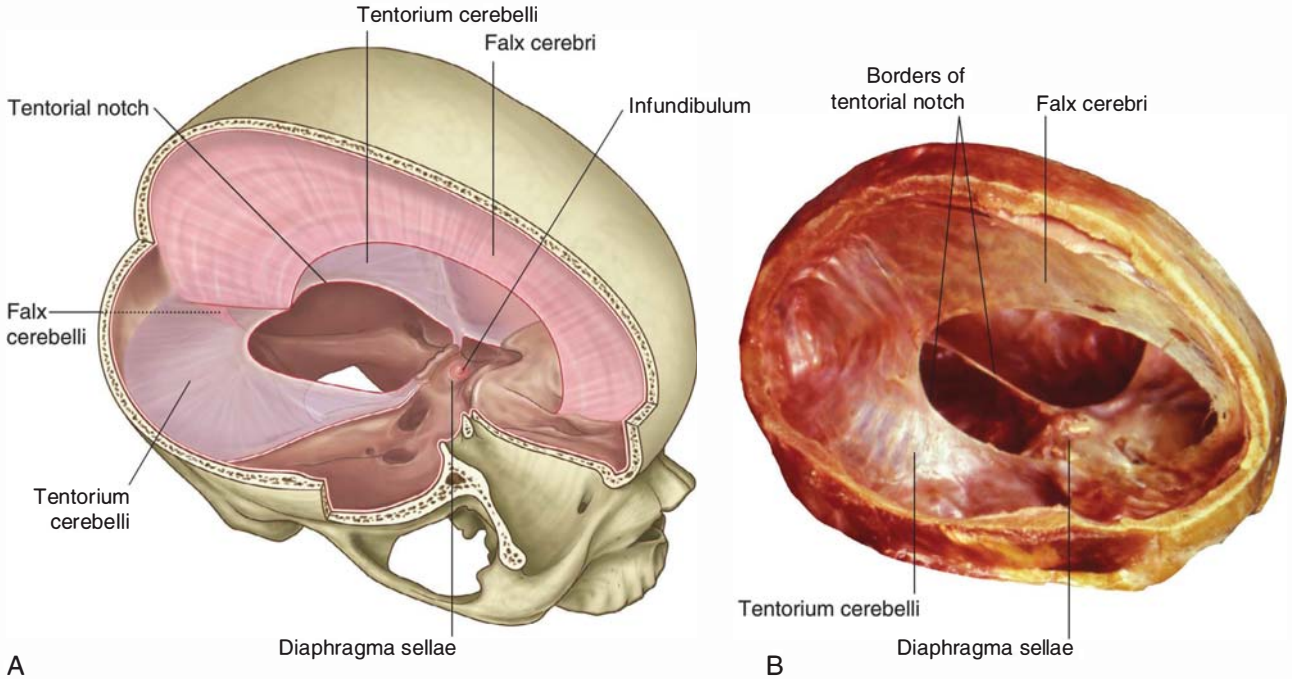


Fig. 8.32 Dural partitions. **A.** Diagram. **B.** Dissection.

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Arterial supply

The arterial supply to the dura mater (Fig. 8.33) travels in the outer periosteal layer of the dura and consists of:

- **anterior meningeal arteries** in the anterior cranial fossa,
- the **middle** and **accessory meningeal arteries** in the middle cranial fossa, and
- the **posterior meningeal artery** and other meningeal branches in the posterior cranial fossa.

All are small arteries except for the middle meningeal artery, which is much larger and supplies the greatest part of the dura.

The anterior meningeal arteries are branches of the **ethmoidal arteries**.

The middle meningeal artery is a branch of the maxillary artery. It enters the middle cranial fossa through the foramen spinosum and divides into anterior and posterior branches:

- The anterior branch passes in an almost vertical direction to reach the vertex of the skull, crossing the pterion during its course.
- The posterior branch passes in a posterosuperior direction, supplying this region of the middle cranial fossa.

The accessory meningeal artery is usually a small branch of the maxillary artery that enters the middle cranial fossa through the foramen ovale and supplies areas medial to this foramen.

The posterior meningeal artery and other meningeal branches supplying the dura mater in the posterior cranial fossa come from several sources (Fig. 8.33):

- The posterior meningeal artery, the terminal branch of the **ascending pharyngeal artery**, enters the posterior cranial fossa through the jugular foramen.
- A meningeal branch from the ascending pharyngeal artery enters the posterior cranial fossa through the hypoglossal canal.
- Meningeal branches from the **occipital artery** enter the posterior cranial fossa through the jugular foramen and the mastoid foramen.
- A meningeal branch from the **vertebral artery** arises as the vertebral artery enters the posterior cranial fossa through the foramen magnum.

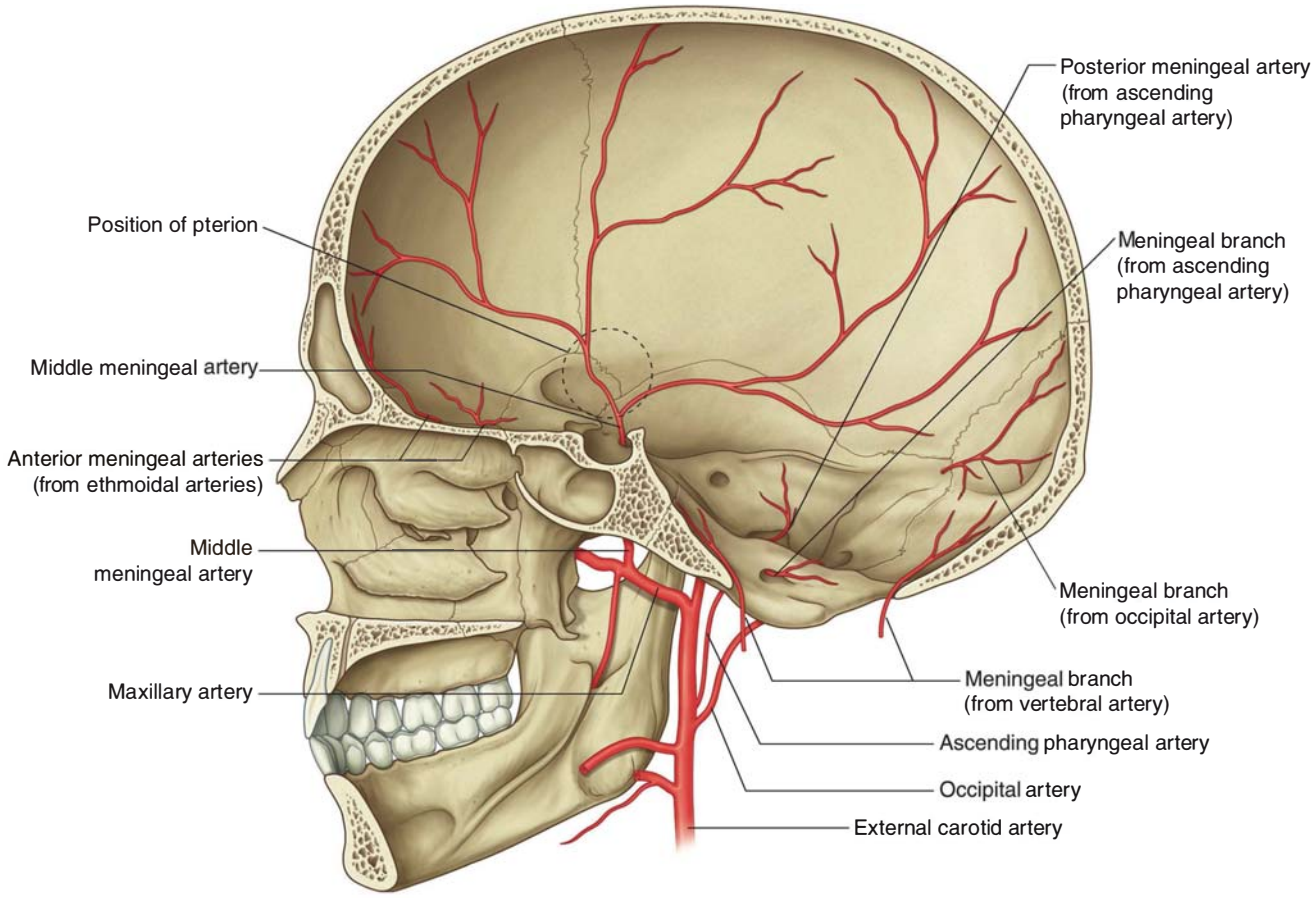


Fig. 8.33 Dural arterial supply.

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Innervation

Innervation of the dura mater (Fig. 8.34) is by small meningeal branches of all three divisions of the trigeminal nerve [V_1, V_2, and V_3], the vagus nerve [X], and the first, second, and, sometimes, third cervical nerves. (Possible involvement of the glossopharyngeal [IX] and hypoglossal nerves [XII] in the posterior cranial fossa has also been reported.)

In the anterior cranial fossa meningeal branches from the ethmoidal nerves, which are branches of the ophthalmic nerve [V_1], supply the floor and the anterior part of the falx cerebri.

Additionally, a meningeal branch of the ophthalmic nerve [V_1] turns and runs posteriorly, supplying the tentorium cerebelli and the posterior part of the falx cerebri.

The middle cranial fossa is supplied medially by meningeal branches from the maxillary nerve [V_2] and laterally, along the distribution of the middle meningeal artery, by meningeal branches from the mandibular nerve [V_3].

The posterior cranial fossa is supplied by meningeal branches from the first, second, and, sometimes, third cervical nerves, which enter the fossa through the foramen magnum, the hypoglossal canal, and the jugular foramen. Meningeal branches of the vagus nerve [X] have also been described. (Possible contributions from the glossopharyngeal [IX] and hypoglossal [XII] nerves have also been reported.)

Arachnoid mater

The arachnoid mater is a thin, avascular membrane that lines, but is not adherent to, the inner surface of the dura mater (Fig. 8.35). From its inner surface thin processes or trabeculae extend downward, cross the subarachnoid space, and become continuous with the pia mater.

Unlike the pia, the arachnoid does not enter the grooves or fissures of the brain, except for the longitudinal fissure between the two cerebral hemispheres.

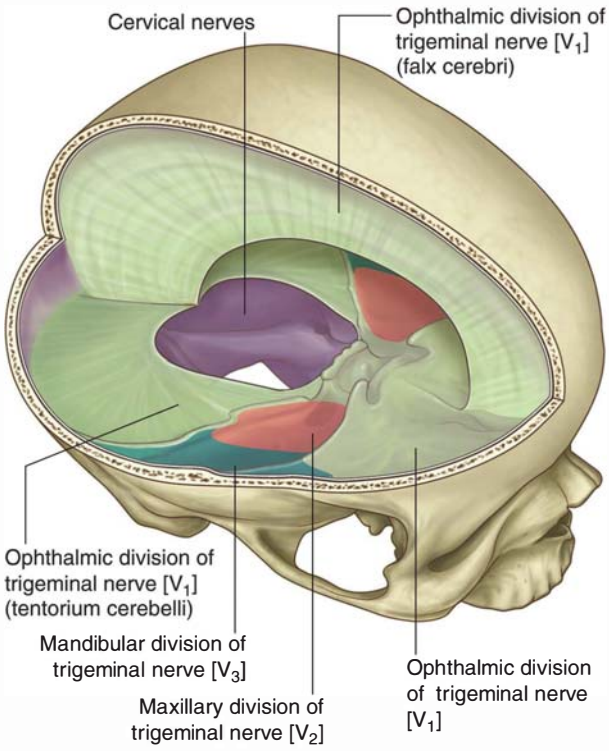


Fig. 8.34 Dural innervation.

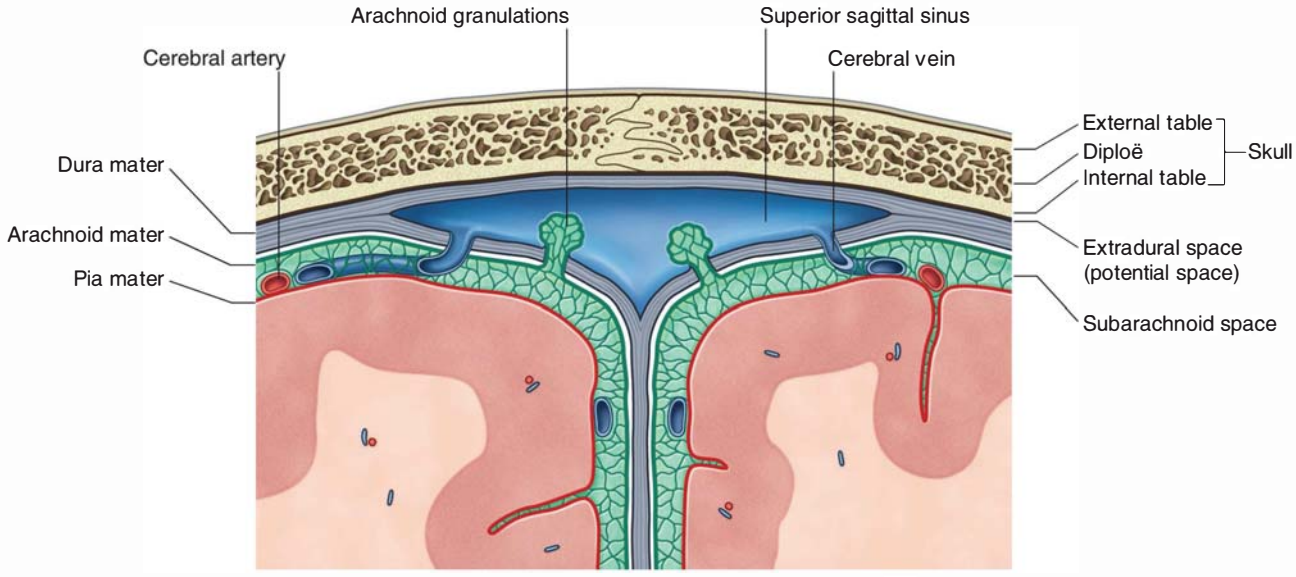


Fig. 8.35 Arrangement of the meninges and spaces.

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Pia mater

The pia mater is a thin, delicate membrane that closely invests the surface of the brain (Fig. 8.35). It follows the contours of the brain, entering the grooves and fissures on its surface, and is closely applied to the roots of the cranial nerves at their origins.

Arrangement of meninges and spaces

There is a unique arrangement of meninges coupled with real and potential spaces within the cranial cavity (Fig. 8.35).

A potential space is related to the dura mater, while a real space exists between the arachnoid mater and the pia mater.

Extradural space

The potential space between dura mater and bone is the **extradural space** (Fig. 8.35). Normally, the outer or periosteal layer of dura mater is firmly attached to the bones surrounding the cranial cavity.

This potential space between dura and bone can become a fluid-filled actual space when a traumatic event results in a vascular hemorrhage. Bleeding into the extradural space due to rupture of a meningeal artery or a torn dural venous sinus results in an extradural hematoma.

Subdural space

Anatomically, a true subdural space does not exist. Blood collecting in this region (subdural hematoma) due to injury represents a dissection of the dural border cell layer, which is the innermost lining of the meningeal dura. Dural border cells are flattened cells surrounded by extracellular spaces filled with amorphous material. While very infrequent, an occasional cell junction may be seen between these cells and the underlying arachnoid layer. Bleeding due to the tearing of a cerebral vein as it crosses through the dura to enter a dural venous sinus can result in a subdural hematoma.

Subarachnoid space

Deep to the arachnoid mater is the only normally occurring fluid-filled space associated with the meninges, the **subarachnoid space** (Fig. 8.35). It occurs because the arachnoid mater clings to the inner surface of the dura mater and does not follow the contour of the brain, while the pia mater, being against the surface of the brain, closely follows the grooves and fissures on the surface of the brain. The narrow subarachnoid space is therefore created between these two membranes (Fig. 8.35).

The subarachnoid space surrounds the brain and spinal cord and in certain locations it enlarges into expanded areas (subarachnoid **cisterns**). It contains cerebrospinal fluid (CSF) and blood vessels.

Cerebrospinal fluid is produced by the choroid plexus, primarily in the ventricles of the brain. It is a clear, colorless, cell-free fluid that circulates through the subarachnoid space surrounding the brain and spinal cord.

The CSF returns to the venous system through **arachnoid villi**. These project as clumps (**arachnoid granulations**) into the superior sagittal sinus, which is a dural venous sinus, and its lateral extensions, the **lateral lacunae** (Fig. 8.35).

In the clinic

Hydrocephalus

Hydrocephalus is a dilatation of the cerebral ventricular system, which is due to either an obstruction to the flow of CSF, an overproduction of CSF, or a failure of reabsorption of CSF.

Cerebrospinal fluid is secreted by the choroid plexus within the lateral, third, and fourth ventricles of the brain. As it is produced it passes from the lateral ventricles through the interventricular foramina (the foramina of Monro) to enter the third ventricle. From the third ventricle it passes through the cerebral aqueduct (aqueduct of Sylvius) into the fourth ventricle, and from here it passes into the subarachnoid space via the midline foramen or the two lateral foramina (foramen of Magendie and foramina of Luschka).

The CSF passes around the spinal cord inferiorly, envelops the brain superiorly, and is absorbed through the arachnoid granulations in the walls of the dural venous sinuses. In adults almost half a liter of CSF is produced per day.

In adults the commonest cause of hydrocephalus is an interruption of the normal CSF absorption through the arachnoid granulations. This occurs when blood enters the subarachnoid space after subarachnoid hemorrhage, passes over the brain, and interferes with normal CSF absorption. To prevent severe hydrocephalus it may be necessary to place a small catheter through the brain into the ventricular system to relieve the pressure.

Other causes of hydrocephalus include congenital obstruction of the aqueduct of Sylvius and a variety of

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In the clinic—cont'd

tumors (e.g., a midbrain tumor), where the mass obstructs the aqueduct. Rare causes include choroid plexus tumors that secrete CSF.

In children, hydrocephalus is always dramatic in its later stages. The hydrocephalus increases the size and dimensions of the ventricle, and as a result the brain enlarges. Because the skull sutures are not fused, the head expands. Cranial enlargement in utero may make a vaginal delivery impossible, and delivery then has to be by caesarean section.

Both CT and MRI enable a radiologist to determine the site of obstruction and in most cases the cause of the obstruction. A distinction must be made between ventricular enlargement due to hydrocephalus and that due to a variety of other causes (e.g., cerebral atrophy).

In the clinic

Cerebrospinal fluid leak

Leakage of CSF from the subarachnoid space may occur after any procedure in and around the brain, spinal cord, and meningeal membranes. These procedures include lumbar spine surgery, epidural injection, and CSF aspiration.

In "cerebrospinal fluid leak" syndrome, CSF leaks out of the subarachnoid space and through the dura mater for no apparent reason. The clinical consequences of this include dizziness, nausea, fatigue, and a metallic taste in the mouth. Other effects also include facial nerve weakness and double vision.

In the clinic

Meningitis

Meningitis is a rare infection of the leptomeninges (the **leptomeninges** are a combination of the arachnoid mater and the pia mater). Infection of the meninges typically occurs via a blood-borne route, though in some cases it may be by direct spread (e.g., trauma) or from the nasal cavities through the cribriform plate in the ethmoid bone.

Certain types of bacterial inflammation of the meninges are so virulent that overwhelming inflammation and sepsis with cerebral irritation can cause the patient to rapidly pass into a coma and die.

Meningitis is usually treatable with simple antibiotics.

Certain types of bacteria that produce meningitis produce other effects; for example, subcutaneous hemorrhage (ecchymoses) is a feature of meningococcal meningitis.

The typical history of meningitis is nonspecific at first. The patient may have mild headache, fever, drowsiness, and nausea. As the infection progresses, photophobia (light intolerance) and ecchymosis may ensue. Straight leg raising causes marked neck pain and discomfort (Kernig's sign) and an emergency hospital admission is warranted.

Immediate treatment consists of very-high-dose intravenous antibiotics and supportive management.

In the clinic

Brain tumors

Determination of the anatomical structure from which a tumor arises is of the utmost importance, no more so than when it arises within the cranial vault. Misinterpretation of the location of a lesion and its site of origin may have devastating consequences for the patient.

When assessing any lesion in the brain, it is important to define whether it is intra-axial (within the brain) or extra-axial (outside the brain).

Typical extra-axial tumors include meningiomas (tumors of the meninges) and acoustic neuromas. Meningiomas typically arise from the meninges, with preferred sites including regions at and around the falx cerebri, the free edge of the tentorium cerebelli, and the anterior margin of the middle cranial fossa. Acoustic neuromas are typically at and around the vestibulocochlear nerve [VIII] and in the cerebellopontine angle.

Intra-axial lesions are either primary or secondary. By far the commonest type are the secondary brain lesions, which in most cases are metastatic tumor deposits.

Metastatic tumor lesions are typically found in patients with either breast carcinoma or lung carcinoma, though many other malignancies can give rise to cerebral metastases.

Primary brain lesions are rare and range from benign tumors to extremely aggressive lesions with a poor prognosis. These tumors arise from the different cell lines and include gliomas, oligodendrocytomas, and choroid plexus tumors. Primary brain tumors may occur at any age, though there is a small peak incidence in the first few years of life followed by a later peak in early to middle age.

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BRAIN AND ITS BLOOD SUPPLY

Brain

The brain is a component of the central nervous system.

During development the brain can be divided into five continuous parts (Figs. 8.36 and 8.37). From rostral (or cranial) to caudal they are:

- The **telencephalon** (**cerebrum**) becomes the large cerebral hemispheres. The surface of these hemispheres consists of elevations (gyri) and depressions (sulci), and the hemispheres are partially separated by a deep longitudinal fissure. The cerebrum fills the area of the skull above the tentorium cerebelli and is subdivided into lobes based on position.
- The **diencephalon**, which is hidden from view in the adult brain by the cerebral hemispheres, consists of the thalamus, hypothalamus, and other related structures, and classically is considered to be the most rostral part of the brainstem. (However, in common usage today, the term brainstem usually refers to the midbrain, pons, and medulla.)
- The **mesencephalon** (**midbrain**), which is the first part of the brainstem seen when an intact adult brain is examined, located at the junction between and in both the middle and posterior cranial fossae.
- The **metencephalon**, which gives rise to the cerebellum (consisting of two lateral hemispheres and a midline part in the posterior cranial fossa below the tentorium cerebelli) and the pons (anterior to the cerebellum, and is a bulging part of the brainstem in the most anterior part of the posterior cranial fossa against the clivus and dorsum sellae).
- The **myelencephalon** (**medulla oblongata**), the caudalmost part of the brainstem, ends at the foramen magnum or the uppermost rootlets of the first cervical nerve and to which cranial nerves VI to XII are attached.

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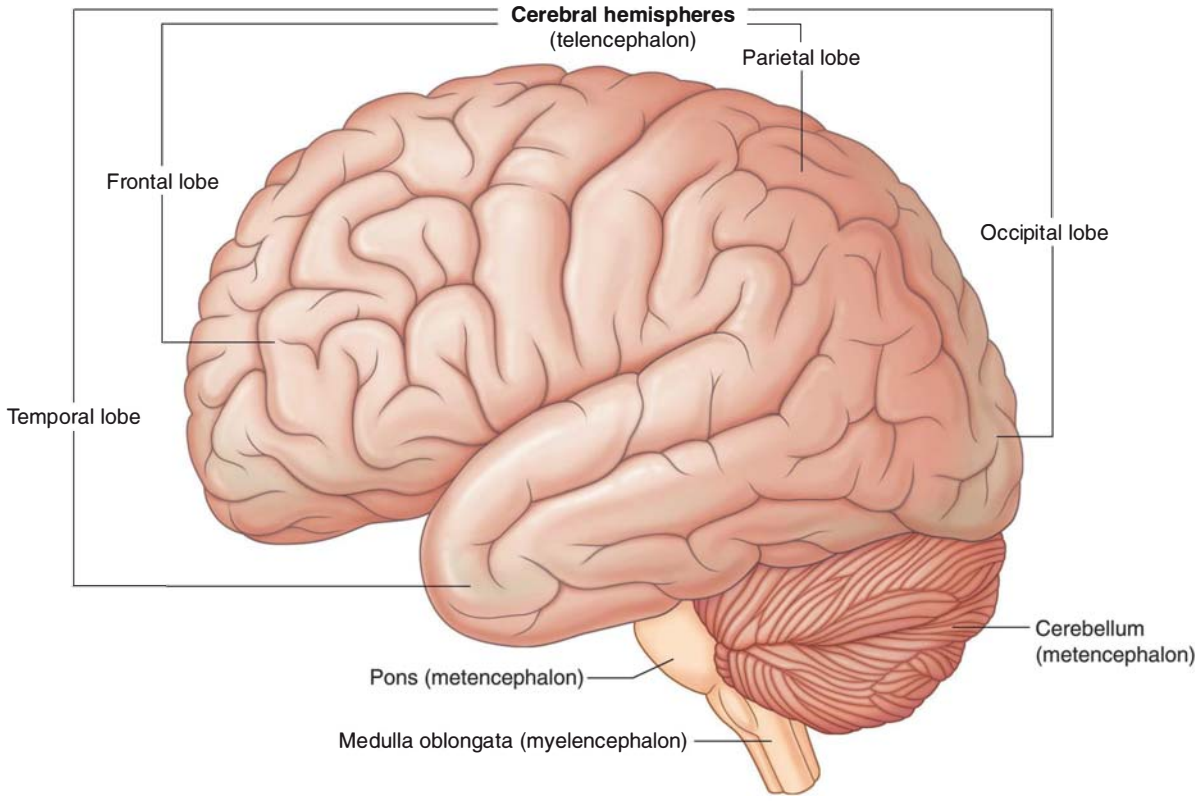


Fig. 8.36 Lateral view of the brain.

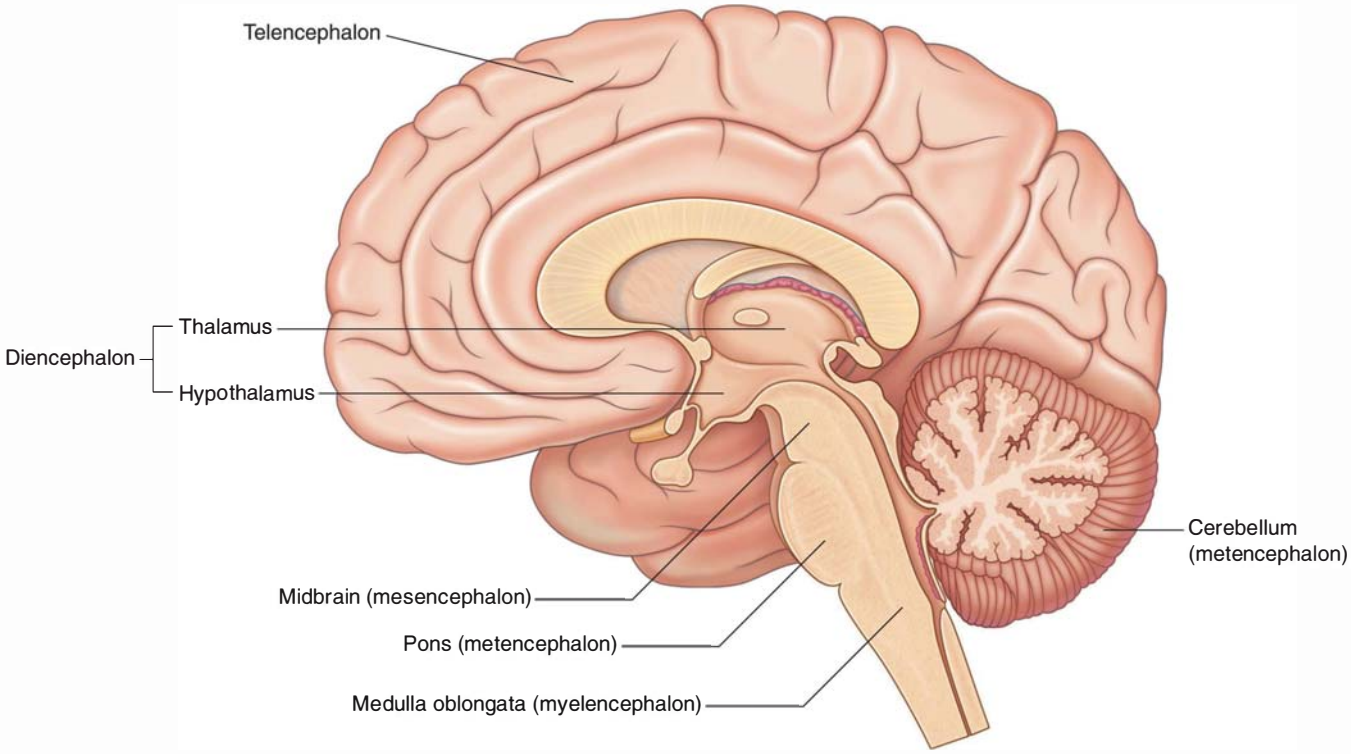


Fig. 8.37 Sagittal section of the brain.

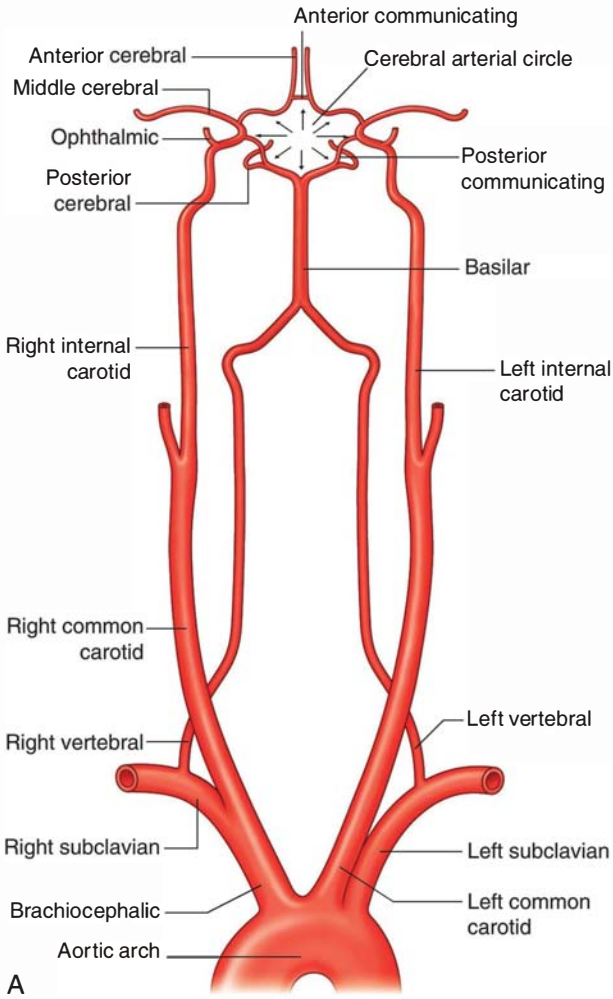
Blood supply

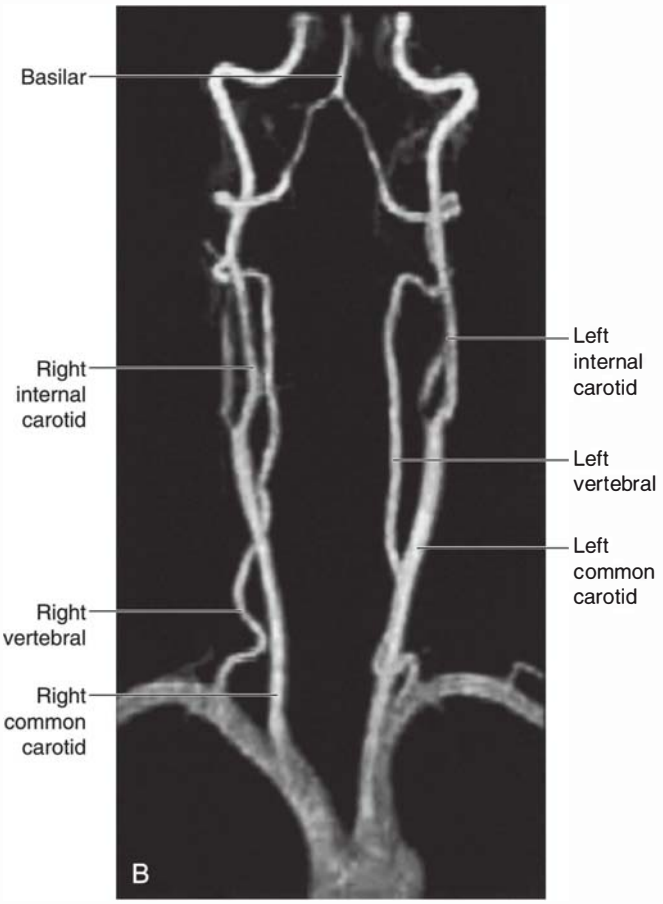
The brain receives its arterial supply from two pairs of vessels, the **vertebral** and **internal carotid arteries** (Fig. 8.38), which are interconnected in the cranial cavity to produce a **cerebral arterial circle** (of Willis).

The two vertebral arteries enter the cranial cavity through the foramen magnum and just inferior to the pons fuse to form the **basilar artery**.

The two internal carotid arteries enter the cranial cavity through the carotid canals on either side.

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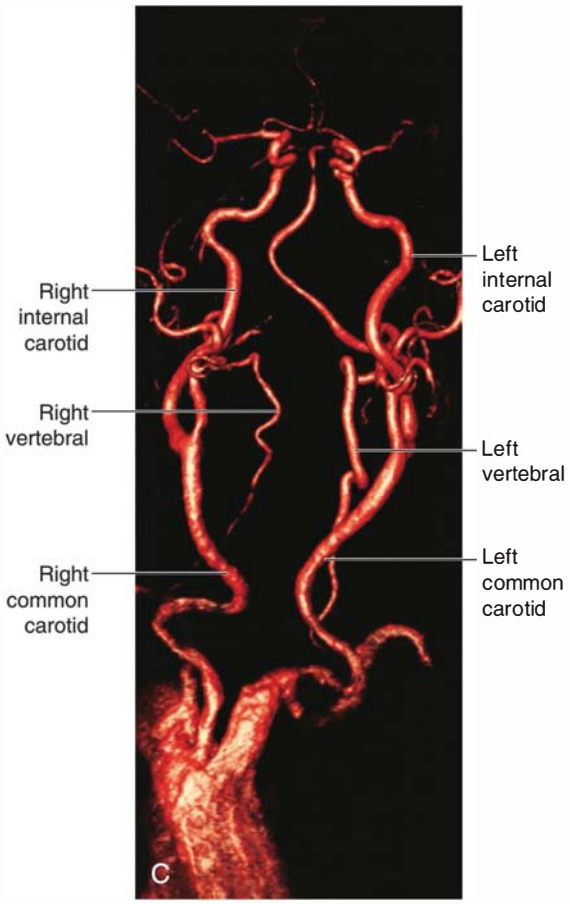


Fig. 8.38 Arterial supply to the brain. **A.** Diagram. **B.** Magnetic resonance angiogram showing normal carotid and vertebral arteries. **C.** Enhanced CT scan of carotid vessels.

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Vertebral arteries

Each vertebral artery arises from the first part of each **subclavian artery** (Fig. 8.38) in the lower part of the neck, and passes superiorly through the transverse foramina of the upper six cervical vertebrae. On entering the cranial cavity through the foramen magnum each vertebral artery gives off a small meningeal branch.

Continuing forward, the vertebral artery gives rise to three additional branches before joining with its companion vessel to form the basilar artery (Figs. 8.38 and 8.39):

- One branch joins with its companion from the other side to form the single **anterior spinal artery**, which then descends in the anterior median fissure of the spinal cord.
- A second branch is the **posterior spinal artery**, which passes posteriorly around the medulla and then descends on the posterior surface of the spinal cord in the area of the attachment of the posterior roots—there are two posterior spinal arteries, one on each side (although the posterior spinal arteries can originate directly from the vertebral arteries, they more commonly branch from the posterior inferior cerebellar arteries).
- Just before the two vertebral arteries join, each gives off a **posterior inferior cerebellar artery**.

The basilar artery travels in a rostral direction along the anterior aspect of the pons (Fig. 8.39). Its branches in a caudal to rostral direction include the **anterior inferior cerebellar arteries**, several small **pontine arteries**, and the **superior cerebellar arteries**. The basilar artery ends as a bifurcation, giving rise to two **posterior cerebral arteries**.

Internal carotid arteries

The two internal carotid arteries arise as one of the two terminal branches of the common carotid arteries (Fig. 8.38). They proceed superiorly to the base of the skull where they enter the carotid canal.

Entering the cranial cavity each internal carotid artery gives off the **ophthalmic artery**, the **posterior communicating artery**, the **middle cerebral artery**, and the **anterior cerebral artery** (Fig. 8.39).

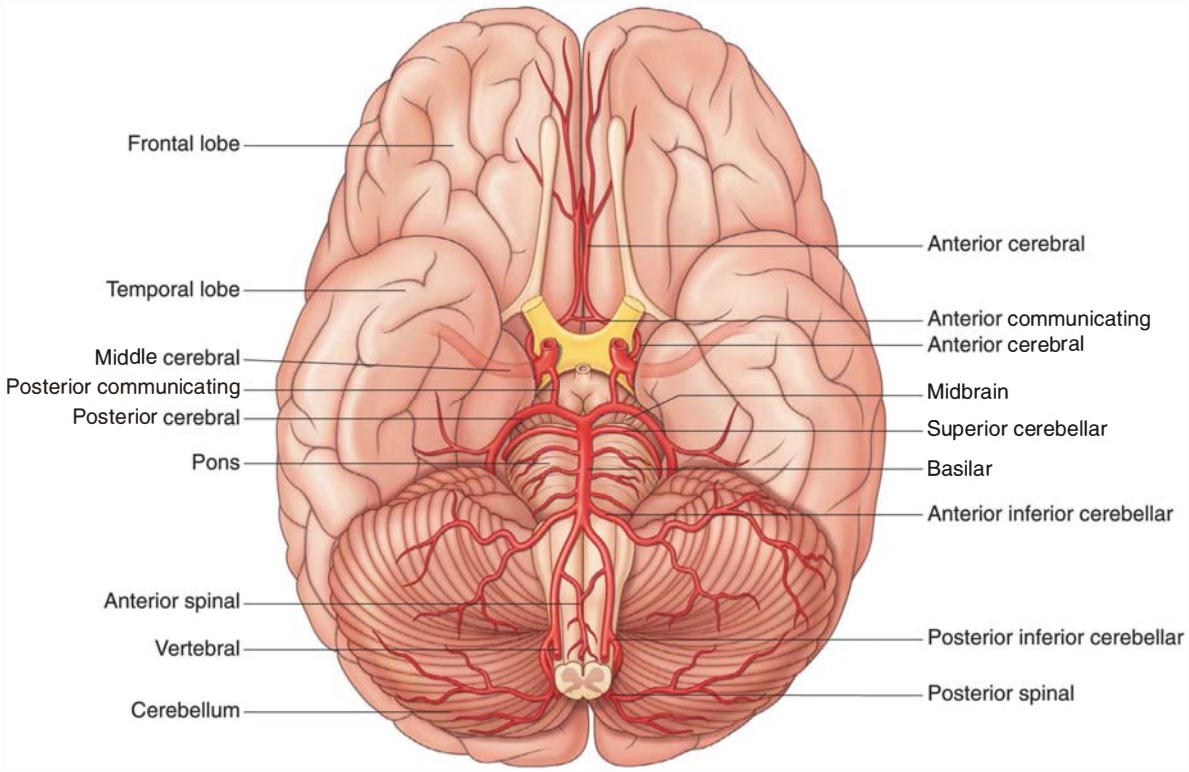


Fig. 8.39 Arteries on the base of the brain.

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Cerebral arterial circle

The cerebral arterial circle (of Willis) is formed at the base of the brain by the interconnecting vertebrobasilar and internal carotid systems of vessels (Fig. 8.38). This anastomotic interconnection is accomplished by:

- an anterior communicating artery connecting the left and right anterior cerebral arteries to each other, and
- two posterior communicating arteries, one on each side, connecting the internal carotid artery with the posterior cerebral artery (Figs. 8.38 and 8.39).

In the clinic

Stroke

A stroke, or cerebrovascular accident (CVA), is defined as the interruption of blood flow to the brain or brainstem resulting in impaired neurological function lasting more than 24 hours. Neurological impairment resolving within 24 hours is known as a transient ischemic attack (TIA) or mini-stroke. Based on their etiology, strokes are broadly classified as either ischemic or hemorrhagic. Ischemic strokes are further divided into those caused by thrombotic or embolic phenomena. The latter is by far the commonest type of stroke and is often caused by emboli that originate from atherosclerotic plaques in the carotid arteries that migrate into and block smaller intracranial vessels. Hemorrhagic strokes are caused by rupture of blood vessels.

The risk factors for stroke are those of cardiovascular disease, such as diabetes, hypertension, and smoking. In younger patients underlying clotting disorders, use of oral contraceptives, and illicit substance abuse (such as cocaine) are additional causes.

The symptoms and signs of a stroke depend on the distribution of impaired brain perfusion. Common presentations include rapid-onset hemiparesis or hemisensory loss, visual field deficits, dysarthria, ataxia, and a decreased level of consciousness.

Stroke is a neurological emergency. It is therefore important to establish the diagnosis as early as possible so that urgent and potentially life-saving treatment can be administered. Potent thrombolytic (blood-thinning) drugs can restore cerebral blood flow and improved patient outcome if administered within 3 to 4.5 hours of onset of the patient's symptoms.

Following initial clinical history taking and neurological examination, all patients with suspected stroke should undergo urgent brain imaging with computed tomography (CT). This is to identify hemorrhagic strokes for which thrombolytic therapy is contraindicated and to exclude an alternative diagnosis such as malignancy. In ischemic stroke, early CT imaging may appear normal or can show a relatively darker area of low density that corresponds to the region of abnormal brain perfusion. Due to subsequent brain edema and swelling, the affected brain also loses its normal sulcal pattern (Fig. 8.40A). If thrombolysis is performed, a 24-hour follow-up CT scan is routinely carried out to evaluate for complications such as intracranial hemorrhage.

Additional diagnostic workup of stroke includes hematological and biochemical blood tests to identify causes such as hypoglycemia or underlying clotting disorders. A toxicology screen may be useful to identify substance intoxication, which can mimic stroke.

The full extent of neurological injury can be evaluated on subsequent magnetic resonance imaging (MRI) of the brain, which has better soft tissue resolution compared to CT. MRI is also useful for identifying strokes that may be too small to detect on a CT scan. MRI scans are produced by using complicated algorithms that create a series of images, also known as sequences. Various sequences can be obtained to assess different anatomical and physiological properties of the brain. A stroke, whether acute or chronic, will appear as a bright region on a sequence that is sensitive to fluid (T2 weighted) (Fig. 8.40B). To identify whether a stroke is acute, further sequences are obtained, known as diffusion-weighted imaging (DWI) (Fig. 8.40C) and the apparent diffusion coefficient (ADC) (Fig. 8.40D) map. These evaluate the diffusion of water molecules in the brain. If the region of abnormality appears bright on the DWI sequence and dark on the ADC map, this is known as restricted diffusion, which is compatible with an acute stroke. These changes can persist for up to a week after the initial insult.

Imaging of the carotid and vertebral arteries is also performed to assess for any treatable atherosclerotic changes and stenosis. This can be done with ultrasound, CT, or less frequently, MRI.

Management of a stroke is multidisciplinary. Supportive treatment to stabilize the patient is a priority. Stroke specialists, speech and language therapists, occupational therapists, and physiotherapists have key roles in patient rehabilitation. Long-term use of antiplatelet drugs such as aspirin and modification of cardiovascular disease risk factors are important in the secondary prevention of stroke.

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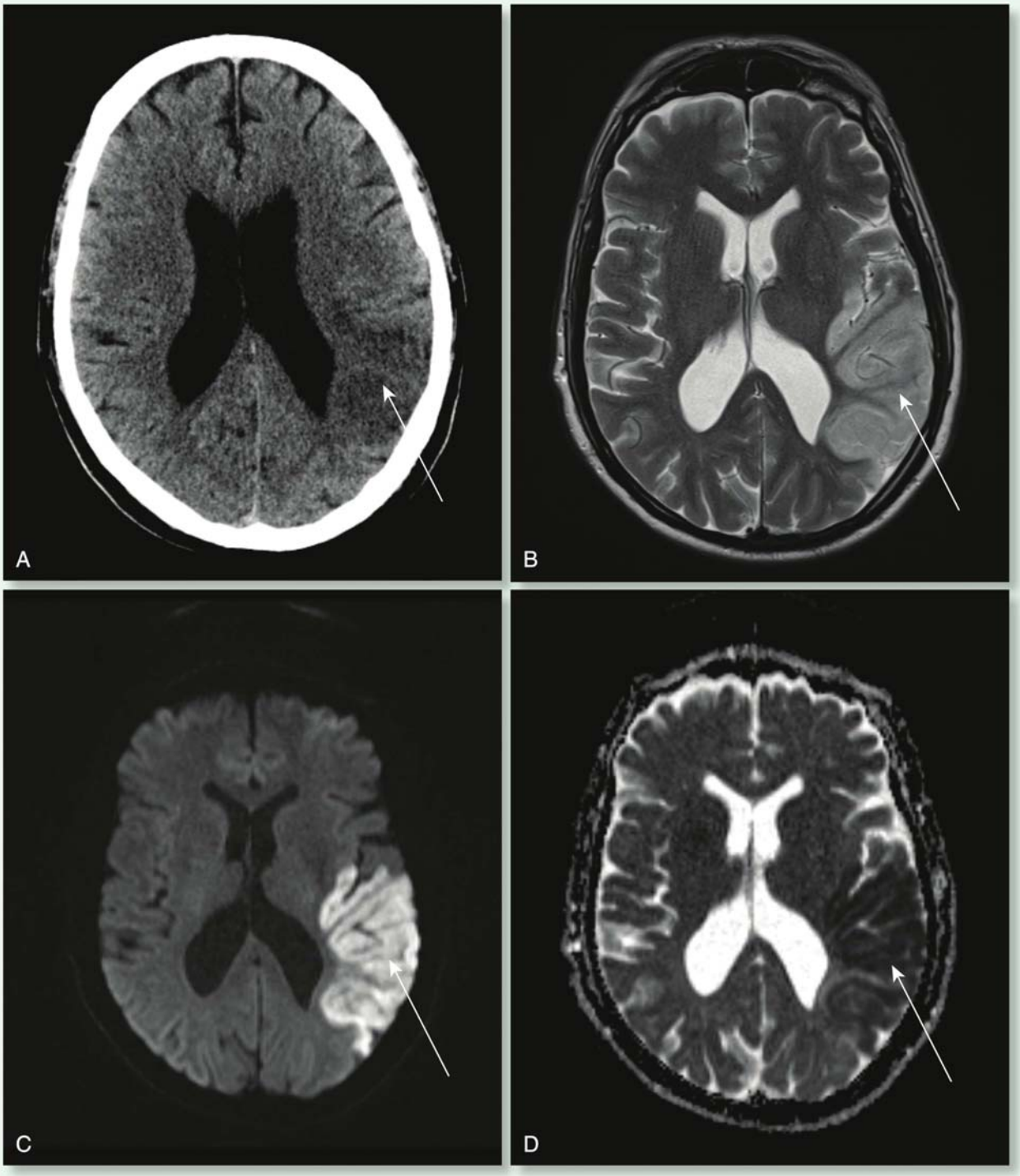


Fig. 8.40 Different imaging modalities used to evaluate a stroke *(arrows)*. **A.** CT scan. **B.** T2- weighted CT. **C.** Diffusion-weighted image (DWI). **D.** Apparent diffusion coefficient image (ADC).

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In the clinic

Endarterectomy

Endarterectomy is a surgical procedure to remove atheromatous plaque from arteries.

Atheromatous plaques occur in the subendothelial layer of vessels and consist of lipid-laden macrophages and cholesterol debris. The developing plaque eventually accumulates fibrous connective tissue and calcifies. Plaque commonly occurs around vessel bifurcations, limiting blood flow, and may embolize to distal organs.

During endarterectomy, plaque is removed and the vessel reopened. In many instances a patch of material is sewn over the hole in the vessel, enabling improved flow and preventing narrowing from the suturing of the vessel.

In the clinic

Intracerebral aneurysms

Cerebral aneurysms arise from the vessels in and around the cerebral arterial circle (of Willis). They typically occur in and around the anterior communicating artery, the posterior communicating artery, the branches of the middle cerebral artery, the distal end of the basilar artery (Fig. 8.41), and the posterior inferior cerebellar artery.

As the aneurysms enlarge, they have a significant risk of rupture. Typically patients have no idea that there is anything wrong. As the aneurysm ruptures, the patient complains of a sudden-onset "thunderclap" headache that produces neck stiffness and may induce vomiting. In a number of patients death ensues, but many patients reach the hospital, where the diagnosis is established. An initial CT scan demonstrates blood within the subarachnoid space, and this may be associated with an intracerebral bleed. Further management usually includes cerebral angiography, which enables the radiologist to determine the site, size, and origin of the aneurysm.

Usually patients undergo complex surgery to ligate the neck of the aneurysm. More recently radiological intervention has superseded the management of some aneurysms in specific sites. This treatment involves cannulation of the femoral artery, and placement of a long catheter through the aorta into the carotid circulation and thence into the cerebral circulation. The tip of the catheter is placed within the aneurysm and is packed with fine microcoils (Fig. 8.42), which seals the rupture.

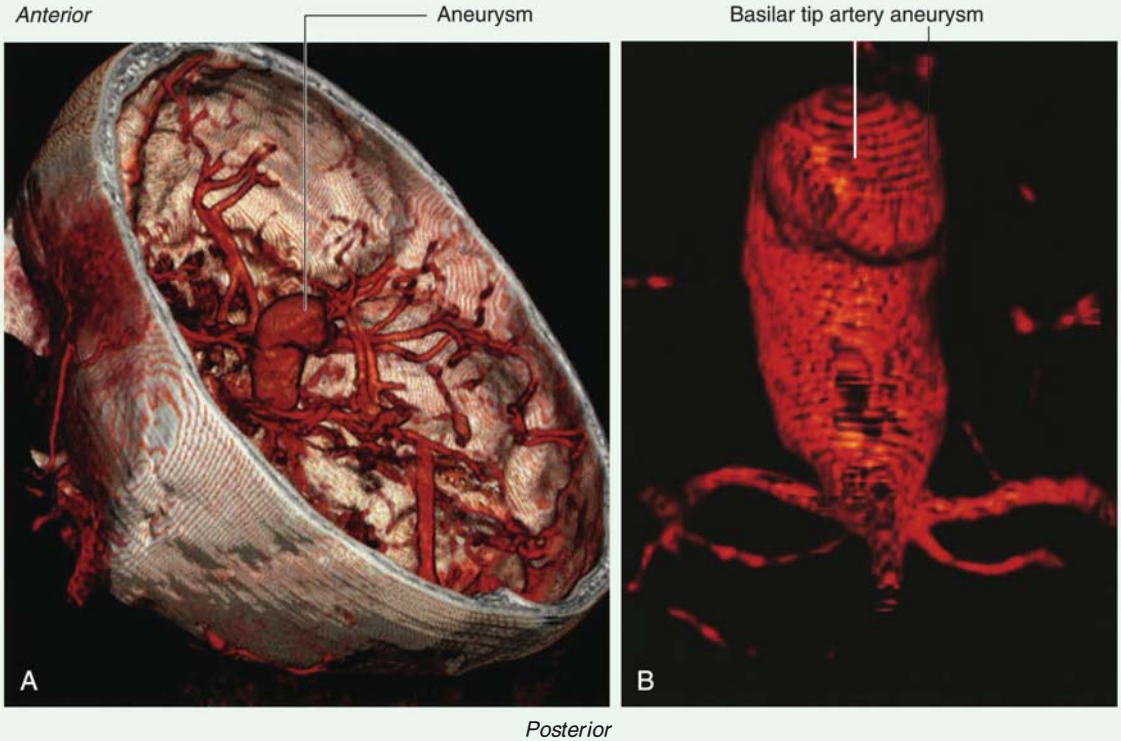


Fig. 8.41 Basilar tip aneurysm. **A.** Three-dimensional cranial cutaway CT scan. **B.** Magnified view of aneurysm.

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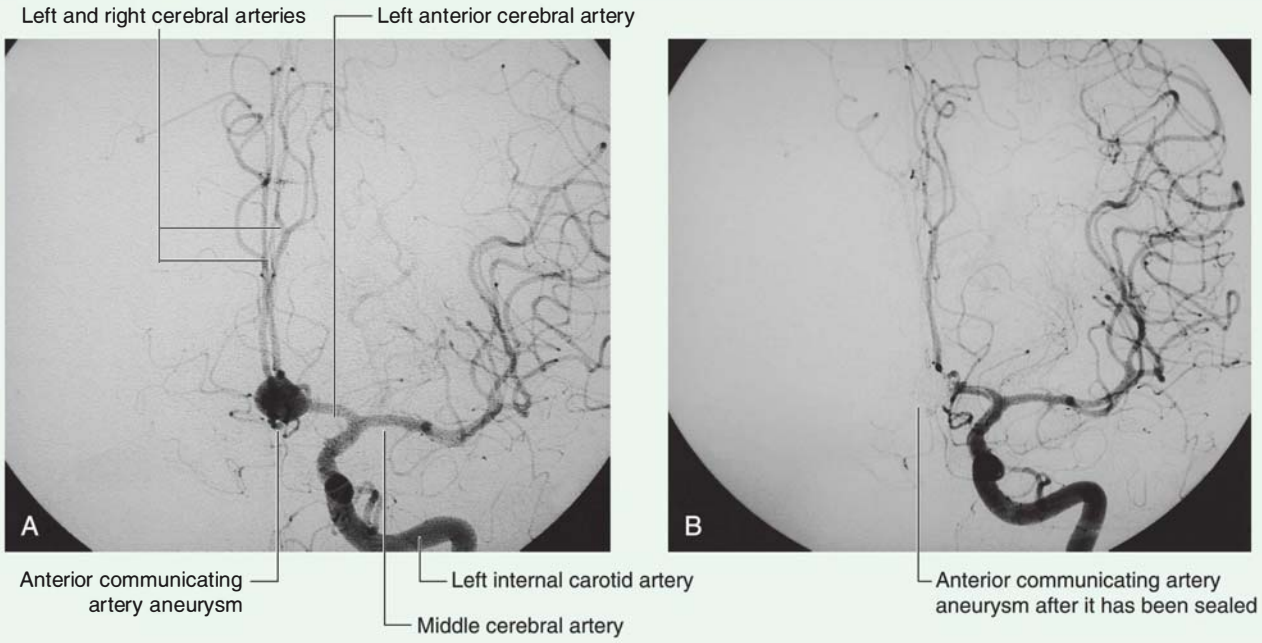


Fig. 8.42 Anterior communicating aneurysm. **A.** Left carotid angiogram. **B.** Left carotid angiogram after embolization.

Venous drainage

Venous drainage of the brain begins internally as networks of small venous channels lead to larger cerebral veins, cerebellar veins, and veins draining the brainstem, which eventually empty into **dural venous sinuses**. The dural venous sinuses are endothelial-lined spaces between the outer periosteal and the inner meningeal layers of the dura mater, and eventually lead to the **internal jugular veins**.

Also emptying into the dural venous sinuses are **diploic veins**, which run between the internal and external tables of compact bone in the roof of the cranial cavity, and **emissary veins**, which pass from outside the cranial cavity to the dural venous sinuses (Fig. 8.43).

The emissary veins are important clinically because they can be a conduit through which infections can enter the cranial cavity because they have no valves.

Dural venous sinuses

The dural venous sinuses include the superior sagittal, inferior sagittal, straight, transverse, sigmoid, and occipital sinuses, the confluence of sinuses, and the cavernous, sphenoparietal, superior petrosal, inferior petrosal, and basilar sinuses (Fig. 8.44, Table 8.3).

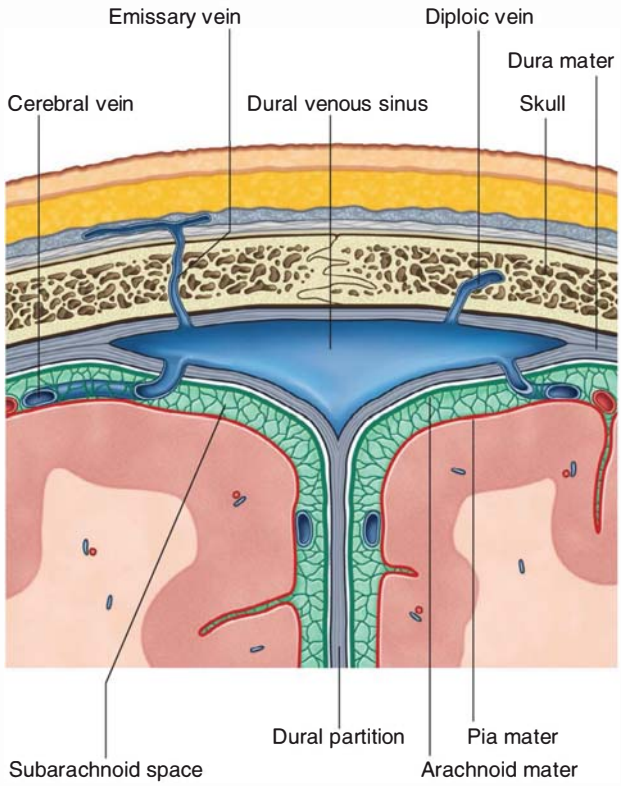


Fig. 8.43 Dural venous sinuses.

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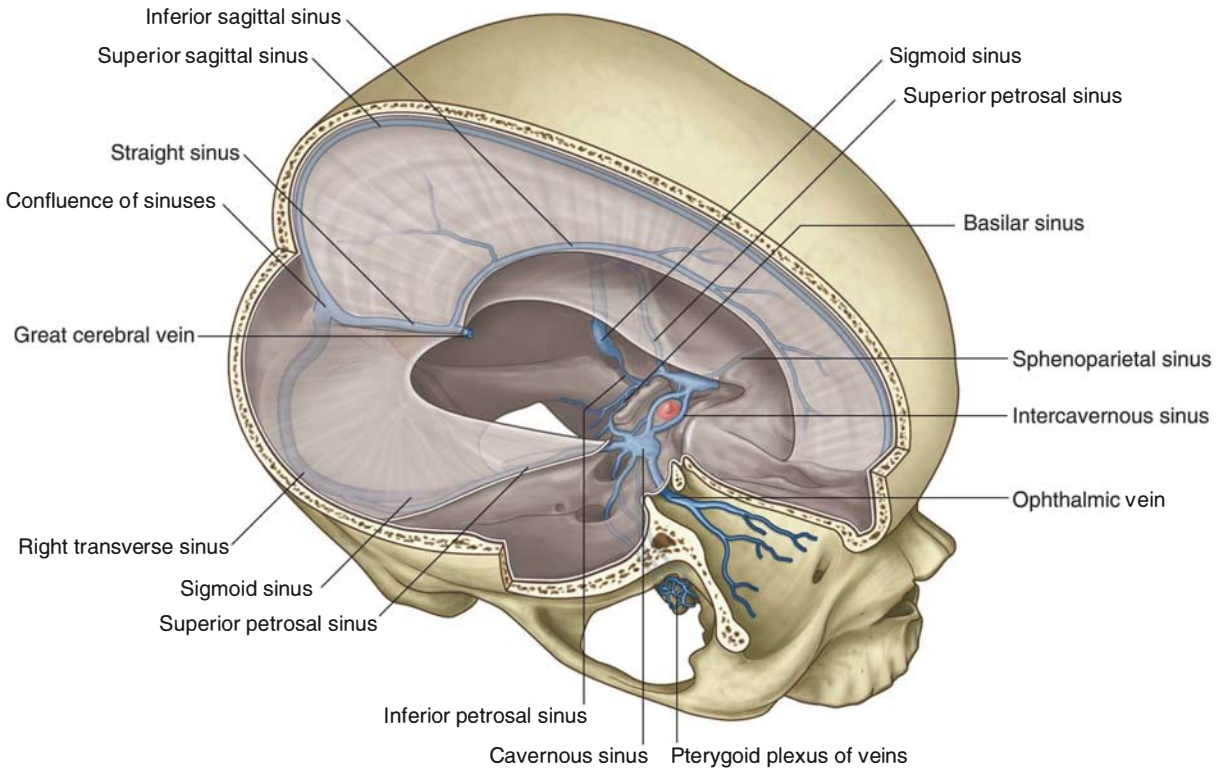


Fig. 8.44 Veins, meninges, and dural venous sinuses.

Table 8.3 Dural venous sinuses

Dural sinus	Location	Receives
Superior sagittal	Superior border of falx cerebri	Superior cerebral, diploic, and emissary veins and CSF
Inferior sagittal	Inferior margin of falx cerebri	A few cerebral veins and veins from the falx cerebri
Straight	Junction of falx cerebri and tentorium cerebelli	Inferior sagittal sinus, great cerebral vein, posterior cerebral veins, superior cerebellar veins, and veins from the falx cerebri
Occipital	In falx cerebelli against occipital bone	Communicates inferiorly with vertebral plexus of veins
Confluence of sinuses	Dilated space at the internal occipital protuberance	Superior sagittal, straight, and occipital sinuses
Transverse (right and left)	Horizontal extensions from the confluence of sinuses along the posterior and lateral attachments of the tentorium cerebelli	Drainage from confluence of sinuses (right—transverse and usually superior sagittal sinuses; left—transverse and usually straight sinuses); also superior petrosal sinus, and inferior cerebral, cerebellar, diploic, and emissary veins
Sigmoid (right and left)	Continuation of transverse sinuses to internal jugular vein; groove of parietal, temporal, and occipital bones	Transverse sinuses, and cerebral, cerebellar, diploic, and emissary veins
Cavernous (paired)	Lateral aspect of body of sphenoid	Cerebral and ophthalmic veins, sphenoparietal sinuses, and emissary veins from pterygoid plexus of veins
Intercavernous	Crossing sella turcica	Interconnect cavernous sinuses
Sphenoparietal (paired)	Inferior surface of lesser wings of sphenoid	Diploic and meningeal veins
Superior petrosal (paired)	Superior margin of petrous part of temporal bone	Cavernous sinus, and cerebral and cerebellar veins
Inferior petrosal (paired)	Groove between petrous part of temporal bone and occipital bone ending in internal jugular vein	Cavernous sinus, cerebellar veins, and veins from the internal ear and brainstem
Basilar	Clivus, just posterior to sella turcica of sphenoid	Connect bilateral inferior petrosal sinuses and communicate with vertebral plexus of veins

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Superior sagittal sinus

The superior sagittal sinus is in the superior border of the falx cerebri (Fig. 8.44). It begins anteriorly at the foramen cecum, where it may receive a small emissary vein from the nasal cavity, and ends posteriorly in the confluence of sinuses, usually bending to the right to empty into the right transverse sinus. The superior sagittal sinus communicates with lateral extensions (lateral lacunae) of the sinus containing numerous arachnoid granulations.

The superior sagittal sinus usually receives cerebral veins from the superior surface of the cerebral hemispheres, diploic and emissary veins, and veins from the falx cerebri.

Inferior sagittal and straight sinuses

The inferior sagittal sinus is in the inferior margin of the falx cerebri (Fig. 8.44). It receives a few cerebral veins and veins from the falx cerebri, and ends posteriorly at the anterior edge of the tentorium cerebelli, where it is joined by the great cerebral vein and together with the great cerebral vein forms the straight sinus (Fig. 8.44).

The straight sinus continues posteriorly along the junction of the falx cerebri and the tentorium cerebelli and ends in the confluence of sinuses, usually bending to the left to empty into the left transverse sinus.

The straight sinus usually receives blood from the inferior sagittal sinus, cerebral veins (from the posterior part of the cerebral hemispheres), the great cerebral vein (draining deep areas of the cerebral hemispheres), superior cerebellar veins, and veins from the falx cerebri.

Confluence of sinuses, transverse and sigmoid sinuses

The superior sagittal and straight sinuses, and the occipital sinus (in the falx cerebelli) empty into the confluence of sinuses, which is a dilated space at the internal occipital protuberance (Fig. 8.44) and is drained by the right and left transverse sinuses.

The paired transverse sinuses extend in horizontal directions from the confluence of sinuses where the tentorium cerebelli joins the lateral and posterior walls of the cranial cavity.

The right transverse sinus usually receives blood from the superior sagittal sinus and the left transverse sinus usually receives blood from the straight sinus.

The transverse sinuses also receive blood from the superior petrosal sinus, veins from the inferior parts of the cerebral hemispheres and the cerebellum, and diploic and emissary veins.

As the transverse sinuses leave the surface of the occipital bone, they become the sigmoid sinuses (Fig. 8.44), which turn inferiorly, grooving the parietal, temporal, and occipital bones, before ending at the beginning of the internal jugular veins. The sigmoid sinuses also receive blood from cerebral, cerebellar, diploic, and emissary veins.

Cavernous sinuses

The paired cavernous sinuses are against the lateral aspect of the body of the sphenoid bone on either side of the sella turcica (Fig. 8.45). They are of great clinical importance because of their connections and the structures that pass through them.

The cavernous sinuses receive blood not only from cerebral veins but also from the ophthalmic veins (from the orbit) and emissary veins (from the pterygoid plexus of veins in the infratemporal fossa). These connections provide pathways for infections to pass from extracranial sites into intracranial locations. In addition, because structures pass through the cavernous sinuses and are located in the walls of these sinuses they are vulnerable to injury due to inflammation.

Structures passing through each cavernous sinus are:

- the internal carotid artery, and
- the abducent nerve [VI].

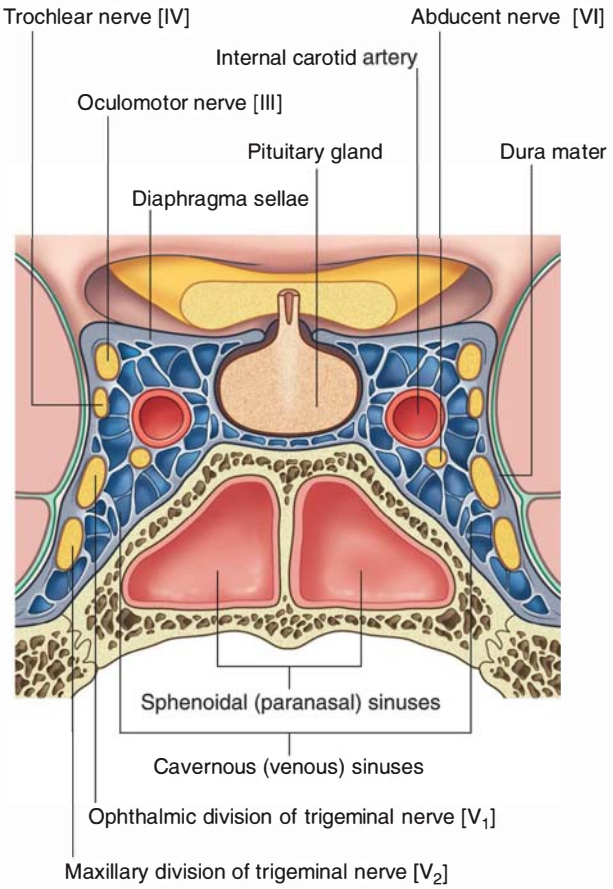


Fig. 8.45 Cavernous sinuses.

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Structures in the lateral wall of each cavernous sinus are, from superior to inferior:

- the oculomotor nerve [III],
- the trochlear nerve [IV],
- the ophthalmic nerve [V_1], and
- the maxillary nerve [V_2].

Connecting the right and left cavernous sinuses are the intercavernous sinuses on the anterior and posterior sides of the pituitary stalk (Fig. 8.44).

Sphenoparietal sinuses drain into the anterior ends of each cavernous sinus. These small sinuses are along the inferior surface of the lesser wings of the sphenoid and receive blood from the diploic and meningeal veins.

Superior and inferior petrosal sinuses

The superior petrosal sinuses drain the cavernous sinuses into the transverse sinuses. Each superior petrosal sinus begins at the posterior end of the cavernous sinus, passes posterolaterally along the superior margin of the petrous part of each temporal bone, and connects to the transverse sinus (Fig. 8.44). The superior petrosal sinuses also receive cerebral and cerebellar veins.

The inferior petrosal sinuses also begin at the posterior ends of the cavernous sinuses. These bilateral sinuses pass posteroinferiorly in a groove between the petrous part of the temporal bone and the basal part of the occipital bone, ending in the internal jugular veins. They assist in draining the cavernous sinuses and also receive blood from cerebellar veins and veins from the internal ear and brainstem.

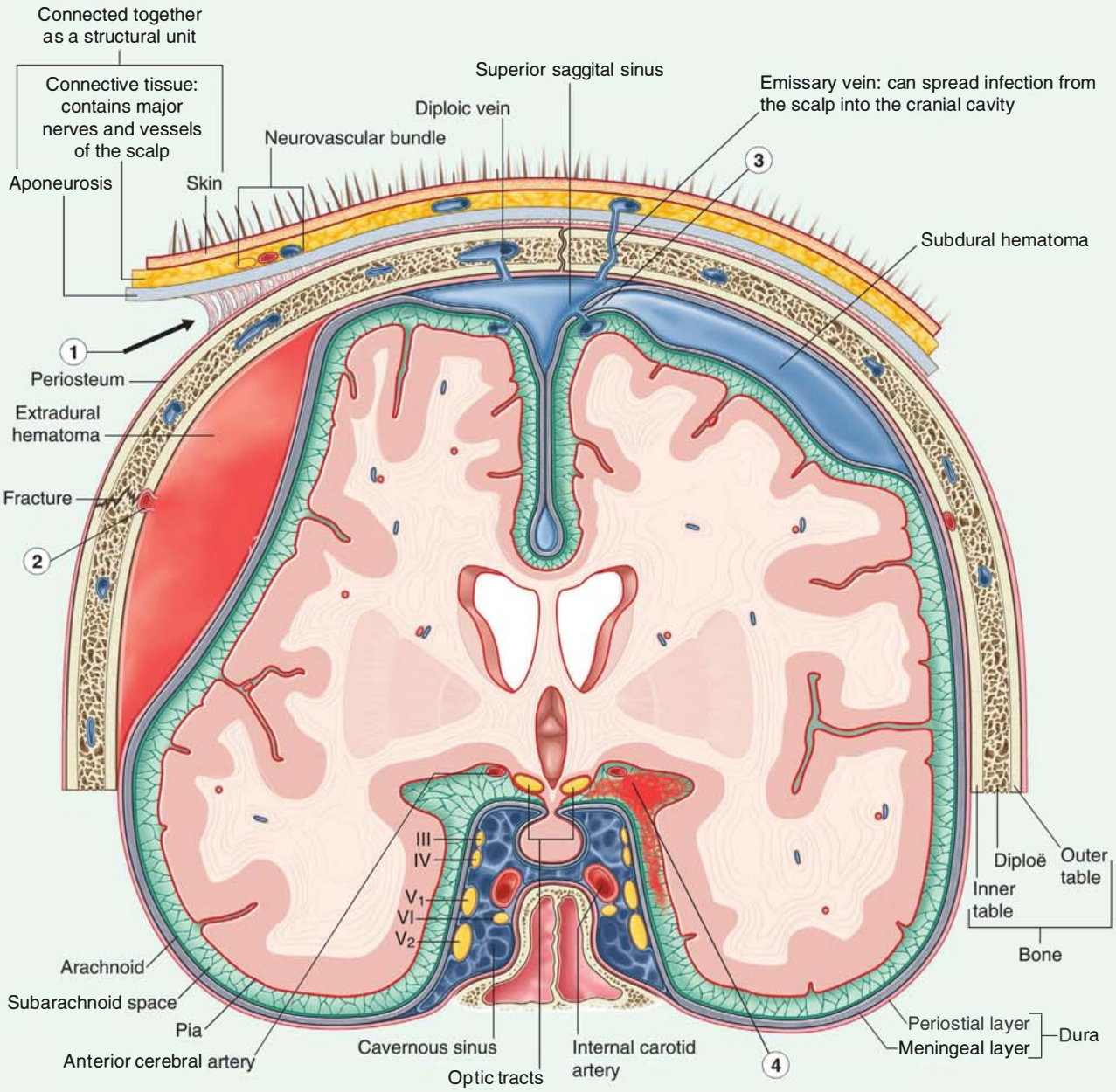
Basilar sinuses connect the inferior petrosal sinuses to each other and to the vertebral plexus of veins. They are on the clivus, just posterior to the sella turcica of the sphenoid bone (Fig. 8.44).

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In the clinic

Scalp and meninges

Summary of relationships and clinical significance of the scalp and meninges (Fig. 8.46).



1 Loose connective tissue (danger area)
- In scalping injuries, this is the layer in which separation occurs.
- Infection can easily spread in this layer.
- Blunt trauma can result in hemorrhage in this layer (blood can spread forward into the face, resulting in "black eyes").

2 Rupture of the middle meningeal artery (branches) by fracture of the inner table of bone results in extradural hematoma. Under pressure, the blood progressively separates dura from the bone.

3 Tear to cerebral vein where it crosses dura to enter cranial venous sinus can result in subdural hematoma. The tear separates a thin layer of meningeal dura from that which remains attached to the periosteal layer. As a result, the hematoma is covered by an inner limiting membrane derived from part of the meningeal dura.

4 Aneurysm
- Ruptured aneurysms of vessels of the cerebral arterial circle hemorrhage directly into the subarachnoid space and CSF.

Fig. 8.46 Scalp and meninges.

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In the clinic

Head injury

Head trauma is a common injury and is a significant cause of morbidity and death. It typically occurs in young male adults and accounts for approximately 10% of all deaths within this age group. Head injury may occur in isolation, but often the patient has other injuries; it should always be suspected in patients with multiple injuries. Among patients with multiple trauma, 50% die from the head injury.

At the time of the initial head injury two processes take place.

- First the primary brain injury may involve primary axonal and cellular damage, which results from the shearing deceleration forces within the brain. These injuries are generally not repairable. Further primary brain injuries include intracerebral hemorrhage and penetrating injuries, which may directly destroy gray and white matter.
- The secondary injuries are sequelae of the initial trauma. They include scalp laceration, fracture of the cranial vault, disruption of intracerebral arteries and veins, intracerebral edema, and infection. In most cases these can be treated if diagnosed early, and rapid and effective treatment will significantly improve the patient's recovery and prognosis.

In the clinic

Types of intracranial hemorrhage

Primary brain hemorrhage

The many causes of a primary brain hemorrhage include aneurysm rupture, hypertension (intracerebral hematoma secondary to high blood pressure), and bleeding after cerebral infarction.

Extradural hemorrhage

An extradural hemorrhage (Fig. 8.47) is caused by arterial damage and results from tearing of the branches of the middle meningeal artery, which typically occurs in the region of the pterion. Blood collects between the periosteal layer of the dura and the calvaria and under arterial pressure slowly expands.

The typical history is of a blow to the head (often during a sporting activity) that produces a minor loss of consciousness. Following the injury the patient usually regains consciousness and has a lucid interval for a period of hours. After this rapid drowsiness and unconsciousness ensue, which may lead to death.

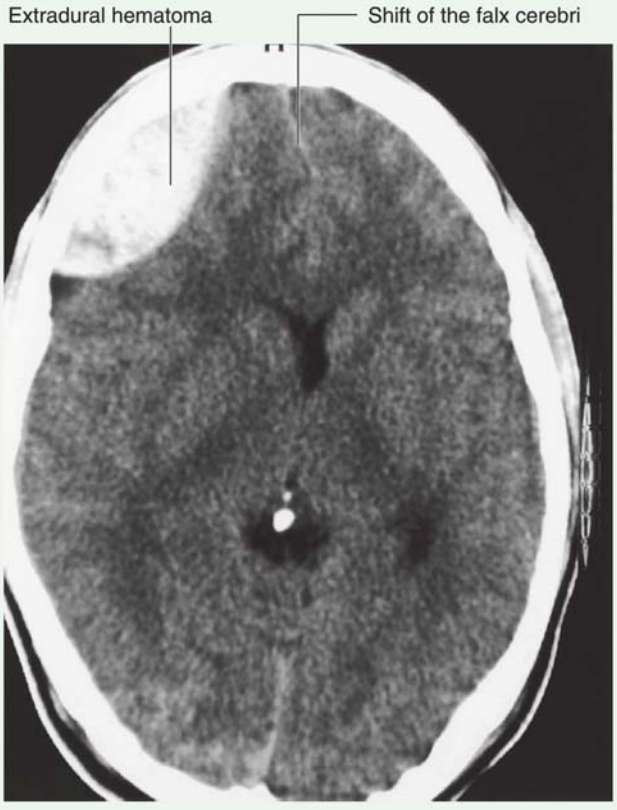


Fig. 8.47 Extradural hematoma. Axial CT scan of brain.

(continues)

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In the clinic—cont'd

Subdural hematoma

A subdural hematoma (Fig. 8.48) results from venous bleeding, usually from torn cerebral veins where they enter the superior sagittal sinus. The tear and resulting seepage of blood separates the thin layer of dural border cells from the rest of the dura as the hematoma develops.

Patients at most risk of developing a subdural hematoma are the young and elderly. The increased CSF space in patients with cerebral atrophy results in a greater than normal stress on the cerebral veins entering the sagittal sinus. The clinical history usually includes a trivial injury followed by an insidious loss of consciousness or alteration of personality.

Subarachnoid hemorrhage

Subarachnoid hemorrhage (Fig. 8.49) may occur in patients who have undergone significant cerebral trauma, but typically it results from a ruptured intracerebral aneurysm arising from the vessels supplying and around the arterial circle (of Willis).



Fig. 8.48 Chronic (low-density) subdural hematoma. Axial CT scan of brain.

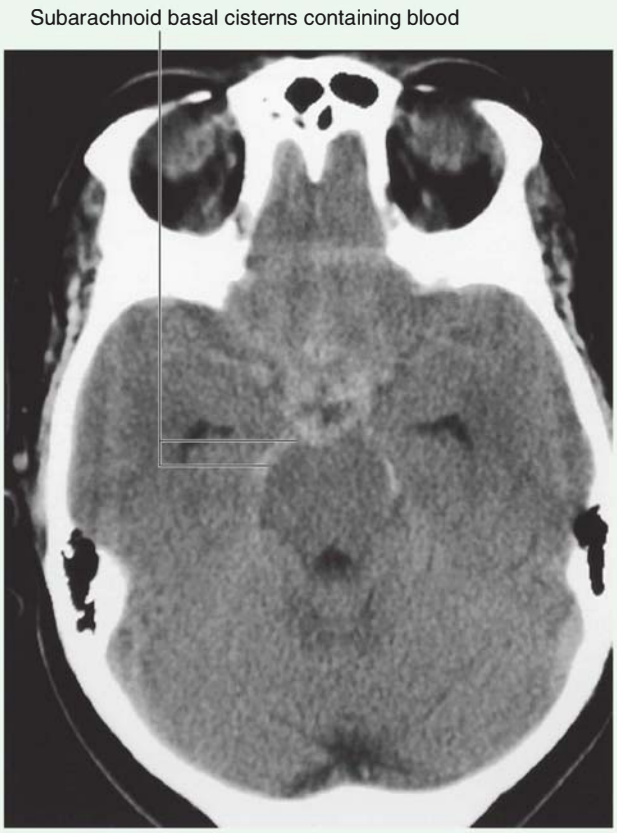


Fig. 8.49 Subarachnoid hemorrhage. Axial CT scan of brain.

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In the clinic

Emissary veins

Emissary veins connect extracranial veins with intracranial veins and are important clinically because they can be a conduit through which infections can enter the cranial cavity. Emissary veins lack valves, as do the majority of veins in the head and neck.

In the clinic

Concussion

Concussion (mild traumatic brain injury [MTBI]) is the most common type of traumatic brain injury. The injury typically results from a rapid deceleration of the head or by a rotation of the brain within the cranial cavity. General symptoms of MTBI can include posttraumatic amnesia, confusion, loss of consciousness, headache, dizziness, vomiting, lack of motor coordination, and light sensitivity. The diagnosis of concussion, MTBI, is based on the event, the current neurological status, and the state of consciousness of the patient.

In the clinic

Clinical assessment of patients with head injury

Clinical assessment of patients with head injury always appears relatively straightforward. In reality it is usually far from straightforward.

Patients may have a wide spectrum of modes of injury from a simple fall to complex multiple trauma. The age of the patient and ability to communicate about the injuries are important factors.

The circumstances in which the injury may have occurred should be documented because some head injuries result from a serious assault, and the physician may be required to give evidence to a court of law.

Determining the severity of head injury may be difficult because some injuries occur as a result of or in association with alcohol intoxication.

Even when the diagnosis has been made and the correct management has been instigated, the circumstances in which the injury occurred and the environment to which the patient will return after treatment need to be reviewed to prevent further injuries (e.g., an elderly person tripping on loose carpet on a staircase).

A thorough clinical examination includes all systems, but with a special focus on the central and peripheral nervous systems. The level of consciousness must also be assessed and accurately documented using the Glasgow coma scale, which allows clinicians to place a numerical value upon the level of consciousness so that any deterioration or improvement can be measured and quantified.

Glasgow coma scale

The Glasgow coma scale was proposed in 1974 and is now widely accepted throughout the world. There is a total score of 15 points, such that 15/15 indicates that the patient is alert and fully oriented, whereas 3/15 indicates a severe and deep coma. The points score comprises a best motor response (total of 6 points), best verbal response (total of 5 points), and best eye movement response (total of 4 points).

In the clinic

Treatment of head injury

Treatment of primary brain injury is extremely limited. Axonal disruption and cellular death are generally irrecoverable. Whenever the brain is injured, like most tissues, it swells. Because the brain is encased within a fixed space (the skull), swelling impairs cerebral function and has two other important effects.

- First, the swelling compresses the blood supply into the skull, resulting in a physiologically dramatic increase in blood pressure.
- Second, the cerebral swelling may be diffuse, eventually squeezing the brain and brainstem through the foramen magnum (**coning**). This compression and disruption of the brainstem may lead to a loss of basic cardiorespiratory function, and death will ensue. Focal cerebral edema may cause one side of the brain to herniate beneath the falx cerebri (**falcine herniation**).

Simple measures to prevent the swelling include hyperventilation (which alters the intracerebral acid–base balance and decreases swelling) and intravenous corticosteroids (though their action is often delayed).

Extracerebral hematoma may be removed surgically.

The outcome of patients with head injury depends on how the secondary injury is managed. Even with a severe primary injury, patients may recover to lead a normal life.

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CRANIAL NERVES

The 12 pairs of cranial nerves are part of the peripheral nervous system (PNS) and pass through foramina or fissures in the cranial cavity. All nerves except one, the accessory nerve [XI], originate from the brain.

In addition to having somatic and visceral components similar to those of spinal nerves, some cranial nerves also contain special sensory and motor components (Tables 8.4 and 8.5).

The special sensory components are associated with hearing, seeing, smelling, balancing, and tasting.

Special motor components include those that innervate skeletal muscles derived embryologically from the pharyngeal arches and not from somites.

In human embryology, six pharyngeal arches are designated, but the fifth pharyngeal arch never develops. Each of the pharyngeal arches that does develop is associated with a developing cranial nerve or one of its branches.

Table 8.4 Cranial nerve functional components

Functional component	Abbreviation	General function	Cranial nerves containing component
General somatic afferent	GSA	Perception of touch, pain, temperature	Trigeminal nerve [V]; facial nerve [VII]; glossopharyngeal nerve [IX]; vagus nerve [X]
General visceral afferent	GVA	Sensory input from viscera	Glossopharyngeal nerve [IX]; vagus nerve [X]
Special afferent*	SA	Smell, taste, vision, hearing, and balance	Olfactory nerve [I]; optic nerve [II]; facial nerve [VII]; vestibulocochlear nerve [VIII]; glossopharyngeal nerve [IX]; vagus nerve [X]
General somatic efferent	GSE	Motor innervation to skeletal (voluntary) muscles	Oculomotor nerve [III]; trochlear nerve [IV]; abducent nerve [VI]; accessory nerve [XI]; hypoglossal nerve [XII]
General visceral efferent	GVE	Motor innervation to smooth muscle, heart muscle, and glands	Oculomotor nerve [III]; facial nerve [VII]; glossopharyngeal nerve [IX]; vagus nerve [X]
Branchial efferent**	BE	Motor innervation to skeletal muscles derived from pharyngeal arch mesoderm	Trigeminal nerve [V]; facial nerve [VII]; glossopharyngeal nerve [IX]; vagus nerve [X]

Other terminology used when describing functional components:
*Special sensory, or special visceral afferent (SVA): smell, taste. Special somatic afferent (SSA): vision, hearing, balance.
**Special visceral efferent (SVE) or branchial motor.

Table 8.5 Cranial nerves (see Table 8.4 for abbreviations)

	COMPONENT			
Nerve	**Afferent**	**Efferent**	**Exit from skull**	**Function**
Olfactory nerve [I]	SA		Cribriform plate of ethmoid bone	Smell
Optic nerve [II]	SA		Optic canal	Vision
Oculomotor nerve [III]		GSE, GVE	Superior orbital fissure	GSE—innervates levator palpebrae superioris, superior rectus, inferior rectus, medial rectus, and inferior oblique muscles GVE—innervates sphincter pupillae for pupillary constriction; ciliary muscles for accommodation of the lens for near vision
Trochlear nerve [IV]		GSE	Superior orbital fissure	Innervates superior oblique muscle
Trigeminal nerve [V]	GSA	BE	Superior orbital fissure—ophthalmic division [V_1] Foramen rotundum—maxillary nerve [V_2] Foramen ovale—mandibular division [V_3]	GSA—sensory from: ophthalmic division [V_1]—eyes, conjunctiva, orbital contents, nasal cavity, frontal sinus, ethmoidal cells, upper eyelid, dorsum of nose, anterior part of scalp, dura in anterior cranial fossa, superior part of tentorium cerebelli; maxillary nerve [V_2]—dura in middle cranial fossa, nasopharynx, palate, nasal cavity, upper teeth, maxillary sinus, skin covering the side of the nose, lower eyelid, cheek, upper lip; mandibular division [V_3]—skin of lower face, cheek, lower lip, anterior part of external ear, part of external acoustic meatus, temporal fossa, anterior two-thirds of tongue, lower teeth, mastoid air cells, mucous membranes of cheek, mandible, dura in middle cranial fossa BE—innervates temporalis, masseter, medial and lateral pterygoids, tensor tympani, tensor veli palatini, anterior belly of digastric, and mylohyoid muscles

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Table 8.5 Cranial nerves (see Table 8.4 for abbreviations)—cont'd

	COMPONENT			
Nerve	**Afferent**	**Efferent**	**Exit from skull**	**Function**
Abducent nerve [VI]		GSE	Superior orbital fissure	Innervates lateral rectus muscle
Facial nerve [VII]	GSA, SA	GVE, BE	Stylomastoid foramen (nerve leaves cranial cavity through internal acoustic meatus and gives rise to branches in the facial canal of the temporal bone prior to exiting through the stylomastoid foramen; these branches leave the skull through other fissures and canals.)	GSA—sensory from part of external acoustic meatus and deeper parts of auricle SA—taste from anterior two-thirds of tongue GVE—innervates lacrimal gland, submandibular and sublingual salivary glands, and mucous membranes of nasal cavity, hard and soft palates BE—innervates muscles of face (muscles of facial expression) and scalp derived from the second pharyngeal arch, and stapedius, posterior belly of digastric, stylohyoid muscles
Vestibulocochlear nerve [VIII]	SA		(Nerve leaves cranial cavity through internal acoustic meatus)	Vestibular division—balance Cochlear division—hearing
Glossopharyngeal nerve [IX]	GVA, SA, GSA	GVE, BE	Jugular foramen	GVA—sensory from carotid body and sinus GSA—posterior one-third of tongue, palatine tonsils, oropharynx, and mucosa of middle ear, pharyngotympanic tube, and mastoid air cells SA—taste from posterior one-third of tongue GVE—innervates parotid salivary gland BE—innervates stylopharyngeus muscle
Vagus nerve [X]	GSA, GVA, SA	GVE, BE	Jugular foramen	GSA—sensory from larynx, laryngopharynx, deeper parts of auricle, part of external acoustic meatus, and dura in posterior cranial fossa GVA—sensory from aortic body chemoreceptors and aortic arch baroreceptors, esophagus, bronchi, lungs, heart, and abdominal viscera of the foregut and midgut SA—taste from the epiglottis and pharynx GVE—innervates smooth muscle and glands in the pharynx, larynx, thoracic viscera, and abdominal viscera of the foregut and midgut BE—innervates one tongue muscle (palatoglossus), muscles of soft palate (except tensor veli palatini), pharynx (except stylopharyngeus), and larynx
Accessory nerve [XI]		GSE	Jugular foramen	Innervates sternocleidomastoid and trapezius muscles
Hypoglossal nerve [XII]		GSE	Hypoglossal canal	Innervates hyoglossus, genioglossus, and styloglossus muscles and all intrinsic muscles of the tongue

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These cranial nerves carry efferent fibers that innervate the musculature derived from the pharyngeal arch.

Innervation of the musculature derived from the five pharyngeal arches that do develop is as follows:

- first arch—trigeminal nerve [V_3],
- second arch—facial nerve [VII],
- third arch—glossopharyngeal nerve [IX],
- fourth arch—superior laryngeal branch of the vagus nerve [X],
- sixth arch—recurrent laryngeal branch of the vagus nerve [X].

Olfactory nerve [I]

The **olfactory nerve [I]** carries special afferent (SA) fibers for the sense of smell. Its sensory neurons have:

- peripheral processes that act as receptors in the nasal mucosa, and
- central processes that return information to the brain.

The receptors are in the roof and upper parts of the nasal cavity, and the central processes, after joining into small bundles, enter the cranial cavity by passing through the cribriform plate of the ethmoid bone (Fig. 8.50). They terminate by synapsing with secondary neurons in the olfactory bulbs (Fig. 8.51).

Optic nerve [II]

The **optic nerve [II]** carries SA fibers for vision. These fibers return information to the brain from photoreceptors in the retina. Neuronal processes leave the retinal

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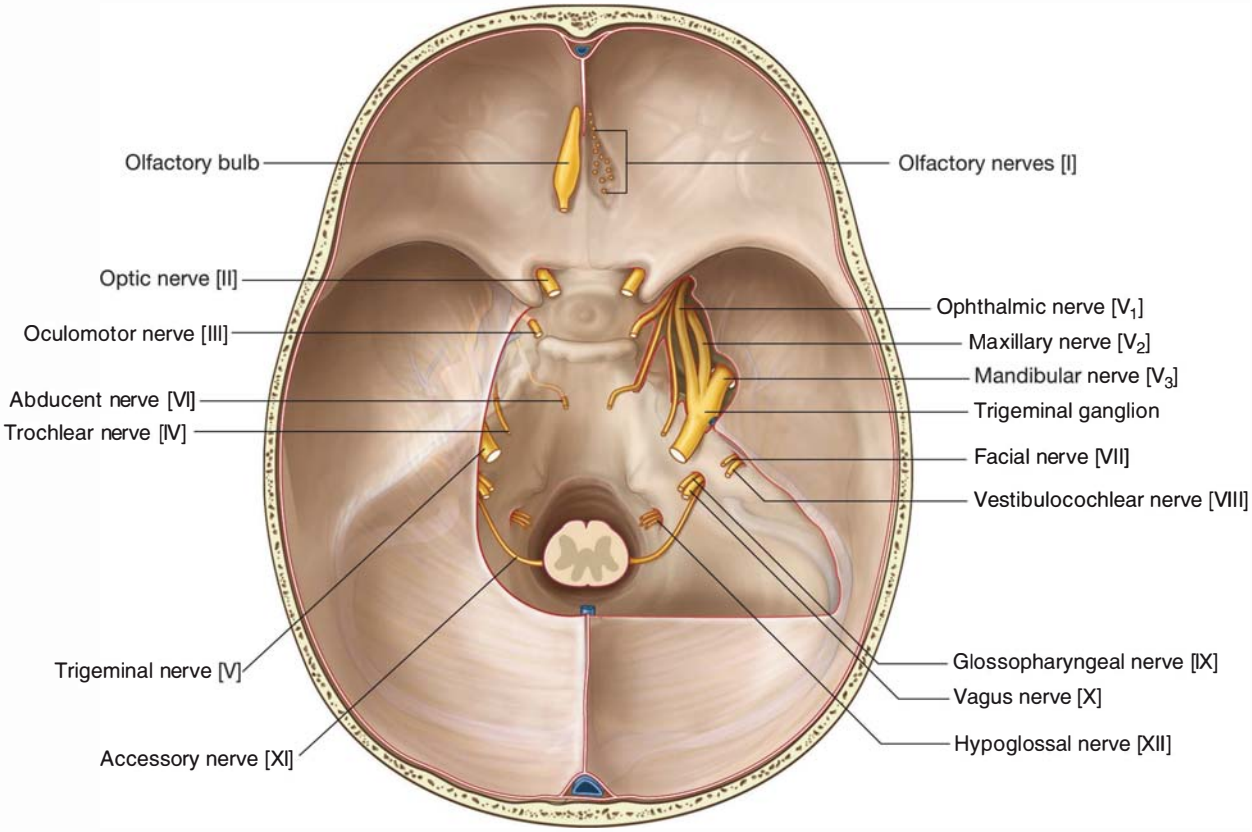


Fig. 8.50 Cranial nerves exiting the cranial cavity.

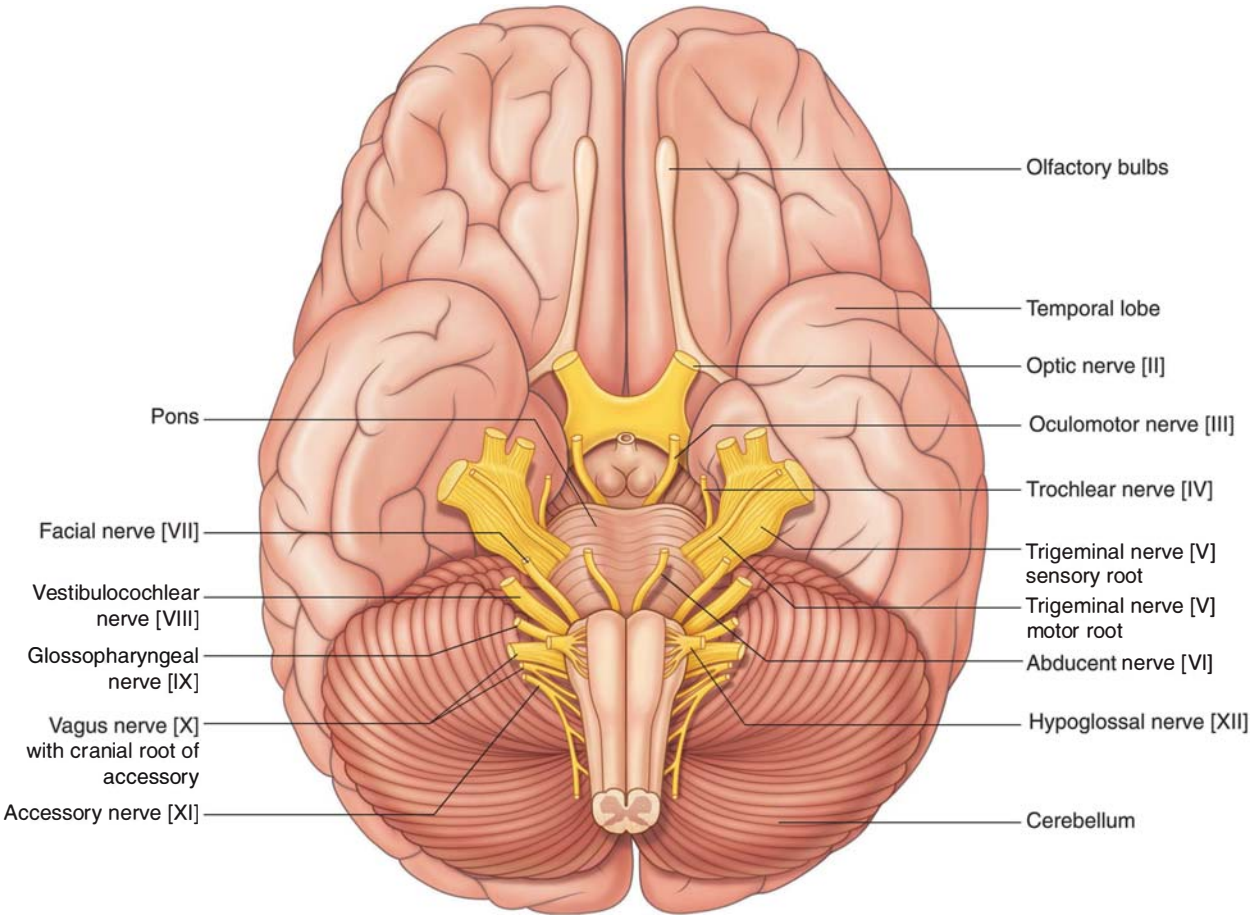


Fig. 8.51 Cranial nerves on the base of the brain.

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receptors, join into small bundles, and are carried by the optic nerves to other components of the visual system in the brain. The optic nerves enter the cranial cavity through the optic canals (Fig. 8.50).

Oculomotor nerve [III]

The **oculomotor nerve [III]** carries two types of fibers:

- General somatic efferent (GSE) fibers innervate most of the extra-ocular muscles.
- General visceral efferent (GVE) fibers are part of the parasympathetic part of the autonomic division of the PNS.

The oculomotor nerve [III] leaves the anterior surface of the brainstem between the midbrain and the pons (Fig. 8.51). It enters the anterior edge of the tentorium cerebelli, continues in an anterior direction in the lateral wall of the cavernous sinus (Figs. 8.50 and 8.51; see Fig. 8.45), and leaves the cranial cavity through the superior orbital fissure.

In the orbit, the GSE fibers in the oculomotor nerve innervate levator palpebrae superioris, superior rectus, inferior rectus, medial rectus, and inferior oblique muscles.

The GVE fibers are preganglionic parasympathetic fibers that synapse in the ciliary ganglion and ultimately innervate the sphincter pupillae muscle, responsible for pupillary constriction, and the ciliary muscles, responsible for accommodation of the lens for near vision.

Trochlear nerve [IV]

The **trochlear nerve [IV]** is a cranial nerve that carries GSE fibers to innervate the superior oblique muscle, an extra-ocular muscle in the orbit. It arises in the midbrain and is the only cranial nerve to exit from the posterior surface of the brainstem (Fig. 8.51). After curving around the midbrain, it enters the inferior surface of the free edge of the tentorium cerebelli, continues in an anterior direction in the lateral wall of the cavernous sinus (Figs. 8.50 and 8.51; see Fig. 8.45), and enters the orbit through the superior orbital fissure.

Trigeminal nerve [V]

The **trigeminal nerve [V]** is the major general sensory nerve of the head and also innervates muscles that move the lower jaw. It carries general somatic afferent (GSA) and branchial efferent (BE) fibers:

- The GSA fibers provide sensory input from the face, anterior one-half of the scalp, mucous membranes of the oral and nasal cavities and the paranasal sinuses, the nasopharynx, part of the ear and external acoustic meatus, part of the tympanic membrane, the orbital contents and conjunctiva, and the dura mater in the anterior and middle cranial fossae.
- The BE fibers innervate the muscles of mastication; the tensor tympani, tensor veli palatini, and mylohyoid muscles; and the anterior belly of the digastric muscle.

The trigeminal nerve exits from the anterolateral surface of the pons as a large sensory root and a small motor root (Fig. 8.51). These roots continue forward out of the posterior cranial fossa and into the middle cranial fossa by passing over the medial tip of the petrous part of the temporal bone (Fig. 8.50).

In the middle cranial fossa the sensory root expands into the **trigeminal ganglion** (Fig. 8.50), which contains cell bodies for the sensory neurons in the trigeminal nerve and is comparable to a spinal ganglion. The ganglion is in a depression (the trigeminal depression) on the anterior surface of the petrous part of the temporal bone, in a dural cave (the **trigeminal cave**). The motor root is below and completely separate from the sensory root at this point.

Arising from the anterior border of the trigeminal ganglion are the three terminal divisions of the trigeminal nerve, which in descending order are:

- the **ophthalmic nerve (ophthalmic division [V_1])**.
- the **maxillary nerve (maxillary division [V_2])**, and
- the **mandibular nerve (mandibular division [V_3])**.

Ophthalmic nerve [V_1]

The ophthalmic nerve [V_1] passes forward in the dura of the lateral wall of the cavernous sinus (see Fig. 8.45), leaves the cranial cavity, and enters the orbit through the superior orbital fissure (Fig. 8.50).

The ophthalmic nerve [V_1] carries sensory branches from the eyes, conjunctiva, and orbital contents, including the lacrimal gland. It also receives sensory branches from the nasal cavity, frontal sinus, ethmoidal cells, falx cerebri, dura in the anterior cranial fossa and superior parts of the tentorium cerebelli, upper eyelid, dorsum of the nose, and the anterior part of the scalp.

Maxillary nerve [V_2]

The maxillary nerve [V_2] passes forward in the dura mater of the lateral wall of the cavernous sinus just inferior to the ophthalmic nerve [V_1] (see Fig. 8.45), leaves the cranial cavity through the foramen rotundum (Fig. 8.50), and enters the pterygopalatine fossa.

The maxillary nerve [V_2] receives sensory branches from the dura in the middle cranial fossa, the nasopharynx, the palate, the nasal cavity, teeth of the upper jaw, maxillary sinus, and skin covering the side of the nose, the lower eyelid, the cheek, and the upper lip.

Mandibular nerve [V_3]

The mandibular nerve [V_3] leaves the inferior margin of the trigeminal ganglion and leaves the skull through the foramen ovale (Fig. 8.50).

The motor root of the trigeminal nerve also passes through the foramen ovale and unites with the sensory component of the mandibular nerve [V_3] outside the skull. Thus the mandibular nerve [V_3] is the only division of the trigeminal nerve that contains a motor component.

Outside the skull the motor fibers innervate the four muscles of mastication (temporalis, masseter, and medial and lateral pterygoids), as well as the tensor tympani muscle, the tensor veli palatini muscle, the anterior belly of the digastric muscle, and the mylohyoid muscle.

The mandibular nerve [V_3] also receives sensory branches from the skin of the lower face, cheek, lower lip, anterior part of the external ear, part of the external acoustic meatus and the temporal region, the anterior two-thirds of the tongue, the teeth of the lower jaw, the mastoid air cells, the mucous membranes of the cheek, the mandible, and dura in the middle cranial fossa.

Abducent nerve [VI]

The **abducent nerve [VI]** carries GSE fibers to innervate the lateral rectus muscle in the orbit. It arises from the brainstem between the pons and medulla and passes forward, piercing the dura covering the clivus (Figs. 8.50 and 8.51). Continuing upward in a dural canal, it crosses the superior edge of the petrous temporal bone, enters and crosses the cavernous sinus (see Fig. 8.45) just inferolateral to the internal carotid artery, and enters the orbit through the superior orbital fissure.

Facial nerve [VII]

The **facial nerve [VII]** carries GSA, SA, GVE, and BE fibers:

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- The GSA fibers provide sensory input from part of the external acoustic meatus and deeper parts of the auricle.
- The SA fibers are for taste from the anterior two-thirds of the tongue.
- The GVE fibers are part of the parasympathetic part of the autonomic division of the PNS and stimulate secretomotor activity in the lacrimal gland, submandibular and sublingual salivary glands, and glands in the mucous membranes of the nasal cavity, and hard and soft palates.
- The BE fibers innervate the muscles of the face (muscles of facial expression) and scalp derived from the second pharyngeal arch, and the stapedius muscle, the posterior belly of the digastric muscle, and the stylohyoid muscle.

The facial nerve [VII] attaches to the lateral surface of the brainstem, between the pons and medulla oblongata (Fig. 8.51). It consists of a large motor root and a smaller sensory root (the **intermediate nerve**):

- The intermediate nerve contains the SA fibers for taste, the parasympathetic GVE fibers, and the GSA fibers.
- The larger motor root contains the BE fibers.

The motor and sensory roots cross the posterior cranial fossa and leave the cranial cavity through the internal acoustic meatus (Fig. 8.50). After entering the facial canal in the petrous part of the temporal bone, the two roots fuse and form the facial nerve [VII]. Near this point the nerve enlarges as the **geniculate ganglion**, which is similar to a spinal ganglion containing cell bodies for sensory neurons.

At the geniculate ganglion the facial nerve [VII] turns and gives off the **greater petrosal nerve**, which carries mainly preganglionic parasympathetic (GVE) fibers (Table 8.6).

The facial nerve [VII] continues along the bony canal, giving off the **nerve to the stapedius** and the **chorda tympani**, before exiting the skull through the stylomastoid foramen.

The chorda tympani carries taste (SA) fibers from the anterior two-thirds of the tongue and preganglionic parasympathetic (GVE) fibers destined for the submandibular ganglion (Table 8.6).

Vestibulocochlear nerve [VIII]

The vestibulocochlear nerve [VIII] carries SA fibers for hearing and balance, and consists of two divisions:

- a vestibular component for balance, and
- a cochlear component for hearing.

The vestibulocochlear nerve [VIII] attaches to the lateral surface of the brainstem, between the pons and medulla, after emerging from the internal acoustic meatus and crossing the posterior cranial fossa (Figs. 8.50 and 8.51). The two divisions combine into the single nerve seen in the posterior cranial fossa within the substance of the petrous part of the temporal bone.

Glossopharyngeal nerve [IX]

The glossopharyngeal nerve [IX] carries GVA, GSA, SA, GVE, and BE fibers:

- The GVA fibers provide sensory input from the carotid body and sinus.
- The GSA fibers provide sensory input from the posterior one-third of the tongue, palatine tonsils, oropharynx, and mucosa of the middle ear, pharyngotympanic tube, and mastoid air cells.
- The SA fibers are for taste from the posterior one-third of the tongue.
- The GVE fibers are part of the parasympathetic part of the autonomic division of the PNS and stimulate secretomotor activity in the parotid salivary gland.
- The BE fibers innervate the muscle derived from the third pharyngeal arch (the stylopharyngeus muscle).

Table 8.6 Parasympathetic ganglia of the head

Ganglion	Cranial nerve origin of preganglionic fibers	Branch supplying preganglionic fibers to ganglion	Function
Ciliary	Oculomotor nerve [III]	Branch to ciliary ganglion	Innervation of sphincter pupillae muscle for pupillary constriction, and ciliary muscles for accommodation of the lens for near vision
Pterygopalatine	Facial nerve [VII]	Greater petrosal nerve	Innervation of lacrimal gland, and mucous glands of nasal cavity, maxillary sinus, and palate
Otic	Glossopharyngeal nerve [IX]	Lesser petrosal nerve	Innervation of parotid gland
Submandibular	Facial nerve [VII]	Chorda tympani to lingual	Innervation of submandibular and sublingual glands

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The glossopharyngeal nerve [IX] arises as several rootlets on the anterolateral surface of the upper medulla oblongata (Fig. 8.51). The rootlets cross the posterior cranial fossa and enter the jugular foramen (Fig. 8.50). Within the jugular foramen, and before exiting from it, the rootlets merge to form the glossopharyngeal nerve.

Within or immediately outside the jugular foramen are two ganglia (the **superior** and **inferior ganglia**), which contain the cell bodies of the sensory neurons in the glossopharyngeal nerve [IX].

Tympanic nerve

Branching from the glossopharyngeal nerve [IX] either within or immediately outside the jugular foramen is the **tympanic nerve**. This branch reenters the temporal bone, enters the middle ear cavity, and participates in the formation of the **tympanic plexus**. Within the middle ear cavity it provides sensory innervation to the mucosa of the cavity, pharyngotympanic tube, and mastoid air cells.

The tympanic nerve also contributes GVE fibers, which leave the tympanic plexus in the **lesser petrosal nerve**—a small nerve that exits the temporal bone, enters the middle cranial fossa, and descends through the foramen ovale to exit the cranial cavity carrying preganglionic parasympathetic fibers to the otic ganglion (Table 8.6).

Vagus nerve [X]

The vagus nerve [X] carries GSA, GVA, SA, GVE, and BE fibers:

- The GSA fibers provide sensory input from the larynx, laryngopharynx, deeper parts of the auricle, part of the external acoustic meatus, and the dura mater in the posterior cranial fossa.
- The GVA fibers provide sensory input from the aortic body chemoreceptors and aortic arch baroreceptors, and the esophagus, bronchi, lungs, heart, and abdominal viscera in the foregut and midgut.
- The SA fibers are for taste around the epiglottis and pharynx.
- The GVE fibers are part of the parasympathetic part of the autonomic division of the PNS and stimulate smooth muscle and glands in the pharynx, larynx, thoracic viscera, and abdominal viscera of the foregut and midgut.
- The BE fibers innervate one muscle of the tongue (palatoglossus), the muscles of the soft palate (except the tensor veli palatini), pharynx (except the stylopharyngeus), and larynx.

The vagus nerve arises as a group of rootlets on the anterolateral surface of the medulla oblongata just inferior to the rootlets arising to form the glossopharyngeal nerve [IX] (Fig. 8.51). The rootlets cross the posterior cranial fossa and enter the jugular foramen (Fig. 8.50). Within this foramen, and before exiting from it, the rootlets merge to form the vagus nerve [X]. Within or immediately outside the jugular foramen are two ganglia, the **superior** (jugular) and **inferior** (nodose) **ganglia**, which contain the cell bodies of the sensory neurons in the vagus nerve [X].

Accessory nerve [XI]

The accessory nerve [XI] is a cranial nerve that carries GSE fibers to innervate the sternocleidomastoid and trapezius muscles. It is a unique cranial nerve because its roots arise from motor neurons in the upper five segments of the cervical spinal cord. These fibers leave the lateral surface of the spinal cord and, joining together as they ascend, enter the cranial cavity through the foramen magnum (Fig. 8.51). The accessory nerve [XI] continues through the posterior cranial fossa and exits through the jugular foramen (Fig. 8.50). It then descends in the neck to innervate the sternocleidomastoid and trapezius muscles from their deep surfaces.

Cranial root of the accessory nerve

Some descriptions of the accessory nerve [XI] refer to a few rootlets arising from the caudal part of the medulla oblongata on the anterolateral surface just inferior to the rootlets arising to form the vagus nerve [X] as the "cranial" root of the accessory nerve (Fig. 8.51). Leaving the medulla, the cranial roots course with the "spinal" roots of the accessory nerve [XI] into the jugular foramen, at which point the cranial roots join the vagus nerve [X]. As part of the vagus nerve [X], they are distributed to the pharyngeal musculature innervated by the vagus nerve [X] and are therefore described as being part of the vagus nerve [X].

Hypoglossal nerve [XII]

The hypoglossal nerve [XII] carries GSE fibers to innervate all intrinsic muscles and most of the extrinsic muscles of the tongue. It arises as several rootlets from the anterior surface of the medulla (Fig. 8.51), passes laterally across the posterior cranial fossa, and exits through the hypoglossal canal (Fig. 8.50). This nerve innervates the hyoglossus, styloglossus, and genioglossus muscles and all intrinsic muscles of the tongue.

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In the clinic

Cranial nerve lesions

Cranial Nerve	Clinical Findings	Example of Lesion
Olfactory nerve [I]	Loss of smell (anosmia)	Injury to the cribriform plate; congenital absence
Optic nerve [II]	Blindness/visual field abnormalities, loss of pupillary constriction	Direct trauma to the orbit; disruption of the optic pathway
Oculomotor nerve [III]	Dilated pupil, ptosis, loss of normal pupillary reflex, eye moves down inferiorly and laterally (down and out)	Pressure from an aneurysm arising from the posterior communicating, posterior cerebral, or superior cerebellar artery; pressure from a herniating cerebral uncus (false localizing sign); cavernous sinus mass or thrombosis
Trochlear nerve [IV]	Inability to look inferiorly when the eye is adducted (down and in)	Along the course of the nerve around the brainstem; orbital fracture
Trigeminal nerve [V]	Loss of sensation and pain in the region supplied by the three divisions of the nerve over the face; loss of motor function of the muscles of mastication on the side of the lesion	Typically, in the region of the trigeminal ganglion, though local masses around the foramina through which the divisions pass can produce symptoms
Abducent nerve [VI]	Inability of lateral eye movement	Brain lesion or cavernous sinus lesion extending onto the orbit
Facial nerve [VII]	Paralysis of facial muscles Abnormal taste sensation from the anterior two-thirds of the tongue and dry conjunctivae Paralysis of contralateral facial muscles below the eye	Damage to the branches within the parotid gland Injury to temporal bone; viral inflammation of nerve Brainstem injury
Vestibulocochlear nerve [VIII]	Progressive unilateral hearing loss and tinnitus (ringing in the ear)	Tumor at the cerebellopontine angle
Glossopharyngeal nerve [IX]	Loss of taste to the posterior one-third of the tongue and sensation of the soft palate	Brainstem lesion; penetrating neck injury
Vagus nerve [X]	Soft palate deviation with deviation of the uvula to the normal side; vocal cord paralysis	Brainstem lesion; penetrating neck injury
Accessory nerve [XI]	Paralysis of sternocleidomastoid and trapezius muscles	Penetrating injury to the posterior triangle of the neck
Hypoglossal nerve [XII]	Atrophy of ipsilateral muscles of the tongue and deviation toward the affected side; speech disturbance	Penetrating injury to the neck and skull base pathology

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In the clinic

Overview of cranial nerves

Cranial nerve reflexes

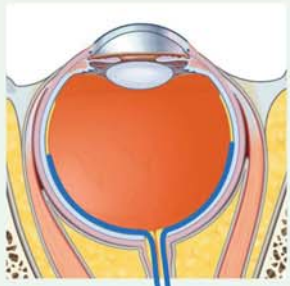
Corneal (blink) reflex
- Afferent—Trigeminal nerve (CN V)
- Efferent—Facial nerve (CN VII)

Gag reflex
- Afferent—Glossopharyngeal nerve (CN IX)
- Efferent—Vagus nerve (CN X)

Pupillary (light) reflex
- Afferent—optic nerve (CN II)
- Efferent—oculomotor nerve (CN III)

(continues)

In the clinic—cont'd

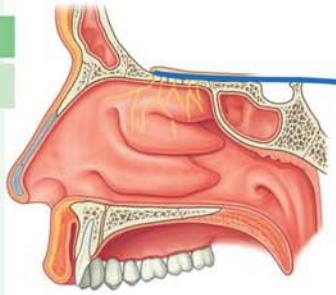


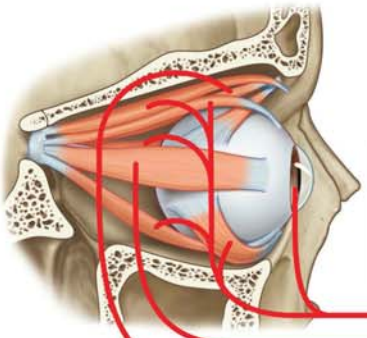
Optic nerve [II]

Special sensory – vision

Olfactory nerve [I]

Special sensory – smell





Oculomotor nerve [III]

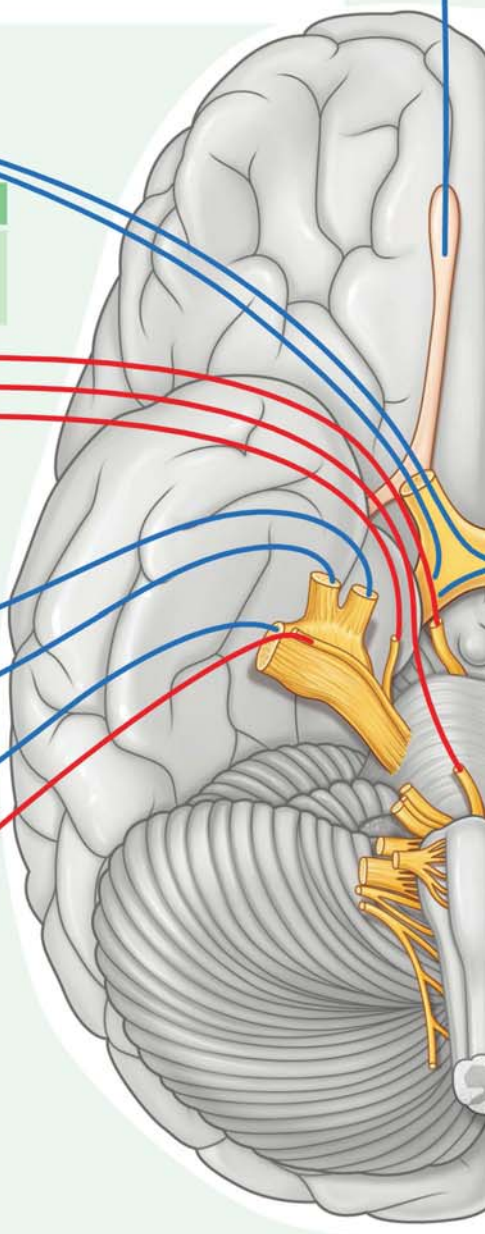
Somatic motor – five extra-ocular muscles (superior rectus, medial rectus, inferior oblique, inferior rectus, and levator palpebrae superioris)
Visceral motor – ciliary muscles and sphincter pupillae muscles

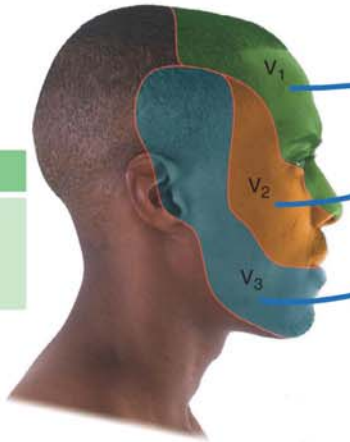
Trochlear nerve [IV]

Somatic motor – one extra-ocular muscle (superior oblique)

Abducent nerve [VI]

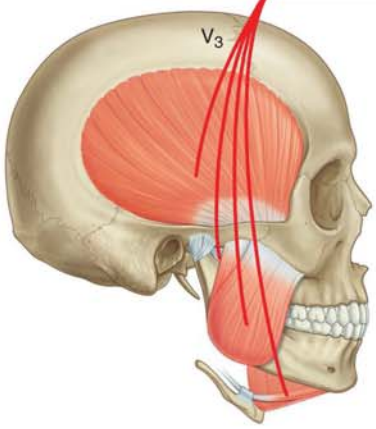
Somatic motor – one extra-ocular muscle (lateral rectus)





Trigeminal nerve [V] sensory root

Somatic sensory – eyes, orbital contents, face, sinuses, teeth, nasal cavities, oral cavity, nasopharynx, dura, anterior part of external ear, and part of external acoustic meatus



Trigeminal nerve [V] motor root

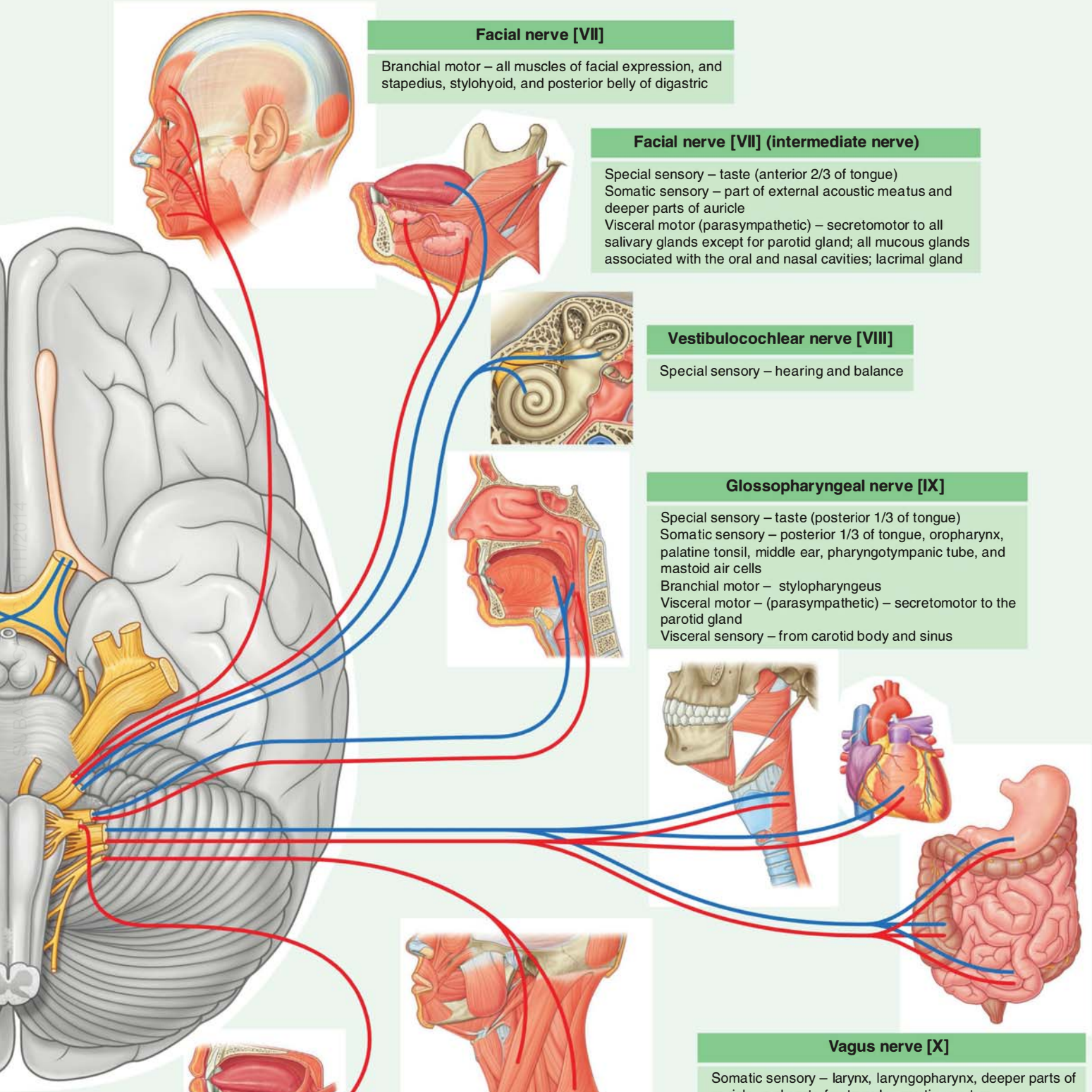
Branchial motor – the four muscles of mastication (medial pterygoid, lateral pterygoid, masseter, temporalis) and mylohyoid, anterior belly of digastric, tensor tympani, and tensor veli palatini

Efferent (motor) fibers

Afferent (sensory) fibers

Fig. 8.52 Overview of cranial nerves.

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Facial nerve [VII]

Branchial motor – all muscles of facial expression, and stapedius, stylohyoid, and posterior belly of digastric

Facial nerve [VII] (intermediate nerve)

Special sensory – taste (anterior 2/3 of tongue)
Somatic sensory – part of external acoustic meatus and deeper parts of auricle
Visceral motor (parasympathetic) – secretomotor to all salivary glands except for parotid gland; all mucous glands associated with the oral and nasal cavities; lacrimal gland

Vestibulocochlear nerve [VIII]

Special sensory – hearing and balance

Glossopharyngeal nerve [IX]

Special sensory – taste (posterior 1/3 of tongue)
Somatic sensory – posterior 1/3 of tongue, oropharynx, palatine tonsil, middle ear, pharyngotympanic tube, and mastoid air cells
Branchial motor – stylopharyngeus
Visceral motor – (parasympathetic) – secretomotor to the parotid gland
Visceral sensory – from carotid body and sinus

Vagus nerve [X]

Somatic sensory – larynx, laryngopharynx, deeper parts of auricle, and part of external acoustic meatus
Special sensory – taste from epiglottis and pharynx
Branchial motor – all muscles of pharynx except for stylopharyngeus; all muscles of the soft palate except for tensor veli palatini, all intrinsic muscles of larynx
Visceral motor – (parasympathetic) – thoracic viscera and abdominal viscera to end of midgut
Visceral sensory – thoracic viscera and abdominal viscera to end of midgut, chemo- and baroreceptors (and in some cases carotid body)

Hypoglossal nerve [XII]

Somatic motor – all muscles of the tongue except palatoglossus

Accessory nerve [XI]

Somatic motor – sternocleidomastoid and trapezius

FACE

A face-to-face meeting is an important initial contact between individuals. Part of this exchange is the use of facial expressions to convey emotions. In fact, a physician can gain important information about an individual's general health by observing a patient's face.

Thus an understanding of the unique organization of the various structures between the superciliary arches superiorly, the lower edge of the mandible inferiorly, and as far back as the ears on either side, the area defined as the face, is particularly useful in the practice of medicine.

Muscles

The muscles of the face (Fig. 8.53) develop from the second pharyngeal arch and are innervated by branches of the

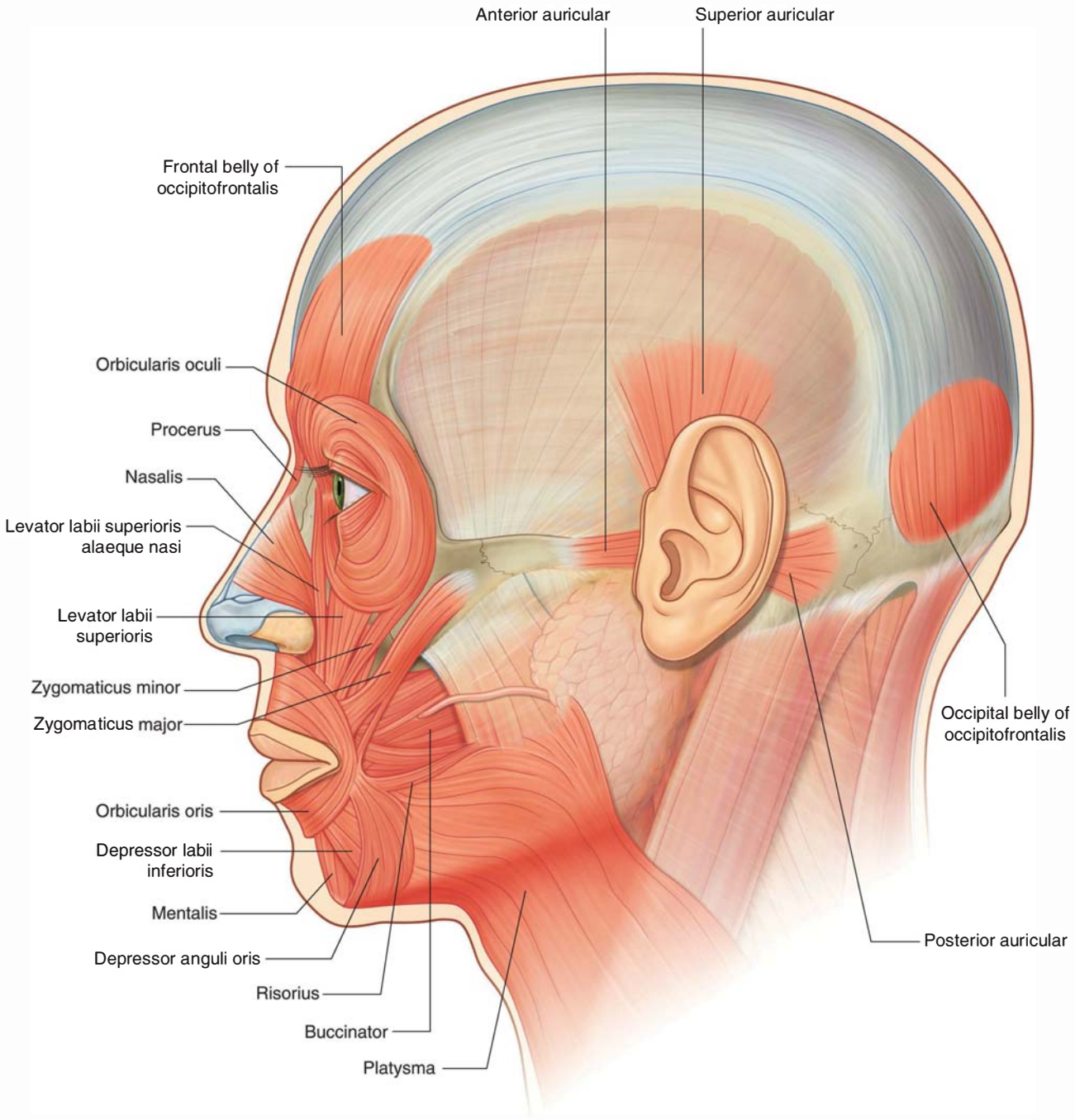


Fig. 8.53 Facial muscles.

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facial nerve [VII]. They are in the superficial fascia, with origins from either bone or fascia, and insertions into the skin.

Because these muscles control expressions of the face, they are sometimes referred to as muscles of "facial expression." They also act as sphincters and dilators of the orifices of the face (i.e., the orbits, nose, and mouth). This organizational arrangement into functional groups provides a logical approach to understanding these muscles (Table 8.7).

Orbital group

Two muscles are associated with the orbital group—the orbicularis oculi and the corrugator supercilii.

Table 8.7 Muscles of the face

Muscle	Origin	Insertion	Innervation	Function
ORBITAL GROUP				
Orbicularis oculi				
—Palpebral part	Medial palpebral ligament	Lateral palpebral raphe	Facial nerve [VII]	Closes the eyelids gently
—Orbital part	Nasal part of frontal bone; frontal process of maxilla; medial palpebral ligament	Fibers form an uninterrupted ellipse around orbit	Facial nerve [VII]	Closes the eyelids forcefully
Corrugator supercilii	Medial end of the superciliary arch	Skin of the medial half of eyebrow	Facial nerve [VII]	Draws the eyebrows medially and downward
NASAL GROUP				
Nasalis				
—Transverse part	Maxilla just lateral to nose	Aponeurosis across dorsum of nose with muscle fibers from the other side	Facial nerve [VII]	Compresses nasal aperture
—Alar part	Maxilla over lateral incisor	Alar cartilage of nose	Facial nerve [VII]	Draws cartilage downward and laterally, opening nostril
Procerus	Nasal bone and upper part of lateral nasal cartilage	Skin of lower forehead between eyebrows	Facial nerve [VII]	Draws down medial angle of eyebrows, producing transverse wrinkles over bridge of nose
Depressor septi	Maxilla above medial incisor	Mobile part of the nasal septum	Facial nerve [VII]	Pulls nose inferiorly
ORAL GROUP				
Depressor anguli oris	Oblique line of mandible below canine, premolar, and first molar teeth	Skin at the corner of mouth and blending with orbicularis oris	Facial nerve [VII]	Draws corner of mouth downward and laterally
Depressor labii inferioris	Anterior part of oblique line of mandible	Lower lip at midline; blends with muscle from opposite side	Facial nerve [VII]	Draws lower lip downward and laterally
Mentalis	Mandible inferior to incisor teeth	Skin of chin	Facial nerve [VII]	Raises and protrudes lower lip as it wrinkles skin on chin
Risorius	Fascia over masseter muscle	Skin at the corner of the mouth	Facial nerve [VII]	Retracts corner of mouth

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Table 8.7 Muscles of the face—cont'd

Muscle	Origin	Insertion	Innervation	Function
Zygomaticus major	Posterior part of lateral surface of zygomatic bone	Skin at the corner of the mouth	Facial nerve [VII]	Draws the corner of the mouth upward and laterally
Zygomaticus minor	Anterior part of lateral surface of zygomatic bone	Upper lip just medial to corner of mouth	Facial nerve [VII]	Draws the upper lip upward
Levator labii superioris	Infra-orbital margin of maxilla	Skin of upper lateral half of upper lip	Facial nerve [VII]	Raises upper lip; helps form nasolabial furrow
Levator labii superioris alaeque nasi	Frontal process of maxilla	Alar cartilage of nose and upper lip	Facial nerve [VII]	Raises upper lip and opens nostril
Levator anguli oris	Maxilla below infra-orbital foramen	Skin at the corner of mouth	Facial nerve [VII]	Raises corner of mouth; helps form nasolabial furrow
Orbicularis oris	From muscles in area; maxilla and mandible in midline	Forms ellipse around mouth	Facial nerve [VII]	Closes lips; protrudes lips
Buccinator	Posterior parts of maxilla and mandible; pterygomandibular raphe	Blends with orbicularis oris and into lips	Facial nerve [VII]	Presses the cheek against teeth; compresses distended cheeks
OTHER MUSCLES OR GROUPS				
Anterior auricular	Anterior part of temporal fascia	Into helix of ear	Facial nerve [VII]	Draws ear upward and forward
Superior auricular	Epicranial aponeurosis on side of head	Upper part of auricle	Facial nerve [VII]	Elevates ear
Posterior auricular	Mastoid process of temporal bone	Convexity of concha of ear	Facial nerve [VII]	Draws ear upward and backward
Occipitofrontalis —Frontal belly	Skin of eyebrows	Into galea aponeurotica	Facial nerve [VII]	Wrinkles forehead; raises eyebrows
—Occipital belly	Lateral part of superior nuchal line of occipital bone and mastoid process of temporal bone	Into galea aponeurotica	Facial nerve [VII]	Draws scalp backward

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Orbicularis oculi

The **orbicularis oculi** is a large muscle that completely surrounds each orbital orifice and extends into each eyelid (Fig. 8.54). It closes the eyelids. It has two major parts:

- The outer **orbital part** is a broad ring that encircles the orbital orifice and extends outward beyond the orbital rim.
- The inner **palpebral part** is in the eyelids and consists of muscle fibers originating in the medial corner of the eye that arch across each lid to attach laterally.

The orbital and palpebral parts have specific roles to play during eyelid closure. The palpebral part closes the eye gently, whereas the orbital part closes the eye more forcefully and produces some wrinkling on the forehead.

An additional small lacrimal part of the orbicularis oculi muscle is deep, medial in position, and attaches to bone posterior to the lacrimal sac of the lacrimal apparatus in the orbit.

Corrugator supercilii

The second muscle in the orbital group is the much smaller **corrugator supercilii** (Fig. 8.54), which is deep to the eyebrows and the orbicularis oculi muscle and is active when frowning. It arises from the medial end of the superciliary arch, passing upward and laterally to insert into the skin of the medial half of the eyebrow. It draws the eyebrows toward the midline, causing vertical wrinkles above the nose.

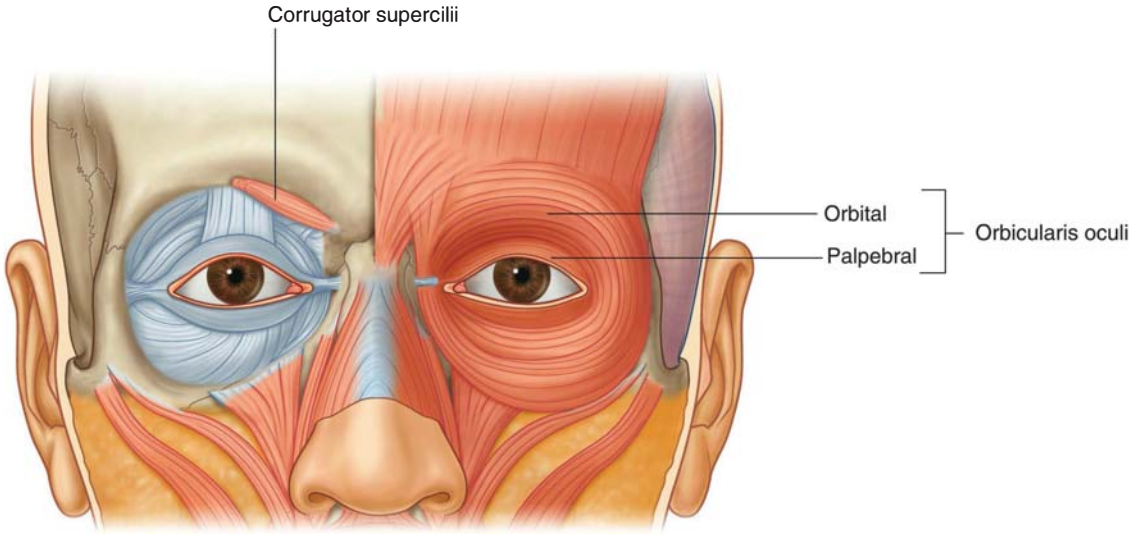


Fig. 8.54 Orbital group of facial muscles.

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Nasal group

Three muscles are associated with the nasal group—the nasalis, the procerus, and the depressor septi nasi (Fig. 8.55).

Nasalis

The largest and best developed of the muscles of the nasal group is the **nasalis**, which is active when the nares are flared (Fig. 8.55). It consists of a transverse part (the compressor naris) and an alar part (the dilator naris):

- The **transverse part** of the nasalis compresses the nares—it originates from the maxilla and its fibers pass upward and medially to insert, along with fibers from the same muscle on the opposite side, into an aponeurosis across the dorsum of the nose.
- The **alar part** of the nasalis draws the alar cartilages downward and laterally, so opening the nares—it originates from the maxilla, below and medial to the transverse part, and inserts into the alar cartilage.

Procerus

The **procerus** is a small muscle superficial to the nasal bone and is active when an individual frowns (Fig. 8.55).

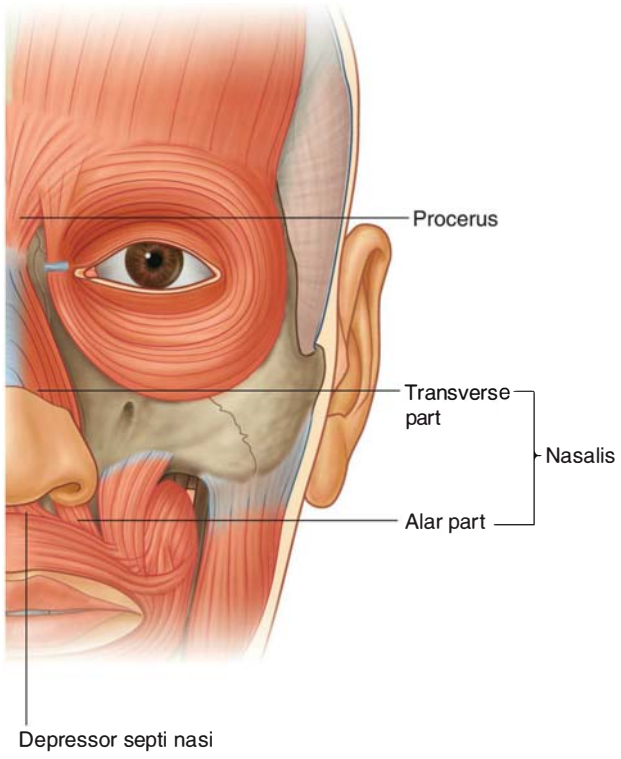


Fig. 8.55 Nasal group of facial muscles.

It arises from the nasal bone and the upper part of the lateral nasal cartilage and inserts into the skin over the lower part of the forehead between the eyebrows. It may be continuous with the frontal belly of the occipitofrontalis muscle of the scalp.

The procerus draws the medial border of the eyebrows downward to produce transverse wrinkles over the bridge of the nose.

Depressor septi nasi

The final muscle in the nasal group is the **depressor septi nasi**, another muscle that assists in widening the nares (Fig. 8.55). Its fibers arise from the maxilla above the central incisor tooth and ascend to insert into the lower part of the nasal septum.

The depressor septi nasi pulls the nose inferiorly, so assisting the alar part of the nasalis in opening the nares.

Oral group

The muscles in the oral group move the lips and cheek. They include the orbicularis oris and buccinator muscles, and a lower and upper group of muscles (Fig. 8.56). Many of these muscles intersect just lateral to the corner of the mouth on each side at a structure termed the modiolus.

Orbicularis oris

The **orbicularis oris** is a complex muscle consisting of fibers that completely encircle the mouth (Fig. 8.56). Its function is apparent when pursing the lips, as occurs during whistling. Some of its fibers originate near the midline from the maxilla superiorly and the mandible inferiorly, whereas other fibers are derived from both the buccinator, in the cheek, and the numerous other muscles

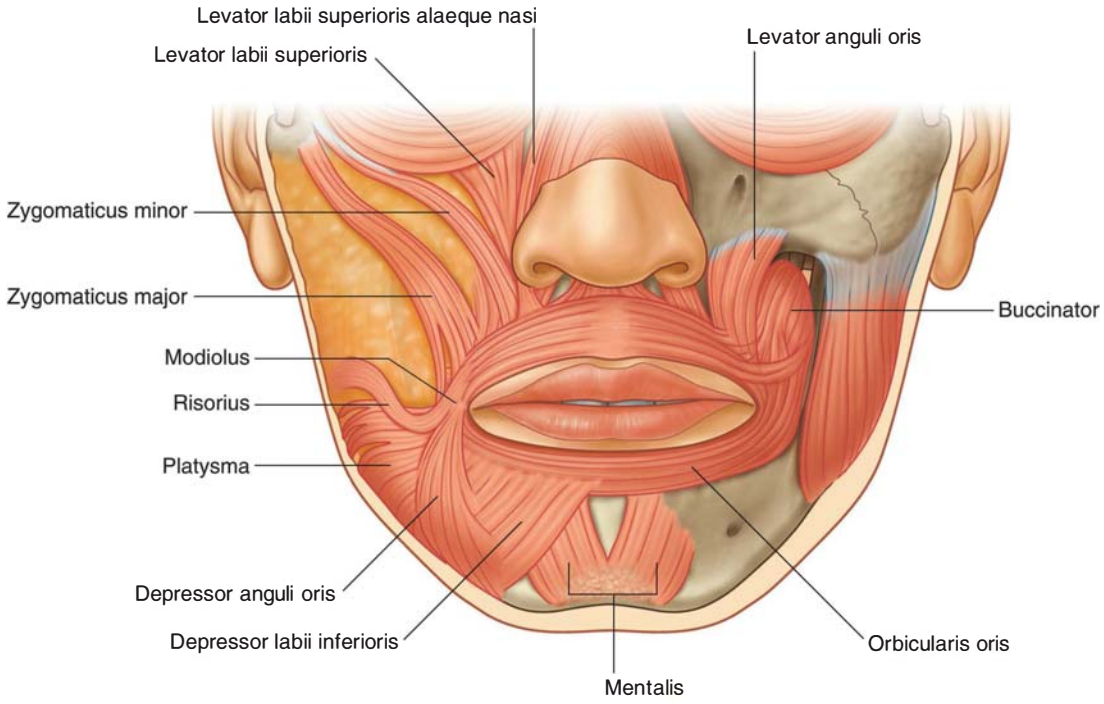


Fig. 8.56 Oral group of facial muscles.

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acting on the lips. It inserts into the skin and mucous membrane of the lips, and into itself.

Contraction of the orbicularis oris narrows the mouth and closes the lips.

Buccinator

The buccinator forms the muscular component of the cheek and is used every time air expanding the cheeks is forcefully expelled (Figs. 8.56 and 8.57). It is in the space between the mandible and the maxilla, deep to the other facial muscles in the area.

The buccinator arises from the posterior part of the maxilla and mandible opposite the molar teeth and the **pterygomandibular raphe**, which is a tendinous band between the pterygoid hamulus superiorly and the mandible inferiorly and is a point of attachment for the buccinator and superior pharyngeal constrictor muscles.

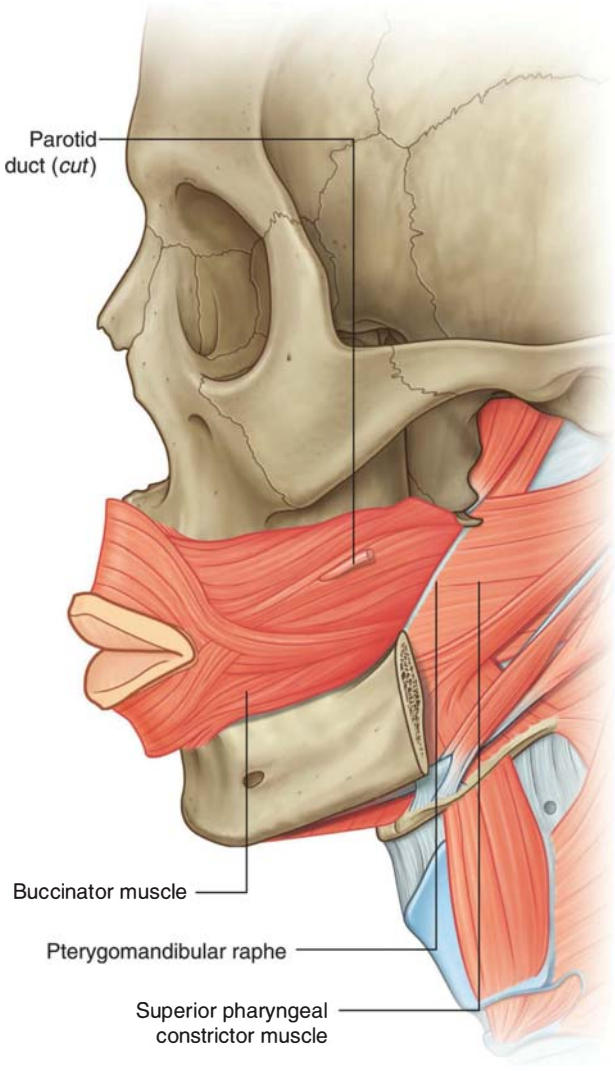


Fig. 8.57 Buccinator muscle.

The fibers of the buccinator pass toward the corner of the mouth to insert into the lips, blending with fibers from the orbicularis oris in a unique fashion. Central fibers of the buccinator cross so that lower fibers enter the upper lip and upper fibers enter the lower lip (Fig. 8.57). The highest and lowest fibers of the buccinator do not cross and enter the upper and lower lips, respectively.

Contraction of the buccinator presses the cheek against the teeth. This keeps the cheek taut and aids in mastication by preventing food from accumulating between the teeth and the cheek. The muscle also assists in the forceful expulsion of air from the cheeks.

Lower group of oral muscles

The muscles in the lower group consist of the depressor anguli oris, depressor labii inferioris. and mentalis (Fig. 8.56).

- The **depressor anguli oris** is active during frowning. It arises along the side of the mandible below the canine, premolar, and first molar teeth and inserts into skin and the upper part of the orbicularis oris near the corner of the mouth. It depresses the corner of the mouth.
- The **depressor labii inferioris** arises from the front of the mandible, deep to the depressor anguli oris. Its fibers move superiorly and medially, some merging with fibers from the same muscle on the opposite side and fibers from the orbicularis oris before inserting into the lower lip. It depresses the lower lip and moves it laterally.
- The **mentalis** helps position the lip when drinking from a cup or when pouting. It is the deepest muscle of the lower group arising from the mandible just inferior to the incisor teeth, with its fibers passing downward and medially to insert into the skin of the chin. It raises and protrudes the lower lip as it wrinkles the skin of the chin.

Upper group of oral muscles

The muscles of the upper group of oral muscles consist of the risorius, zygomaticus major, zygomaticus minor, levator labii superioris, levator labii superioris alaeque nasi, and levator anguli oris (Fig. 8.56).

- The **risorius** helps produce a grin (Fig. 8.56). It is a thin, superficial muscle that extends laterally from the corner of the mouth in a slightly upward direction. Contraction of its fibers pulls the corner of the mouth laterally and upward.
- The **zygomaticus major** and **zygomaticus minor** help produce a smile (Fig. 8.56). The zygomaticus major

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is a superficial muscle that arises deep to the orbicularis oculi along the posterior part of the lateral surface of the zygomatic bone, and passes downward and forward, blending with the orbicularis oris and inserting into skin at the corner of the mouth. The zygomaticus minor arises from the zygomatic bone anterior to the origin of the zygomaticus major, parallels the path of the zygomaticus major, and inserts into the upper lip medial to the corner of the mouth. Both zygomaticus muscles raise the corner of the mouth and move it laterally.

- The **levator labii superioris** deepens the furrow between the nose and the corner of the mouth during sadness (Fig. 8.56). It arises from the maxilla just superior to the infra-orbital foramen, and its fibers pass downward and medially to blend with the orbicularis oris and insert into the skin of the upper lip.
- The **levator labii superioris alaeque nasi** is medial to the levator labii superioris, arises from the maxilla next to the nose, and inserts into both the alar cartilage of the nose and skin of the upper lip (Fig. 8.56). It may assist in flaring the nares.
- The **levator anguli oris** is more deeply placed and covered by the other two levators and the zygomaticus muscles (Fig. 8.56). It arises from the maxilla, just inferior to the infra-orbital foramen and inserts into the skin at the corner of the mouth. It elevates the corner of the mouth and may help deepen the furrow between the nose and the corner of the mouth during sadness.

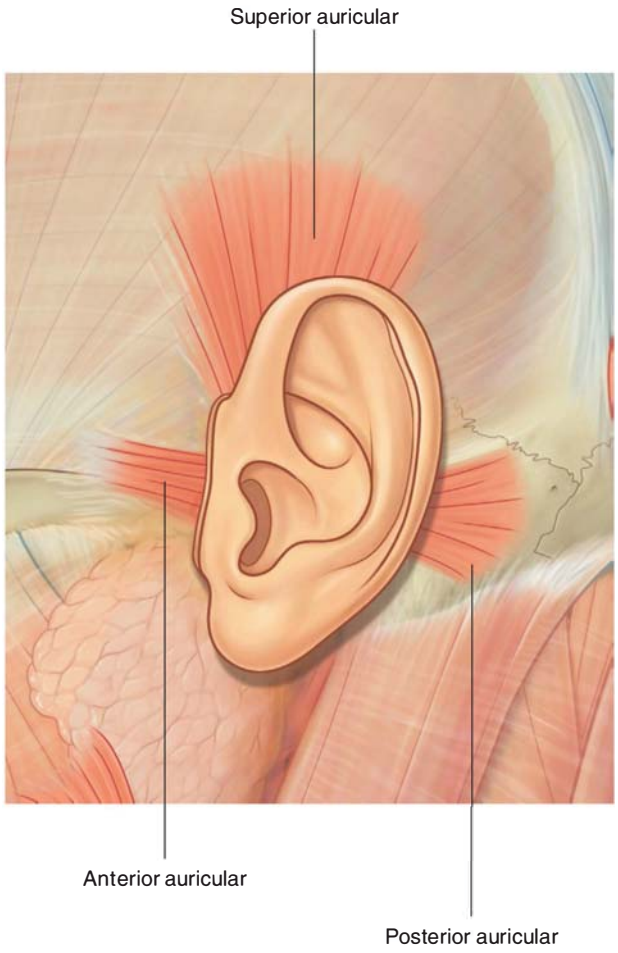


Fig. 8.58 Auricular muscles.

Other muscles or muscle groups

Several additional muscles or groups of muscles not in the area defined as the face, but derived from the second pharyngeal arch and innervated by the facial nerve [VII], are considered muscles of facial expression. They include the platysma, auricular, and occipitofrontalis muscles (see Fig. 8.53).

Platysma

The **platysma** is a large, thin sheet of muscle in the superficial fascia of the neck. It arises below the clavicle in the upper part of the thorax and ascends through the neck to the mandible. At this point, the more medial fibers insert on the mandible, whereas the lateral fibers join with muscles around the mouth.

The platysma tenses the skin of the neck and can move the lower lip and corners of the mouth down.

Auricular muscles

Three of these muscles, "other muscles of facial expression," are associated with the ear—the anterior, superior, and posterior **auricular muscles** (Fig. 8.58):

- The anterior muscle is anterolateral and pulls the ear upward and forward.
- The superior muscle is superior and elevates the ear.
- The posterior muscle is posterior and retracts and elevates the ear.

Occipitofrontalis

The **occipitofrontalis** is the final muscle in this category of "other muscles of facial expression" and is associated with the scalp (see Fig. 8.53). It consists of a frontal belly anteriorly and an occipital belly posteriorly. An aponeurotic tendon connects the two:

- The frontal belly covers the forehead and is attached to the skin of the eyebrows.
- The occipital belly arises from the posterior aspect of the skull and is smaller than the frontal belly.

The occipitofrontalis muscles move the scalp and wrinkle the forehead.

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Parotid gland

The **parotid glands** are the largest of the three pairs of main salivary glands in the head and numerous structures pass through them. They are anterior to and below the lower half of the ear, superficial, posterior, and deep to the ramus of the mandible (Fig. 8.59). They extend down to the lower border of the mandible and up to the zygomatic arch. Posteriorly they cover the anterior part of the sternocleidomastoid muscle and continue anteriorly to halfway across the masseter muscle.

The **parotid duct** leaves the anterior edge of the parotid gland midway between the zygomatic arch and the corner of the mouth (Fig. 8.59). It crosses the face in a transverse direction and, after crossing the medial border of the masseter muscle, turns deeply into the buccal fat pad and pierces the buccinator muscle. It opens into the oral cavity near the second upper molar tooth.

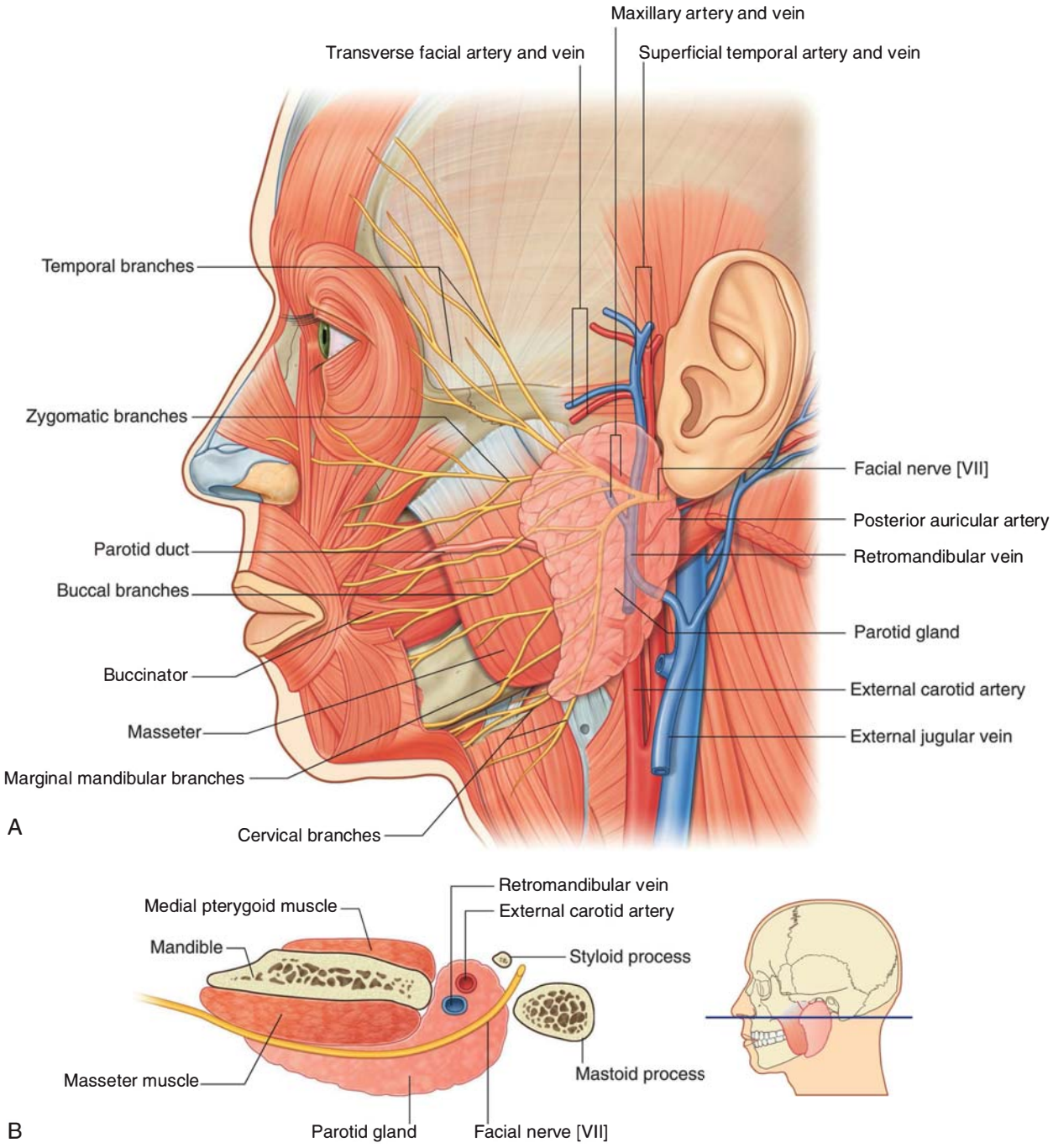


Fig. 8.59 Parotid gland. **A.** Lateral view. **B.** Cross section.

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Important relationships

Several major structures enter and pass through or pass just deep to the parotid gland. These include the facial nerve [VII], the external carotid artery and its branches, and the retromandibular vein and its tributaries (Fig. 8.59).

Facial nerve

The facial nerve [VII] exits the skull through the stylomastoid foramen and then passes into the parotid gland, where it usually divides into upper and lower trunks. These pass through the substance of the parotid gland, where there may be further branching and anastomosing of the nerves.

Five terminal groups of branches of the facial nerve [VII]—the **temporal**, **zygomatic**, **buccal**, **marginal mandibular**, and **cervical branches**—emerge from the upper, anterior, and lower borders of the parotid gland (Fig. 8.59).

The intimate relationships between the facial nerve [VII] and the parotid gland mean that surgical removal of the parotid gland is a difficult dissection if all branches of the facial nerve [VII] are to be spared.

External carotid artery and its branches

The external carotid artery enters into or passes deep to the inferior border of the parotid gland (Fig. 8.59). As it continues in a superior direction, it gives off the **posterior auricular artery** before dividing into its two terminal branches (the **maxillary** and **superficial temporal arteries**) near the lower border of the ear:

- The maxillary artery passes horizontally, deep to the mandible.
- The superficial temporal artery continues in a superior direction and emerges from the upper border of the gland after giving off the **transverse facial artery**.

Retromandibular vein and its tributaries

The retromandibular vein is formed in the substance of the parotid gland when the **superficial temporal** and **maxillary veins** join together (Fig. 8.59), and passes inferiorly in the substance of the parotid gland. It usually divides into anterior and posterior branches just below the inferior border of the gland.

Arterial supply

The parotid gland receives its arterial supply from the numerous arteries that pass through its substance.

Innervation

Sensory innervation of the parotid gland is provided by the **auriculotemporal nerve**, which is a branch of the mandibular nerve [V_3]. This division of the trigeminal nerve exits the skull through the foramen ovale.

The auriculotemporal nerve also carries secretomotor fibers to the parotid gland. These postganglionic parasympathetic fibers have their origin in the otic ganglion associated with the mandibular nerve [V_3] and are just inferior to the foramen ovale. Preganglionic parasympathetic fibers to the otic ganglion come from the glossopharyngeal nerve [IX].

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In the clinic

Parotid gland

The parotid gland is the largest of the paired salivary glands and is enclosed within the split investing layer of deep cervical fascia.

The parotid gland produces a watery saliva and salivary amylase, which are necessary for food bolus formation, oral digestion, and smooth passage of the bolus into the upper gastrointestinal tract.

Tumors of the parotid gland

The commonest tumors of the parotid gland (Fig. 8.60) are benign and typically involve the superficial gland. These include pleomorphic adenoma and adenolymphoma. Their importance is in relation to their anatomical position. The relationship of any tumor to the branches of the facial nerve [VII] must be defined because resection of the tumor may damage the nerve.

Parotid gland stones

It is not uncommon for stones to develop within the parotid gland. They typically occur within the main confluence of the ducts and within the main parotid duct. The patient usually complains of intense pain when salivating and tends to avoid foods that produce this symptom. The pain can be easily reproduced in the clinic by squirting lemon juice into the patient's mouth.

Surgery depends upon where the stone is. If it is within the anterior aspect of the duct, a simple incision in the buccal mucosa with a sphincterotomy may allow removal. If the stone is farther back within the main duct, complete gland excision may be necessary.



Fig. 8.60 Tumor in parotid gland. Axial CT scan.

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Innervation

During development a cranial nerve becomes associated with each of the pharyngeal arches. Because the face is primarily derived from the first and second pharyngeal arches, innervation of neighboring facial structures is as follows:

- The trigeminal nerve [V] innervates facial structures derived from the first arch.
- The facial nerve [VII] innervates facial structures derived from the second arch.

Sensory innervation

Because the face is derived developmentally from a number of structures originating from the first pharyngeal arch, cutaneous innervation of the face is by branches of the trigeminal nerve [V].

The trigeminal nerve [V] divides into three major divisions—the ophthalmic [V_1], maxillary [V_2], and mandibular [V_3] nerves—before leaving the middle cranial fossa (Fig. 8.61). Each of these divisions passes out of the cranial cavity to innervate a part of the face, so most of the skin covering the face is innervated solely by branches of the trigeminal nerve [V]. The exception is a small area covering the angle and lower border of the ramus of the mandible and parts of the ear, where the facial [VII], vagus [X], and cervical nerves contribute to the innervation.

Ophthalmic nerve [V_1]

The ophthalmic nerve [V_1] exits the skull through the superior orbital fissure and enters the orbit. Its branches (Fig. 8.61) that innervate the face include:

- the **supra-orbital** and **supratrochlear nerves**, which leave the orbit superiorly and innervate the upper eyelid, forehead, and scalp;

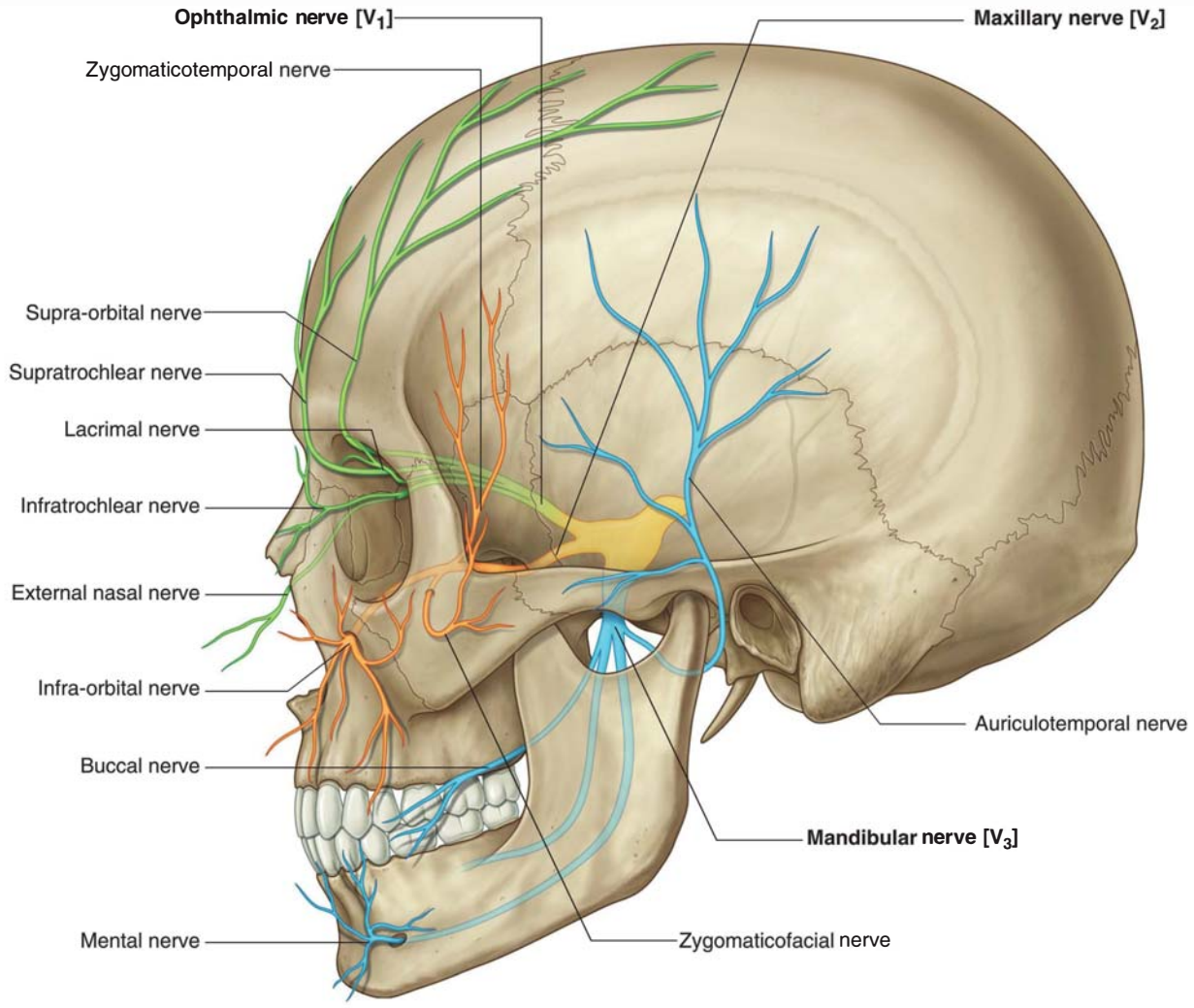


Fig. 8.61 Trigeminal nerve [V] leaving the skull.

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- the **infratrochlear nerve**, which exits the orbit in the medial angle to innervate the medial half of the upper eyelid, the skin in the area of the medial angle, and the side of the nose;
- the **lacrimal nerve**, which exits the orbit in the lateral angle to innervate the lateral half of the upper eyelid and the skin in the area of the lateral angle; and
- the **external nasal nerve**, which supplies the anterior part of the nose (Fig. 8.62).

Maxillary nerve [V_2]

The maxillary nerve [V_2] exits the skull through the foramen rotundum. Branches (Fig. 8.61) that innervate the face include:

- a small **zygomaticotemporal branch**, which exits the zygomatic bone and supplies a small area of the anterior temple above the zygomatic arch;
- a small **zygomaticofacial branch**, which exits the zygomatic bone and supplies a small area of skin over the zygomatic bone; and
- the large **infra-orbital nerve**, which exits the maxilla through the infra-orbital foramen and immediately divides into multiple branches to supply the lower eyelid, cheek, side of the nose, and upper lip (Fig. 8.62).

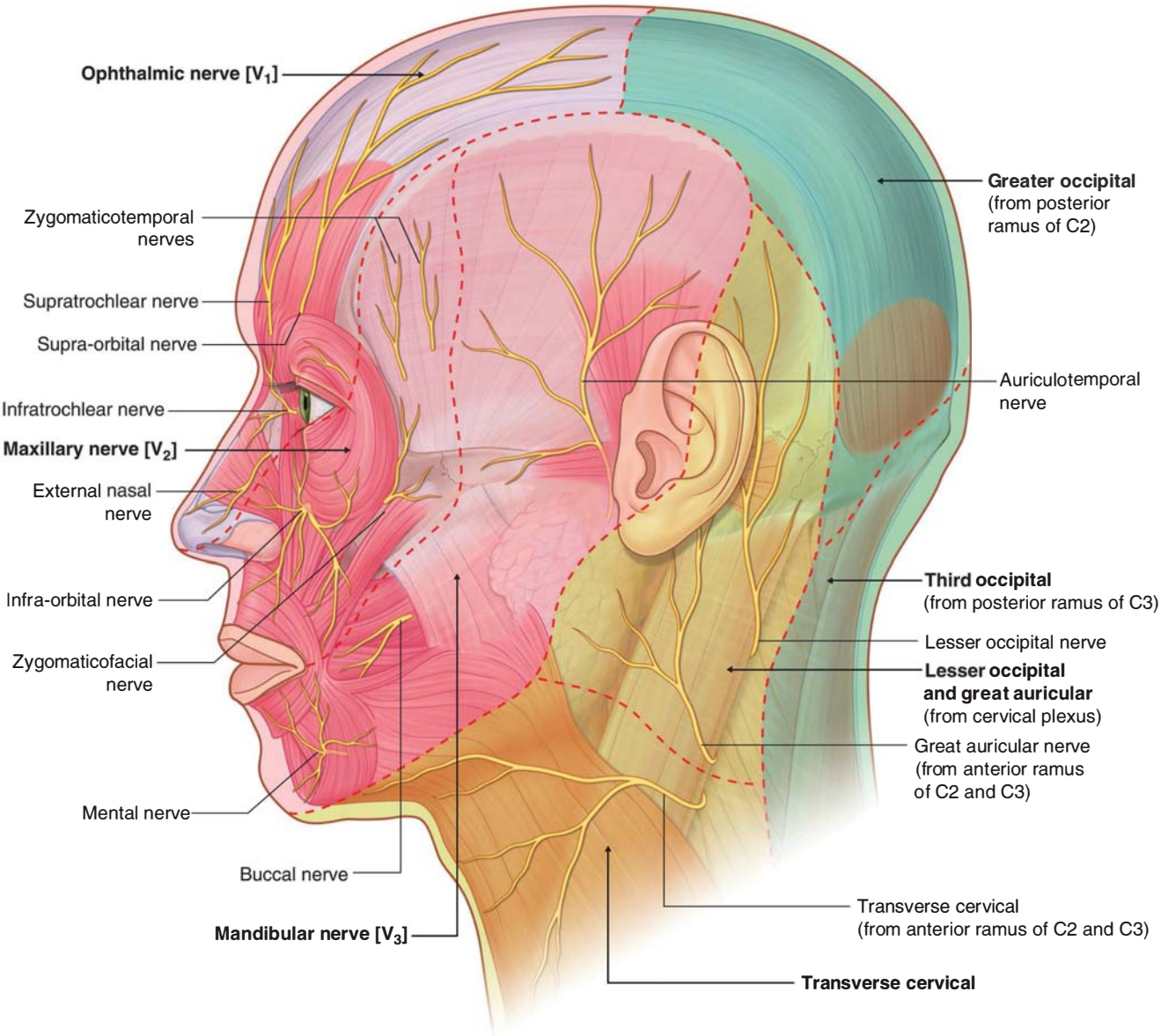


Fig. 8.62 Cutaneous distribution of the trigeminal nerve [V].

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Mandibular nerve [V_3]

The mandibular nerve [V_3] exits the skull through the foramen ovale. Branches (Fig. 8.62) innervating the face include:

- the **auriculotemporal nerve**, which enters the face just posterior to the temporomandibular joint, passes through the parotid gland, and ascends just anterior to the ear to supply the external acoustic meatus, the surface of the tympanic membrane (eardrum), and a large area of the temple;
- the **buccal nerve**, which is on the surface of the buccinator muscle supplying the cheek; and
- the **mental nerve**, which exits the mandible through the mental foramen and immediately divides into multiple branches to supply the skin and mucous membrane of the lower lip and skin of the chin (Fig. 8.62).

Motor innervation

The muscles of the face, as well as those associated with the external ear and the scalp, are derived from the second pharyngeal arch. The cranial nerve associated with this arch is the facial nerve [VII] and therefore branches of the facial nerve [VII] innervate all these muscles.

The facial nerve [VII] exits the posterior cranial fossa through the internal acoustic meatus. It passes through the temporal bone, giving off several branches, and emerges from the base of the skull through the stylomastoid foramen (Fig. 8.63). At this point it gives off the **posterior auricular nerve**. This branch passes upward, behind the ear, to supply the occipital belly of the occipitofrontalis muscle of the scalp and the posterior auricular muscle of the ear.

The main stem of the facial nerve [VII] then gives off another branch, which innervates the posterior belly of the digastric muscle and the stylohyoid muscle. At this point, the facial nerve [VII] enters the deep surface of the parotid gland (Fig. 8.63B).

Once in the parotid gland, the main stem of the facial nerve [VII] usually divides into upper (temporofacial) and lower (cervicofacial) branches. As these branches pass through the substance of the parotid gland they may branch further or take part in an anastomotic network (the parotid plexus).

Whatever types of interconnections occur, five terminal groups of branches of the facial nerve [VII]—the temporal, zygomatic, buccal, marginal mandibular, and cervical branches—emerge from the parotid gland (Fig. 8.63A).

Although there are variations in the pattern of distribution of the five terminal groups of branches, the basic pattern is as follows:

- Temporal branches exit from the superior border of the parotid gland to supply muscles in the area of the temple, forehead, and supra-orbital area.
- Zygomatic branches emerge from the anterosuperior border of the parotid gland to supply muscles in the infra-orbital area, the lateral nasal area, and the upper lip.
- Buccal branches emerge from the anterior border of the parotid gland to supply muscles in the cheek, the upper lip, and the corner of the mouth.
- Marginal mandibular branches emerge from the anteroinferior border of the parotid gland to supply muscles of the lower lip and chin.
- Cervical branches emerge from the inferior border of the parotid gland to supply the platysma.

Vessels

The arterial supply to the face is primarily from branches of the external carotid artery, though there is some limited supply from a branch of the internal carotid artery.

Similarly, most of the venous return is back to the internal jugular vein, though some important connections from the face result in venous return through a clinically relevant intracranial pathway involving the cavernous sinus.

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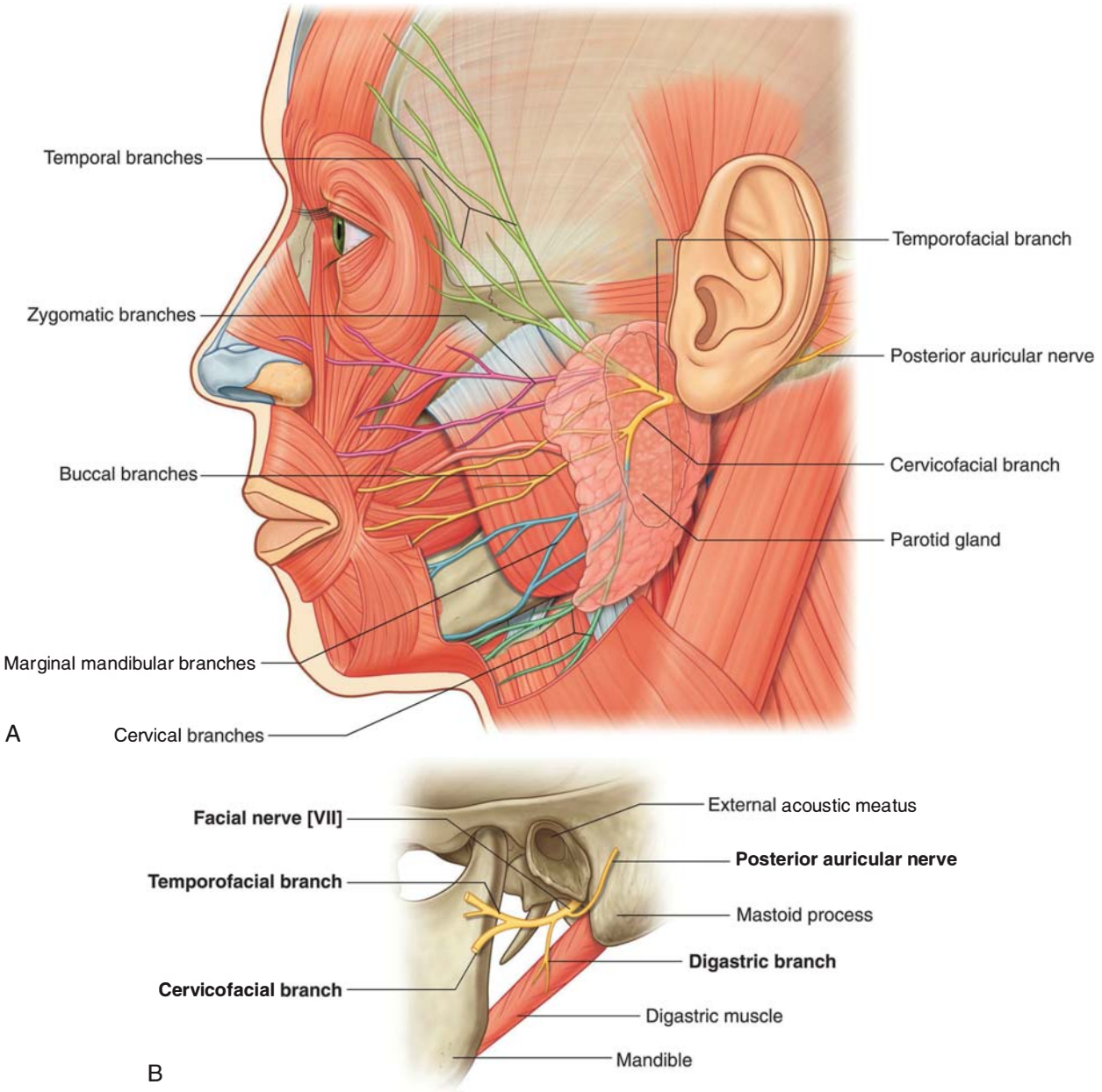


Fig. 8.63 Facial nerve [VII] on the face. **A.** Terminal branches. **B.** Branches before entering the parotid gland.

Arteries

Facial artery

The facial artery is the major vessel supplying the face (Fig. 8.64). It branches from the anterior surface of the external carotid artery, passes up through the deep structures of the neck, and appears at the lower border of the mandible after passing posterior to the submandibular gland. Curving around the inferior border of the mandible just anterior to the masseter, where its pulse can be felt, the facial artery then enters the face. From this point the facial artery runs upward and medially in a tortuous course. It passes along the side of the nose and terminates as the **angular artery** at the medial corner of the eye.

Along its path the facial artery is deep to the platysma, risorius, and zygomaticus major and minor, superficial to the buccinator and levator anguli oris, and may pass superficially to or through the levator labii superioris.

Branches of the facial artery include the superior and inferior labial branches and the lateral nasal branch (Fig. 8.64).

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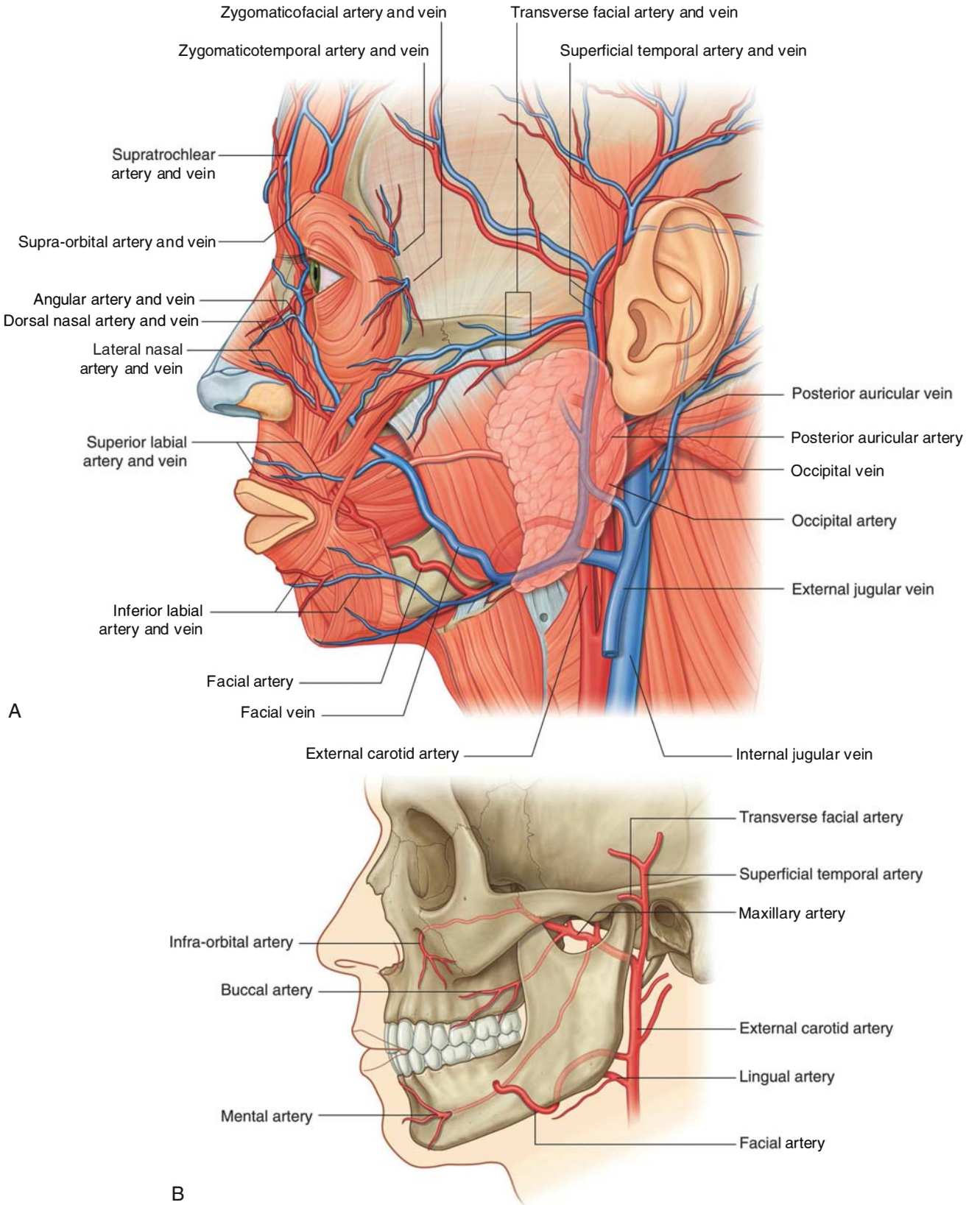


Fig. 8.64 Vasculature of the face. **A.** Lateral view. **B.** Branches of the maxillary artery.

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The labial branches arise near the corner of the mouth:

- The **inferior labial branch** supplies the lower lip.
- The **superior labial branch** supplies the upper lip, and also provides a branch to the nasal septum.

Near the midline, the superior and inferior labial branches anastomose with their companion arteries from the opposite side of the face. This provides an important connection between the facial arteries and the external carotid arteries of opposite sides.

The **lateral nasal branch** is a small branch arising from the facial artery as it passes along the side of the nose. It supplies the lateral surface and dorsum of the nose.

Transverse facial artery

Another contributor to the vascular supply of the face is the transverse facial artery (Fig. 8.64), which is a branch of the superficial temporal artery (the smaller of the two terminal branches of the external carotid artery).

The transverse facial artery arises from the superficial temporal artery within the substance of the parotid gland, passes through the gland, and crosses the face in a transverse direction. Lying on the superficial surface of the masseter muscle, it is between the zygomatic arch and the parotid duct.

Branches of the maxillary artery

The maxillary artery, the larger of the two terminal branches of the external carotid artery, gives off several small branches which contribute to the arterial supply to the face:

- The **infra-orbital artery** enters the face through the infra-orbital foramen and supplies the lower eyelid, upper lip, and the area between these structures.
- The **buccal artery** enters the face on the superficial surface of the buccinator muscle and supplies structures in this area.
- The **mental artery** enters the face through the mental foramen and supplies the chin.

Branches of the ophthalmic artery

Three small arteries from the internal carotid artery also contribute to the arterial supply of the face. These vessels arise from the **ophthalmic artery**, a branch of the internal carotid artery, after the ophthalmic artery enters the orbit:

- The **zygomaticofacial** and **zygomaticotemporal arteries** come from the lacrimal branch of the ophthalmic artery (Fig. 8.64), enter the face through the zygomaticofacial and zygomaticotemporal foramina, and supply the area of the face over the zygomatic bone.
- The **dorsal nasal artery**, a terminal branch of the ophthalmic artery, exits the orbit in the medial corner, and supplies the dorsum of the nose.

Other branches of the ophthalmic artery (the supra-orbital and supratrochlear arteries) supply the anterior scalp.

Veins

Facial vein

The facial vein is the major vein draining the face (Fig. 8.64). Its point of origin is near the medial corner of the orbit as the **supratrochlear** and **supra-orbital veins** come together to form the **angular vein**. This vein becomes the facial vein as it proceeds inferiorly and assumes a position just posterior to the facial artery. The facial vein descends across the face with the facial artery until it reaches the inferior border of the mandible. Here the artery and vein part company and the facial vein passes superficial to the submandibular gland to enter the internal jugular vein.

Throughout its course the facial vein receives tributaries from veins draining the eyelids, external nose, lips, cheek, and chin that accompany the various branches of the facial artery.

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Transverse facial vein

The transverse facial vein is a small vein that accompanies the transverse facial artery in its journey across the face (Fig. 8.64). It empties into the superficial temporal vein within the substance of the parotid gland.

Intracranial venous connections

As it crosses the face, the facial vein has numerous connections with venous channels passing into deeper regions of the head (Fig. 8.65):

- near the medial corner of the orbit, it communicates with ophthalmic veins;
- in the area of the cheek it communicates with veins passing into the infra-orbital foramen;
- it also communicates with veins passing into deeper regions of the face (i.e., the deep facial vein connecting with the pterygoid plexus of veins).

All these venous channels have interconnections with the intracranial cavernous sinus through emissary veins that connect intracranial with extracranial veins. There are no valves in the facial vein or any other venous channels in the head, so blood can move in any direction. Because of the interconnections between the veins, infections of the face, primarily above the mouth (i.e., the "danger area") should be handled with great care to prevent the dissemination of infectious material in an intracranial direction.

Lymphatic drainage

Lymphatic drainage from the face primarily moves toward three groups of lymph nodes (Fig. 8.66):

- **submental nodes** inferior and posterior to the chin, which drain lymphatics from the medial part of the lower lip and chin bilaterally;
- **submandibular nodes** superficial to the submandibular gland and inferior to the body of the mandible, which drain the lymphatics from the medial corner of the orbit, most of the external nose the medial part of the cheek, the upper lip, and the lateral part of the lower lip that follows the course of the facial artery;
- **pre-auricular and parotid** nodes anterior to the ear, which drain lymphatics from most of the eyelids, a part of the external nose, and the lateral part of the cheek.

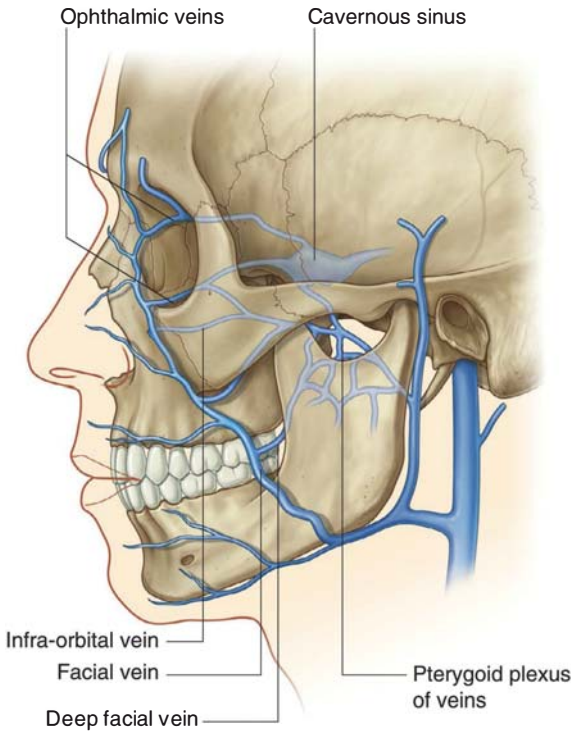


Fig. 8.65 Intracranial venous connections.

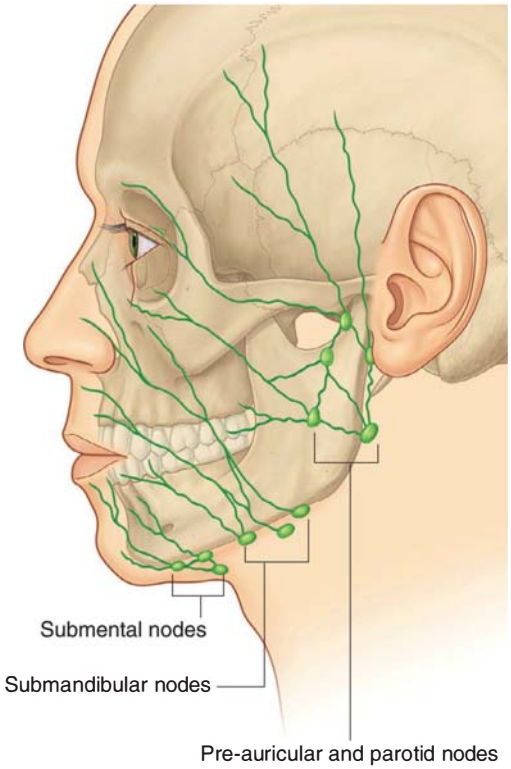


Fig. 8.66 Lymphatic drainage of the face.

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In the clinic

Facial nerve [VII] palsy (Bell's palsy)

The complexity of the facial nerve [VII] is demonstrated by the different pathological processes and sites at which these processes occur.

The facial nerve [VII] is formed from the nuclei within the brainstem emerging at the junction of the pons and the medulla. It enters the internal acoustic meatus, passes to the geniculate ganglion (which gives rise to further branches), and emerges from the skull base after a complex course within the temporal bone, leaving through the stylomastoid foramen. It enters the parotid gland and gives rise to five terminal groups of branches that supply muscles in the face and a number of additional branches that supply deeper or more posterior muscles. A series of lesions may affect the nerve along its course, and it is possible, with good clinical expertise, to determine the exact site of the lesion in relation to the course of the nerve.

Central lesions

A primary brainstem lesion affecting the motor nucleus of the facial nerve [VII] would lead to ipsilateral (same side) weakness of the whole face. However, because the upper part of the nucleus receives motor input from the left and right cerebral hemispheres a lesion occurring above the nucleus leads to contralateral lower facial weakness. In this example, motor innervation to the upper face is spared because the upper part of the nucleus receives input from both hemispheres. Preservation and loss of the special functions are determined by the extent of the lesion.

Lesions at and around the geniculate ganglion

Typically lesions at and around the geniculate ganglion are accompanied by loss of motor function on the whole of the ipsilateral (same) side of the face. Taste to the anterior two-thirds of the tongue, lacrimation, and some salivation also are likely to be affected because the lesion is proximal to the greater petrosal and chorda tympani branches of the nerve.

Lesions at and around the stylomastoid foramen

Lesions at and around the stylomastoid foramen are the commonest abnormality of the facial nerve [VII] and usually result from a viral inflammation of the nerve within the bony canal before exiting through the stylomastoid foramen. Typically the patient has an ipsilateral loss of motor function of the whole side of the face. Not only does this produce an unusual appearance, but it also complicates chewing of food. Lacrimation and taste may not be affected if the lesion remains distal to the greater petrosal and chorda tympani branches that originate deep in the temporal bone.

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In the clinic

Trigeminal neuralgia

Trigeminal neuralgia (tic douloureux) is a complex sensory disorder of the sensory root of the trigeminal nerve. Typically the pain is in the region of the mandibular [V_3] and maxillary [V_2] nerves, and is usually of sudden onset, is excruciating in nature, and may be triggered by touching a sensitive region of skin.

The etiology of trigeminal neuralgia is unknown, although anomalous blood vessels lying adjacent to the sensory route of the maxillary [V_2] and mandibular [V_3] nerves may be involved.

If symptoms persist and are unresponsive to medical care, surgical exploration of the trigeminal nerve (which is not without risk) may be necessary to remove any aberrant vessels.

SCALP

The scalp is the part of the head that extends from the superciliary arches anteriorly to the external occipital protuberance and superior nuchal lines posteriorly. Laterally it continues inferiorly to the zygomatic arch.

The scalp is a multilayered structure with layers that can be defined by the word itself:

- S—skin,
- C—connective tissue (dense),
- A—aponeurotic layer,
- L—loose connective tissue, and
- P—pericranium (Fig. 8.67).

Layers

Examining the layers of the scalp reveals that the first three layers are tightly held together, forming a single unit. This unit is sometimes referred to as the scalp proper and is the tissue torn away during serious "scalping" injuries.

Skin

The skin is the outer layer of the scalp (Figs. 8.67 and 8.68). It is similar structurally to skin throughout the body with the exception that hair is present on a large amount of it.

Connective tissue (dense)

Deep to the skin is dense connective tissue. This layer anchors the skin to the third layer and contains the arteries, veins, and nerves supplying the scalp. When the scalp is cut, the dense connective tissue surrounding the vessels tends to hold cut vessels open. This results in profuse bleeding.

Aponeurotic layer

The deepest layer of the first three layers is the aponeurotic layer. Firmly attached to the skin by the dense connective

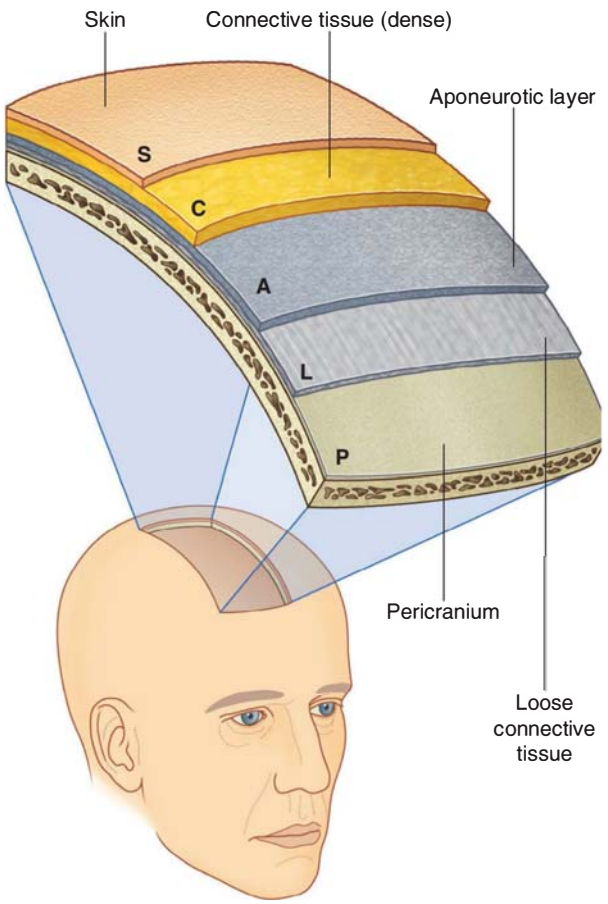


Fig. 8.67 SCALP.

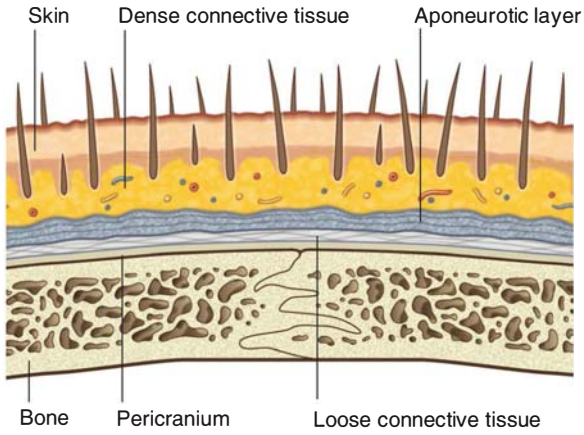


Fig. 8.68 Layers of the scalp.

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tissue of the second layer, this layer consists of the occipitofrontalis muscle, which has a frontal belly anteriorly, an occipital belly posteriorly, and an aponeurotic tendon—the **epicranial aponeurosis** (**galea aponeurotica**)—connecting the two (Fig. 8.69).

The frontal belly of the occipitofrontalis begins anteriorly where it is attached to the skin of the eyebrows. It passes upward, across the forehead, to become continuous with the aponeurotic tendon.

Posteriorly, each occipital belly of the occipitofrontalis arises from the lateral part of the superior nuchal line of the occipital bone and the mastoid process of the temporal bone. It also passes superiorly to attach to the aponeurotic tendon.

The occipitofrontalis muscles move the scalp, wrinkle the forehead, and raise the eyebrows. The frontal belly is innervated by temporal branches of the facial nerve [VII] and the posterior belly by the posterior auricular branch.

Loose connective tissue

A layer of loose connective tissue separates the aponeurotic layer from the pericranium and facilitates movement of the scalp proper over the calvaria (Figs. 8.67 and 8.69). Because of its consistency, infections tend to localize and spread through the loose connective tissue (also see "In the clinic" on p. 890).

Pericranium

The pericranium is the deepest layer of the scalp and is the periosteum on the outer surface of the calvaria. It is attached to the bones of the calvaria but is removable, except in the area of the sutures.

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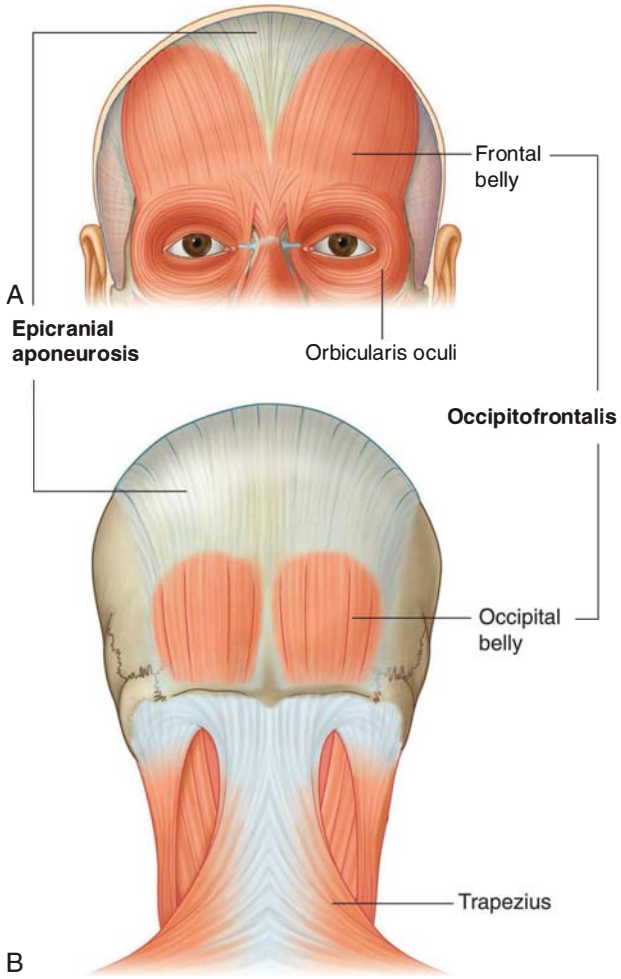


Fig. 8.69 Occipitofrontalis muscle. **A.** Frontal belly. **B.** Occipital belly.

Innervation

Sensory innervation of the scalp is from two major sources, cranial nerves or cervical nerves, depending on whether it is anterior or posterior to the ears and the vertex of the head (Fig. 8.70), The occipitofrontalis muscle is innervated by branches of the facial nerve [VII].

Anterior to the ears and the vertex

Branches of the trigeminal nerve [V] supply the scalp anterior to the ears and the vertex of the head (Fig. 8.70). These branches are the supratrochlear, supra-orbital, zygomaticotemporal, and auriculotemporal nerves:

- The **supratrochlear nerve** exits the orbit, passes through the frontalis muscle, continues superiorly across the front of the forehead, and supplies the front of the forehead near the midline.
- The **supra-orbital nerve** exits the orbit through the supra-orbital notch or foramen, passes through the frontalis muscle, and continues superiorly across the scalp as far back as the vertex of the head.
- The **zygomaticotemporal nerve** exits the skull through a foramen in the zygomatic bone and supplies the scalp over a small anterior area of the temple.
- The **auriculotemporal nerve** exits from the skull, deep to the parotid gland, passes just anterior to the ear, continues superiorly anterior to the ear until nearly reaching the vertex of the head, and supplies the scalp over the temporal region and anterior to the ear to near the vertex.

Posterior to the ears and the vertex

Posterior to the ears and vertex, sensory innervation of the scalp is by cervical nerves, specifically branches from spinal

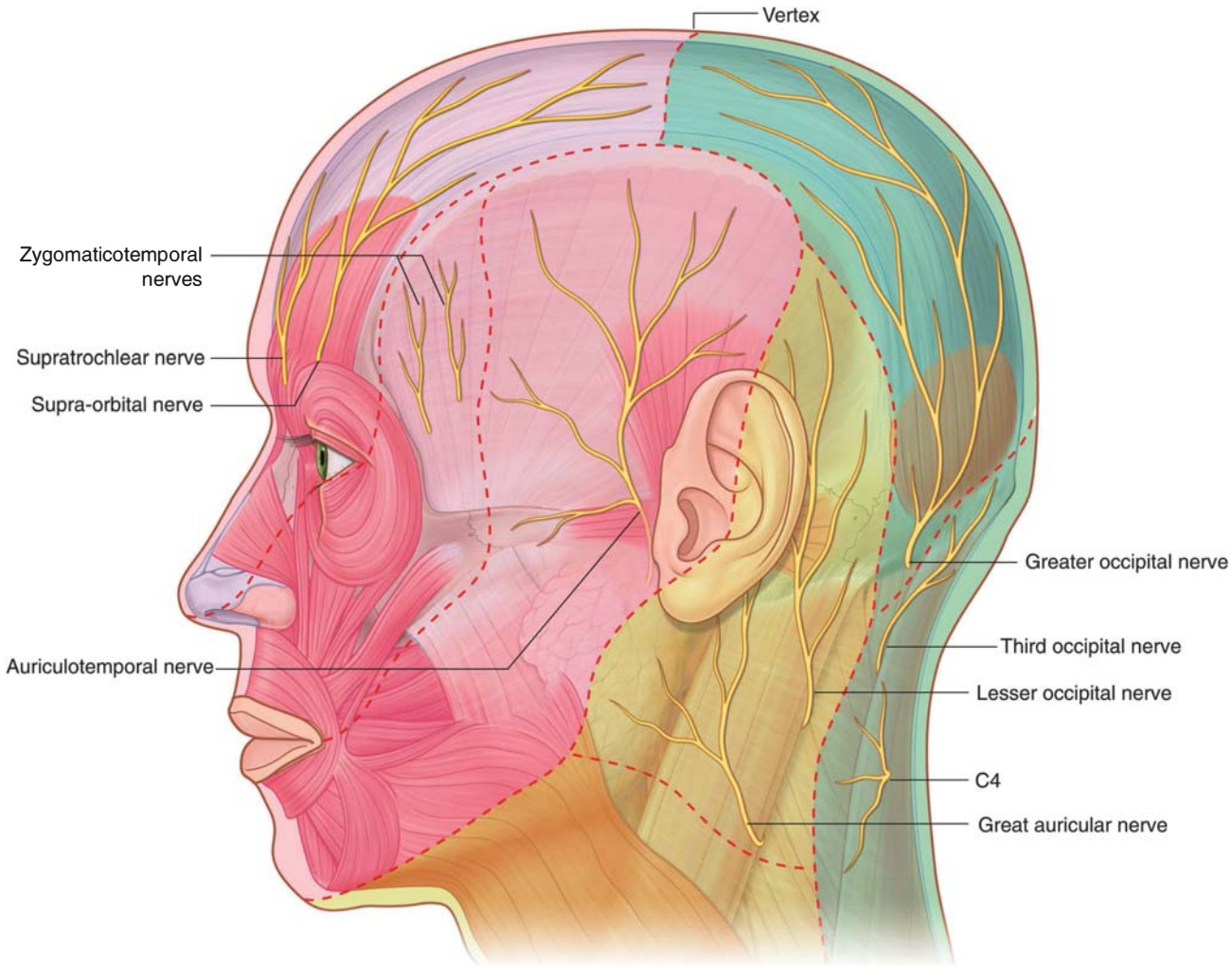


Fig. 8.70 Innervation of the scalp.

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cord levels C2 and C3 (Fig. 8.70). These branches are the great auricular, the lesser occipital, the greater occipital, and the third occipital nerves:

- The **great auricular nerve** is a branch of the cervical plexus, arises from the anterior rami of the C2 and C3 spinal nerves, ascends on the surface of the sternocleidomastoid muscle, and innervates a small area of the scalp just posterior to the ear.
- The **lesser occipital nerve** is also a branch of the cervical plexus, arises from the anterior ramus of the C2 spinal nerve, ascends on the posterior border of the sternocleidomastoid muscle, and supplies an area of the scalp posterior and superior to the ear.
- The **greater occipital nerve** is a branch of the posterior ramus of the C2 spinal nerve, emerges just inferior to the obliquus capitis inferior muscle, ascends superficial to the suboccipital triangle, pierces the semispinalis capitis and trapezius muscles, and then spreads out to supply a large part of the posterior scalp as far superiorly as the vertex.
- The **third occipital nerve** is a branch of the posterior ramus of the C3 spinal nerve, pierces the semispinalis capitis and trapezius muscles, and supplies a small area of the lower part of the scalp.

Vessels

Arteries

Arteries supplying the scalp (Fig. 8.71) are branches of either the external carotid artery or the ophthalmic artery, which is a branch of the internal carotid artery.

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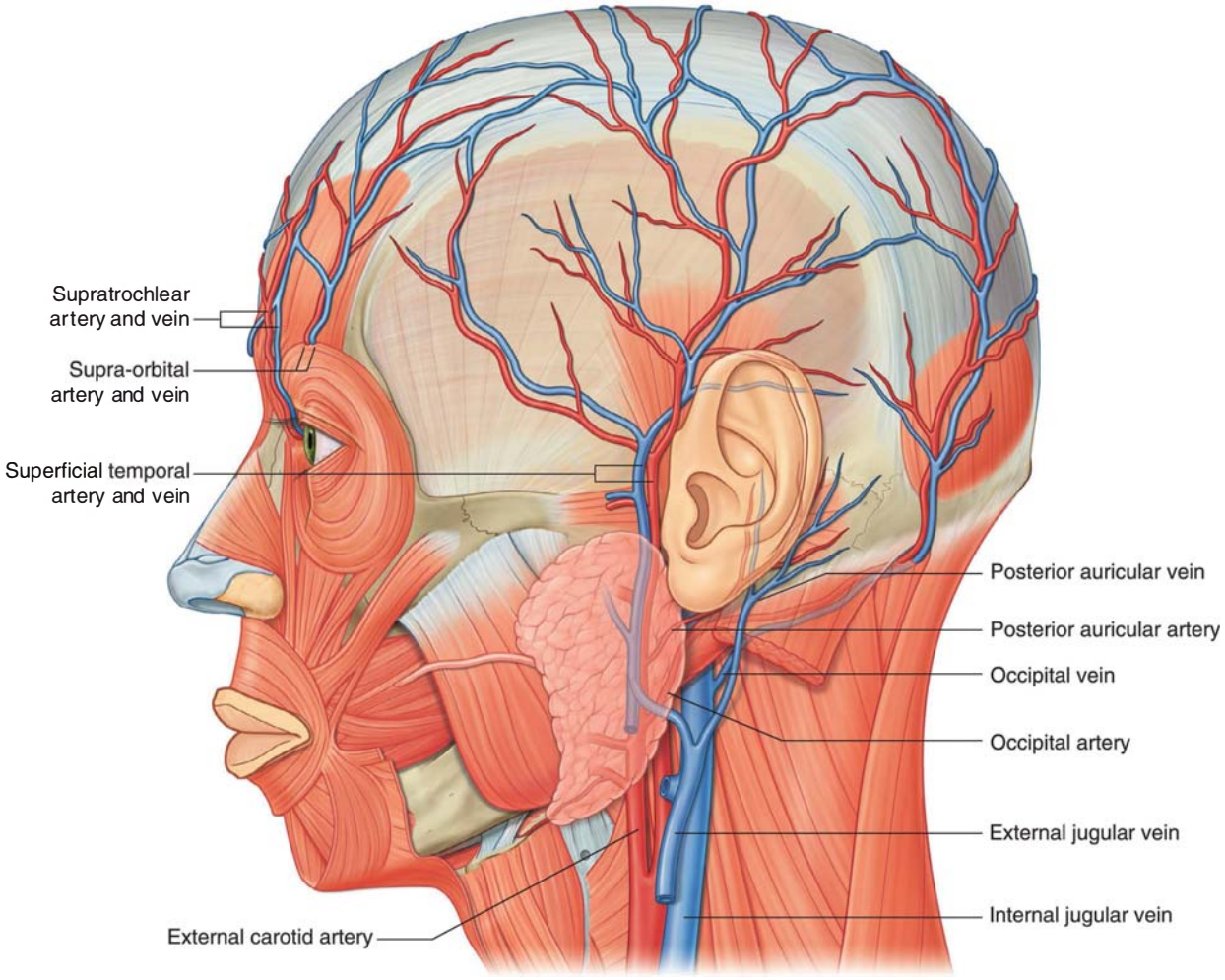


Fig. 8.71 Vasculature of the scalp.

Branches from the ophthalmic artery

The supratrochlear and supra-orbital arteries supply the anterior and superior aspects of the scalp. They branch from the ophthalmic artery while it is in the orbit, continue through the orbit, and exit onto the forehead in association with the supratrochlear and supra-orbital nerves. Like the nerves, the arteries ascend across the forehead to supply the scalp as far posteriorly as the vertex of the head.

Branches from the external carotid artery

Three branches of the external carotid artery supply the largest part of the scalp—the superficial temporal, posterior auricular, and occipital arteries supply the lateral and posterior aspects of the scalp (Fig. 8.71):

- The smallest branch (the **posterior auricular artery**) leaves the posterior aspect of the external carotid artery, passes through deeper structures, and emerges to supply an area of the scalp posterior to the ear.
- Also arising from the posterior aspect of the external carotid artery is the **occipital artery**, which ascends in a posterior direction, passes through several layers of back musculature, and emerges to supply a large part of the posterior aspect of the scalp.
- The third arterial branch supplying the scalp is the **superficial temporal artery**, a terminal branch of the external carotid artery that passes superiorly, just anterior to the ear, divides into anterior and posterior branches, and supplies almost the entire lateral aspect of the scalp.

Veins

Veins draining the scalp follow a pattern similar to the arteries:

- The **supratrochlear** and **supra-orbital veins** drain the anterior part of the scalp from the superciliary arches to the vertex of the head (Fig. 8.71), pass inferior to the superciliary arches, communicate with the ophthalmic veins in the orbit, and continue inferiorly to participate in the formation of the angular vein, which is the upper tributary to the facial vein.
- The **superficial temporal vein** drains the entire lateral area of the scalp before passing inferiorly to join in the formation of the retromandibular vein.
- The **posterior auricular vein** drains the area of the scalp posterior to the ear and eventually empties into a tributary of the retromandibular vein.
- The **occipital vein** drains the posterior aspect of the scalp from the external occipital protuberance and superior nuchal lines to the vertex of the head; deeper, it passes through the musculature in the posterior neck to join in the formation of the plexus of veins in the suboccipital triangle.

In the clinic

Scalp laceration

The scalp has an extremely rich blood supply from the external carotid arteries, so lacerations of the scalp tend to bleed profusely. Importantly, scalp bleeding is predominantly arterial, because of two reasons. First, in the erect position the venous pressure is extremely low. Second, the vessels do not retract and close when lacerated because the connective tissue in which they are found holds them open.

Lymphatic drainage

Lymphatic drainage of the scalp generally follows the pattern of arterial distribution.

The lymphatics in the occipital region initially drain to occipital nodes near the attachment of the trapezius muscle at the base of the skull (Fig. 8.72). Further along

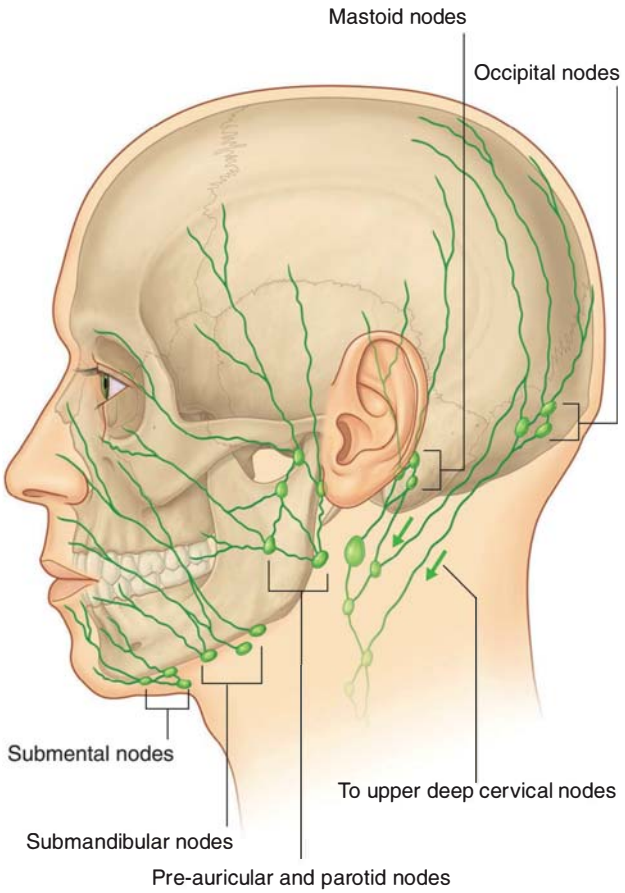


Fig. 8.72 Lymphatic drainage of the scalp.

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the pathway occipital nodes drain into upper deep cervical nodes. There is also some direct drainage to upper deep cervical nodes from this part of the scalp.

Lymphatics from the upper part of the scalp drain in two directions:

- Posterior to the vertex of the head they drain to **mastoid nodes** (retro-auricular/posterior auricular nodes) posterior to the ear near the mastoid process of the temporal bone, and efferent vessels from these nodes drain into upper deep cervical nodes.
- Anterior to the vertex of the head they drain to pre-auricular and parotid nodes anterior to the ear on the surface of the parotid gland.

Finally, there may be some lymphatic drainage from the forehead to the submandibular nodes through efferent vessels that follow the facial artery.

ORBIT

The orbits are bilateral structures in the upper half of the face below the anterior cranial fossa and anterior to the middle cranial fossa that contain the eyeball, the optic nerve, the extra-ocular muscles, the lacrimal apparatus, adipose tissue, fascia, and the nerves and vessels that supply these structures.

Bony orbit

Seven bones contribute to the framework of each orbit (Fig. 8.73). They are the maxilla, zygomatic, frontal, ethmoid, lacrimal, sphenoid, and palatine bones. Together they give the bony orbit the shape of a pyramid, with its wide base opening anteriorly onto the face and its apex extending in a posteromedial direction. Completing the pyramid configuration are medial, lateral, superior, and inferior walls.

The apex of the pyramid-shaped bony orbit is the optic foramen, whereas the base (the orbital rim) is formed:

- superiorly by the frontal bone,
- medially by the frontal process of the maxilla,
- inferiorly by the zygomatic process of the maxilla and the zygomatic bone, and
- laterally by the zygomatic bone, the frontal process of the zygomatic bone, and the zygomatic process of the frontal bone.

Roof

The **roof** (**superior wall**) of the bony orbit is made up of the orbital part of the frontal bone with a small contribution from the sphenoid bone (Fig. 8.73). This thin plate of bone separates the contents of the orbit from the brain in the anterior cranial fossa.

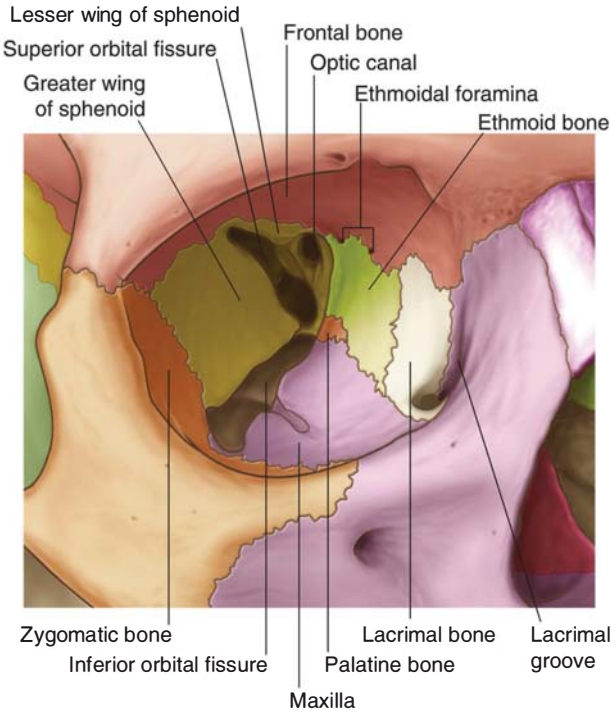


Fig. 8.73 Bones of the orbit.

Unique features of the superior wall include:

- anteromedially, the possible intrusion of part of the frontal sinus and the trochlear fovea, for the attachment of a pulley through which the superior oblique muscle passes;
- anterolaterally, a depression (the lacrimal fossa) for the orbital part of the lacrimal gland.

Posteriorly, the lesser wing of the sphenoid bone completes the roof.

Medial wall

The **medial walls** of the paired bony orbits are parallel to each other and each consists of four bones—the maxilla, lacrimal, ethmoid, and sphenoid bones (Fig. 8.73).

The largest contributor to the medial wall is the orbital plate of the ethmoid bone. This part of the ethmoid bone contains collections of ethmoidal cells, which are clearly visible in a dried skull.

Also visible, at the junction between the roof and the medial wall, usually associated with the frontoethmoidal suture, are the **anterior** and **posterior ethmoidal foramina**. The anterior and posterior ethmoidal nerves and vessels leave the orbit through these openings.

Anterior to the ethmoid bone is the small lacrimal bone, and completing the anterior part of the medial wall is the

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frontal process of the maxilla. These two bones participate in the formation of the **lacrimal groove**, which contains the lacrimal sac and is bound by the **posterior lacrimal crest** (part of the lacrimal bone) and the **anterior lacrimal crest** (part of the maxilla).

Posterior to the ethmoid bone the medial wall is completed by a small part of the sphenoid bone, which forms a part of the medial wall of the optic canal.

Floor

The **floor** (**inferior wall**) of the bony orbit, which is also the roof of the maxillary sinus, consists primarily of the orbital surface of the maxilla (Fig. 8.73), with small contributions from the zygomatic and palatine bones.

Beginning posteriorly and continuing along the lateral boundary of the floor of the bony orbit is the inferior orbital fissure. Beyond the anterior end of the fissure the zygomatic bone completes the floor of the bony orbit.

Posteriorly, the orbital process of the palatine bone makes a small contribution to the floor of the bony orbit near the junction of the maxilla, ethmoid, and sphenoid bones.

Lateral wall

The **lateral wall** of the bony orbit consists of contributions from two bones—anteriorly, the zygomatic bone and posteriorly, the greater wing of the sphenoid bone (Fig. 8.73). The superior orbital fissure is between the greater wing of the sphenoid and the lesser wing of the sphenoid that forms part of the roof.

Eyelids

The upper and lower eyelids are anterior structures that, when closed, protect the surface of the eyeball.

The space between the eyelids, when they are open, is the **palpebral fissure**.

The layers of the eyelids, from anterior to posterior, consist of skin, subcutaneous tissue, voluntary muscle, the orbital septum, the tarsus, and conjunctiva (Fig. 8.74).

The upper and lower eyelids are basically similar in structure except for the addition of two muscles in the upper eyelid.

Skin and subcutaneous tissue

The skin of the eyelids is not particularly substantial, and only a thin layer of connective tissue separates the skin from the underlying voluntary muscle layer (Fig. 8.74). The thin layer of connective tissue and its loose arrangement account for the accumulation of fluid (blood) when an injury occurs.

Orbicularis oculi

The muscle fibers encountered next in an anteroposterior direction through the eyelid belong to the **palpebral part** of the **orbicularis oculi** (Fig. 8.74). This muscle is part of the larger orbicularis oculi muscle, which consists primarily of two parts—an **orbital part**, which surrounds the orbit, and the palpebral part, which is in the eyelids. The orbicularis oculi is innervated by the facial nerve [VII] and closes the eyelids.

In the clinic

Orbital fracture

Fractures of the orbit are not uncommon and may involve the orbital margins with extension into the maxilla, frontal, and zygomatic bones. These fractures are often part of complex facial fractures. Fractures within the orbit frequently occur within the floor and the medial wall; however, superior and lateral wall fractures also occur. Inferior orbital floor fractures are one of the commonest types of injuries. These fractures may drag the inferior rectus muscle and associated tissues into the fracture line. In these instances, patients may have upward gaze failure (upward gaze diplopia) in the affected eye. Medial wall fractures characteristically show air within the orbit in radiographs. This is due to fracture of the ethmoidal labyrinth, permitting direct continuity between the orbit and the ethmoidal paranasal sinuses. Occasionally, patients feel a full sensation within the orbit when blowing the nose.

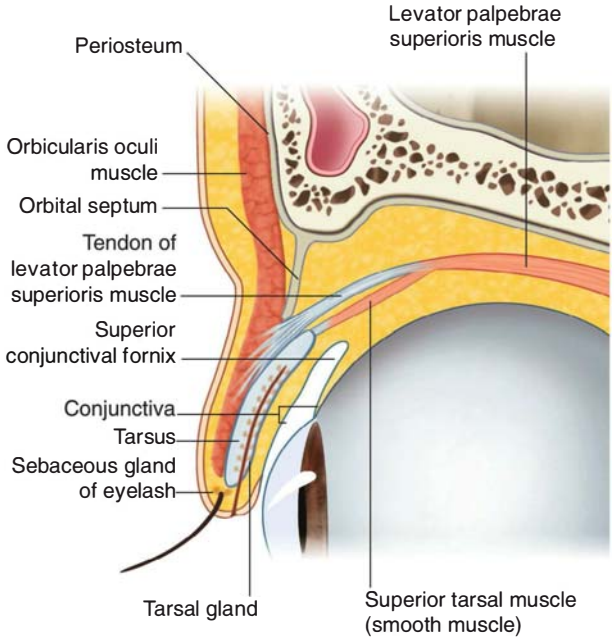


Fig. 8.74 Eyelids.

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The palpebral part is thin and anchored medially by the **medial palpebral ligament** (Fig. 8.75), which attaches to the anterior lacrimal crest and laterally blends with fibers from the muscle in the lower eyelid at the **lateral palpebral ligament** (Fig. 8.75).

A third part of the orbicularis oculi muscle that can be identified consists of fibers on the medial border, which pass deeply to attach to the posterior lacrimal crest. These fibers form the lacrimal part of the orbicularis oculi, which may be involved in the drainage of tears.

Orbital septum

Deep to the palpebral part of the orbicularis oculi is an extension of periosteum into both the upper and lower eyelids from the margin of the orbit (Fig. 8.76). This is the **orbital septum**, which extends downward into the upper eyelid and upward into the lower eyelid and is continuous with the periosteum outside and inside the orbit (Fig. 8.76). The orbital septum attaches to the tendon of the levator palpebrae superioris muscle in the upper eyelid and attaches to the tarsus in the lower eyelid.

Tarsus and levator palpebrae superioris

Providing major support for each eyelid is the tarsus (Fig. 8.77). There is a large **superior tarsus** in the upper eyelid and a smaller **inferior tarsus** in the lower eyelid (Fig. 8.77). These plates of dense connective tissue are attached medially to the anterior lacrimal crest of the maxilla by the medial palpebral ligament and laterally to the orbital tubercle on the zygomatic bone by the lateral palpebral ligament.

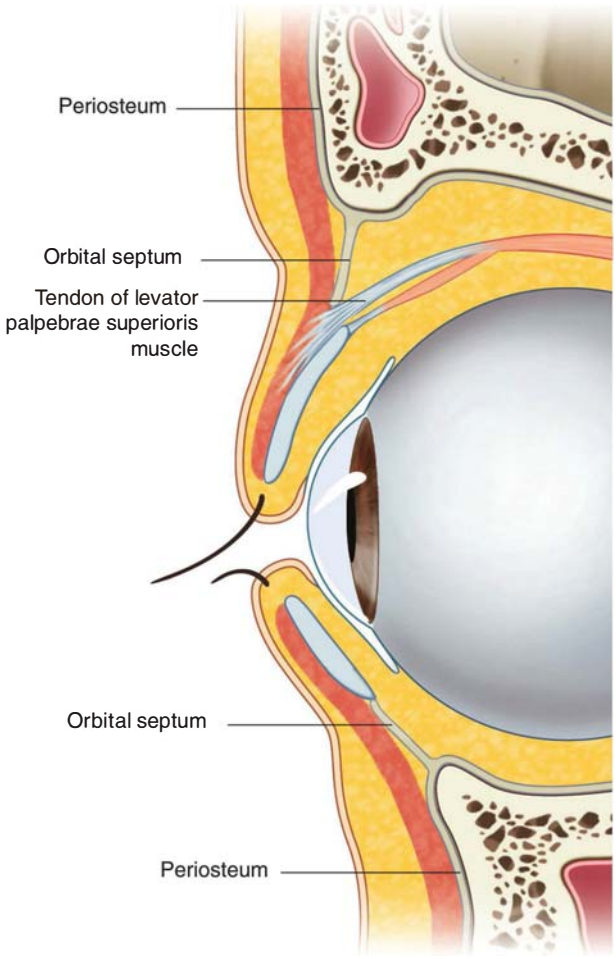


Fig. 8.76 Orbital septum.

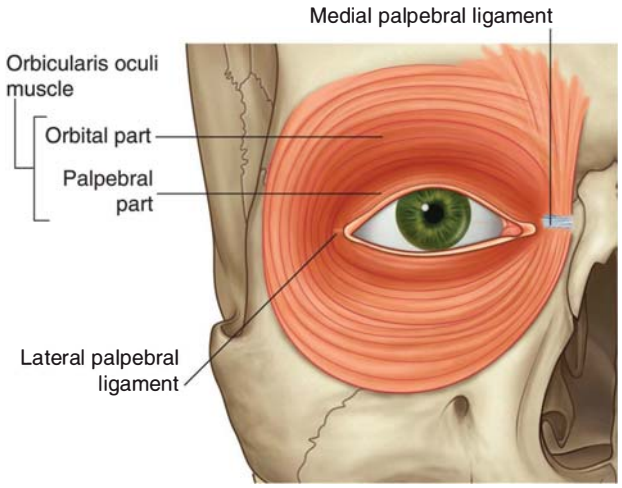


Fig. 8.75 Orbicularis oculi muscle.

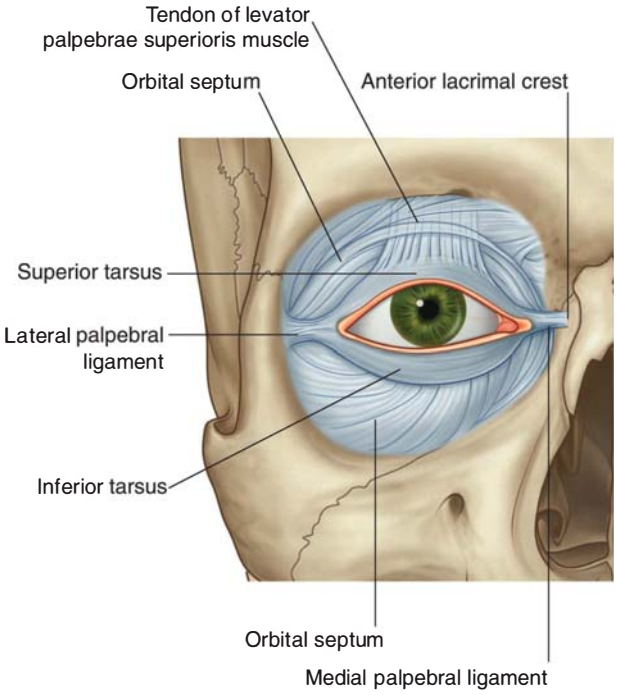


Fig. 8.77 Tarsal plates.

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Although the tarsal plates in the upper and lower eyelids are generally similar in structure and function, there is one unique difference. Associated with the tarsus in the upper eyelid is the **levator palpebrae superioris muscle** (Fig. 8.77), which raises the eyelid. Its origin is from the posterior part of the roof of the orbit, just superior to the optic foramen, and it inserts into the anterior surface of the superior tarsus, with the possibility of a few fibers attaching to the skin of the upper eyelid. It is innervated by the oculomotor nerve [III].

In companion with the levator palpebrae superioris muscle is a collection of smooth muscle fibers passing from the inferior surface of the levator to the upper edge of the superior tarsus (see Fig. 8.74). Innervated by postganglionic sympathetic fibers from the superior cervical ganglion, this muscle is the **superior tarsal muscle**.

Loss of function of either the levator palpebrae superioris muscle or the superior tarsal muscle results in a ptosis or drooping of the upper eyelid.

Conjunctiva

The structure of the eyelid is completed by a thin membrane (the **conjunctiva**), which covers the posterior surface of each eyelid (see Fig. 8.74). This membrane covers the full extent of the posterior surface of each eyelid before reflecting onto the outer surface (**sclera**) of the eyeball. It attaches to the eyeball at the junction between the sclera and the cornea. With this membrane in place, a **conjunctival sac** is formed when the eyelids are closed, and the upper and lower extensions of this sac are the **superior** and **inferior conjunctival fornices** (Fig. 8.74).

Glands

Embedded in the tarsal plates are tarsal glands (see Fig. 8.74), which empty onto the free margin of each eyelid. These glands are modified sebaceous glands and secrete an oily substance that increases the viscosity of the tears and decreases the rate of evaporation of tears from the surface of the eyeball. Blockage and inflammation of a tarsal gland is a **chalazion** and is on the inner surface of the eyelid.

The tarsal glands are not the only glands associated with the eyelids. Associated with the eyelash follicles are sebaceous and sweat glands (see Fig. 8.74). Blockage and inflammation of either of these is a **stye** and is on the edge of the eyelid.

Vessels

The arterial supply to the eyelids is from the numerous vessels in the area (Fig. 8.78). They include:

- the supratrochlear, supra-orbital, lacrimal, and dorsal nasal arteries from the ophthalmic artery;
- the angular artery from the facial artery;
- the transverse facial artery from the superficial temporal artery; and
- branches from the superficial temporal artery itself.

Venous drainage follows an external pattern through veins associated with the various arteries and an internal pattern moving into the orbit through connections with the ophthalmic veins.

Lymphatic drainage is primarily to the parotid nodes, with some drainage from the medial corner of the eye along lymphatic vessels associated with the angular and facial arteries to the submandibular nodes.

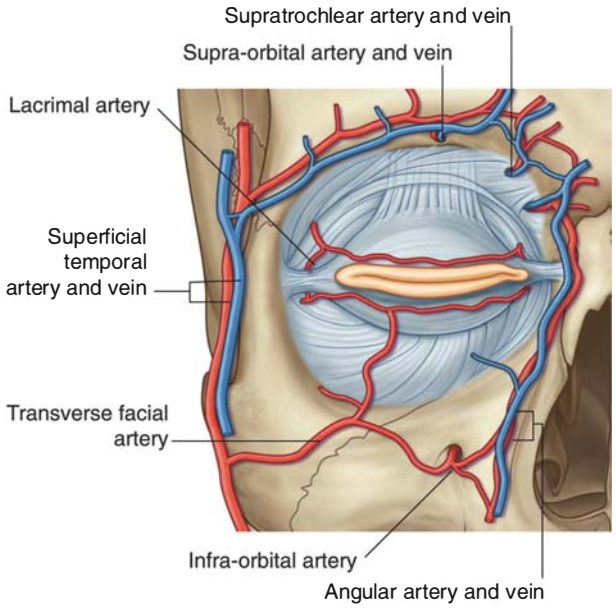


Fig. 8.78 Vasculature of the eyelids.

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Innervation

Innervation of the eyelids includes both sensory and motor components.

The sensory nerves are all branches of the trigeminal nerve [V] (Fig. 8.79). Palpebral branches arise from:

- the supra-orbital, supratrochlear, infratrochlear, and lacrimal branches of the ophthalmic nerve [V_1]; and
- the infra-orbital branch of the maxillary nerve [V_2].

Motor innervation is from:

- the facial nerve [VII], which innervates the palpebral part of the orbicularis oculi;
- the oculomotor nerve [III], which innervates the levator palpebrae superioris; and
- sympathetic fibers, which innervate the superior tarsal muscle.

Loss of innervation of the orbicularis oculi by the facial nerve [VII] causes an inability to close the eyelids tightly and the lower eyelid droops away, resulting in a spillage of tears.

Loss of innervation of the levator palpebrae superioris by the oculomotor nerve causes an inability to open the superior eyelid voluntarily, producing a complete ptosis.

Loss of innervation of the superior tarsal muscle by sympathetic fibers causes a constant partial ptosis.

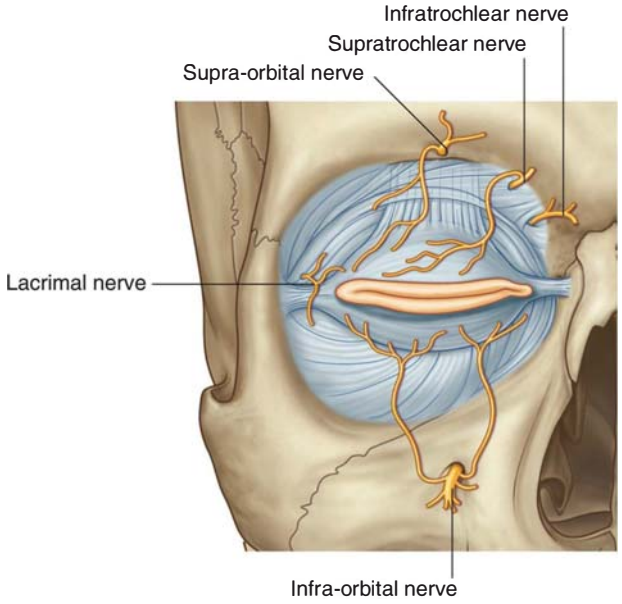


Fig. 8.79 Innervation of the eyelids.

In the clinic

Horner's syndrome

Horner's syndrome is caused by any lesion that leads to a loss of sympathetic function in the head. It is characterized by three typical features:

- pupillary constriction due to paralysis of the dilator pupillae muscle,
- partial ptosis (drooping of the upper eyelid) due to paralysis of the superior tarsal muscle, and
- absence of sweating on the ipsilateral side of the face and the neck due to absence of innervation of the sweat glands.

Secondary changes may also include:

- ipsilateral vasodilation due to loss of the normal sympathetic control of the subcutaneous blood vessels, and
- enophthalmos (sinking of the eye)—believed to result from paralysis of the orbitalis muscle, although this is an uncommon feature of Horner's syndrome.

The orbitalis muscle spans the inferior orbital fissure and helps maintain the forward position of orbital contents.

The commonest cause for Horner's syndrome is a tumor eroding the cervicothoracic ganglion, which is typically an apical lung tumor.

Surgically induced Horner's syndrome

A surgically induced Horner's syndrome may be necessary for patients who suffer severe hyperhidrosis (sweating). This often debilitating condition may be so severe that patients are confined to their home for fear of embarrassment. Treatment is relatively straightforward. The patient is anesthetized and a bifurcate endotracheal tube is placed into the left and right main bronchi. A small incision is made in the intercostal space on the appropriate side, and a surgically induced pneumothorax is created. The patient is ventilated through the contralateral lung.

Using an endoscope the apex of the thoracic cavity can be viewed from inside and the cervicothoracic ganglion readily identified. Obliterative techniques include thermocoagulation and surgical excision. After the ganglion has been destroyed, the endoscope is removed, the lung is reinflated, and the small hole is sutured.

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Lacrimal apparatus

The lacrimal apparatus is involved in the production, movement, and drainage of fluid from the surface of the eyeball. It is made up of the **lacrimal gland** and its ducts, the **lacrimal canaliculi**, the **lacrimal sac**, and the **nasolacrimal duct**.

The lacrimal gland is anterior in the superolateral region of the orbit (Fig. 8.80) and is divided into two parts by the levator palpebrae superioris (Fig. 8.81):

- The larger **orbital part** is in a depression, the lacrimal fossa, in the frontal bone.
- The smaller **palpebral part** is inferior to the levator palpebrae superioris in the superolateral part of the eyelid.

Numerous ducts empty the glandular secretions into the lateral part of the superior fornix of the conjunctiva.

Fluid is continually being secreted by the lacrimal gland and moved across the surface of the eyeball from lateral to medial as the eyelids blink.

The fluid accumulates medially in the **lacrimal lake** and is drained from the lake by the lacrimal canaliculi, one canaliculus associated with each eyelid (Fig. 8.80). The **lacrimal punctum** is the opening through which fluid enters each canaliculus.

Passing medially, the lacrimal canaliculi eventually join the lacrimal sac between the anterior and posterior lacrimal crests, posterior to the medial palpebral ligament and anterior to the lacrimal part of the orbicularis oculi muscle (Figs. 8.82 and 8.83). When the orbicularis oculi muscle contracts during blinking, the small lacrimal part of the muscle may dilate the lacrimal sac and draw tears into it through the canaliculi from the conjunctival sac.

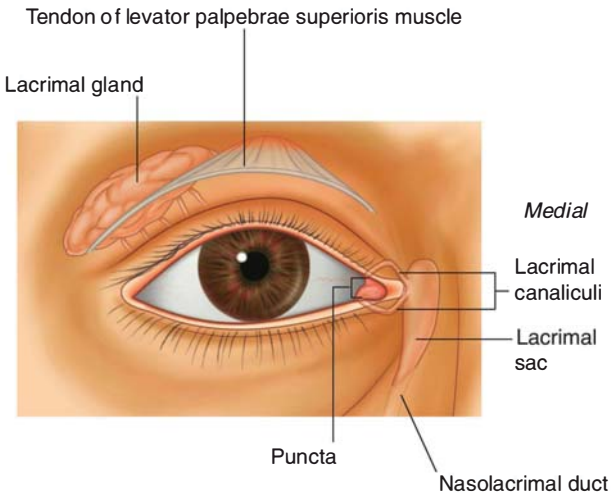


Fig. 8.80 Lacrimal gland, anterior view.

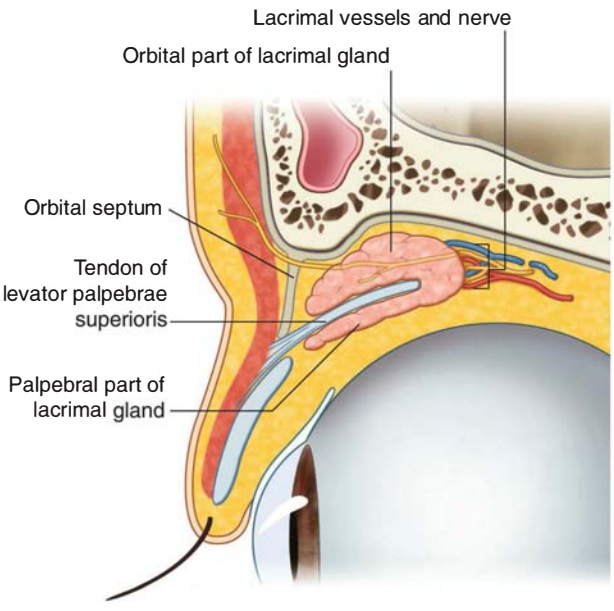


Fig. 8.81 Lacrimal gland and levator palpebrae superioris.

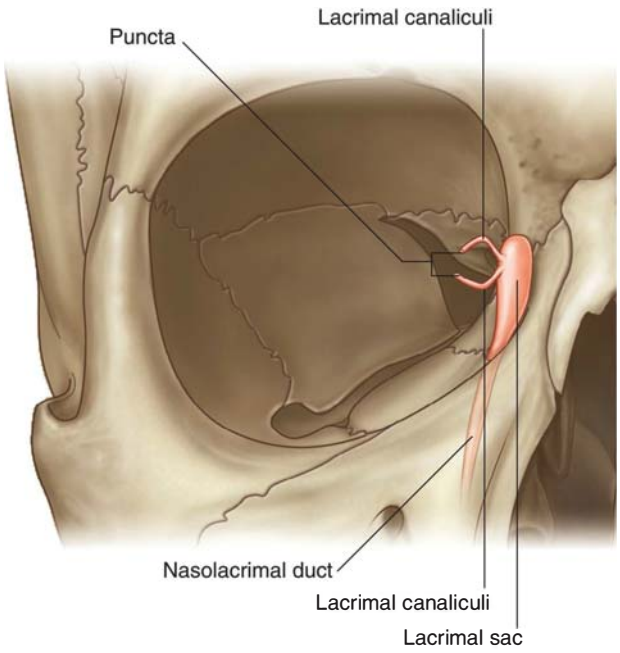


Fig. 8.82 The lacrimal sac.

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Innervation

The innervation of the lacrimal gland involves three different components (Fig. 8.84).

Sensory innervation

Sensory neurons from the lacrimal gland return to the CNS through the lacrimal branch of the ophthalmic nerve [V_1].

Secretomotor (parasympathetic) innervation

Secretomotor fibers from the parasympathetic part of the autonomic division of the PNS stimulate fluid secretion from the lacrimal gland. These preganglionic parasympathetic neurons leave the CNS in the facial nerve [VII], enter the greater petrosal nerve (a branch of the facial nerve [VII]), and continue with this nerve until it becomes the **nerve of the pterygoid canal** (Fig. 8.84).

The nerve of the pterygoid canal eventually joins the pterygopalatine ganglion where the preganglionic parasympathetic neurons synapse on postganglionic parasympathetic neurons. The postganglionic neurons join the maxillary nerve [V_2] and continue with it until the zygomatic nerve branches from it, and travel with the zygomatic nerve until it gives off the zygomaticotemporal nerve, which eventually distributes postganglionic parasympathetic fibers in a small branch that joins the lacrimal nerve. The lacrimal nerve passes to the lacrimal gland.

Sympathetic innervation

Sympathetic innervation of the lacrimal gland follows a similar path as parasympathetic innervation. Postganglionic sympathetic fibers originating in the superior cervical ganglion travel along the plexus surrounding the internal carotid artery (Fig. 8.84). They leave this plexus as the deep petrosal nerve and join the parasympathetic fibers in the nerve of the pterygoid canal. Passing through the pterygopalatine ganglion, the sympathetic fibers from this point onward follow the same path as the parasympathetic fibers to the lacrimal gland.

Vessels

The arterial supply to the lacrimal gland is by branches from the ophthalmic artery and venous drainage is through the ophthalmic veins.

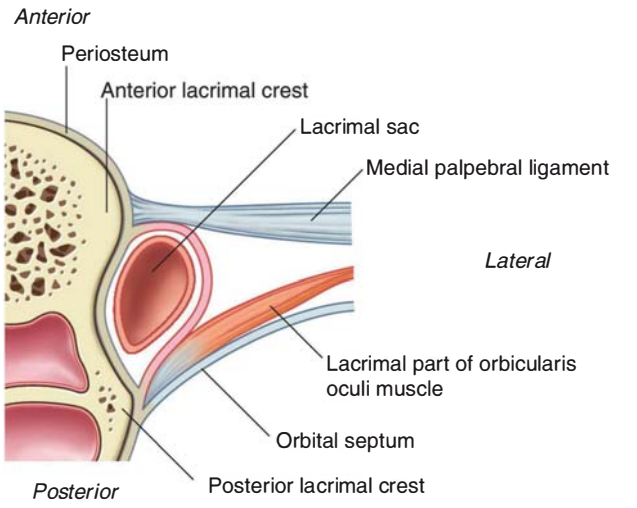


Fig. 8.83 Position of lacrimal sac.

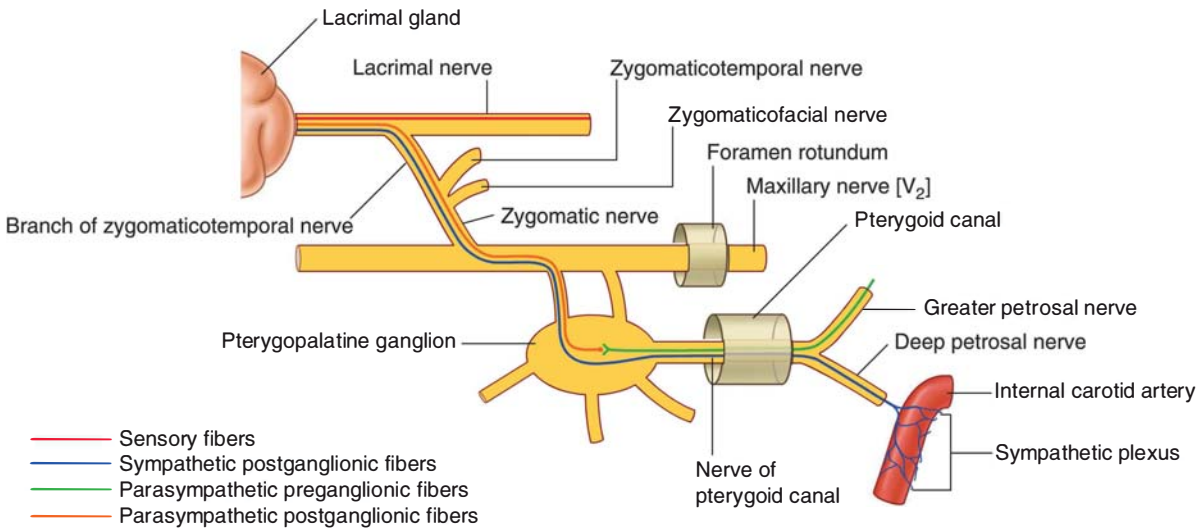


Fig. 8.84 Innervation of the lacrimal gland.

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Fissures and foramina

Numerous structures enter and leave the orbit through a variety of openings (Fig. 8.85).

Optic canal

When the bony orbit is viewed from an anterolateral position, the round opening at the apex of the pyramidal-shaped orbit is the optic canal, which opens into the middle cranial fossa and is bounded medially by the body of the sphenoid and laterally by the lesser wing of the sphenoid. Passing through the optic canal are the optic nerve and the ophthalmic artery (Fig. 8.86).

Superior orbital fissure

Just lateral to the optic canal is a triangular-shaped gap between the roof and lateral wall of the bony orbit. This is the superior orbital fissure and allows structures to pass between the orbit and the middle cranial fossa (Fig. 8.85).

Passing through the superior orbital fissure are the superior and inferior branches of the oculomotor nerve [III], the trochlear nerve [IV], the abducent nerve [VI], the lacrimal, frontal, and nasociliary branches of the ophthalmic nerve [V_1], and the superior ophthalmic vein (Fig. 8.86).

Inferior orbital fissure

Separating the lateral wall of the orbit from the floor of the orbit is a longitudinal opening, the inferior orbital fissure (Fig. 8.85). Its borders are the greater wing of the sphenoid

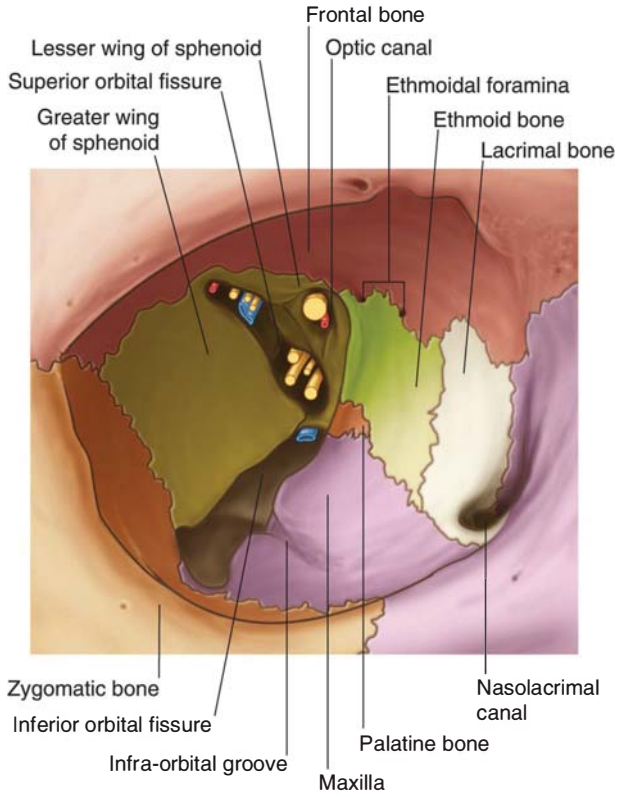


Fig. 8.85 Openings into the bony orbit.

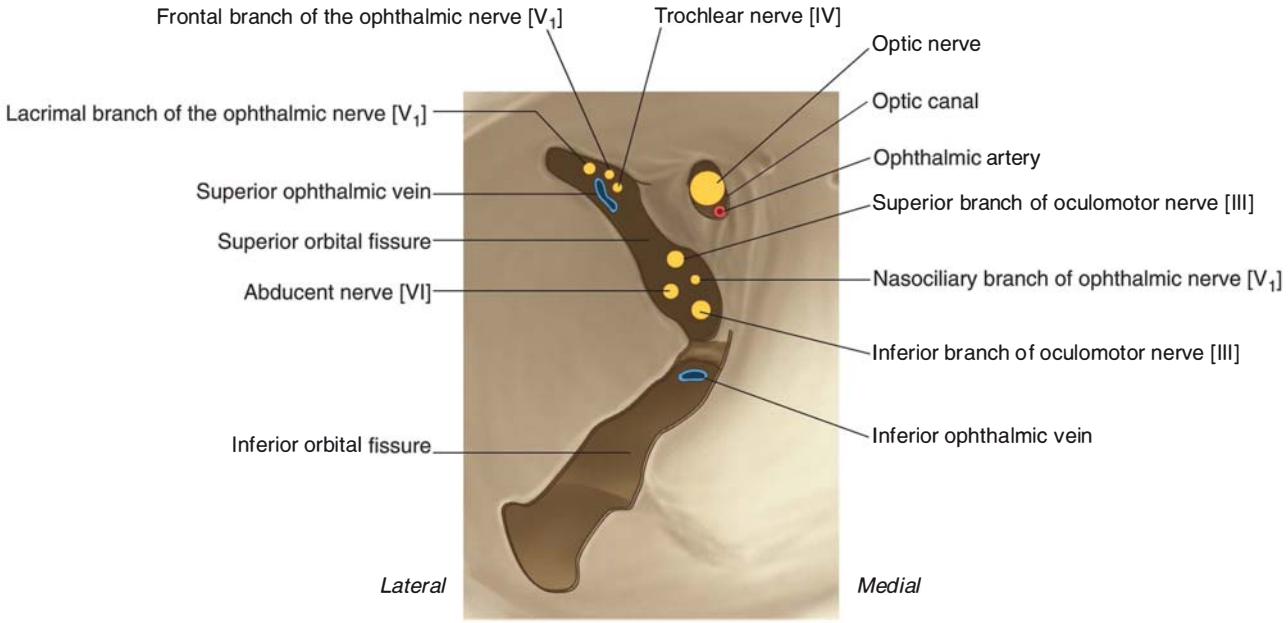


Fig. 8.86 Optic canal and superior orbital fissure.

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and the maxilla, palatine, and zygomatic bones. This long fissure allows communication between:

- the orbit and the pterygopalatine fossa posteriorly,
- the orbit and the infratemporal fossa in the middle, and
- the orbit and the temporal fossa posterolaterally.

Passing through the inferior orbital fissure are the maxillary nerve [V_2] and its zygomatic branch, the infra-orbital vessels, and a vein communicating with the pterygoid plexus of veins.

Infra-orbital foramen

Beginning posteriorly and crossing about two-thirds of the inferior orbital fissure, a groove (the **infra-orbital groove**) is encountered, which continues anteriorly across the floor of the orbit (Fig. 8.85). This groove connects with the **infra-orbital canal** that opens onto the face at the **infra-orbital foramen**.

The infra-orbital nerve, part of the maxillary nerve [V_2], and vessels pass through this structure as they exit onto the face.

Other openings

Associated with the medial wall of the bony orbit are several smaller openings (Fig. 8.85).

The **anterior** and **posterior ethmoidal foramina** are at the junction between the superior and medial walls. These openings provide exits from the orbit into the ethmoid bone for the anterior and posterior ethmoidal nerves and vessels.

Completing the openings on the medial wall is a canal in the lower part of the wall anteriorly. Clearly visible is the depression for the lacrimal sac formed by the lacrimal bone and the frontal process of the maxilla. This depression is continuous with the nasolacrimal canal, which leads to the inferior nasal meatus. Contained within the nasolacrimal canal is the nasolacrimal duct, a part of the lacrimal apparatus.

Fascial specializations

Periorbita

The periosteum lining the bones that form the orbit is the **periorbita** (Fig. 8.87A). It is continuous at the margins of the orbit with the periosteum on the outer surface of the skull and sends extensions into the upper and lower eyelids (the **orbital septa**).

At the various openings where the orbit communicates with the cranial cavity the periorbita is continuous with the periosteal layer of dura mater. In the posterior part of the orbit, the periorbita thickens around the optic canal and the central part of the superior orbital fissure. This is the point of origin of the four rectus muscles and is the **common tendinous ring**.

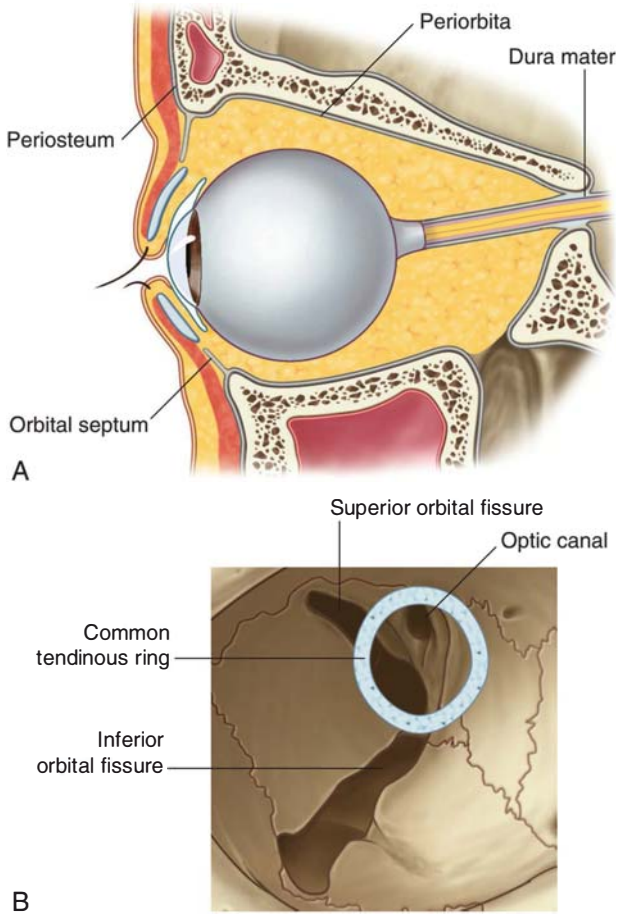


Fig. 8.87 Periorbita. **A.** Lateral view. **B.** Common tendinous ring.

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Fascial sheath of the eyeball

The **fascial sheath of the eyeball** (bulbar sheath) is a layer of fascia that encloses a major part of the eyeball (Figs. 8.88 and 8.89):

- Posteriorly, it is firmly attached to the sclera (the white part of the eyeball) around the point of entrance of the optic nerve into the eyeball.
- Anteriorly, it is firmly attached to the sclera near the edge of the cornea (the clear part of the eyeball).
- Additionally, as the muscles approach the eyeball, the investing fascia surrounding each muscle blends with the fascial sheath of the eyeball as the muscles pass through and continue to their point of attachment.

A specialized lower part of the fascial sheath of the eyeball is the **suspensory ligament** (Figs. 8.88 and 8.89), which supports the eyeball. This "sling-like" structure is made up of the fascial sheath of the eyeball and contributions from the two inferior ocular muscles and the medial and lateral ocular muscles.

Check ligaments of the medial and lateral rectus muscles

Other fascial specializations in the orbit are the check ligaments (Fig. 8.89). These are expansions of the investing fascia covering the medial and lateral rectus muscles, which attach to the medial and lateral walls of the bony orbit:

- The medial check ligament is an extension from the fascia covering the medial rectus muscle and attaches immediately posterior to the posterior lacrimal crest of the lacrimal bone.
- The lateral check ligament is an extension from the fascia covering the lateral rectus muscle and is attached to the orbital tubercle of the zygomatic bone.

Functionally, the positioning of these ligaments seems to restrict the medial and lateral rectus muscles, thus the names of the fascial specializations.

Muscles

There are two groups of muscles within the orbit:

- **extrinsic muscles of eyeball** (**extra-ocular muscles**) involved in movements of the eyeball or raising upper eyelids, and
- intrinsic muscles within the eyeball, which control the shape of the lens and size of the pupil.

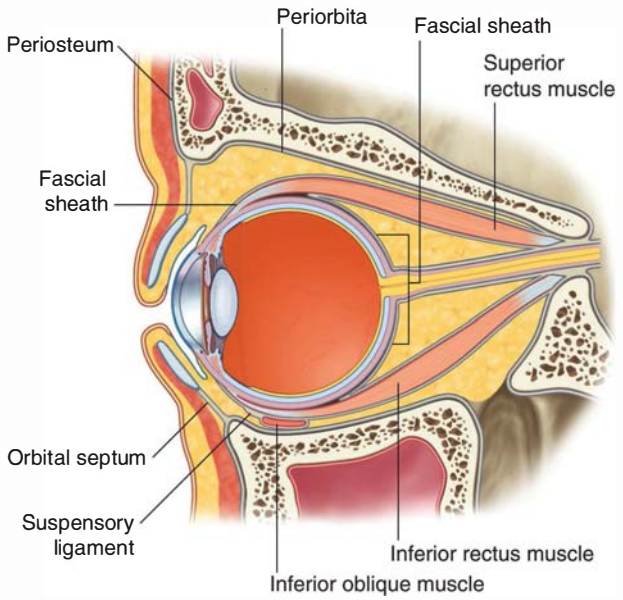


Fig. 8.88 Fascial sheath of the eyeball.

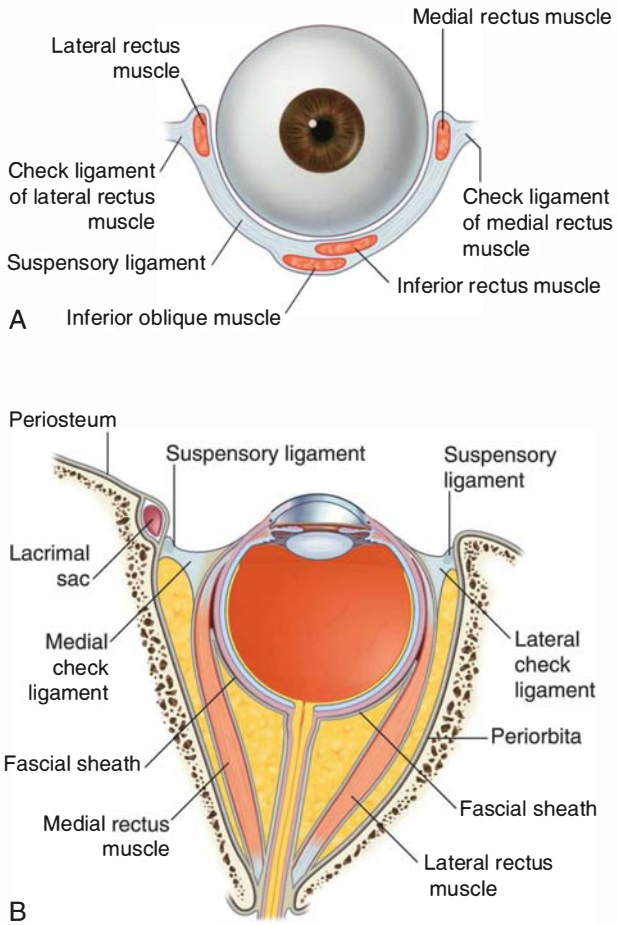


Fig. 8.89 Check ligaments. **A.** Anterior view. **B.** Superior view.

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Table 8.8 Extrinsic (extra-ocular) muscles

Muscle	Origin	Insertion	Innervation	Function
Levator palpebrae superioris	Lesser wing of sphenoid anterior to optic canal	Anterior surface of tarsal plate; a few fibers to skin and superior conjunctival fornix	Oculomotor nerve [III]—superior branch	Elevation of upper eyelid
Superior rectus	Superior part of common tendinous ring	Anterior half of eyeball superiorly	Oculomotor nerve [III]—superior branch	Elevation, adduction, medial rotation of eyeball
Inferior rectus	Inferior part of common tendinous ring	Anterior half of eyeball inferiorly	Oculomotor nerve [III]—inferior branch	Depression, adduction, lateral rotation of eyeball
Medial rectus	Medial part of common tendinous ring	Anterior half of eyeball medially	Oculomotor nerve [III]—inferior branch	Adduction of eyeball
Lateral rectus	Lateral part of common tendinous ring	Anterior half of eyeball laterally	Abducent nerve [VI]	Abduction of eyeball
Superior oblique	Body of sphenoid, superior and medial to optic canal	Outer posterior quadrant of eyeball (superior surface)	Trochlear nerve [IV]	Depression, abduction, medial rotation of eyeball
Inferior oblique	Medial floor of orbit posterior to rim; maxilla lateral to nasolacrimal groove	Outer posterior quadrant of eyeball (inferior surface)	Oculomotor nerve [III]—inferior branch	Elevation, abduction, lateral rotation of eyeball

The extrinsic muscles include the levator palpebrae superioris, superior rectus, inferior rectus, medial rectus, lateral rectus, superior oblique, and inferior oblique.

The intrinsic muscles include the ciliary muscle, the sphincter pupillae, and the dilator pupillae.

Extrinsic muscles

Of the seven muscles in the extrinsic group of muscles, one raises the eyelids, whereas the other six move the eyeball itself (Table 8.8).

The movements of the eyeball, in three dimensions, (Fig. 8.90) are:

- elevation—moving the pupil superiorly,
- depression—moving the pupil inferiorly,
- abduction—moving the pupil laterally,
- adduction—moving the pupil medially,
- internal rotation (intorsion)—rotating the upper part of the pupil medially (or toward the nose), and
- external rotation (extorsion)—rotating the upper part of the pupil laterally (or toward the temple).

The axis of each orbit is directed slightly laterally from back to front, but each eyeball is directed anteriorly (Fig. 8.91). Therefore the pull of some muscles has multiple effects on the movement of the eyeball, whereas that of others has a single effect.

Levator palpebrae superioris

Levator palpebrae superioris raises the upper eyelid (Table 8.8). It is the most superior muscle in the orbit, originating from the roof, just anterior to the optic canal on the inferior

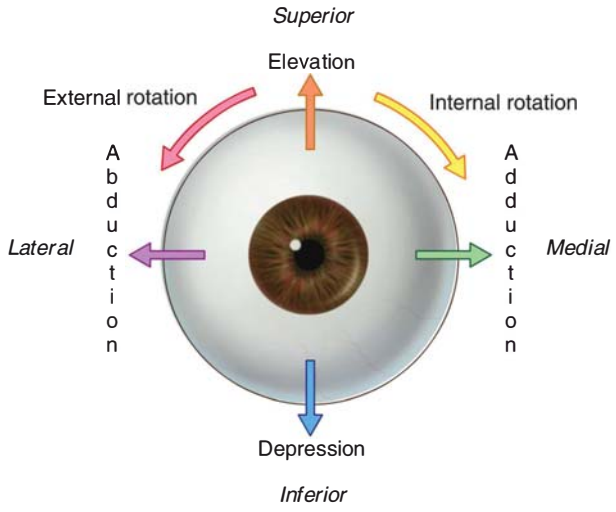


Fig. 8.90 Movements of the eyeball.

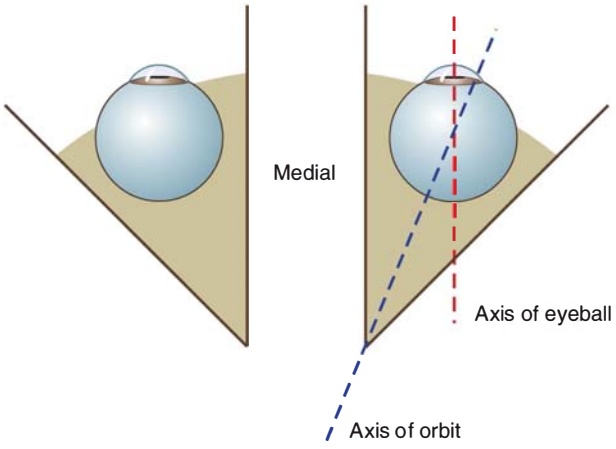


Fig. 8.91 Axes of the eyeball and orbit.

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surface of the lesser wing of the sphenoid (Fig. 8.92B). Its primary point of insertion is into the anterior surface of the superior tarsus, but a few fibers also attach to the skin of the upper eyelid and the superior conjunctival fornix.

Innervation is by the superior branch of the oculomotor nerve [III].

Contraction of the levator palpebrae superioris raises the upper eyelid.

A unique feature of the levator palpebrae superioris is that a collection of smooth muscle fibers passes from its inferior surface to the upper edge of the superior tarsus (see Fig. 8.74). This group of smooth muscle fibers (the superior tarsal muscle) help maintain eyelid elevation and are innervated by postganglionic sympathetic fibers from the superior cervical ganglion.

Loss of oculomotor nerve [III] function results in complete ptosis or drooping of the superior eyelid, whereas loss of sympathetic innervation to the superior tarsal muscle results in partial ptosis.

Rectus muscles

Four rectus muscles occupy medial, lateral, inferior, and superior positions as they pass from their origins posteriorly to their points of attachment on the anterior half of the eyeball (Fig. 8.92 and Table 8.8). They originate as a group from a common tendinous ring at the apex of the orbit and form a cone of muscles as they pass forward to their attachment on the eyeball.

Superior and inferior rectus muscles

The superior and inferior rectus muscles have complicated actions because the apex of the orbit, where the muscles originate, is medial to the central axis of the eyeball when looking directly forward:

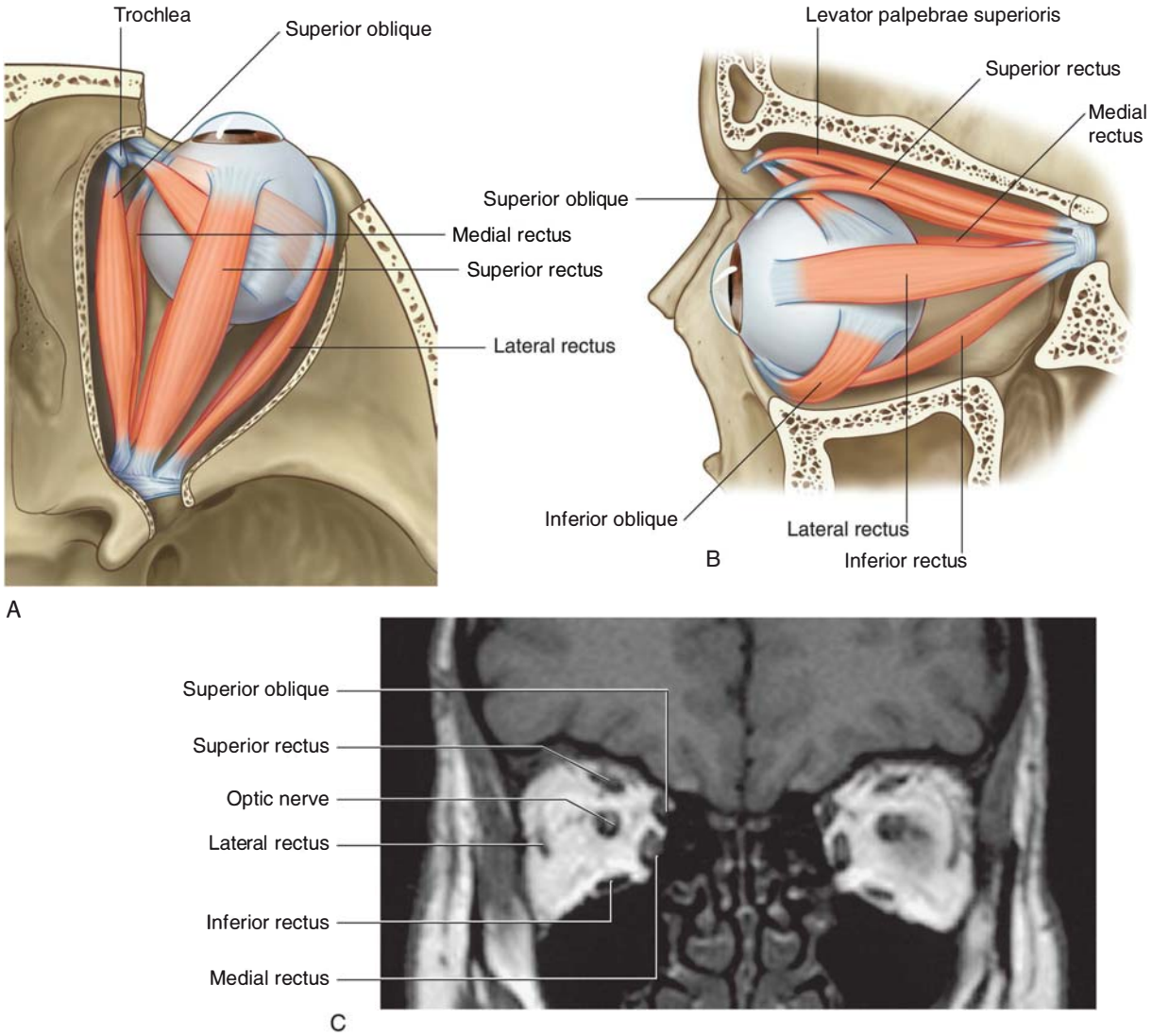


Fig. 8.92 Muscles of the eyeball. **A.** Superior view. **B.** Lateral view. **C.** Coronal magnetic resonance image through the eye.

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- The **superior rectus** originates from the superior part of the common tendinous ring above the optic canal.
- The **inferior rectus** originates from the inferior part of the common tendinous ring below the optic canal (Fig. 8.93).

As these muscles pass forward in the orbit to attach to the anterior half of the eyeball, they are also directed laterally (Fig. 8.92). Because of these orientations:

- Contraction of the superior rectus elevates, adducts, and internally rotates the eyeball (Fig. 8.94A).
- Contraction of the inferior rectus depresses, adducts, and externally rotates the eyeball (Fig. 8.94A).

The **superior branch** of the oculomotor nerve [III] innervates the superior rectus, and the **inferior branch** of the oculomotor nerve [III] innervates the inferior rectus.

To isolate the function of and to test the superior and inferior rectus muscles, a patient is asked to track a physician's finger laterally and then either upward or downward (Fig. 8.94B). The first movement brings the axis of the eyeball into alignment with the long axis of the superior and inferior rectus muscles. Moving the finger upward tests the superior rectus muscle and moving it downward tests the inferior rectus muscle (Fig. 8.94B).

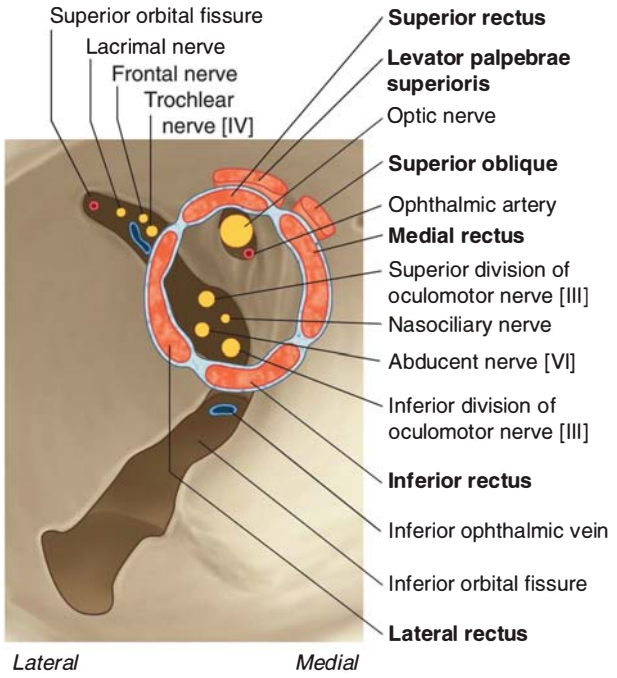


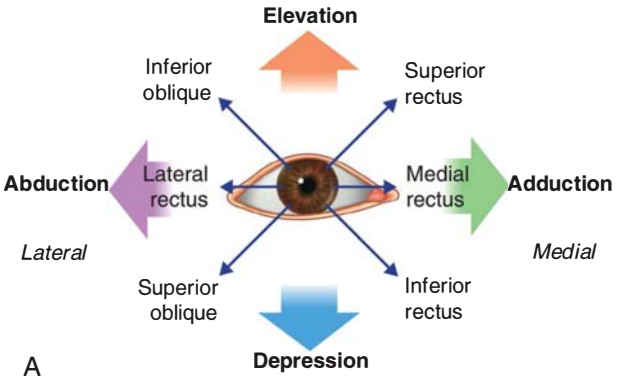
Fig. 8.93 Origins of muscles of the eyeball, coronal view.

Medial and lateral rectus muscles

The orientation and actions of the medial and lateral rectus muscles are more straightforward than those of the superior and inferior rectus muscles.

The **medial rectus** originates from the medial part of the common tendinous ring medial to and below the optic canal, whereas the **lateral rectus** originates from the lateral part of the common tendinous ring as the common tendinous ring bridges the superior orbital fissure (Fig. 8.93).

The medial and lateral rectus muscles pass forward and attach to the anterior half of the eyeball (Fig. 8.92).



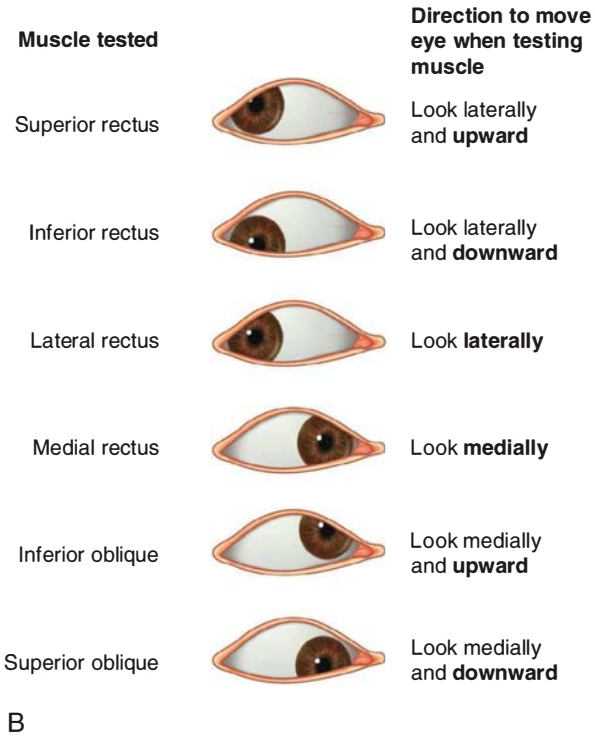


Fig. 8.94 Actions of muscles of the eyeball. **A.** Action of individual muscles (anatomical action). **B.** Movement of eye when testing specific muscle (clinical testing).

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Contraction of medial rectus adducts the eyeball, whereas contraction of lateral rectus abducts the eyeball (Fig. 8.94A).

The inferior branch of the oculomotor nerve [III] innervates the medial rectus, and the abducent nerve [VI] innervates the lateral rectus.

To isolate the function of and test the medial and lateral rectus muscles, a patient is asked to track a physician's finger medially and laterally, respectively, in the horizontal plane (Fig. 8.94B).

Oblique muscles

The oblique muscles are in the superior and inferior parts of the orbit, do not originate from the common tendinous ring, are angular in their approaches to the eyeball, and, unlike the rectus muscles, attach to the posterior half of the eyeball (Table 8.8).

Superior oblique

The superior oblique arises from the body of the sphenoid, superior and medial to the optic canal and medial to the origin of the levator palpebrae superioris (Figs. 8.92 and 8.93). It passes forward, along the medial border of the roof of the orbit, until it reaches a fibrocartilaginous pulley (the **trochlea**), which is attached to the trochlear fovea of the frontal bone.

The tendon of the superior oblique passes through the trochlea and turns laterally to cross the eyeball in a posterolateral direction. It continues deep to the superior rectus muscle and inserts into the outer posterior quadrant of the eyeball.

Contraction of the superior oblique therefore directs the pupil down and out (Fig. 8.94A).

The trochlear nerve [IV] innervates the superior oblique along its superior surface.

To isolate the function of and to test the superior oblique muscle, a patient is asked to track a physician's finger medially to bring the axis of the tendon of the muscle into alignment with the axis of the eyeball, and then to look down, which tests the muscle (Fig. 8.94B).

Inferior oblique

The inferior oblique is the only extrinsic muscle that does not take origin from the posterior part of the orbit. It arises from the medial side of the floor of the orbit, just posterior to the orbital rim, and is attached to the orbital surface of the maxilla just lateral to the nasolacrimal groove (Fig. 8.92).

The inferior oblique crosses the floor of the orbit in a posterolateral direction between the inferior rectus and the floor of the orbit, before inserting into the outer posterior quadrant just under the lateral rectus.

Contraction of the inferior oblique directs the pupil up and out (Fig. 8.94A).

The inferior branch of the oculomotor nerve innervates the inferior oblique.

To isolate the function of and to test the inferior oblique muscle, a patient is asked to track a physician's finger medially to bring the axis of the eyeball into alignment with the axis of the muscle and then to look up, which tests the muscle (Fig. 8.94B).

Extrinsic muscles and eyeball movements

Six of the seven extrinsic muscles of the orbit are directly involved in movements of the eyeball.

For each of the rectus muscles, the medial, lateral, inferior, and superior, and the superior and inferior obliques, a specific action or group of actions can be described (Table 8.8). However, these muscles do not act in isolation. They work as teams of muscles in the coordinated movement of the eyeball to position the pupil as needed.

For example, although the lateral rectus is the muscle primarily responsible for moving the eyeball laterally, it is assisted in this action by the superior and inferior oblique muscles.

In the clinic

Examination of the eye

Examination of the eye includes assessment of the visual capabilities, the extrinsic musculature and its function, and disease processes that may affect the eye in isolation or as part of the systemic process.

Examination of the eye includes tests for visual acuity, astigmatism, visual fields, and color interpretation (to exclude color blindness) in a variety of circumstances. The physician also assesses the retina, the optic nerve and its coverings, the lens, and the cornea.

The extrinsic muscles are supplied by the abducent nerve [VI], the trochlear nerve [IV], and the oculomotor nerve [III].

The extrinsic muscles work synergistically to provide appropriate and conjugate eye movement:

- lateral rectus—abducent nerve [VI],
- superior oblique—trochlear nerve [IV], and
- remainder—oculomotor nerve [III].

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In the clinic—cont'd

The eye may be affected in systemic diseases. Diabetes mellitus typically affects the eye and may cause cataracts, macular disease, and retinal hemorrhage, all impairing vision.

Occasionally unilateral paralysis of the extra-ocular muscles occurs and is due to brainstem injury or direct nerve injury, which may be associated with tumor compression or trauma. The paralysis of a muscle is easily demonstrated when the patient attempts to move the eye in the direction associated with normal action of that muscle. Typically the patient complains of double vision (diplopia).

Loss of innervation of the muscles around the eye

Loss of innervation of the orbicularis oculi by the facial nerve [VII] causes an inability to close the eyelids tightly, allowing the lower eyelid to droop away causing spillage of tears. This loss of tears allows drying of the conjunctiva, which may ulcerate, so allowing secondary infection.

Loss of innervation of the levator palpebrae superioris by oculomotor nerve [III] damage causes an inability of the superior eyelid to elevate, producing a ptosis. Usually, oculomotor nerve [III] damage is caused by severe head injury.

Loss of innervation of the superior tarsal muscle by sympathetic fibers causes a constant partial ptosis. Any lesion along the sympathetic trunk can induce this. An apical pulmonary malignancy should always be suspected because the ptosis may be part of Horner's syndrome (see "In the clinic" on p. 931).

Vessels

Arteries

The arterial supply to the structures in the orbit, including the eyeball, is by the ophthalmic artery (Fig. 8.95). This vessel is a branch of the internal carotid artery, given off immediately after the internal carotid artery leaves the cavernous sinus. The ophthalmic artery passes into the orbit through the optic canal with the optic nerve.

In the orbit the ophthalmic artery initially lies inferior and lateral to the optic nerve (Fig. 8.95). As it passes forward in the orbit, it crosses superior to the optic nerve and proceeds anteriorly on the medial side of the orbit.

In the orbit the ophthalmic artery gives off numerous branches as follows:

- the **lacrimal artery**, which arises from the ophthalmic artery on the lateral side of the optic nerve, and passes anteriorly on the lateral side of the orbit, supplying the lacrimal gland, muscles, the anterior ciliary branch to the eyeball, and the lateral sides of the eyelid;
- the **central retinal artery**, which enters the optic nerve, proceeds down the center of the nerve to the retina, and is clearly seen when viewing the retina with an ophthalmoscope—occlusion of this vessel or of the parent artery leads to blindness;
- the **long** and **short posterior ciliary arteries**, which are branches that enter the eyeball posteriorly, piercing the sclera, and supplying structures inside the eyeball;
- the **muscular arteries**, which are branches supplying the intrinsic muscles of the eyeball;
- the **supra-orbital artery**, which usually arises from the ophthalmic artery immediately after it has crossed the optic nerve, proceeds anteriorly, and exits the orbit through the supra-orbital foramen with the supra-orbital nerve—it supplies the forehead and scalp as it passes across these areas to the vertex of the skull;
- the **posterior ethmoidal artery**, which exits the orbit through the posterior ethmoidal foramen to supply the ethmoidal cells and nasal cavity;
- the **anterior ethmoidal artery**, which exits the orbit through the anterior ethmoidal foramen, enters the

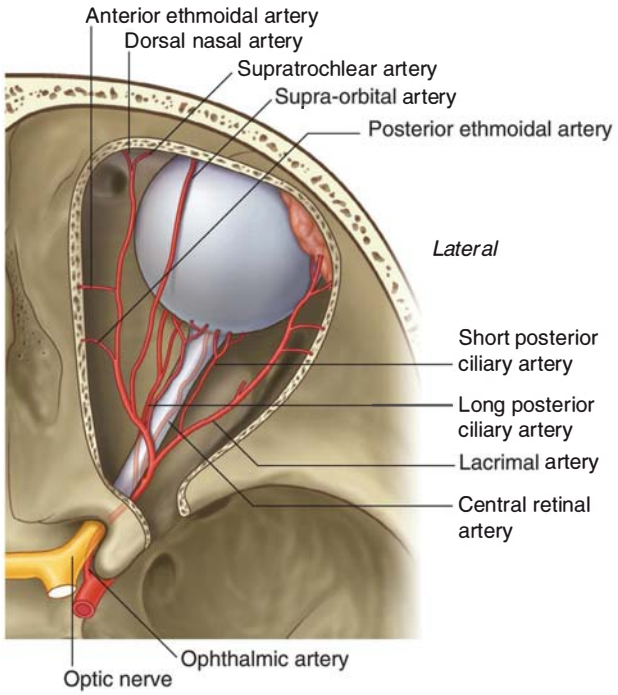


Fig. 8.95 Arterial supply to the orbit and eyeball.

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cranial cavity giving off the anterior meningeal branch, and continues into the nasal cavity supplying the septum and lateral wall, and ending as the dorsal nasal artery;

- the **medial palpebral arteries**, which are small branches supplying the medial area of the upper and lower eyelids;
- the **dorsal nasal artery**, which is one of the two terminal branches of the ophthalmic artery, leaves the orbit to supply the upper surface of the nose; and
- the **supratrochlear artery**, which is the other terminal branch of the ophthalmic artery and leaves the orbit with the supratrochlear nerve, supplying the forehead as it passes across it in a superior direction.

Veins

There are two venous channels in the orbit, the superior and inferior ophthalmic veins (Fig. 8.96).

The **superior ophthalmic vein** begins in the anterior area of the orbit as connecting veins from the supra-orbital vein and the angular vein join together. It passes across the superior part of the orbit, receiving tributaries from the companion veins to the branches of the ophthalmic artery and veins draining the posterior part of the eyeball. Posteriorly, it leaves the orbit through the superior orbital fissure and enters the cavernous sinus.

The **inferior ophthalmic vein** is smaller than the superior ophthalmic vein, begins anteriorly, and passes across the inferior part of the orbit. It receives various tributaries from muscles and the posterior part of the eyeball as it crosses the orbit.

The inferior ophthalmic vein leaves the orbit posteriorly by:

- joining with the superior ophthalmic vein,
- passing through the superior orbital fissure on its own to join the cavernous sinus, or
- passing through the inferior orbital fissure to join with the pterygoid plexus of veins in the infratemporal fossa.

Because the ophthalmic veins communicate with the cavernous sinus, they act as a route by which infections can spread from outside to inside the cranial cavity.

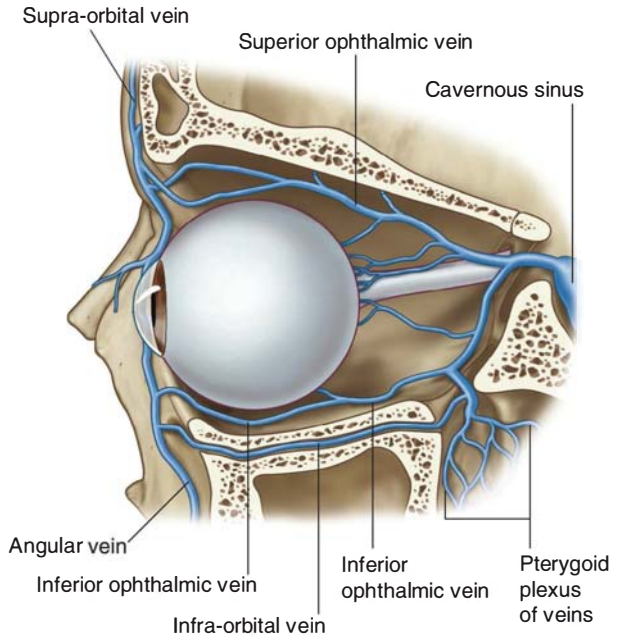


Fig. 8.96 Venous drainage of the orbit and eyeball.

Innervation

Numerous nerves pass into the orbit and innervate structures within its bony walls. They include the optic nerve [II], the oculomotor nerve [III], the trochlear nerve [IV], the abducent nerve [VI], and autonomic nerves. Other nerves such as the ophthalmic nerve [V_1] innervate orbital structures and then travel out of the orbit to innervate other regions.

Optic nerve

The optic nerve [II] is not a true cranial nerve, but rather an extension of the brain carrying afferent fibers from the retina of the eyeball to the visual centers of the brain. The optic nerve is surrounded by the cranial meninges, including the subarachnoid space, which extend as far forward as the eyeball.

Any increase in intracranial pressure therefore results in increased pressure in the subarachnoid space surrounding the optic nerve. This may impede venous return along the retinal veins, causing edema of the optic disc (papilledema), which can be seen when the retina is examined using an ophthalmoscope.

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The optic nerve leaves the orbit through the optic canal (Fig. 8.97). It is accompanied in the optic canal by the ophthalmic artery.

Oculomotor nerve

The oculomotor nerve [III] leaves the anterior surface of the brainstem between the midbrain and the pons. It passes forward in the lateral wall of the cavernous sinus.

Just before entering the orbit the oculomotor nerve [III] divides into superior and inferior branches (Fig. 8.98). These branches enter the orbit through the superior orbital fissure, lying within the common tendinous ring (Fig. 8.97).

Inside the orbit the small superior branch passes upward over the lateral side of the optic nerve to innervate the superior rectus and levator palpebrae superioris muscles (Fig. 8.98).

The large inferior branch divides into three branches:

- one passing below the optic nerve as it passes to the medial side of the orbit to innervate the medial rectus muscle,
- a second descending to innervate the inferior rectus muscle, and
- the third descending as it runs forward along the floor of the orbit to innervate the inferior oblique muscle (Fig. 8.98).

As the third branch descends, it gives off the **branch to the ciliary ganglion**. This is the parasympathetic root to the ciliary ganglion and carries preganglionic parasympathetic fibers that will synapse in the ciliary ganglion with postganglionic parasympathetic fibers. The postganglionic fibers are distributed to the eyeball through short ciliary nerves and innervate the sphincter pupillae and ciliary muscles.

Trochlear nerve

The trochlear nerve [IV] arises from the posterior surface of the midbrain, and passes around the midbrain to enter the edge of the tentorium cerebelli. It continues on an intradural path arriving in and passing through the lateral wall of the cavernous sinus just below the oculomotor nerve [III].

Just before entering the orbit, the trochlear nerve ascends, passing across the oculomotor nerve [III] and entering the orbit through the superior orbital fissure above the common tendinous ring (Fig. 8.97). In the orbit

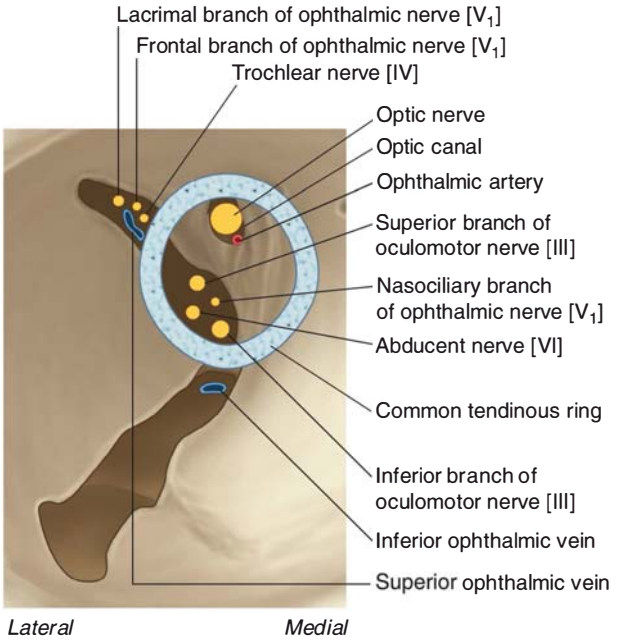


Fig. 8.97 Innervation of the orbit and eyeball.

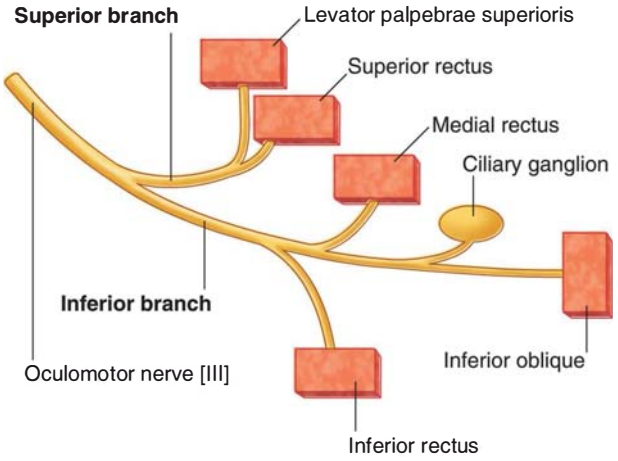


Fig. 8.98 Oculomotor nerve [III] and its divisions.

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the trochlear nerve [IV] ascends and turns medially, crossing above the levator palpebrae superioris muscle to enter the upper border of the superior oblique muscle (Fig. 8.99).

Abducent nerve

The abducent nerve [VI] arises from the brainstem between the pons and medulla. It enters the dura covering the clivus and continues in a dural canal until it reaches the cavernous sinus.

The abducent nerve enters the cavernous sinus and runs through the sinus lateral to the internal carotid artery. It passes out of the sinus and enters the orbit through the superior orbital fissure within the common tendinous ring (Fig. 8.97). Once in the orbit it courses laterally to supply the lateral rectus muscle.

Postganglionic sympathetic fibers

Preganglionic sympathetic fibers arise from the upper segments of the thoracic spinal cord, mainly T1. They enter the sympathetic chain through white rami communicantes, and ascend to the **superior cervical ganglion** where they synapse with postganglionic sympathetic fibers.

The postganglionic fibers are distributed along the internal carotid artery and its branches.

The postganglionic sympathetic fibers destined for the orbit travel with the ophthalmic artery. Once in the orbit the fibers are distributed to the eyeball either by:

- passing through the ciliary ganglion, without synapsing, and joining the short ciliary nerves, which pass from the ganglion to the eyeball; or
- passing through long ciliary nerves to reach the eyeball.

In the eyeball postganglionic sympathetic fibers innervate the dilator pupillae muscle.

Ophthalmic nerve [V_1]

The ophthalmic nerve [V_1] is the smallest and most superior of the three divisions of the trigeminal nerve. This purely sensory nerve receives input from structures in the orbit and from additional branches on the face and scalp.

Leaving the trigeminal ganglion, the ophthalmic nerve [V_1] passes forward in the lateral wall of the cavernous sinus inferior to the trochlear [IV] and oculomotor [III] nerves. Just before it enters the orbit it divides into three branches—the nasociliary, lacrimal, and frontal nerves (Fig. 8.100). These branches enter the orbit through the superior orbital fissure with the frontal and lacrimal nerves outside the common tendinous ring, and the nasociliary nerve within the common tendinous ring (Fig. 8.97).

Lacrimal nerve

The lacrimal nerve is the smallest of the three branches of the ophthalmic nerve [V_1]. Once in the orbit it passes forward along the upper border of the lateral rectus muscle

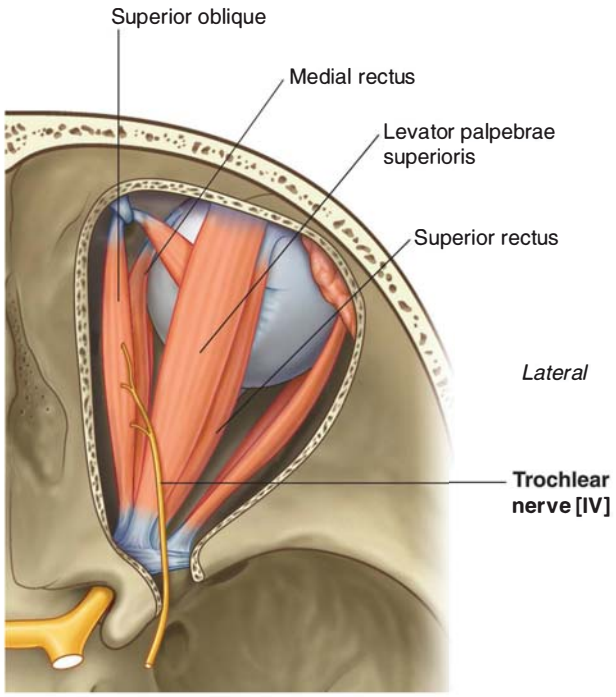


Fig. 8.99 Trochlear nerve [IV] in the orbit.

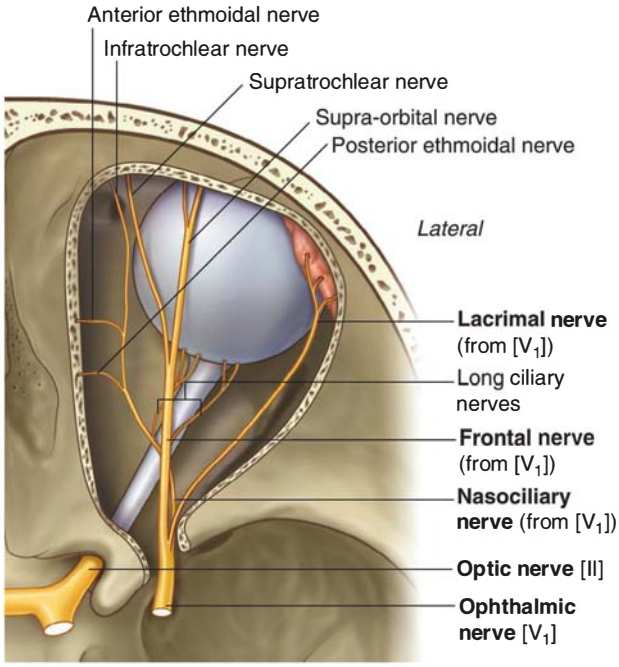


Fig. 8.100 Ophthalmic nerve [V_1] and its divisions.

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(Fig. 8.101). It receives a branch from the zygomaticotemporal nerve, which carries parasympathetic and sympathetic postganglionic fibers for distribution to the lacrimal gland.

Reaching the anterolateral aspect of the orbit, the lacrimal nerve supplies the lacrimal gland, conjunctiva, and lateral part of the upper eyelid.

Frontal nerve

The frontal nerve is the largest branch of the ophthalmic nerve [V_1] and receives sensory input from areas outside the orbit. Exiting the superior orbital fissure, this branch passes forward between the levator palpebrae superioris and the periorbita on the roof of the orbit (Fig. 8.97). About midway across the orbit it divides into its two terminal branches—the supra-orbital and supratrochlear nerves (Figs. 8.100 and 8.101):

- The **supratrochlear nerve** continues forward in an anteromedial direction, passing above the trochlea, exits the orbit medial to the supra-orbital foramen, and supplies the conjunctiva and skin of the upper eyelid and the skin on the lower medial part of the forehead.
- The **supra-orbital nerve** is the larger of the two branches, continues forward, passing between the levator palpebrae superioris muscle and the periorbita covering the roof of the orbit (Fig. 8.101), exits the orbit through the supra-orbital notch and ascends across the forehead and scalp, supplying the upper eyelid and conjunctiva, the forehead, and as far posteriorly as the middle of the scalp.

Nasociliary nerve

The nasociliary nerve is intermediate in size between the frontal and lacrimal nerves and is usually the first branch from the ophthalmic nerve (Fig. 8.100). It is most deeply placed in the orbit, entering the area within the common tendinous ring between the superior and inferior branches of the oculomotor nerve [III] (see Fig. 8.97).

Once in the orbit, the nasociliary nerve crosses the superior surface of the optic nerve as it passes in a medial direction below the superior rectus muscle (Figs. 8.100 and 8.102). Its first branch, the **communicating branch**

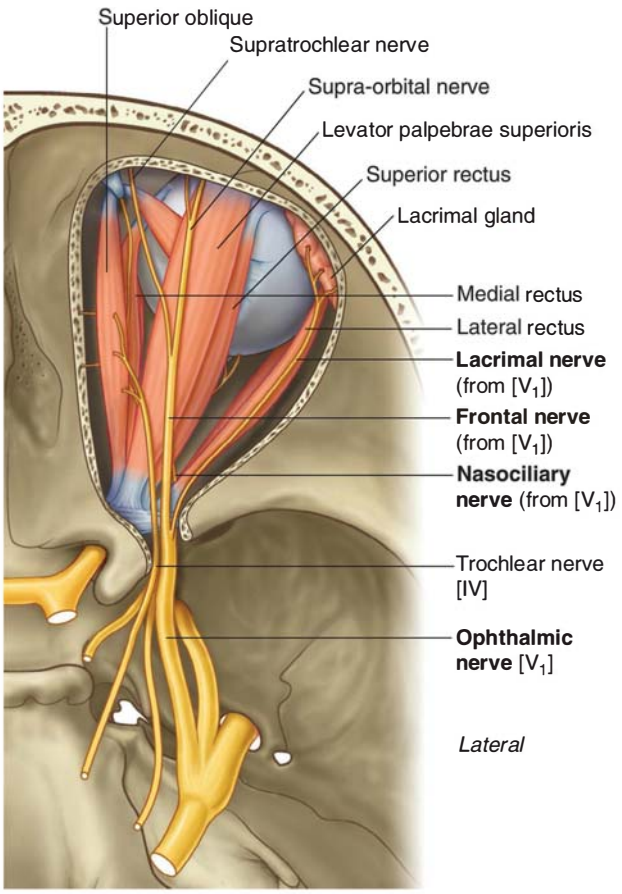


Fig. 8.101 Relationship of the ophthalmic nerve [V_1] and its divisions to the muscles of the eyeball.

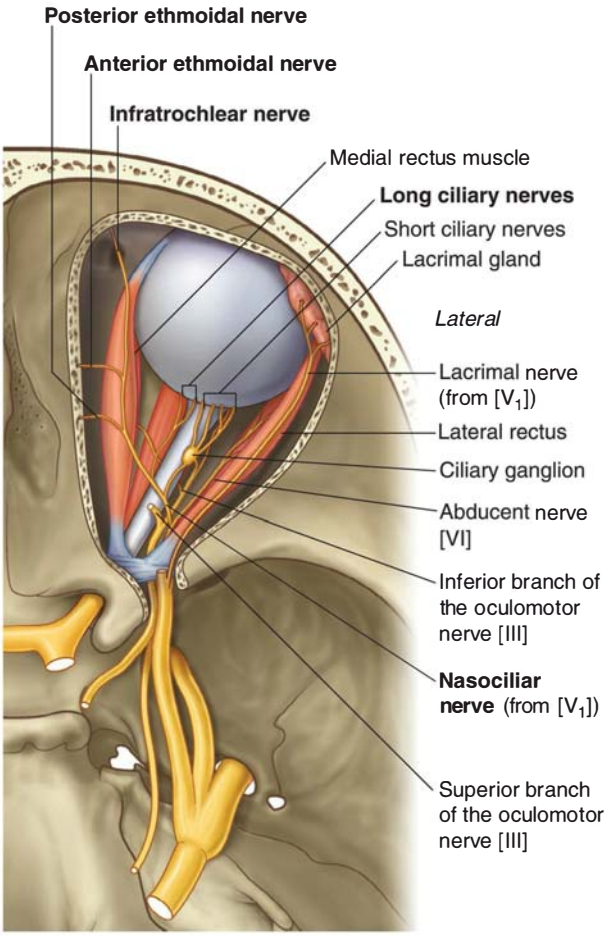


Fig. 8.102 Course of the nasociliary nerve (from [V_1]) in the orbit.

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with the ciliary ganglion (**sensory root to the ciliary ganglion**), is given off early in its path through the orbit.

The nasociliary nerve continues forward along the medial wall of the orbit, between the superior oblique and the medial rectus muscles, giving off several branches (Fig. 8.102). These include:

- the **long ciliary nerves**, which are sensory to the eyeball but may also contain sympathetic fibers for pupillary dilation;
- the **posterior ethmoidal nerve**, which exits the orbit through the posterior ethmoidal foramen to supply posterior ethmoidal cells and the sphenoidal sinus;
- the **infratrochlear nerve**, which distributes to the medial part of the upper and lower eyelids, the lacrimal sac, and skin of the upper half of the nose; and
- the **anterior ethmoidal nerve**, which exits the orbit through the anterior ethmoidal foramen to supply the anterior cranial fossa, nasal cavity, and skin of the lower half of the nose (Fig. 8.102).

Ciliary ganglion

The ciliary ganglion is a parasympathetic ganglion of the oculomotor nerve [III]. It is associated with the nasociliary branch of the ophthalmic nerve [V_1] and is the site where preganglionic and postganglionic parasympathetic neurons synapse as fibers from this part of the autonomic division of the PNS make their way to the eyeball. The ciliary ganglion is also traversed by postganglionic sympathetic fibers and sensory fibers as they travel to the eyeball.

The ciliary ganglion is a very small ganglion, in the posterior part of the orbit immediately lateral to the optic nerve and between the optic nerve and the lateral rectus muscle (Fig. 8.102). It is usually described as receiving at least two, and possibly three, branches or roots from other nerves in the orbit.

Parasympathetic root

As the inferior branch of the oculomotor nerve [III] passes the area of the ciliary ganglion, it sends a branch to the ganglion (the parasympathetic root). The parasympathetic branch carries preganglionic parasympathetic fibers, which enter the ganglion and synapse with postganglionic parasympathetic fibers within the ganglion (Fig. 8.103).

The postganglionic parasympathetic fibers leave the ganglion through short ciliary nerves, which enter the posterior aspect of the eyeball around the optic nerve.

In the eyeball the parasympathetic fibers innervate:

- the **sphincter pupillae muscle**, responsible for pupillary constriction, and

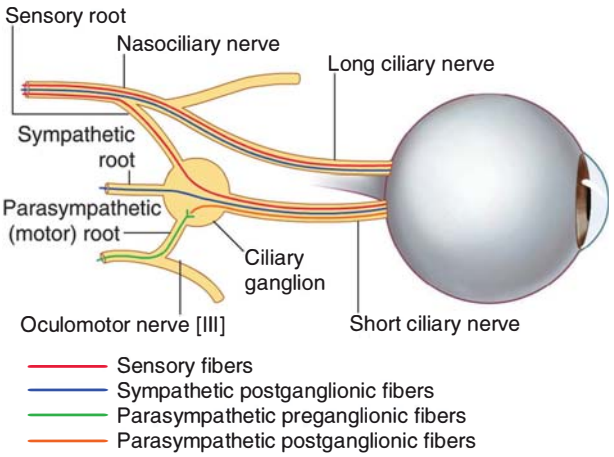


Fig. 8.103 Ciliary ganglion.

- the **ciliary muscle**, responsible for accommodation of the lens of the eye for near vision.

Sensory root

A second branch (the sensory root), passes from the nasociliary nerve to the ganglion (Fig. 8.103). This branch enters the posterosuperior aspect of the ganglion, and carries sensory fibers, which pass through the ganglion and continue along the short ciliary nerves to the eyeball. These fibers are responsible for sensory innervation to all parts of the eyeball; however, the sympathetic fibers also may take alternative routes to the eyeball.

Sympathetic root

The third branch to the ciliary ganglion is the most variable. This branch, when present, is the sympathetic root and contains postganglionic sympathetic fibers from the superior cervical ganglion (Fig. 8.103). These fibers travel up the internal carotid artery, leave the plexus surrounding the artery in the cavernous sinus, and enter the orbit through the common tendinous ring. In the orbit they enter the posterior aspect of the ciliary ganglion, cross the ganglion, and continue along the short ciliary nerves to the eyeball; however, the sympathetic fibers also may take alternative routes to the eyeball.

Sympathetic fibers to the eyeball may not enter the ganglion as a separate sympathetic root. Rather, the postganglionic sympathetic fibers may leave the plexus associated with the internal carotid artery in the cavernous sinus, join the ophthalmic nerve [V_1], and course into the ciliary ganglion in the sensory root from the nasociliary nerve. In addition, the sympathetic fibers carried in the nasociliary nerve may not enter the ganglion at all and may course directly into the eyeball in the long ciliary nerves (Fig.

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8.103). Whatever their path, postganglionic sympathetic fibers reach the eyeball and innervate the dilator pupillae muscle.

Eyeball

The globe-shaped eyeball occupies the anterior part of the orbit. Its rounded shape is disrupted anteriorly, where it bulges outward. This outward projection represents about one-sixth of the total area of the eyeball and is the transparent cornea (Fig. 8.104).

Posterior to the cornea and in order from front to back are the anterior chamber, the iris and pupil, the posterior chamber, the lens, the postremal (vitreous) chamber, and the retina.

Anterior and posterior chambers

The **anterior chamber** is the area directly posterior to the cornea and anterior to the colored part of the eye (**iris**). The central opening in the iris is the **pupil**. Posterior to the iris and anterior to the lens is the smaller **posterior chamber**.

The anterior and posterior chambers are continuous with each other through the pupillary opening. They are filled with a fluid (**aqueous humor**), which is secreted into the posterior chamber, flows into the anterior chamber through the pupil, and is absorbed into the **scleral venous sinus** (the canal of Schlemm), which is a circular venous channel at the junction between the cornea and the iris (Fig. 8.104).

The aqueous humor supplies nutrients to the avascular cornea and lens and maintains the intra-ocular pressure. If the normal cycle of its production and absorption is disturbed so that the amount of fluid increases, intra-ocular pressure will increase. This condition (glaucoma) can lead to a variety of visual problems.

Lens and vitreous humor

The **lens** separates the anterior one-fifth of the eyeball from the posterior four-fifths (Fig. 8.104). It is a transparent, biconvex elastic disc attached circumferentially to muscles associated with the outer wall of the eyeball. This lateral attachment provides the lens with the ability to change its refractive ability to maintain visual acuity. The clinical term for opacity of the lens is a cataract.

The posterior four-fifths of the eyeball, from the lens to the retina, is occupied by the postremal (vitreous) chamber

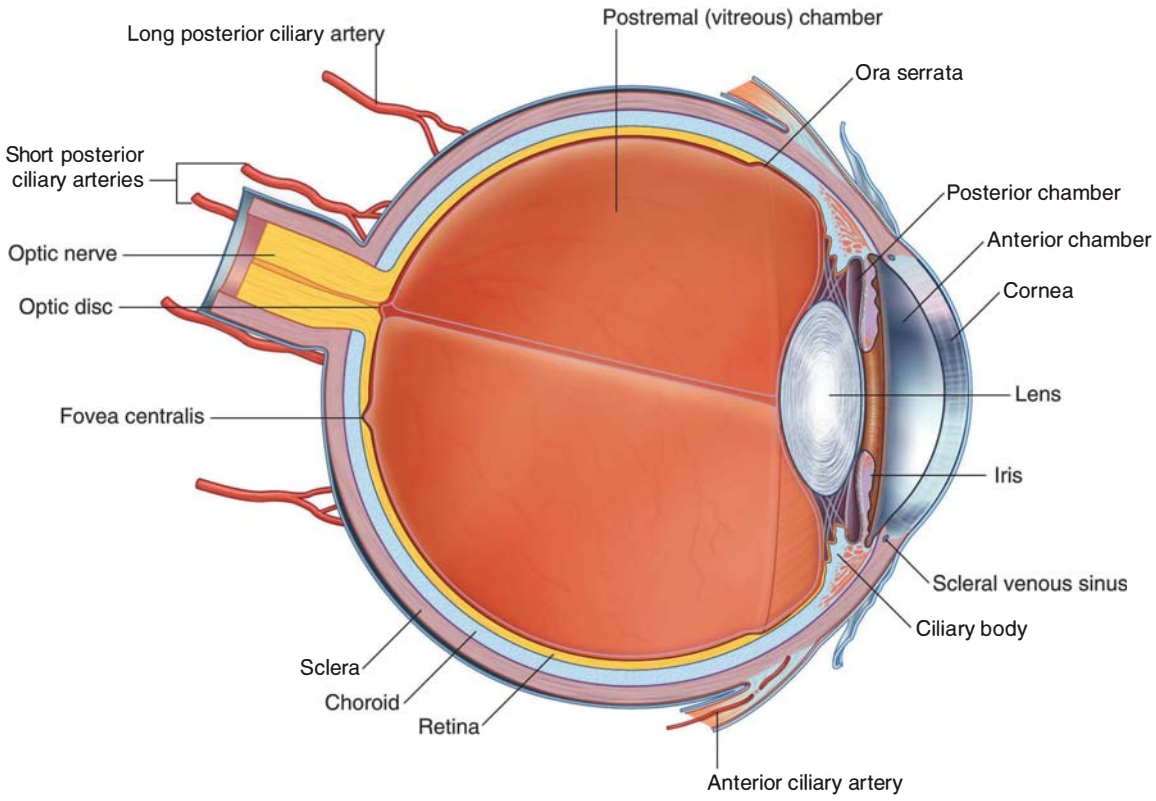


Fig. 8.104 Eyeball.

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(Fig. 8.104). This segment is filled with a transparent, gelatinous substance—the **vitreous body** (**vitreous humor**). This substance, unlike aqueous humor, cannot be replaced.

Walls of the eyeball

Surrounding the internal components of the eyeball are the walls of the eyeball. They consist of three layers: an outer fibrous layer, a middle vascular layer, and an inner retinal layer (Fig. 8.104).

- The outer fibrous layer consists of the sclera posteriorly and the cornea anteriorly.
- The middle vascular layer consists of the **choroid** posteriorly and is continuous with the ciliary body and iris anteriorly.
- The inner layer consists of the optic part of the **retina** posteriorly and the nonvisual retina that covers the internal surface of the ciliary body and iris anteriorly.

Vessels

Arterial supply

The arterial supply to the eyeball is from several sources:

- The short posterior ciliary arteries are branches from the ophthalmic artery that pierce the sclera around the optic nerve and enter the choroid layer (Fig. 8.104).
- The long posterior ciliary arteries, usually two, enter the sclera on the medial and lateral sides of the optic nerve and proceed anteriorly in the choroid layer to anastomose with the anterior ciliary arteries.
- The anterior ciliary arteries are branches of the arteries supplying the muscles (Fig. 8.104)—as the muscles attach to the sclera, these arteries pierce the sclera to anastomose with the long posterior ciliary arteries in the choroid layer.
- The central retinal artery that has traversed the optic nerve and enters the area of the retina at the optic disc.

Venous drainage

Venous drainage of the eyeball is primarily related to drainage of the choroid layer. Four large veins (the **vorticose veins**) are involved in this process. They exit through the sclera from each of the posterior quadrants of the eyeball and enter the superior and inferior ophthalmic veins. There is also a central retinal vein accompanying the central retinal artery.

In the clinic

Glaucoma

Intraocular pressure will rise if the normal cycle of aqueous humor fluid production and absorption is disturbed so that the amount of fluid increases. This condition is glaucoma and can lead to a variety of visual problems including blindness, which results from compression of the retina and its blood supply.

In the clinic

Cataracts

With increasing age and in certain disease states the lens of the eye becomes opaque. Increasing opacity results in increasing visual impairment. A common operation is excision of the cloudy lens and replacement with a new man-made lens.

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In the clinic

Ophthalmoscopy

Direct visualization of the postremal (vitreous) chamber of the eye is possible in most clinical settings. It is achieved using an ophthalmoscope, which is a small battery-operated light with a tiny lens that allows direct visualization of the postremal (vitreous) chamber and the posterior wall of the eye through the pupil and the lens. It is sometimes necessary to place a drug directly onto the eye to dilate the pupil for better visualization.

The optic nerve is easily seen. The typical four branches of the central retinal artery and the fovea are also seen.

Using ophthalmoscopy the physician can look for diseases of the optic nerve, vascular abnormalities, and changes within the retina (Fig. 8.105).

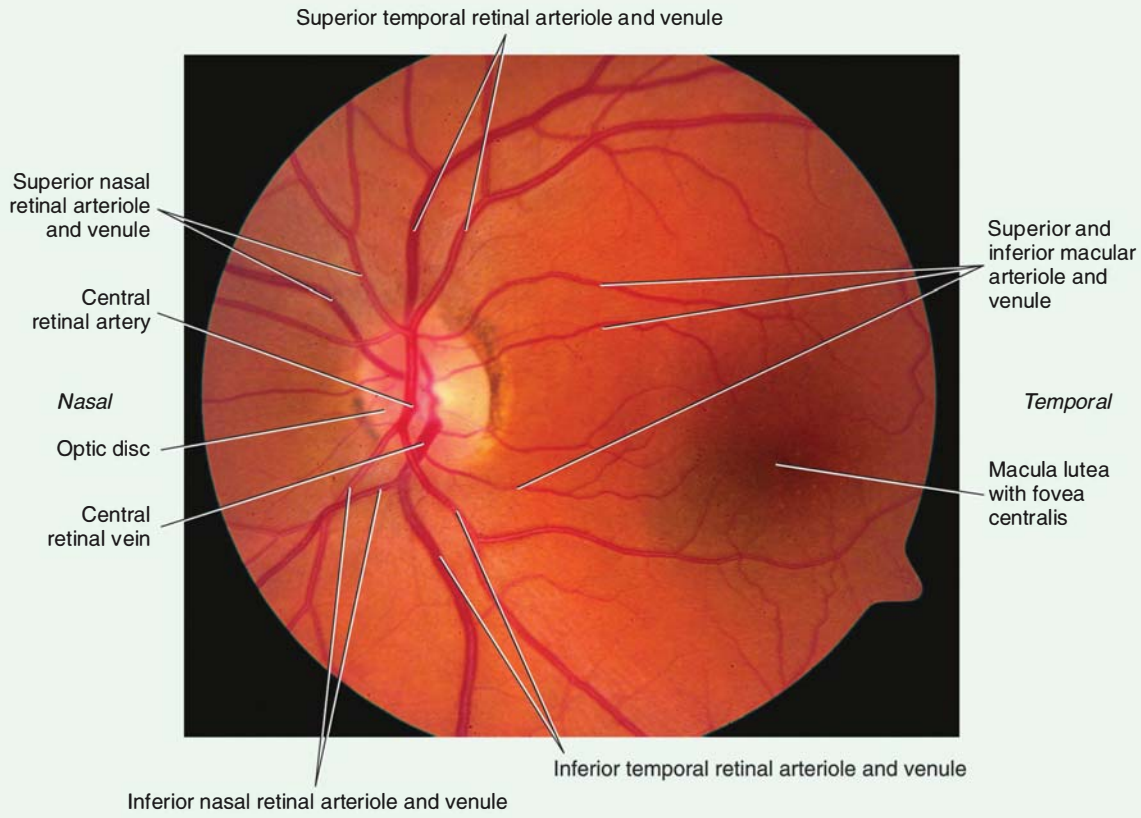


Fig. 8.105 Ophthalmoscopic view of posterior chamber of the right eye.

Fibrous layer of the eyeball

The fibrous layer of the eyeball consists of two components—the sclera covers the posterior and lateral parts of the eyeball, about five-sixths of the surface, and the cornea covers the anterior part (Fig. 8.104).

Sclera

The sclera is an opaque layer of dense connective tissue that can be seen anteriorly through its conjunctival covering as the "white of the eye." It is pierced by numerous vessels and nerves, including the optic nerve posteriorly and provides attachment for the various muscles involved in eyeball movements.

The fascial sheath of the eyeball covers the surface of the sclera externally from the entrance of the optic nerve to the corneoscleral junction while internally the surface of the sclera is loosely attached to the choroid of the vascular layer.

Cornea

Continuous with the sclera anteriorly is the transparent cornea. It covers the anterior one-sixth of the surface of the eyeball and, being transparent, allows light to enter the eyeball.

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Vascular layer of the eyeball

The vascular layer of the eyeball consists of three continuous parts—the choroid, the ciliary body, and the iris from posterior to anterior (Fig. 8.104).

Choroid

The choroid is posterior and represents approximately two-thirds of the vascular layer. It is a thin, highly vascular, pigmented layer consisting of smaller vessels adjacent to the retina and larger vessels more peripherally. It is firmly attached to the retina internally and loosely attached to the sclera externally.

Ciliary body

Extending from the anterior border of the choroid is the ciliary body (Fig. 8.104). This triangular-shaped structure, between the choroid and the iris, forms a complete ring around the eyeball. Its components include the ciliary muscle and the ciliary processes (Fig. 8.106).

The **ciliary muscle** consists of smooth muscle fibers arranged longitudinally, circularly, and radially. Controlled by parasympathetics traveling to the orbit in the oculomotor nerve [III], these muscle fibers, on contraction, decrease the size of the ring formed by the ciliary body.

The **ciliary processes** are longitudinal ridges projecting from the inner surface of the ciliary body (Fig. 8.106). Extending from them are **zonular fibers** attached to the lens of the eyeball, which suspend the lens in its proper position and collectively form the **suspensory ligament of the lens**.

Contraction of the ciliary muscle decreases the size of the ring formed by the ciliary body. This reduces tension on the suspensory ligament of the lens. The lens therefore becomes more rounded (relaxed) resulting in accommodation of the lens for near vision.

Ciliary processes also contribute to the formation of aqueous humor.

Iris

Completing the vascular layer of the eyeball anteriorly is the iris (Fig. 8.104). This circular structure, projecting outward from the ciliary body, is the colored part of the eye with a central opening (the pupil). Controlling the size of the pupil are smooth muscle fibers within the iris (Fig. 8.106):

- Fibers arranged in a circular pattern make up the **sphincter pupillae muscle** (Table 8.9), which is innervated by parasympathetics—contraction of its fibers decreases or constricts the pupillary opening.

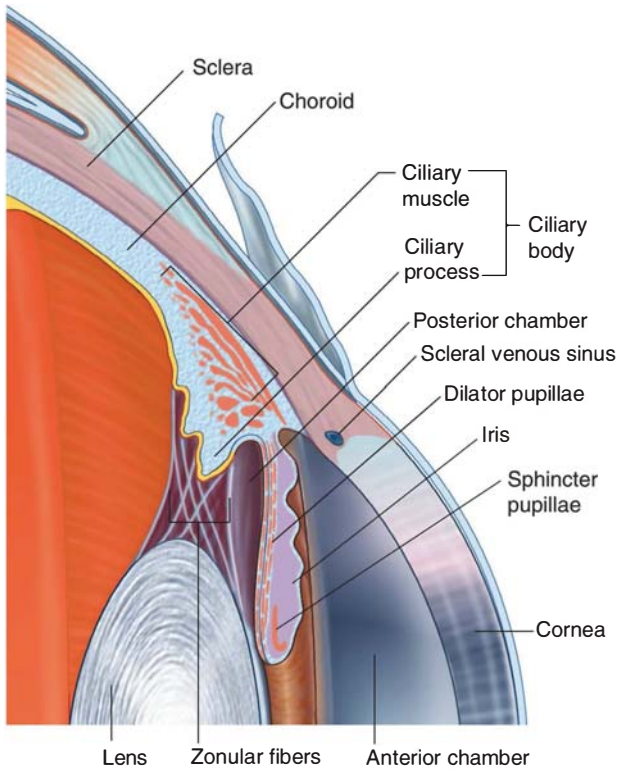


Fig. 8.106 Ciliary body.

Table 8.9 Intrinsic muscles of the eye

Muscle	Location	Innervation	Function
Ciliary	Muscle fibers in the ciliary body	Parasympathetics from the oculomotor nerve [III]	Constricts ciliary body, relaxes tension on lens, lens becomes more rounded
Sphincter pupillae	Circularly arranged fibers in the iris	Parasympathetics from the oculomotor nerve [III]	Constricts pupil
Dilator pupillae	Radially arranged fibers in the iris	Sympathetics from the superior cervical ganglion (T1)	Dilates pupil

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- Fibers arranged in a radial pattern make up the **dilator pupillae muscle**, which is innervated by sympathetics—contraction of its fibers increases or dilates the pupillary opening.

Inner layer of the eyeball

The inner layer of the eyeball is the retina (Fig. 8.104). It consists of two parts. Posteriorly and laterally is the **optic part of the retina**, which is sensitive to light, and anteriorly is the **nonvisual part**, which covers the internal surface of the ciliary body and the iris. The junction between these parts is an irregular line (the **ora serrata**).

Optic part of the retina

The optic part of the retina consists of two layers, an outer pigmented layer and an inner neural layer:

- The **pigmented layer** is firmly attached to the choroid and continues anteriorly over the internal surface of the ciliary body and iris.
- The **neural layer**, which can be further subdivided into its various neural components, is only attached to the pigmented layer around the optic nerve and at the ora serrata.

It is the neural layer that separates in the case of a detached retina.

Several obvious features are visible on the posterior surface of the optic part of the retina.

The **optic disc** is where the optic nerve leaves the retina (Fig. 8.105). It is lighter than the surrounding retina and branches of the central retinal artery spread from this point outward to supply the retina. As there are no light-sensitive receptor cells in the optic disc, it is referred to as a blind spot in the retina.

Lateral to the optic disc a small area with a hint of yellowish coloration is the **macula lutea** with its central depression, the **fovea centralis** (Fig. 8.105). This is the thinnest area of the retina and visual sensitivity here is higher than elsewhere in the retina because it has fewer **rods** (light-sensitive receptor cells that function in dim light and are insensitive to color) and more **cones** (light-sensitive receptor cells that respond to bright light and are sensitive to color).

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In the clinic

High-definition optical coherence tomography

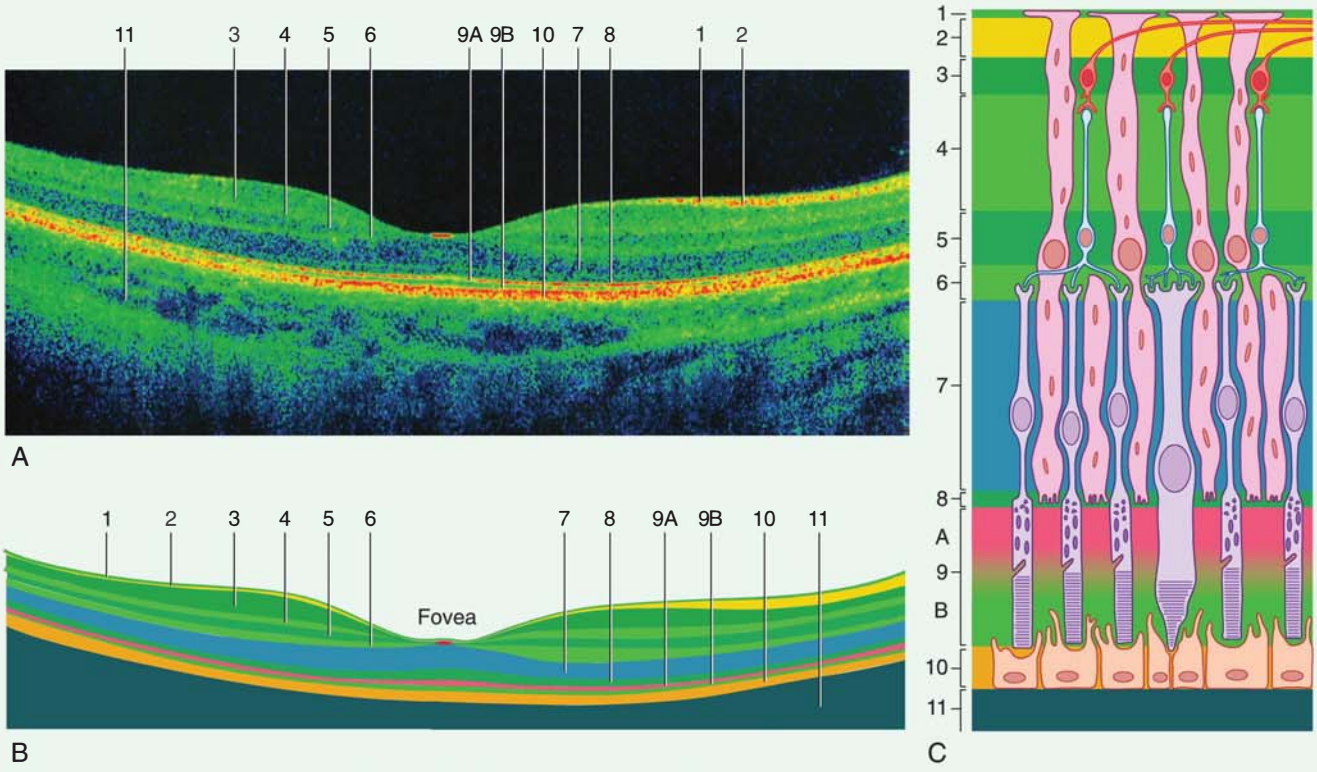
High-definition optical coherence tomography (HD-OCT) (Fig. 8.107) is a procedure used to obtain subsurface images of translucent or opaque materials. It is similar to ultrasound, except that it uses light instead of sound to produce high-resolution cross-sectional images. It is especially useful in the diagnosis and management of optic nerve and retinal diseases.

Epiretinal membrane

An epiretinal membrane (Fig. 8.108) is a thin sheet of fibrous tissue that develops on the surface of the retina in the area of the macula and can cause visual problems. If the visual problems are significant, surgical removal of the membrane may be necessary.

(continues)

In the clinic—cont'd



1 Internal limiting membrane
2 Nerve fiber layer
3 Ganglion cell layer
4 Inner plexiform layer
5 Inner nuclear layer
6 Outer plexiform layer
7 Outer nuclear layer
8 External limiting membrane
9A Photoreceptor inner layer
9B Photoreceptor outer layer
10 Pigment epithelium
11 Choroid

Fig. 8.107 Layers of the retina in a healthy eye. **A.** HD-OCT scan of a healthy eye. **B.** Schematic indicating the layers of the retina on an HD-OCT scan of a healthy eye. **C.** Diagram illustrating the layers of the retina.

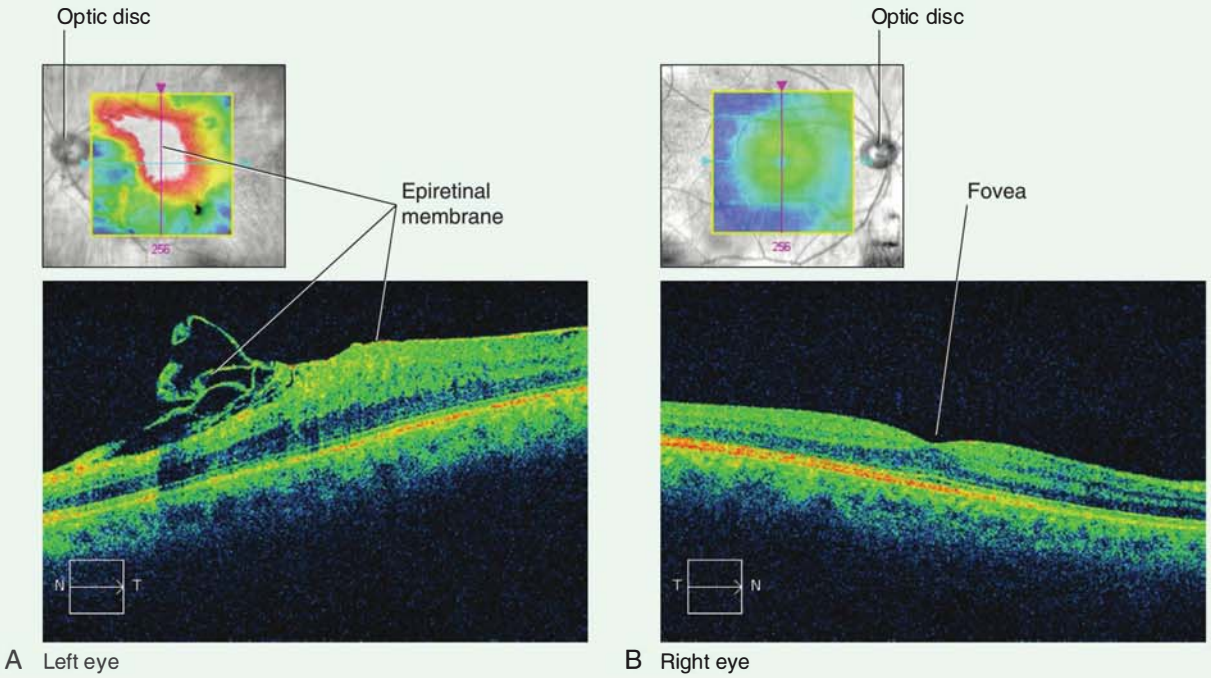


Fig. 8.108 High-definition optical coherence tomography (HD-OCT). **A.** Diseased eye. **B.** Healthy eye.

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EAR

The ear is the organ of hearing and balance. It has three parts (Fig. 8.109):

- The first part is the **external ear** consisting of the part attached to the lateral aspect of the head and the canal leading inward.
- The second part is the **middle ear**—a cavity in the petrous part of the temporal bone bounded laterally, and separated from the external canal, by a membrane and connected internally to the pharynx by a narrow tube.
- The third part is the **internal ear** consisting of a series of cavities within the petrous part of the temporal bone between the middle ear laterally and the internal acoustic meatus medially.

The internal ear converts the mechanical signals received from the middle ear, which start as sound captured by the external ear, into electrical signals to transfer information to the brain. The internal ear also contains receptors that detect motion and position.

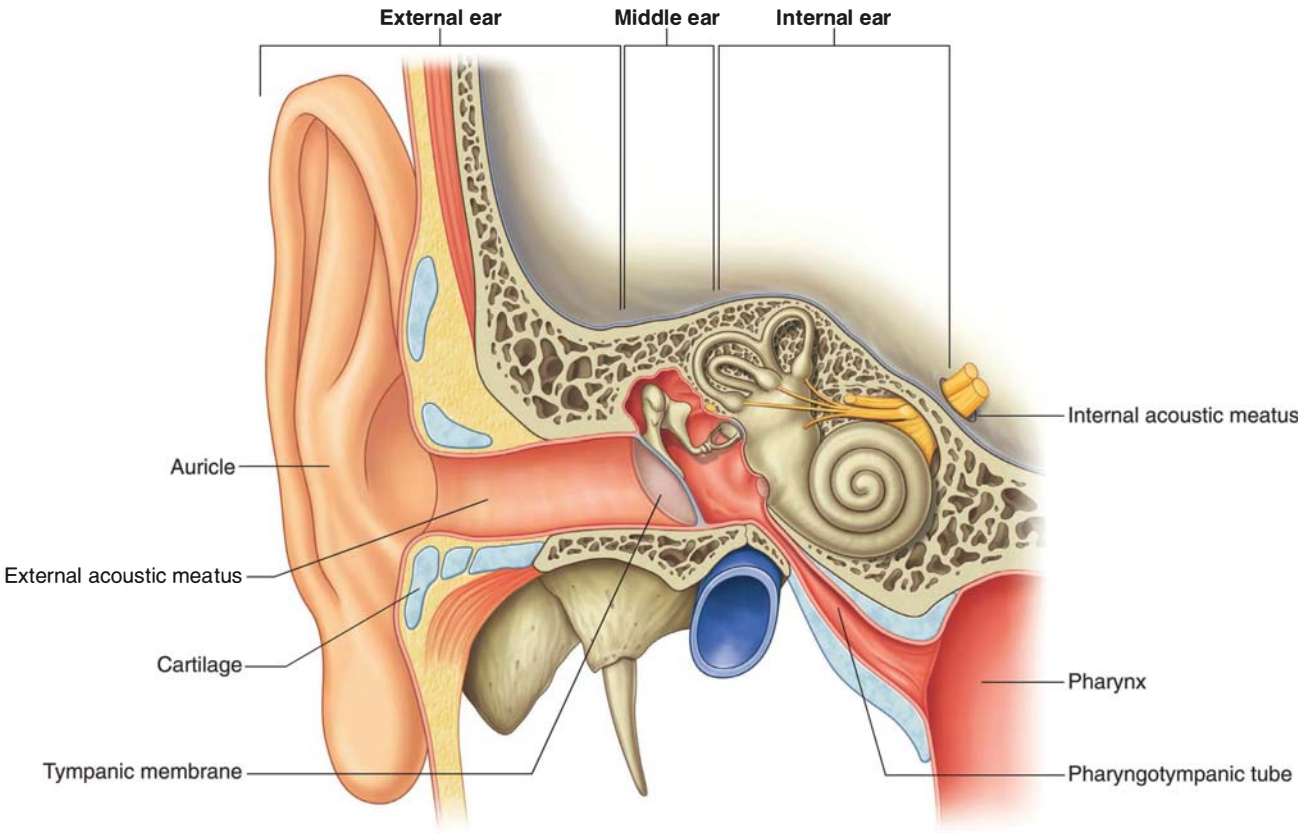


Fig. 8.109 Right ear.

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External ear

The external ear consists of two parts. The part projecting from the side of the head is the **auricle** (**pinna**) and the canal leading inward is the **external acoustic meatus**.

Auricle

The auricle is on the side of the head and assists in capturing sound. It consists of cartilage covered with skin and arranged in a pattern of various elevations and depressions (Fig. 8.110).

The large outside rim of the auricle is the **helix**. It ends inferiorly at the fleshy lobule, the only part of the auricle not supported by cartilage.

The hollow center of the auricle is the **concha of the auricle**. The external acoustic meatus leaves from the depths of this area.

Just anterior to the opening of the external acoustic meatus, in front of the concha, is an elevation (the **tragus**). Opposite the tragus, and above the fleshy **lobule**, is another elevation (the **antitragus**). A smaller curved rim, parallel and anterior to the helix, is the **antihelix**.

Muscles

Numerous intrinsic and extrinsic muscles are associated with the auricle:

- The intrinsic muscles pass between the cartilaginous parts of the auricle and may change the shape of the auricle.
- The extrinsic muscles, the anterior, superior, and posterior auricular muscles, pass from the scalp or skull to the auricle and may also play a role in positioning of the auricle.

Both groups of muscles are innervated by the facial nerve [VII].

Innervation

Sensory innervation of the auricle is from many sources (Fig. 8.111):

- The outer more superficial surfaces of the auricle are supplied by the great auricular nerve (anterior and

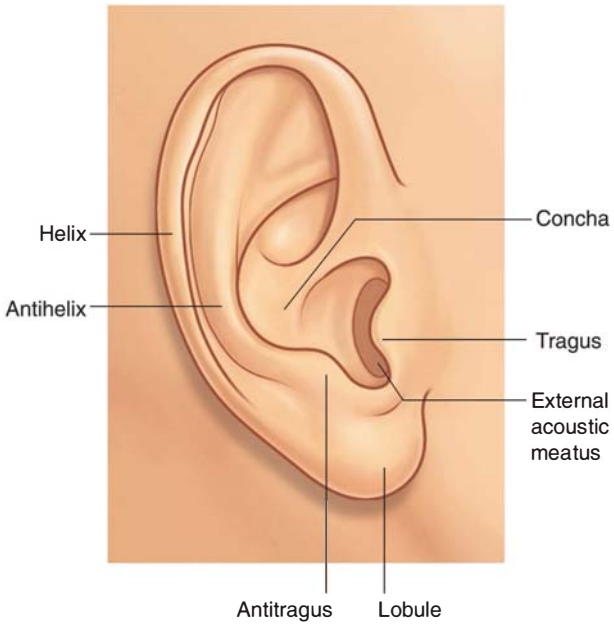


Fig. 8.110 Auricle.

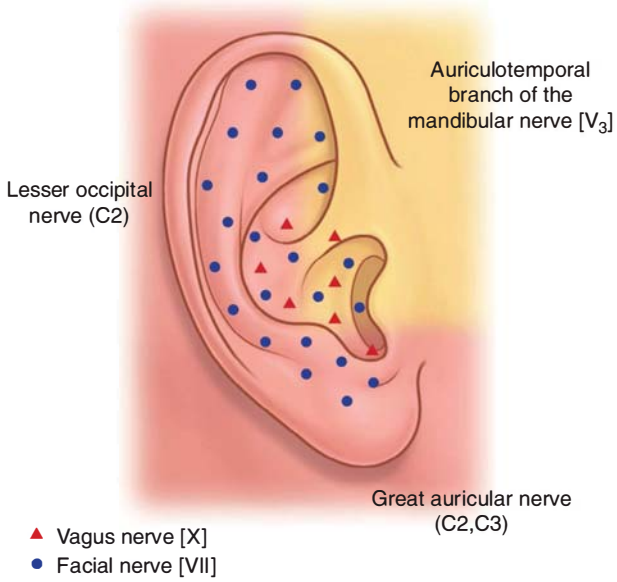


Fig. 8.111 Sensory innervation of the auricle.

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posterior inferior portions) and the lesser occipital nerve (posterosuperior portion) from the cervical plexus and the auriculotemporal branch of the mandibular nerve [V_3] (anterosuperior portion).

- The deeper parts of the auricle are supplied by the vagus nerve [X] (the auricular branch) and the facial nerve [VII] (which sends a branch to the auricular branch of the vagus nerve [X]).

Vessels

The arterial supply to the auricle is from numerous sources. The external carotid artery supplies the posterior auricular artery, the superficial temporal artery supplies anterior auricular branches, and the occipital artery supplies a branch.

Venous drainage is through vessels following the arteries.

Lymphatic drainage of the auricle passes anteriorly into parotid nodes and posteriorly into mastoid nodes, and possibly into the upper deep cervical nodes.

External acoustic meatus

The external acoustic meatus extends from the deepest part of the concha to the **tympanic membrane** (eardrum), a distance of approximately 1 inch (2.5 cm) (Fig. 8.112). Its walls consist of cartilage and bone. The lateral one-third is formed from cartilaginous extensions from some of the auricular cartilages and the medial two-thirds is a bony tunnel in the temporal bone.

Throughout its length the external acoustic meatus is covered with skin, some of which contains hair and modified sweat glands producing **cerumen** (earwax). Its diameter varies, being wider laterally and narrow medially.

The external acoustic meatus does not follow a straight course. From the external opening it passes upward in an anterior direction, then turns slightly posteriorly still passing upward, and finally, turns again in an anterior direction with a slight descent. For examination purposes, observation of the external acoustic meatus and tympanic membrane can be improved by pulling the ear superiorly, posteriorly, and slightly laterally.

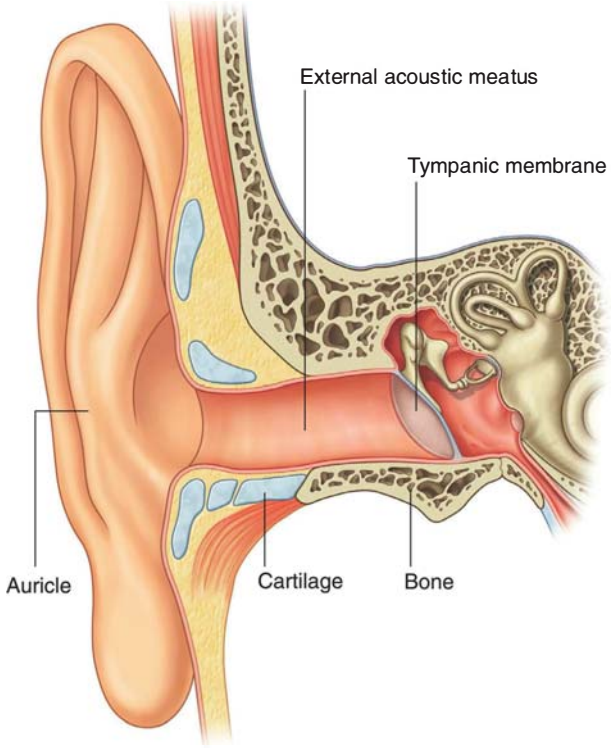


Fig. 8.112 External acoustic meatus.

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Innervation

Sensory innervation of the external acoustic meatus is from several of the cranial nerves. The major sensory input travels through branches of the auriculotemporal nerve, a branch of the mandibular nerve [V_3] (anterior and superior walls), and in the auricular branch of the vagus nerve [X] (posterior and inferior walls). A minor sensory input may also come from a branch of the facial nerve [VII] to the auricular branch of the vagus nerve [X].

Tympanic membrane

The tympanic membrane separates the external acoustic meatus from the middle ear (Figs. 8.113 and 8.114). It is at an angle, sloping medially from top to bottom and posteriorly to anteriorly. Its lateral surface therefore faces inferiorly and anteriorly. It consists of a connective tissue core lined with skin on the outside and mucous membrane on the inside.

Around the periphery of the tympanic membrane a **fibrocartilaginous ring** attaches it to the tympanic part of the temporal bone. At its center, a concavity is produced

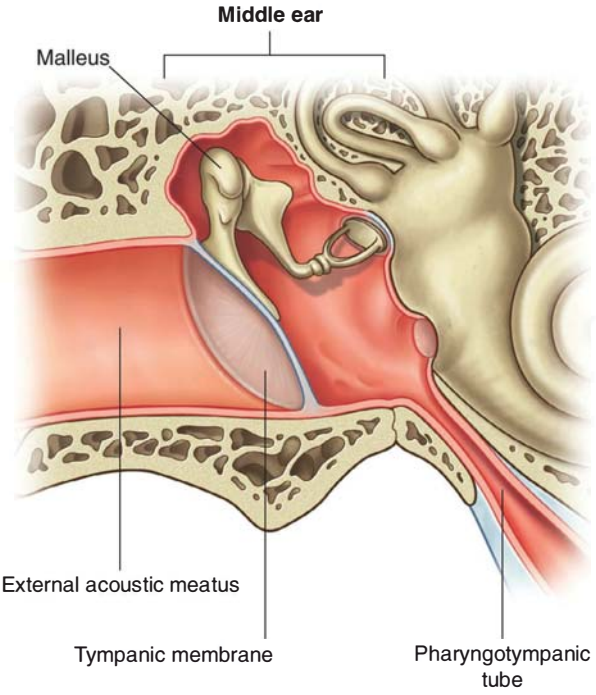


Fig. 8.113 Middle ear.

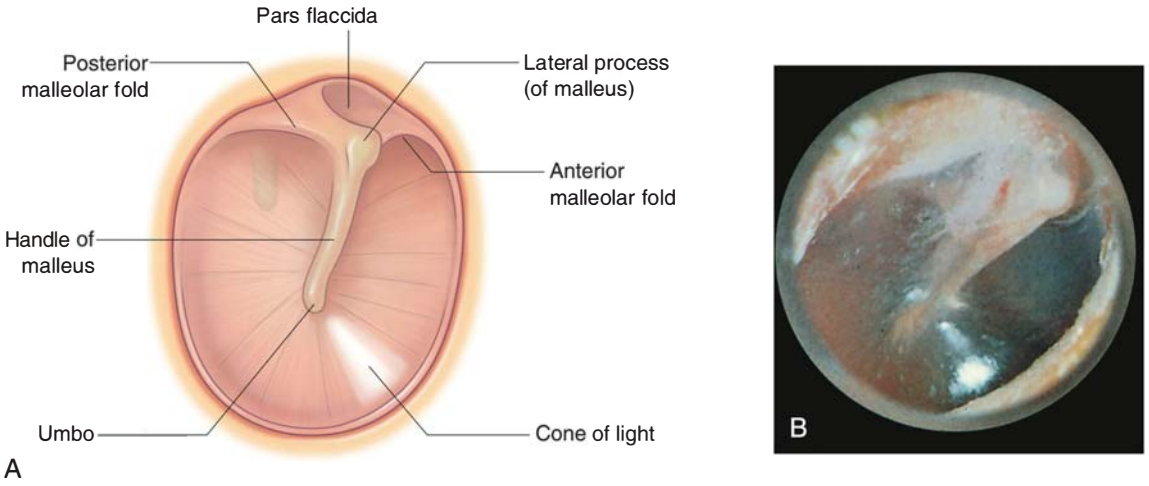


Fig. 8.114 Tympanic membrane (right ear). **A.** Diagram. **B.** Otoscopic view.

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by the attachment on its internal surface of the lower end of the **handle of the malleus**, part of the malleus bone in the middle ear. This point of attachment is the **umbo of the tympanic membrane**.

Anteroinferior to the umbo of the tympanic membrane a bright reflection of light, referred to as the cone of light, is usually visible when examining the tympanic membrane with an otoscope.

Superior to the umbo in an anterior direction is the attachment of the rest of the handle of the malleus (Fig. 8.114). At the most superior extent of this line of attachment a small bulge in the membrane marks the position of the **lateral process** of the malleus as it projects against the internal surface of the tympanic membrane. Extending away from this elevation, on the internal surface of the membrane, are the **anterior** and **posterior malleolar folds**. Superior to these folds the tympanic membrane is thin and slack (the **pars flaccida**), whereas the rest of the membrane is thick and taut (the **pars tensa**).

Innervation

Innervation of the external and internal surfaces of the tympanic membrane is by several cranial nerves:

- Sensory innervation of the skin on the outer surface of the tympanic membrane is primarily by the auriculotemporal nerve, a branch of the mandibular nerve [V_3] with additional participation of the auricular branch of the vagus nerve [X], a small contribution by a branch of the facial nerve [VII] to the auricular branch of the vagus nerve [X], and possibly a contribution from the glossopharyngeal nerve [IX].
- Sensory innervation of the mucous membrane on the inner surface of the tympanic membrane is carried entirely by the glossopharyngeal [IX] nerve.

In the clinic

Examination of the ear

The ear comprises three components—the external, middle, and internal ear.

Clinical examination is carried out to assess hearing and balance. Further examination involves use of an otoscope or other imaging techniques.

External ear

The external ear is easily examined. The external acoustic meatus and the tympanic membrane require otoscopic examination (Fig. 8.114B). An otoscope is a device through which light can be shone and the image magnified to inspect the external acoustic meatus and the tympanic membrane.

The examination begins by grasping the posterosuperior aspect of the ear and gently retracting it to straighten the external auditory meatus. The normal tympanic membrane is relatively translucent and has a gray–reddish tinge. The handle of the malleus is visible near the center of the membrane. In the 5 o'clock position a cone of light is always demonstrated.

Middle and inner ears

The middle ear is investigated by CT and MRI to visualize the malleus, incus, and stapes. The relationship of these bones to the middle ear cavity is determined and any masses identified.

The inner ear is also assessed by CT and MRI.

In the clinic

Swimmer's ear

Swimmer's ear, often called otitis externa, is a painful condition resulting from an infection in the external acoustic meatus. It frequently occurs in swimmers.

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In the clinic

Surfer's ear

Surfer's ear, which is prevalent among individuals who surf or swim in cold water, results from the development of a "bony lump" in the external acoustic meatus. Growth of the lump eventually constricts the meatus and reduces hearing in the affected ear.

In the clinic

Tympanic membrane perforation

Although perforation of the tympanic membrane (eardrum) has many causes, trauma and infection are still the commonest causes today.

Most ruptures of the tympanic membrane tend to heal spontaneously, but surgical intervention may be necessary if the rupture is large.

From time to time it is necessary to enter the middle ear through the tympanic membrane. As the chorda tympani runs in the upper one-third of the tympanic membrane, incisions are always below this level. The richer blood supply to the posterior aspect of the tympanic membrane determines the standard surgical approach in the posteroinferior aspect.

Otitis media (infection of the middle ear) is common and can lead to perforation of the tympanic membrane. The infection can usually be treated with antibiotics. If the infection persists, the chronic inflammatory change may damage the ossicular chain and other structures within the middle ear to produce deafness.

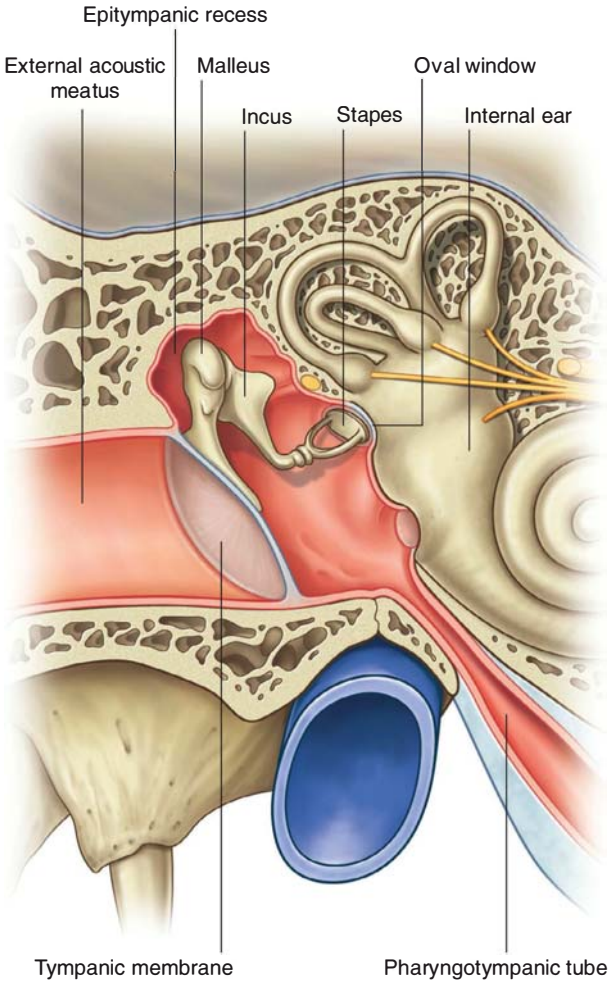


Fig. 8.115 Parts of the middle ear.

Middle ear

The middle ear is an air-filled, mucous membrane–lined space in the temporal bone between the tympanic membrane laterally and the lateral wall of the internal ear medially. It is described as consisting of two parts (Fig. 8.115):

- the **tympanic cavity** immediately adjacent to the tympanic membrane, and
- the **epitympanic recess** superiorly.

The middle ear communicates with the mastoid area posteriorly and the nasopharynx (via the pharyngotympanic tube) anteriorly. Its basic function is to transmit vibrations of the tympanic membrane across the cavity of the middle ear to the internal ear. It accomplishes this through three interconnected but movable bones that bridge the space between the tympanic membrane and the internal ear. These bones are the malleus (connected to the tympanic membrane), the incus (connected to the malleus by a synovial joint), and the stapes (connected to the incus by a synovial joint, and attached to the lateral wall of the internal ear at the oval window).

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Boundaries

The middle ear has a roof and a floor, and anterior, posterior, medial, and lateral walls (Fig. 8.116).

Tegmental wall

The tegmental wall (roof) of the middle ear consists of a thin layer of bone, which separates the middle ear from the middle cranial fossa. This layer of bone is the tegmen tympani on the anterior surface of the petrous part of the temporal bone.

Jugular wall

The jugular wall (floor) of the middle ear consists of a thin layer of bone that separates it from the internal jugular vein. Occasionally, the floor is thickened by the presence of mastoid air cells.

Near the medial border of the floor is a small aperture, through which the tympanic branch from the glossopharyngeal nerve [IX] enters the middle ear.

Membranous wall

The membranous (lateral) wall of the middle ear consists almost entirely of the tympanic membrane, but because the tympanic membrane does not extend superiorly into the epitympanic recess, the upper part of the membranous wall of the middle ear is the bony lateral wall of the epitympanic recess.

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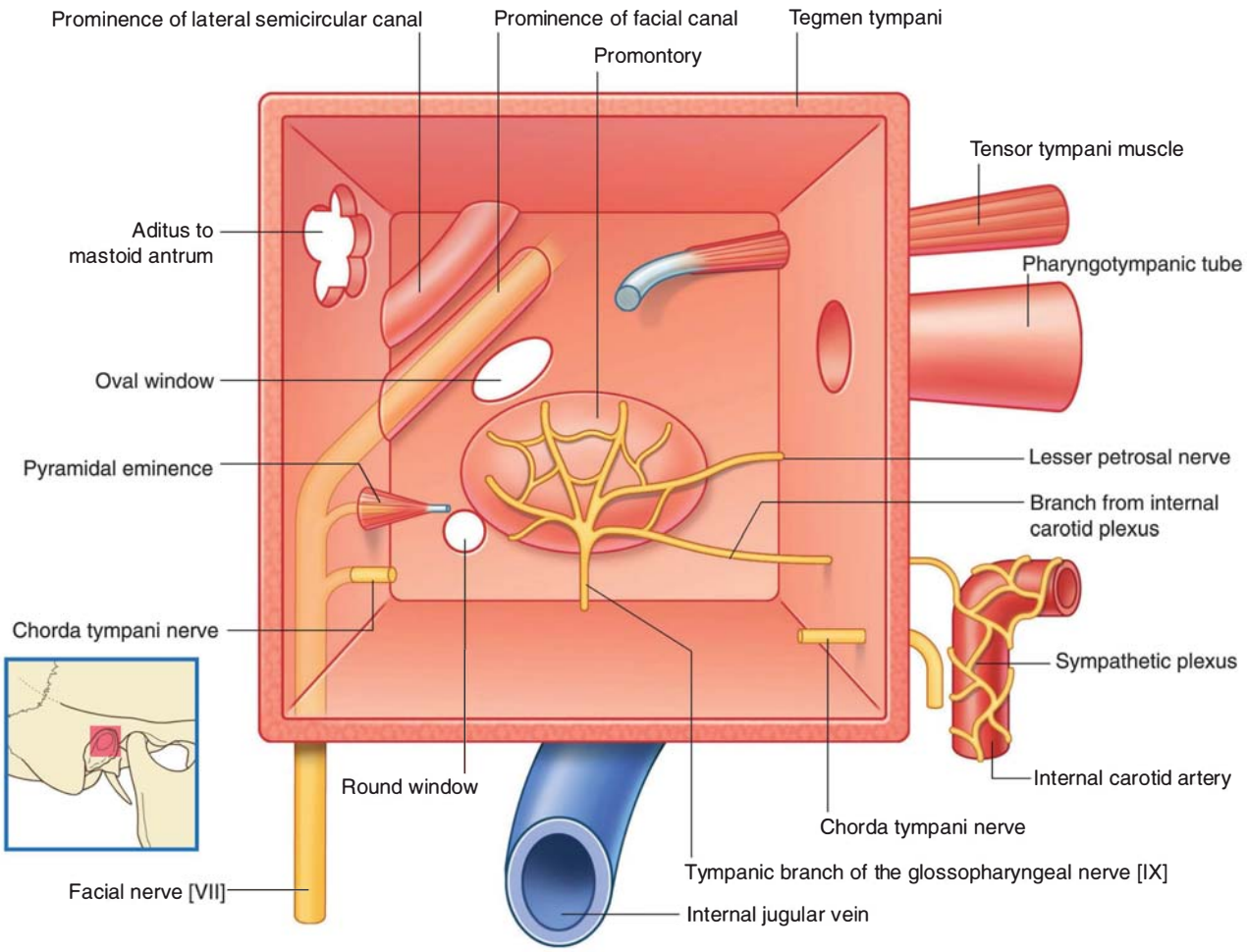


Fig. 8.116 Boundaries of the right middle ear.

Mastoid wall

The mastoid (posterior) wall of the middle ear is only partially complete. The lower part of this wall consists of a bony partition between the tympanic cavity and mastoid air cells. Superiorly, the epitympanic recess is continuous with the **aditus to the mastoid antrum** (Figs. 8.116 and 8.117).

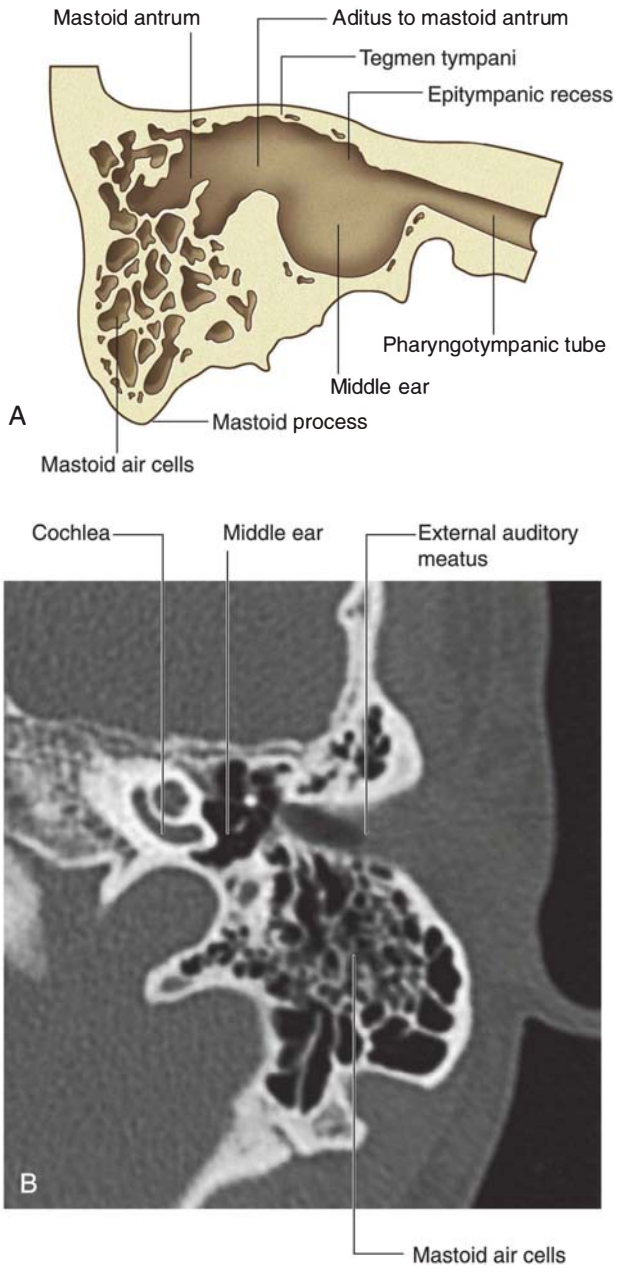


Fig. 8.117 Mastoid antrum and surrounding bone. **A.** Diagram. **B.** High-resolution CT scan of left ear (petrous temporal bone).

Associated with the mastoid wall are:

- the pyramidal eminence, a small elevation through which the tendon of the stapedius muscle enters the middle ear; and
- the opening through which the chorda tympani nerve, a branch of the facial nerve [VII], enters the middle ear.

Anterior wall

The anterior wall of the middle ear is only partially complete. The lower part consists of a thin layer of bone that separates the tympanic cavity from the internal carotid artery. Superiorly, the wall is deficient because of the presence of:

- a large opening for the entrance of the pharyngotympanic tube into the middle ear, and
- a smaller opening for the canal containing the tensor tympani muscle.

The foramen for the exit of the chorda tympani nerve from the middle ear is also associated with this wall (Fig. 8.116).

Labyrinthine wall

The labyrinthine (medial) wall of the middle ear is also the lateral wall of the internal ear. A prominent structure on this wall is a rounded bulge (the **promontory**) produced by the basal coil of the **cochlea**, which is an internal ear structure involved with hearing (Fig. 8.116).

Associated with the mucous membrane covering the promontory is a plexus of nerves (the **tympanic plexus**), which consists primarily of contributions from the tympanic branch of the glossopharyngeal nerve [IX] and branches from the internal carotid plexus. It supplies the mucous membrane of the middle ear, the mastoid area, and the pharyngotympanic tube.

Additionally, a branch of the tympanic plexus (the lesser petrosal nerve) leaves the promontory and the middle ear, travels across the anterior surface of the petrous part of the temporal bone, and leaves the middle cranial fossa through the foramen ovale to enter the otic ganglion. Other structures associated with the labyrinthine wall are two openings, the oval and round windows, and two prominent elevations (Fig. 8.116):

- The **oval window** is posterosuperior to the promontory, is the point of attachment for the **base of the stapes** (**footplate**), and ends the chain of bones that transfer vibrations initiated by the tympanic membrane to the cochlea of the internal ear.

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- The **round window** is posteroinferior to the promontory.
- Posterior and superior to the oval window on the medial wall is the **prominence of the facial canal**, which is a ridge of bone produced by the facial nerve [VII] in its canal as it passes through the temporal bone.
- Just above and posterior to the prominence of the facial canal is a broader ridge of bone (**prominence of the lateral semicircular canal**) produced by the lateral semicircular canal, which is a structure involved in detecting motion.

Mastoid area

Posterior to the epitympanic recess of the middle ear is the aditus to the mastoid antrum, which is the opening to the mastoid antrum (Fig. 8.117).

The **mastoid antrum** is a cavity continuous with collections of air-filled spaces (the **mastoid cells**), throughout the mastoid part of the temporal bone, including the mastoid process. The mastoid antrum is separated from the middle cranial fossa above by only the thin tegmen tympani.

The mucous membrane lining the mastoid air cells is continuous with the mucous membrane throughout the middle ear. Therefore infections in the middle ear can easily spread into the mastoid area.

In the clinic

Mastoiditis

Infection within the mastoid antrum and mastoid cells is usually secondary to infection in the middle ear. The mastoid cells provide an excellent culture medium for infection. Infection of the bone (osteomyelitis) may also develop, spreading into the middle cranial fossa.

Drainage of the pus within the mastoid air cells is necessary and there are numerous approaches for doing this. When undertaking this type of surgery, it is extremely important that care is taken not to damage the mastoid wall of the middle ear to prevent injury to the facial nerve [VII]. Any breach of the inner table of the cranial vault may allow bacteria to enter the cranial cavity and meningitis will ensue.

Pharyngotympanic tube

The pharyngotympanic tube connects the middle ear with the nasopharynx (Fig. 8.118) and equalizes pressure on both sides of the tympanic membrane. Its opening in the middle ear is on the anterior wall, and from here it extends forward, medially, and downward to enter the nasopharynx just posterior to the inferior meatus of the nasal cavity. It consists of:

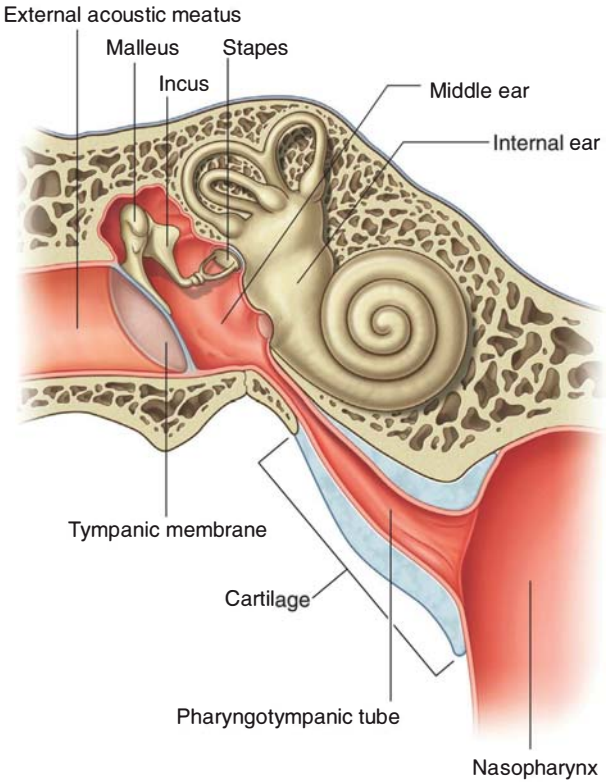


Fig. 8.118 Pharyngotympanic tube.

- a **bony part** (the one-third nearest the middle ear); and
- a **cartilaginous part** (the remaining two-thirds).

The opening of the bony part is clearly visible on the inferior surface of the skull at the junction of the squamous and petrous parts of the temporal bone immediately posterior to the foramen ovale and foramen spinosum.

Vessels

The arterial supply to the pharyngotympanic tube is from several sources. Branches arise from the **ascending pharyngeal artery** (a branch of the external carotid artery) and from two branches of the maxillary artery (the middle meningeal artery and the artery of the pterygoid canal).

Venous drainage of the pharyngotympanic tube is to the pterygoid plexus of veins in the infratemporal fossa.

Innervation

Innervation of the mucous membrane lining the pharyngotympanic tube is primarily from the tympanic plexus because it is continuous with the mucous membrane lining the tympanic cavity, the internal surface of the tympanic membrane, and the mastoid antrum and mastoid cells. This plexus receives its major contribution from the tympanic nerve, a branch of the glossopharyngeal nerve [IX].

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Auditory ossicles

The bones of the middle ear consist of the malleus, incus, and stapes. They form an osseous chain across the middle ear from the tympanic membrane to the oval window of the internal ear (Fig. 8.119).

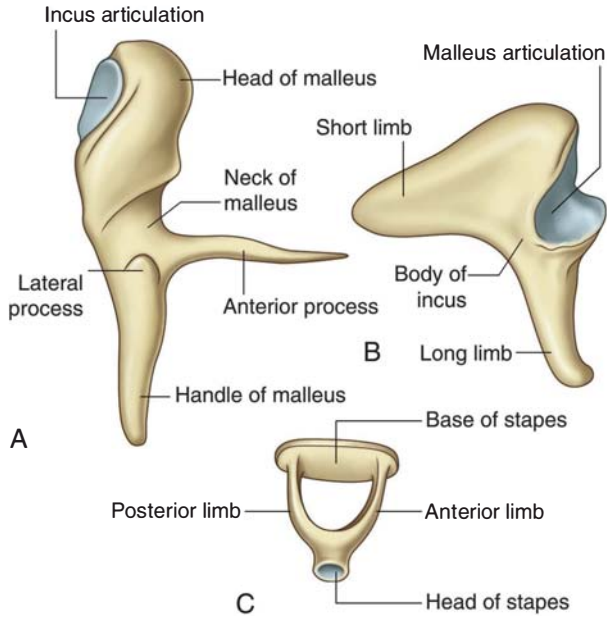


Fig. 8.119 Auditory ossicles. **A.** Malleus. **B.** Incus. **C.** Stapes.

Muscles associated with the auditory ossicles modulate movement during the transmission of vibrations.

Malleus

The malleus is the largest of the auditory ossicles and is attached to the tympanic membrane. Identifiable parts include the **head of the malleus**, **neck of the malleus**, **anterior and lateral processes**, and **handle of the malleus** (Fig. 8.119). The head of the malleus is the rounded upper part of the malleus in the epitympanic recess. Its posterior surface articulates with the incus.

Inferior to the head of the malleus is the constricted neck of the malleus, and below this are the anterior and lateral processes:

- The anterior process is attached to the anterior wall of the middle ear by a ligament.
- The lateral process is attached to the anterior and posterior malleolar folds of the tympanic membrane.

The downward extension of the malleus, below the anterior and lateral processes, is the handle of the malleus, which is attached to the tympanic membrane.

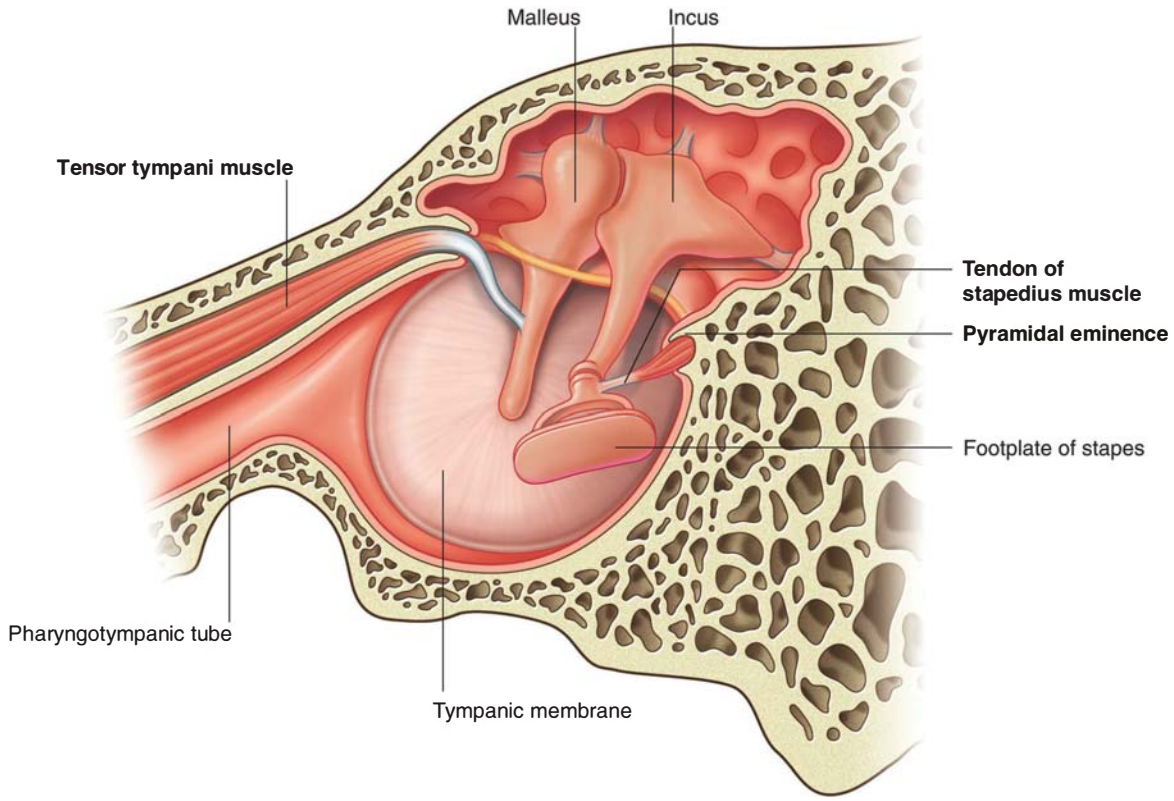


Fig. 8.120 Muscles associated with the auditory ossicles (right ear).

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Incus

The second bone in the series of auditory ossicles is the incus. It consists of the **body of the incus** and **long** and **short limbs** (Fig. 8.119):

- The enlarged body of the incus articulates with the head of the malleus and is in the epitympanic recess.
- The long limb extends downward from the body, paralleling the handle of the malleus, and ends by bending medially to articulate with the stapes.
- The short limb extends posteriorly and is attached by a ligament to the upper posterior wall of the middle ear.

Stapes

The stapes is the most medial bone in the osseous chain and is attached to the oval window. It consists of the **head of the stapes**, **anterior** and **posterior limbs**, and the **base of the stapes** (Fig. 8.119):

- The head of the stapes is directed laterally and articulates with the long process of the incus.
- The two limbs separate from each other and attach to the oval base.
- The base of the stapes fits into the oval window on the labyrinthine wall of the middle ear.

Muscles associated with the ossicles

Two muscles are associated with the bony ossicles of the middle ear—the tensor tympani and stapedius (Fig. 8.120 and Table 8.10).

Tensor tympani

The tensor tympani muscle lies in a bony canal above the pharyngotympanic tube. It originates from the cartilaginous part of the pharyngotympanic tube, the greater wing of the sphenoid, and its own bony canal, and passes through its canal in a posterior direction, ending in a rounded tendon that inserts into the upper part of the handle of the malleus.

Innervation of the tensor tympani is by a branch from the mandibular nerve [V_3].

Contraction of the tensor tympani pulls the handle of the malleus medially. This tenses the tympanic membrane, reducing the force of vibrations in response to loud noises.

Stapedius

The stapedius muscle is a very small muscle that originates from inside the pyramidal eminence, which is a small projection on the mastoid wall of the middle ear (Fig. 8.120). Its tendon emerges from the apex of the pyramidal eminence and passes forward to attach to the posterior surface of the neck of the stapes.

The stapedius is innervated by a branch from the facial nerve [VII].

Contraction of the stapedius muscle, usually in response to loud noises, pulls the stapes posteriorly and prevents excessive oscillation.

Vessels

Numerous arteries supply the structures in the middle ear:

- the two largest branches are the **tympanic branch** of the maxillary artery and the **mastoid branch** of the occipital or posterior auricular arteries;
- smaller branches come from the middle meningeal artery, the ascending pharyngeal artery, the artery of the pterygoid canal, and tympanic branches from the internal carotid artery.

Venous drainage of the middle ear returns to the pterygoid plexus of veins and the superior petrosal sinus.

Innervation

The tympanic plexus innervates the mucous membrane lining the walls and contents of the middle ear, which includes the mastoid area and the pharyngotympanic tube. It is formed by the **tympanic nerve**, a branch of the glossopharyngeal nerve [IX], and from branches of the internal carotid plexus. The tympanic plexus occurs in the

Table 8.10 Muscles of the middle ear

Muscle	Origin	Insertion	Innervation	Function
Tensor tympani	Cartilaginous part of pharyngotympanic tube, greater wing of sphenoid, its own bony canal	Upper part of handle of malleus	Branch from mandibular nerve [V_3]	Contraction pulls handle of malleus medially, tensing tympanic membrane
Stapedius	Attached to inside of pyramidal eminence	Neck of stapes	Branch of facial nerve [VII]	Contraction pulls stapes posteriorly, preventing excessive oscillation

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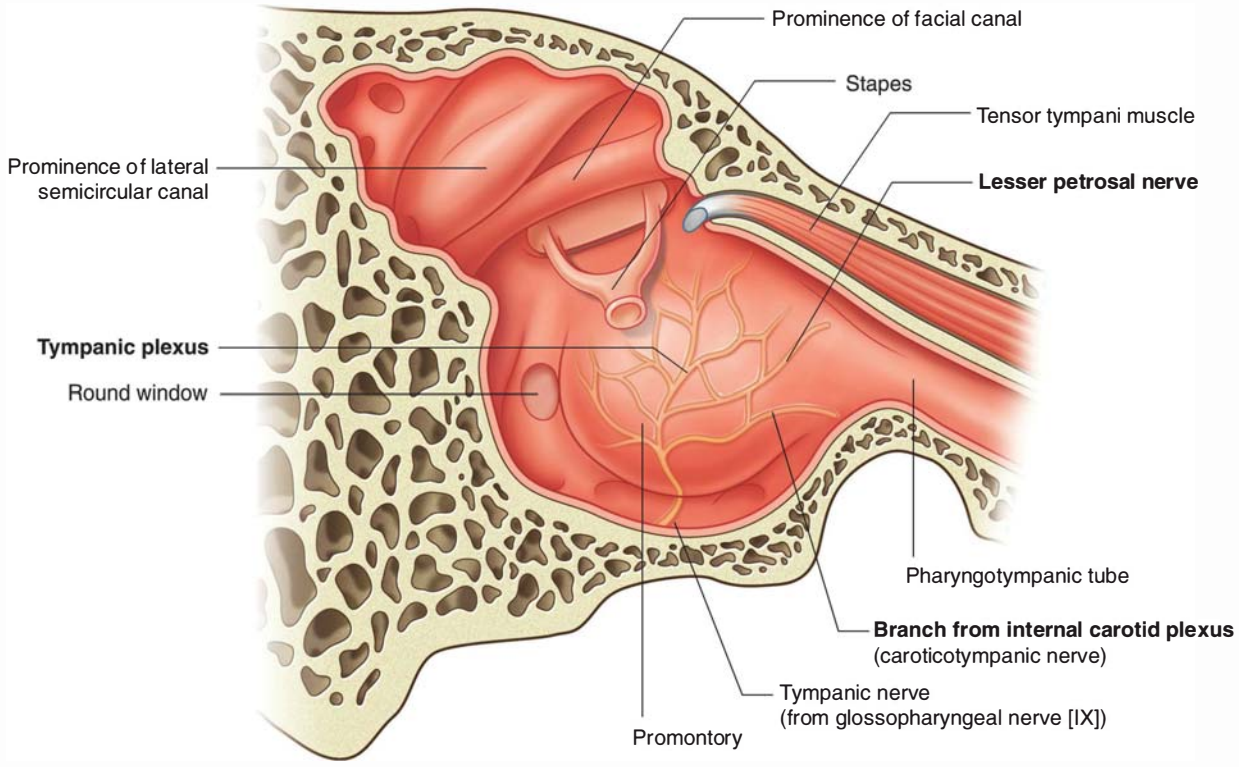


Fig. 8.121 Innervation of the middle ear.

mucous membrane covering the promontory, which is the rounded bulge on the labyrinthine wall of the middle ear (Fig. 8.121).

As the glossopharyngeal nerve [IX] exits the skull through the jugular foramen, it gives off the tympanic nerve. This branch reenters the skull through a small foramen and passes through the bone to the middle ear.

Once in the middle ear, the tympanic nerve forms the **tympanic plexus**, along with branches from the plexus of nerves surrounding the internal carotid artery (**caroticotympanic nerves**). Branches from the tympanic plexus supply the mucous membranes of the middle ear, including the pharyngotympanic tube and the mastoid area.

The tympanic plexus also gives off a major branch (the lesser petrosal nerve), which supplies preganglionic parasympathetic fibers to the otic ganglion (Fig. 8.121).

The lesser petrosal nerve leaves the area of the promontory, exits the middle ear, travels through the petrous part of the temporal bone, and exits onto the anterior surface of the petrous part of the temporal bone through a hiatus just below the hiatus for the greater petrosal nerve (Fig. 8.122). It continues diagonally across the anterior surface of the temporal bone before exiting the middle cranial fossa through the foramen ovale. Once outside the skull it enters the otic ganglion.

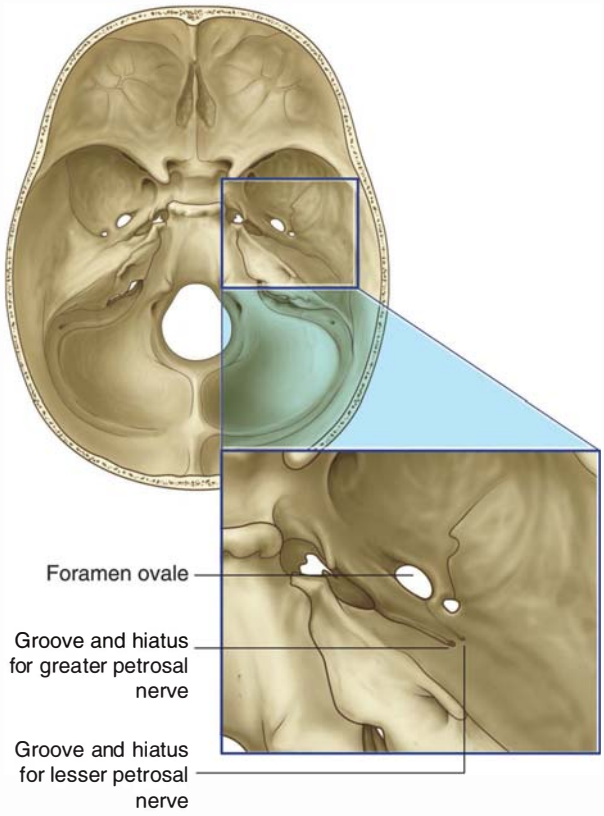


Fig. 8.122 Grooves and hiatuses for the greater and lesser petrosal nerves.

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Internal ear

The internal ear consists of a series of bony cavities (the **bony labyrinth**) and membranous ducts and sacs (the **membranous labyrinth**) within these cavities. All these structures are in the petrous part of the temporal bone between the middle ear laterally and the internal acoustic meatus medially (Figs. 8.123 and 8.124).

The bony labyrinth consists of the **vestibule**, three **semicircular canals**, and the **cochlea** (Fig. 8.124). These bony cavities are lined with periosteum and contain a clear fluid (the **perilymph**).

Suspended within the perilymph but not filling all spaces of the bony labyrinth is the membranous labyrinth, which consists of the **semicircular ducts**, the **cochlear duct**, and two sacs (the **utricle** and the **saccule**). These membranous spaces are filled with **endolymph**.

The structures in the internal ear convey information to the brain about balance and hearing:

- The cochlear duct is the organ of hearing.
- The semicircular ducts, utricle, and saccule are the organs of balance.

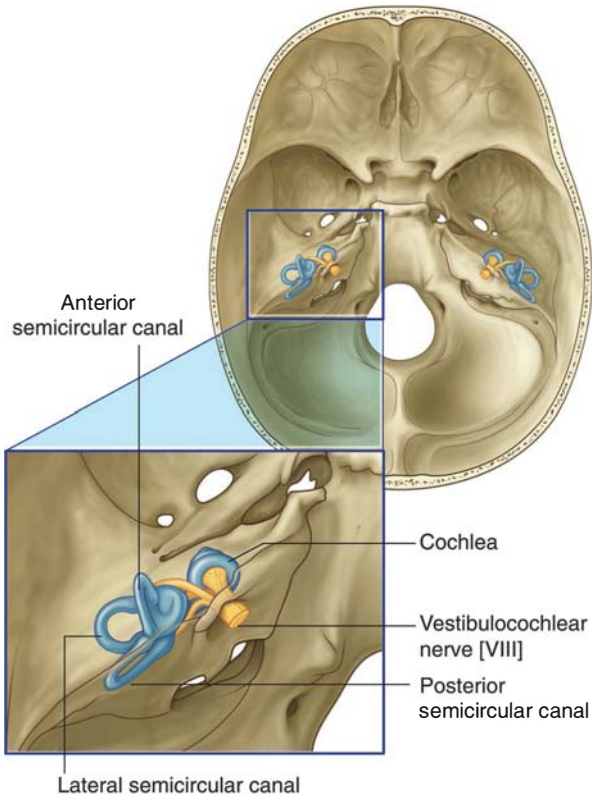


Fig. 8.123 Location of the internal ear in temporal bone.

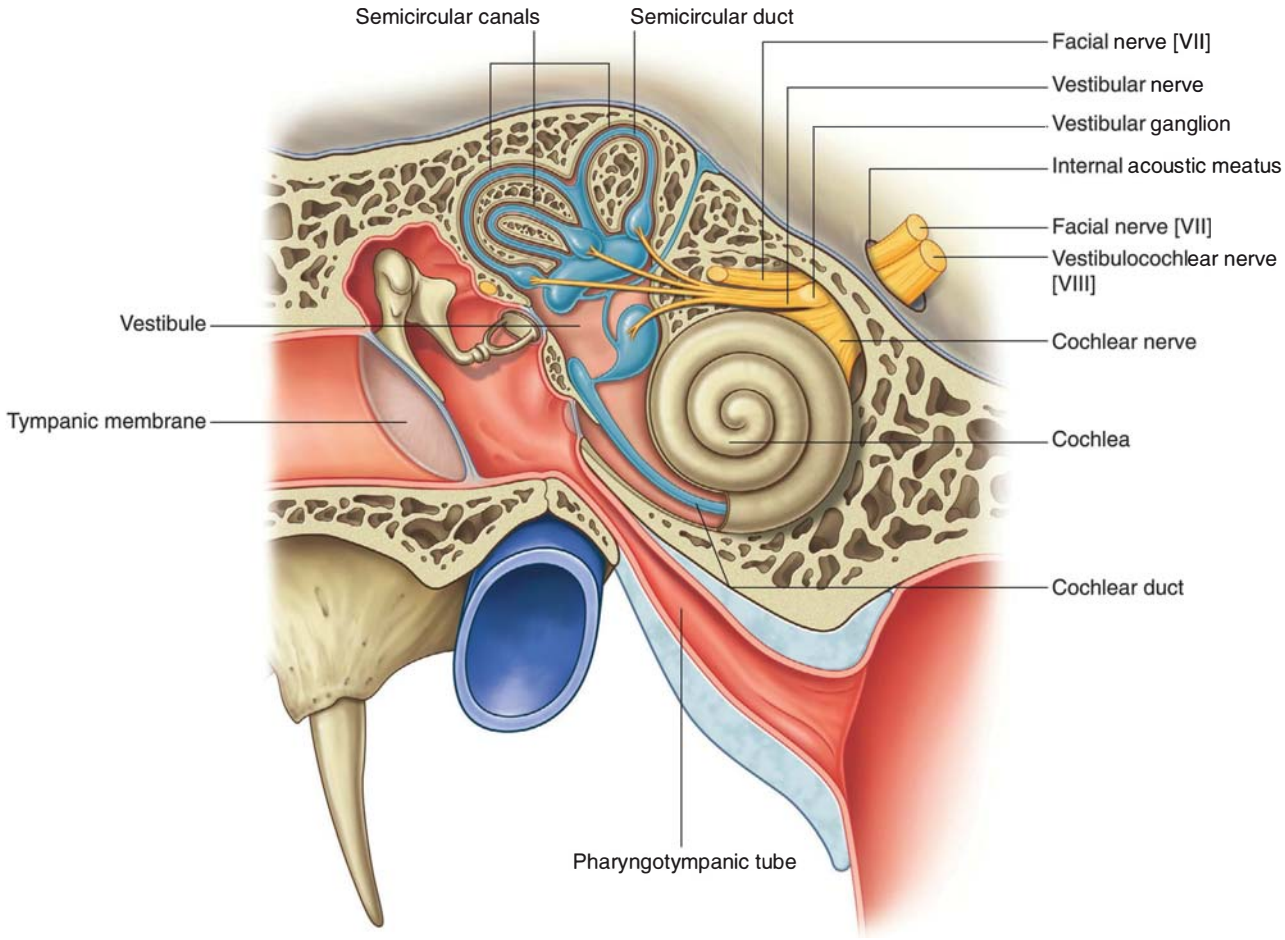


Fig. 8.124 Internal ear.

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The nerve responsible for these functions is the vestibulocochlear nerve [VIII], which divides into vestibular (balance) and cochlear (hearing) parts after entering the internal acoustic meatus (Fig. 8.124).

Bony labyrinth

The vestibule, which contains the oval window in its lateral wall, is the central part of the bony labyrinth (Fig. 8.125). It communicates anteriorly with the cochlea and posterosuperiorly with the semicircular canals.

A narrow canal (the **vestibular aqueduct**) leaves the vestibule, and passes through the temporal bone to open on the posterior surface of the petrous part of the temporal bone.

Semicircular canals

Projecting in a posterosuperior direction from the vestibule are the **anterior**, **posterior**, and **lateral semicircular canals** (Fig. 8.125). Each of these canals forms two-thirds of a circle connected at both ends to the vestibule and with one end dilated to form the **ampulla**. The canals are oriented so that each canal is at right angles to the other two.

Cochlea

Projecting in an anterior direction from the vestibule is the cochlea, which is a bony structure that twists on itself two and one-half to two and three-quarter times around a central column of bone (the **modiolus**). This arrangement produces a cone-shaped structure with a **base of the cochlea** that faces posteromedially and an apex that faces anterolaterally (Fig. 8.126). This positions the wide base of the modiolus near the internal acoustic meatus, where it is entered by branches of the cochlear part of the vestibulocochlear nerve [VIII].

Extending laterally throughout the length of the modiolus is a thin lamina of bone (the **lamina of the modiolus**,

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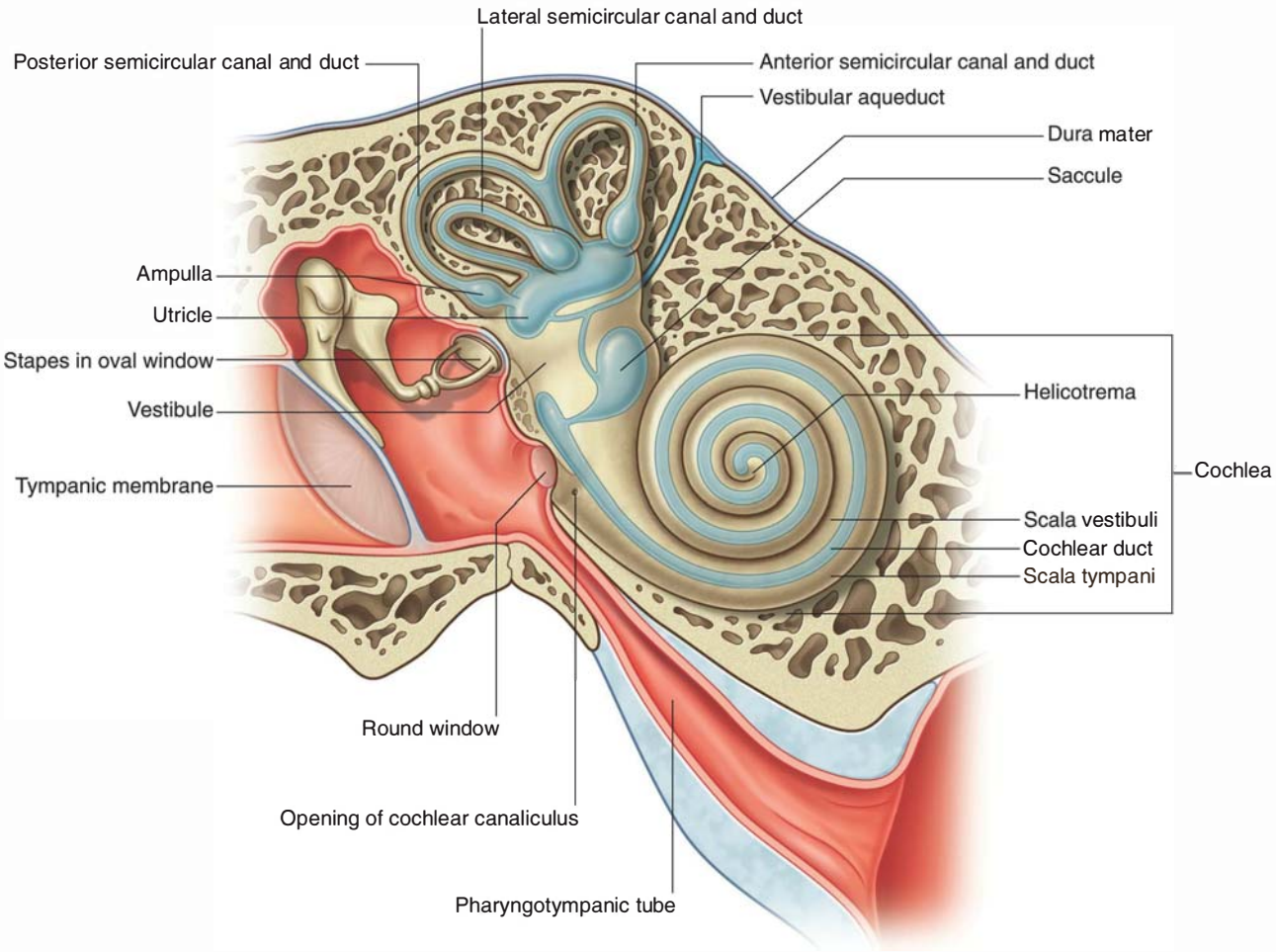


Fig. 8.125 Bony labyrinth.

or **spiral lamina**). Circling around the modiolus, and held in a central position by its attachment to the lamina of the modiolus, is the cochlear duct, which is a component of the membranous labyrinth.

Attached peripherally to the outer wall of the cochlea, the cochlear duct creates two canals (the **scala vestibuli** and the **scala tympani**), which extend throughout the cochlea and are continuous with each other at the apex through a narrow slit (the **helicotrema**):

- The scala vestibuli is continuous with the vestibule.
- The scala tympani is separated from the middle ear by the secondary tympanic membrane covering the round window (Fig. 8.127).

Finally, near the round window is a small channel (the **cochlear canaliculus**), which passes through the temporal bone and opens on its inferior surface into the posterior cranial fossa. This provides a connection between the perilymph-containing cochlea and the subarachnoid space (Fig. 8.127).

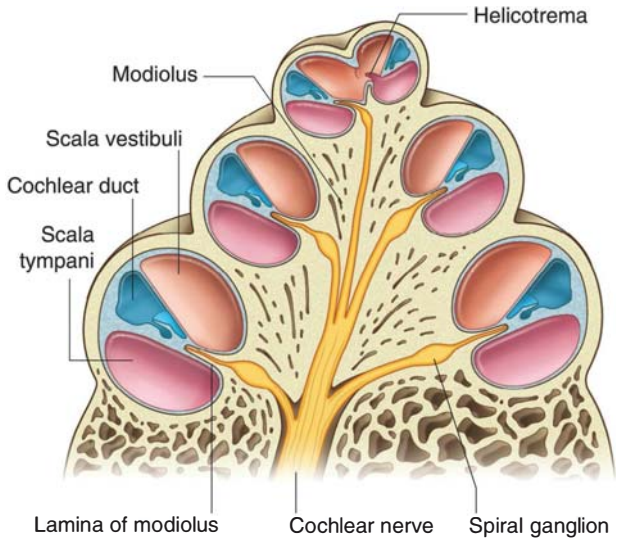


Fig. 8.126 Cochlea.

Membranous labyrinth

The membranous labyrinth is a continuous system of ducts and sacs within the bony labyrinth. It is filled with endolymph and separated from the periosteum that covers the walls of the bony labyrinth by perilymph.

Consisting of two sacs (the utricle and the saccule) and four ducts (the three semicircular ducts and the cochlear duct), the membranous labyrinth has unique functions related to balance and hearing:

- The utricle, saccule, and three semicircular ducts are part of the vestibular apparatus (i.e., organs of balance).
- The cochlear duct is the organ of hearing.

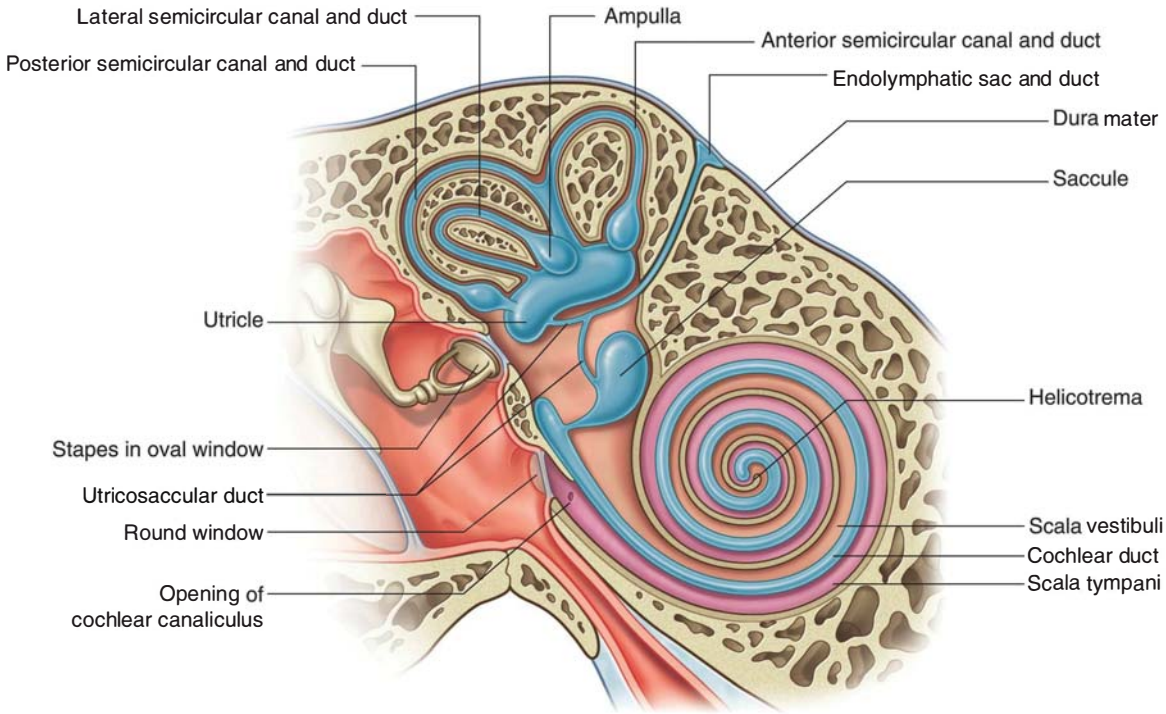


Fig. 8.127 Membranous labyrinth.

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The general organization of the parts of the membranous labyrinth (Fig. 8.127) places:

- the cochlear duct within the cochlea of the bony labyrinth, anteriorly,
- the three semicircular ducts within the three semicircular canals of the bony labyrinth, posteriorly, and
- the saccule and utricle within the vestibule of the bony labyrinth, in the middle.

Organs of balance

Five of the six components of the membranous labyrinth are concerned with balance. These are the two sacs (the utricle and the saccule) and three ducts (the anterior, posterior, and lateral semicircular ducts).

Utricle, saccule, and endolymphatic duct

The utricle is the larger of the two sacs. It is oval, elongated and irregular in shape and is in the posterosuperior part of the vestibule of the bony labyrinth.

The three semicircular ducts empty into the utricle. Each semicircular duct is similar in shape, including a dilated end forming the ampulla, to its complementary bony semicircular canal, only much smaller.

The saccule is a smaller, rounded sac lying in the anteroinferior part of the vestibule of the bony labyrinth (Fig. 8.127). The cochlear duct empties into it.

The utriculosaccular duct establishes continuity between all components of the membranous labyrinth and connects the utricle and saccule. Branching from this small duct is the **endolymphatic duct**, which enters the vestibular aqueduct (a channel through the temporal bone) to emerge onto the posterior surface of the petrous part of the temporal bone in the posterior cranial fossa. Here the endolymphatic duct expands into the **endolymphatic sac**, which is an extradural pouch that functions in resorption of endolymph.

Sensory receptors

Functionally, sensory receptors for balance are organized into unique structures that are located in each of the components of the vestibular apparatus. In the utricle and saccule the sense organ is the **macula of the utricle** and the **macula of the saccule**, respectively, and in the ampulla of each of the three semicircular ducts it is the **crista**.

The utricle responds to centrifugal and vertical acceleration, while the saccule responds to linear acceleration. In contrast, the receptors in the three semicircular ducts respond to movement in any direction.

Organ of hearing

Cochlear duct

The cochlear duct has a central position in the cochlea of the bony labyrinth dividing it into two canals (the scala vestibuli and the scala tympani). It is maintained in this position by being attached centrally to the lamina of the modiolus, which is a thin lamina of bone extending from the modiolus (the central bony core of the cochlea) and peripherally to the outer wall of the cochlea (Fig. 8.128).

Thus, the triangular-shaped cochlear duct has:

- an outer wall against the bony cochlea consisting of thickened, epithelial-lined periosteum (the **spiral ligament**),
- a roof (the **vestibular membrane**), which separates the endolymph in the cochlear duct from the perilymph in the scala vestibuli and consists of a membrane with a connective tissue core lined on either side with epithelium, and
- a floor, which separates the endolymph in the cochlear duct from the perilymph in the scala tympani and consists of the free edge of the lamina of the modiolus, and

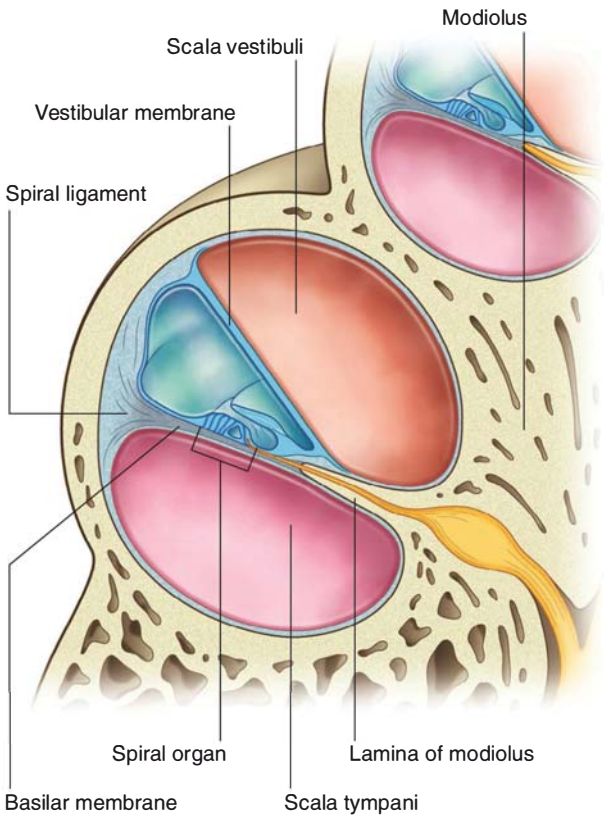


Fig. 8.128 Membranous labyrinth, cross section.

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a membrane (the **basilar membrane**) extending from this free edge of the lamina of the modiolus to an extension of the spiral ligament covering the outer wall of the cochlea.

The **spiral organ** is the organ of hearing, rests on the basilar membrane, and projects into the enclosed, endolymph-filled cochlear duct (Fig. 8.128).

Vessels

The arterial supply to the internal ear is divided between vessels supplying the bony labyrinth and the membranous labyrinth.

The bony labyrinth is supplied by the same arteries that supply the surrounding temporal bone—these include an anterior tympanic branch from the maxillary artery, a stylomastoid branch from the posterior auricular artery, and a petrosal branch from the middle meningeal artery.

The membranous labyrinth is supplied by the **labyrinthine artery**, which either arises from the anteroinferior cerebellar artery or is a direct branch of the basilar artery—whatever its origin, it enters the internal acoustic meatus with the facial [VII] and vestibulocochlear [VIII] nerves and eventually divides into:

- a **cochlear branch**, which passes through the modiolus and supplies the cochlear duct; and
- one or two **vestibular branches**, which supply the vestibular apparatus.

Venous drainage of the membranous labyrinth is through vestibular veins and cochlear veins, which follow the arteries. These come together to form a **labyrinthine vein**, which eventually empties into either the inferior petrosal sinus or the sigmoid sinus.

Innervation

The vestibulocochlear nerve [VIII] carries special afferent fibers for hearing (the cochlear component) and balance (the vestibular component). It enters the lateral surface of the brainstem, between the pons and medulla, after exiting the temporal bone through the internal acoustic meatus and crossing the posterior cranial fossa.

Inside the temporal bone, at the distal end of the internal acoustic meatus, the vestibulocochlear nerve divides to form:

- the **cochlear nerve**, and
- the **vestibular nerve**.

The vestibular nerve enlarges to form the **vestibular ganglion**, before dividing into **superior and inferior parts**, which distribute to the three semicircular ducts and the utricle and saccule (see Fig. 8.124).

The cochlear nerve enters the base of the cochlea and passes upward through the modiolus. The ganglion cells of the cochlear nerve are in the **spiral ganglion** at the base of the lamina of the modiolus as it winds around the modiolus (Fig. 8.126). Branches of the cochlear nerve pass through the lamina of the modiolus to innervate the receptors in the spiral organ.

Facial nerve [VII] in the temporal bone

The facial nerve [VII] is closely associated with the vestibulocochlear nerve [VIII] as it enters the internal acoustic meatus of the temporal bone. Traveling through the temporal bone, its path and several of its branches are directly related to the internal and middle ears.

The facial nerve [VII] enters the internal acoustic meatus in the petrous part of the temporal bone (Fig. 8.129A). The vestibulocochlear nerve and the labyrinthine artery accompany it.

At the distal end of the internal acoustic meatus, the facial nerve [VII] enters the facial canal and continues laterally between the internal and middle ears. At this point the facial nerve [VII] enlarges and bends posteriorly and laterally. The enlargement is the sensory **geniculate ganglion**. As the facial canal continues, the facial nerve [VII] turns sharply downward, and running in an almost vertical direction, it exits the skull through the stylomastoid foramen (Fig. 8.129A).

Branches

Greater petrosal nerve. At the geniculate ganglion, the facial nerve [VII] gives off the greater petrosal nerve (Fig. 8.129A). This is the first branch of the facial nerve [VII]. The greater petrosal nerve leaves the geniculate ganglion, travels anteromedially through the temporal bone, and emerges through the hiatus for the greater petrosal nerve on the anterior surface of the petrous part of the temporal bone (see Fig. 8.122). The greater petrosal nerve carries preganglionic parasympathetic fibers to the pterygopalatine ganglion.

Continuing beyond the bend, the position of the facial nerve [VII] is indicated on the medial wall of the middle ear by a bulge (see Fig. 8.121).

Nerve to stapedius and chorda tympani. Near the beginning of its vertical descent, the facial nerve [VII] gives off a small branch, the nerve to the stapedius (Fig. 8.129), which innervates the stapedius muscle, and just before it exits the skull the facial nerve [VII] gives off the chorda tympani nerve.

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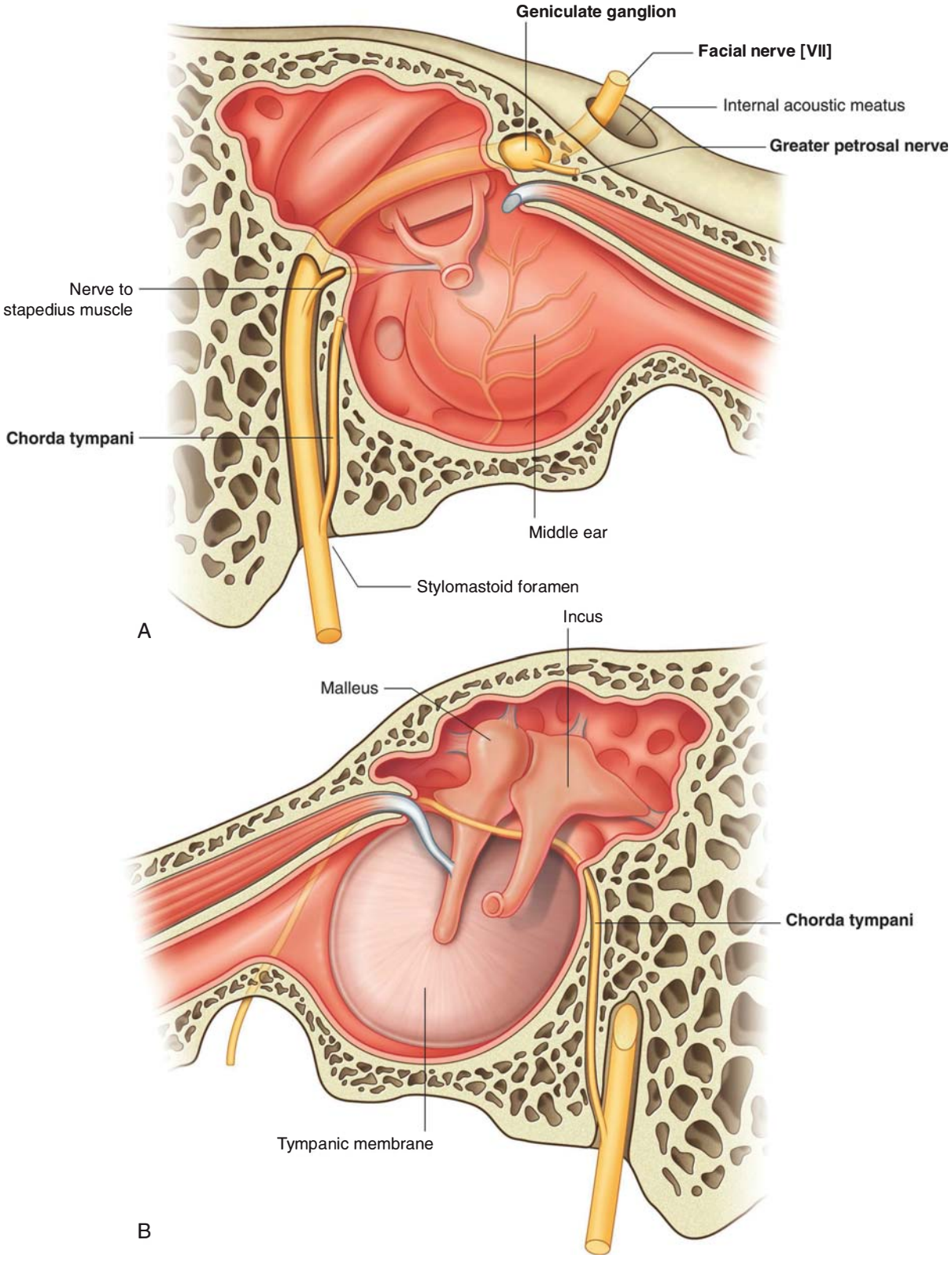


Fig. 8.129 **A.** Facial nerve in the temporal bone. **B.** Chorda tympani in the temporal bone.

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The chorda tympani does not immediately exit the temporal bone, but ascends to enter the middle ear through its posterior wall, passing near the upper aspect of the tympanic membrane between the malleus and incus (Fig. 8.129B). It then exits the middle ear through a canal leading to the **petrotympanic fissure** and exits the skull through this fissure to join the lingual nerve in the infratemporal fossa.

Transmission of sound

A sound wave enters the external acoustic meatus and strikes the tympanic membrane moving it medially (Fig. 8.130). As the handle of the malleus is attached to this membrane, it also moves medially. This moves the head of the malleus laterally. Because the heads of the malleus and incus articulate with each other, the head of the incus is also moved laterally. This pushes the long process of the incus medially. The long process articulates with the stapes, so its movement causes the stapes to move medially. In turn, because the base of the stapes is attached to the oval window, the oval window is also moved medially.

This action completes the transfer of a large-amplitude, low-force, airborne wave that vibrates the tympanic membrane into a small-amplitude, high-force vibration of the oval window, which generates a wave in the fluid-filled scala vestibuli of the cochlea.

The wave established in the perilymph of the scala vestibuli moves through the cochlea and causes an outward bulging of the secondary tympanic membrane covering the round window at the lower end of the scala tympani (Fig. 8.130). This causes the basilar membrane to vibrate, which in turn leads to stimulation of receptor cells in the spiral organ.

The receptor cells send impulses back to the brain through the cochlear part of the vestibulocochlear nerve [VIII] where they are interpreted as sound.

If the sounds are too loud, causing excessive movement of the tympanic membrane, contraction of the tensor tympani muscle (attached to the malleus) and/or the stapedius muscle (attached to the stapes) dampens the vibrations of the ossicles and decreases the force of the vibrations reaching the oval window.

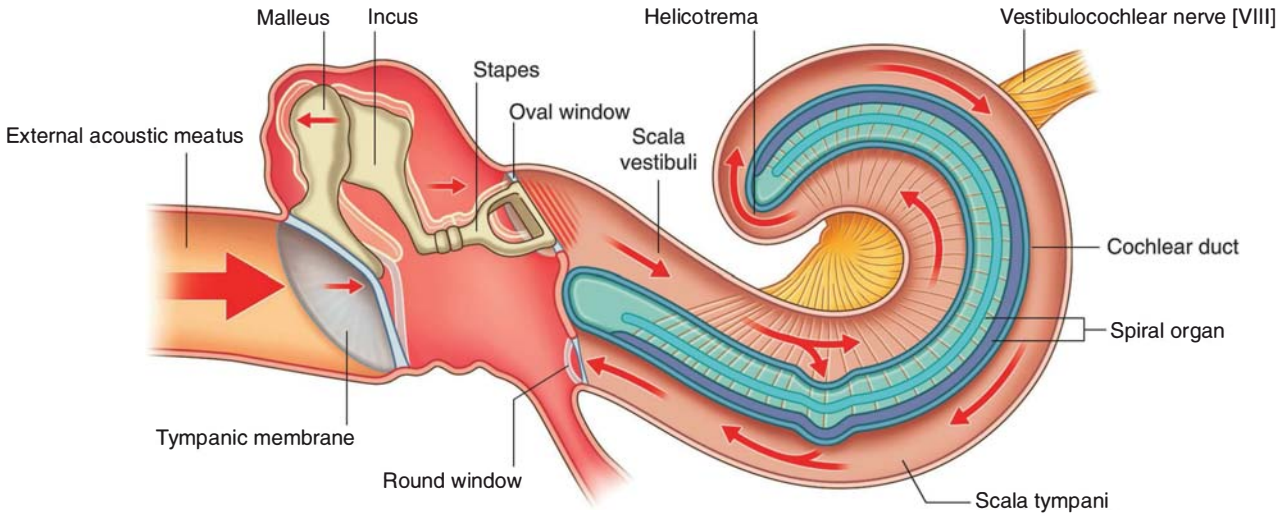


Fig. 8.130 Transmission of sound.

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TEMPORAL AND INFRATEMPORAL FOSSAE

The temporal and infratemporal fossae are interconnected spaces on the lateral side of the head (Fig. 8.131). Their boundaries are formed by bone and soft tissues.

The temporal fossa is superior to the infratemporal fossa, above the zygomatic arch, and communicates with the infratemporal fossa below through the gap between the zygomatic arch and the more medial surface of the skull.

The infratemporal fossa is a wedge-shaped space deep to the masseter muscle and the underlying ramus of the mandible. Structures that travel between the cranial cavity, neck, pterygopalatine fossa, floor of the oral cavity, floor of the orbit, temporal fossa, and superficial regions of the head pass through it.

Of the four muscles of mastication (masseter, temporalis, medial pterygoid, and lateral pterygoid) that move the lower jaw at the temporomandibular joint, one (masseter) is lateral to the infratemporal fossa, two (medial and lateral pterygoid) are in the infratemporal fossa, and one fills the temporal fossa.

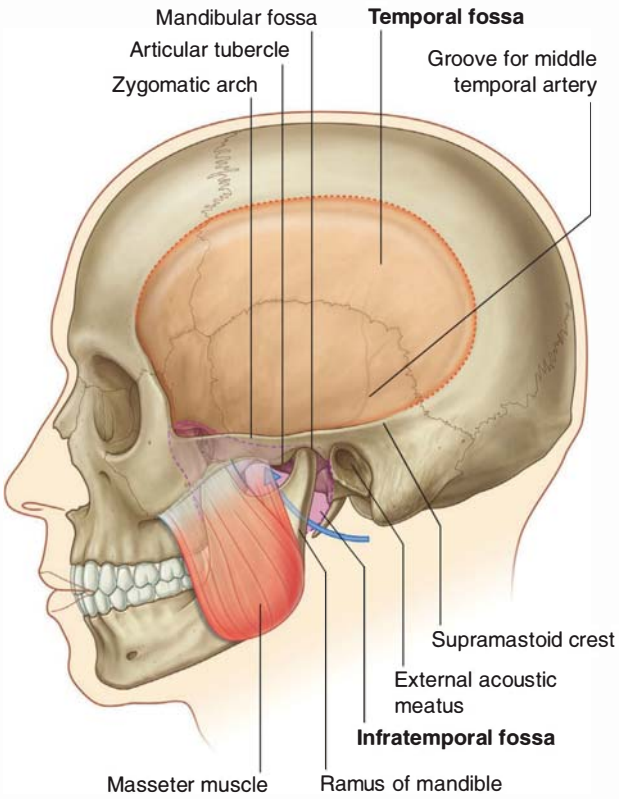


Fig. 8.131 Temporal and infratemporal fossae.

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